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# Engineering Mathematics – 1

## Notes

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### **Prepared by**

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**Note:**

These notes are prepared to build *problem-solving skill* in students. It has lot of worked examples keeping examination in mind. The surrounding exposition is there to make the mathematics behind each method understandable rather than to treat any topic exhaustively.

These are *not* replacement for a textbook or for lectures.

**Help improve these notes.** If you find a mistake — a typing error, a wrong step, a figure that is unclear, or anything that is hard to follow — please let me know. Suggestions for better explanations, examples or problems are equally welcome. Mention the unit, the page number and, where possible, the section, problem or equation number (for example, *Problem 2.5.3* or *equation (3.2.1)*), so the correction can be made quickly.

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# Unit 1 Prerequisites

This unit collects the background that Units 2, 3 and 4 take for granted: sets and number systems, functions and their graphs, limits, continuity and the derivative. Most of it will be familiar from school; it is gathered here so that every later result has its foundations stated in one place.

## 1.1 Sets and Set Operations

Everything that follows in this course is stated in the language of sets, so we begin there.

### Definition 1.1.1: Set

A *set* is a well defined collection of distinct objects. The objects are called the *elements* or *members* of the set. If  $x$  is an element of the set  $A$  we write  $x \in A$ ; if it is not, we write  $x \notin A$ .

A set may be described in two ways.

- **Roster (listing) form:** the elements are listed between braces, as in  $A = \{1, 2, 3, 4\}$ .
- **Set-builder form:** a defining property is given, as in  $A = \{x \in \mathbb{N} \mid x < 5\}$ , read “the set of all natural numbers  $x$  such that  $x$  is less than 5”. (The natural numbers  $\mathbb{N} = \{1, 2, 3, \dots\}$  are recalled formally in [Section 1.2](#).)

Both describe the same set. The vertical bar is read “such that” and is sometimes written as a colon.

### Note:

“Well defined” means that for any object it can be decided, without ambiguity, whether or not it belongs to the set. The collection of all real numbers greater than 2 is a set; the collection of all large real numbers is not.

### 1.1.1 Basic relations between sets

#### Definition 1.1.2: Subset, equality, empty set

Let  $A$  and  $B$  be sets.

- $A$  is a *subset* of  $B$ , written  $A \subseteq B$ , if every element of  $A$  is also an element of  $B$ .
- $A$  is a *proper subset* of  $B$ , written  $A \subset B$ , if  $A \subseteq B$  and  $A \neq B$ .
- $A$  and  $B$  are *equal*, written  $A = B$ , if  $A \subseteq B$  and  $B \subseteq A$ .
- The *empty set*  $\emptyset = \{\}$  contains no elements, and  $\emptyset \subseteq A$  for every set  $A$ .

### 1.1.2 Operations on sets

Throughout,  $U$  denotes a fixed *universal set* containing all the objects under discussion.

**Definition 1.1.3: Union, intersection, difference, complement**

For sets  $A$  and  $B$ :

$$\begin{aligned} A \cup B &= \{x \mid x \in A \text{ or } x \in B\} && (\text{union}), \\ A \cap B &= \{x \mid x \in A \text{ and } x \in B\} && (\text{intersection}), \\ A \setminus B &= \{x \mid x \in A \text{ and } x \notin B\} && (\text{difference}), \\ A^c &= \{x \in U \mid x \notin A\} && (\text{complement}). \end{aligned}$$

Two sets are called *disjoint* if  $A \cap B = \emptyset$ .

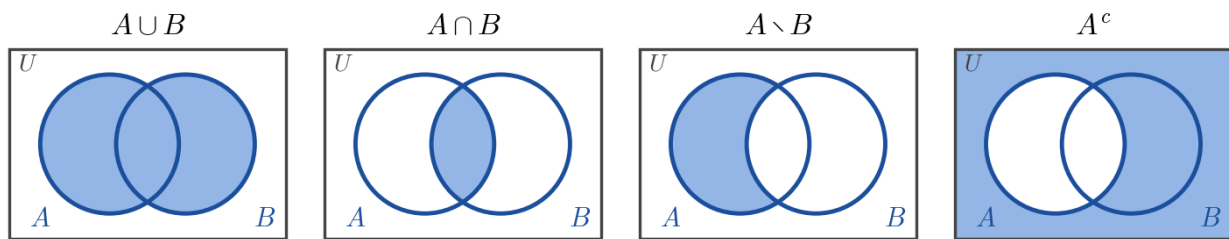


Figure 1.1.1: The four basic operations, shaded. The rectangle is the universal set  $U$ .

**Example 1.1.4: Computing with sets**

Let  $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ ,  $A = \{1, 2, 3, 4\}$  and  $B = \{3, 4, 5, 6\}$ . Then

$$\begin{aligned} A \cup B &= \{1, 2, 3, 4, 5, 6\}, & A \cap B &= \{3, 4\}, \\ A \setminus B &= \{1, 2\}, & B \setminus A &= \{5, 6\}, \\ A^c &= \{5, 6, 7, 8\}, & (A \cup B)^c &= \{7, 8\}. \end{aligned}$$

Note that  $A \setminus B \neq B \setminus A$ : the difference is not commutative, although union and intersection are.

**1.1.3 Properties of the operations**

For all sets  $A$ ,  $B$ ,  $C$  contained in  $U$ :

|               |                                                   |                                                   |
|---------------|---------------------------------------------------|---------------------------------------------------|
| Commutative:  | $A \cup B = B \cup A,$                            | $A \cap B = B \cap A;$                            |
| Associative:  | $A \cup (B \cup C) = (A \cup B) \cup C,$          | $A \cap (B \cap C) = (A \cap B) \cap C;$          |
| Distributive: | $A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$ | $A \cup (B \cap C) = (A \cup B) \cap (A \cup C);$ |
| De Morgan:    | $(A \cup B)^c = A^c \cap B^c,$                    | $(A \cap B)^c = A^c \cup B^c.$                    |

**Note:**

De Morgan's laws are worth reading aloud: “not ( $A$  or  $B$ )” is the same as “(not  $A$ ) and (not  $B$ )”. They can be checked on [Example 1.1.4](#):  $(A \cup B)^c = \{7, 8\}$ , and  $A^c \cap B^c = \{5, 6, 7, 8\} \cap \{1, 2, 7, 8\} = \{7, 8\}$ .

### 1.1.4 The Cartesian product

#### Definition 1.1.5: Cartesian product

The *Cartesian product* of  $A$  and  $B$  is the set of ordered pairs

$$A \times B = \{ (a, b) \mid a \in A, b \in B \}.$$

The pair  $(a, b)$  is *ordered*:  $(a, b) = (c, d)$  precisely when  $a = c$  and  $b = d$ , so  $(1, 2) \neq (2, 1)$ .

#### Example 1.1.6: A small product

If  $A = \{1, 2\}$  and  $B = \{p, q\}$  then

$$A \times B = \{(1, p), (1, q), (2, p), (2, q)\},$$

which has  $2 \times 2 = 4$  elements. In general  $A \times B$  has  $|A| \cdot |B|$  elements when both are finite.

#### Note:

The Cartesian product is what makes coordinates possible: the plane is  $\mathbb{R} \times \mathbb{R}$ , written  $\mathbb{R}^2$ . It is also what makes functions possible, since a function will be described below by the pairs  $(x, f(x))$  it produces.

## 1.2 The Number Systems

Five sets of numbers occur constantly, and each is built from the one before it by supplying something the previous one lacked.

#### Definition 1.2.1: The standard number sets

|                                                                                |                     |
|--------------------------------------------------------------------------------|---------------------|
| $\mathbb{N} = \{1, 2, 3, 4, \dots\}$                                           | natural numbers,    |
| $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$                               | integers,           |
| $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$ | rational numbers,   |
| $\mathbb{R}$ = the set of all points of the number line                        | real numbers,       |
| $\mathbb{Q}' = \mathbb{R} \setminus \mathbb{Q}$                                | irrational numbers, |
| $\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}, i^2 = -1\}$                   | complex numbers.    |

Every real number is either rational or irrational, so  $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'$  with  $\mathbb{Q} \cap \mathbb{Q}' = \emptyset$ . In decimal form, the rationals are exactly the reals whose expansion terminates or recurs.

### 1.2.1 Examples from each set

| Set           | Examples                                              | What it can do that the previous one cannot                                           |
|---------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| $\mathbb{N}$  | 1, 7, 250                                             | counting                                                                              |
| $\mathbb{Z}$  | 0, -3, -128                                           | subtraction always possible: $3 - 5 = -2$                                             |
| $\mathbb{Q}$  | $\frac{1}{2}$ , $-\frac{7}{3}$ , 0.25, $1.\bar{3}$    | division by a non-zero number always possible: $3 \div 5 = \frac{3}{5}$               |
| $\mathbb{Q}'$ | $\sqrt{2}$ , $\pi$ , $e$ , $\sqrt[3]{7}$              | lengths and limits that no fraction can express                                       |
| $\mathbb{R}$  | everything above                                      | every bounded set has a least upper bound; limits of convergent sequences stay inside |
| $\mathbb{C}$  | $i$ , $2 + 3i$ , $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$ | every polynomial equation has a root: $x^2 + 1 = 0$                                   |

### 1.2.2 The inclusion chain

Each set contains the one before it:

$$\boxed{\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}}$$

and each inclusion is *proper* at every stage there is something new.

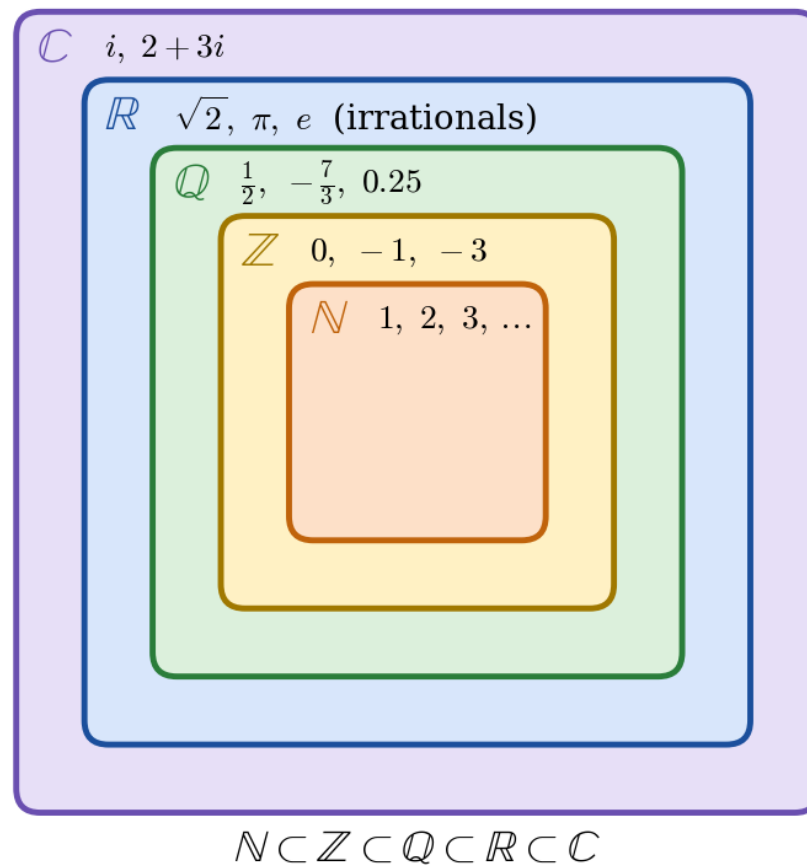


Figure 1.2.1: The number systems nested. Each ring holds numbers the ring inside it cannot express.

Taking the inclusions one at a time:

- (i)  $\mathbb{N} \subset \mathbb{Z}$ . Every natural number is an integer. The inclusion is proper because  $-1 \in \mathbb{Z}$  but  $-1 \notin \mathbb{N}$ . The equation  $x + 3 = 1$  has no solution in  $\mathbb{N}$  but does in  $\mathbb{Z}$ .
- (ii)  $\mathbb{Z} \subset \mathbb{Q}$ . Every integer  $n$  is the fraction  $\frac{n}{1}$ . The inclusion is proper because  $\frac{1}{2} \in \mathbb{Q}$  but  $\frac{1}{2} \notin \mathbb{Z}$ . The equation  $2x = 1$  has no solution in  $\mathbb{Z}$  but does in  $\mathbb{Q}$ .
- (iii)  $\mathbb{Q} \subset \mathbb{R}$ . Every rational number is real. The inclusion is proper because  $\sqrt{2} \in \mathbb{R}$  but  $\sqrt{2} \notin \mathbb{Q}$ . The equation  $x^2 = 2$  has no solution in  $\mathbb{Q}$  but does in  $\mathbb{R}$ .
- (iv)  $\mathbb{R} \subset \mathbb{C}$ . Every real number  $a$  is the complex number  $a + 0i$ . The inclusion is proper because  $i \in \mathbb{C}$  but  $i \notin \mathbb{R}$ . The equation  $x^2 = -1$  has no solution in  $\mathbb{R}$  but does in  $\mathbb{C}$ .

**Note:**

Read the chain as a sequence of repairs. Each set is enlarged exactly far enough to solve an equation the previous one could not, and the new elements are precisely the solutions that were missing.

**Proposition 1.2.2:  $\sqrt{2}$  is irrational**

There is no rational number  $x$  with  $x^2 = 2$ .

*Proof.* Suppose, for contradiction, that  $\sqrt{2} = \frac{p}{q}$  with  $p, q \in \mathbb{Z}$ ,  $q \neq 0$ , and the fraction in lowest terms, so that  $p$  and  $q$  have no common factor. Squaring,

$$2 = \frac{p^2}{q^2} \implies p^2 = 2q^2,$$

so  $p^2$  is even, and therefore  $p$  is even. Write  $p = 2k$ . Substituting,

$$(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies q^2 = 2k^2,$$

so  $q^2$  is even and therefore  $q$  is even. But then  $p$  and  $q$  share the factor 2, contradicting the assumption that the fraction was in lowest terms. Hence no such rational number exists. ■

**Note:**

This one proof is the whole reason  $\mathbb{R}$  is needed: the diagonal of a unit square has a length that no fraction can name.

**1.2.3 Intervals**

Subsets of  $\mathbb{R}$  described by inequalities occur so often that they have their own notation. For  $a < b$ ,

$$\begin{aligned} (a, b) &= \{x \in \mathbb{R} \mid a < x < b\} && \text{open,} \\ [a, b] &= \{x \in \mathbb{R} \mid a \leq x \leq b\} && \text{closed,} \\ [a, b) &= \{x \in \mathbb{R} \mid a \leq x < b\} && \text{half open,} \\ [a, \infty) &= \{x \in \mathbb{R} \mid x \geq a\}, && (-\infty, b) = \{x \in \mathbb{R} \mid x < b\}. \end{aligned}$$

**Note:**

The symbol  $\infty$  is not a real number, so an interval is never closed at an infinite end: one writes  $[a, \infty)$ , never  $[a, \infty]$ .

**1.3 The Geometry of  $\mathbb{R}$ ,  $\mathbb{R}^2$ ,  $\mathbb{R}^3$  and  $\mathbb{R}^n$** 

Repeated Cartesian products of  $\mathbb{R}$  with itself give the spaces in which all the calculus of this course takes place.

**Definition 1.3.1:  $n$ -dimensional real space**

For a positive integer  $n$ ,

$$\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}_{n \text{ times}} = \left\{ (a_1, a_2, \dots, a_n) \mid a_i \in \mathbb{R} \text{ for } i = 1, 2, \dots, n \right\}.$$

An element of  $\mathbb{R}^n$  is called a *point* or a *vector*, and the numbers  $a_1, \dots, a_n$  are its *coordinates* or *components*.

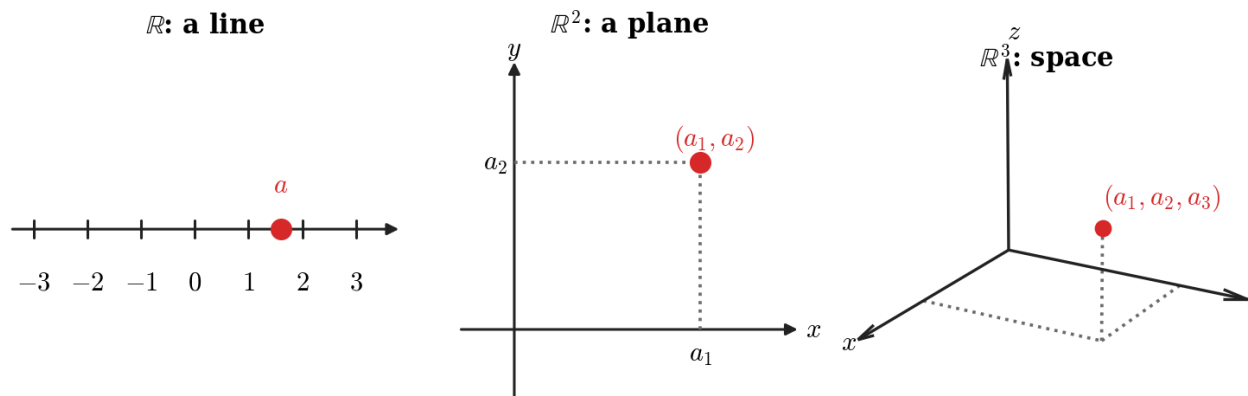


Figure 1.3.1: One coordinate gives a line, two give a plane, three give space. Each new coordinate adds a direction perpendicular to all the previous ones.

### 1.3.1 $\mathbb{R}$ : the number line

$\mathbb{R}^1 = \mathbb{R}$  is pictured as a line. Choosing a point for 0 and a unit of length makes every real number correspond to exactly one point of the line, and every point to exactly one real number. Distance between  $a$  and  $b$  is

$$d(a, b) = |a - b|.$$

Subsets of  $\mathbb{R}$  that arise most often are the intervals of the previous section, which are exactly the “unbroken” pieces of the line.

### 1.3.2 $\mathbb{R}^2$ : the plane

$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$  is pictured as a plane with two perpendicular axes. The point  $(a_1, a_2)$  sits  $a_1$  along the horizontal axis and  $a_2$  along the vertical one. Distance is given by Pythagoras’ theorem,

$$d((a_1, a_2), (b_1, b_2)) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}.$$

Familiar objects in  $\mathbb{R}^2$  include the line  $ax + by = c$  and the circle of radius  $r$  about  $(h, k)$ ,

$$(x - h)^2 + (y - k)^2 = r^2.$$

#### Note:

The graph of a function of one variable lives in  $\mathbb{R}^2$ : it is the set of points  $(x, f(x))$ , a subset of  $\mathbb{R} \times \mathbb{R}$ . That is why the Cartesian product had to come first.

### 1.3.3 $\mathbb{R}^3$ : space

$\mathbb{R}^3$  is pictured as ordinary space with three mutually perpendicular axes  $x, y, z$ . The point  $(a_1, a_2, a_3)$  is located by moving  $a_1$  along the  $x$ -axis, then  $a_2$  parallel to the  $y$ -axis, then  $a_3$  parallel to the  $z$ -axis. Distance is

$$d((a_1, a_2, a_3), (b_1, b_2, b_3)) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}.$$

Familiar objects include the plane  $ax + by + cz = d$  and the sphere

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2.$$

**Note:**

Surfaces  $z = f(x, y)$  live in  $\mathbb{R}^3$ , which is why the two variable calculus of Unit 3 is done there. The three axes are taken in the *right handed* order: curling the fingers of the right hand from the  $x$ -axis to the  $y$ -axis points the thumb along the  $z$ -axis.

**1.3.4  $\mathbb{R}^n$ : the general case**

For  $n > 3$  no picture is available, but the algebra continues unchanged. For  $\mathbf{a} = (a_1, \dots, a_n)$  and  $\mathbf{b} = (b_1, \dots, b_n)$  in  $\mathbb{R}^n$  and  $\alpha \in \mathbb{R}$ :

$$\begin{aligned}\mathbf{a} + \mathbf{b} &= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) && \text{addition,} \\ \alpha \mathbf{a} &= (\alpha a_1, \alpha a_2, \dots, \alpha a_n) && \text{scalar multiplication,} \\ \|\mathbf{a}\| &= \sqrt{a_1^2 + a_2^2 + \dots + a_n^2} && \text{length,} \\ d(\mathbf{a}, \mathbf{b}) &= \|\mathbf{a} - \mathbf{b}\| = \sqrt{\sum_{i=1}^n (a_i - b_i)^2} && \text{distance.}\end{aligned}$$

Setting  $n = 1, 2, 3$  recovers the three formulas above, so the general definition is the pattern the visible cases already follow.

**Note:**

Dimensions beyond three are ordinary in applications: a system with  $n$  unknowns has its solution set in  $\mathbb{R}^n$ , which is exactly the setting of the linear systems in Unit 4. The geometric words point, distance, line, plane keep their meaning even where the picture runs out.

**Example 1.3.2: Distance in three spaces**

$$\begin{aligned}\text{In } \mathbb{R} : \quad d(3, -1) &= |3 - (-1)| = 4, \\ \text{In } \mathbb{R}^2 : \quad d((1, 2), (4, 6)) &= \sqrt{3^2 + 4^2} = 5, \\ \text{In } \mathbb{R}^3 : \quad d((1, 2, 2), (0, 0, 0)) &= \sqrt{1 + 4 + 4} = 3, \\ \text{In } \mathbb{R}^4 : \quad d((1, 1, 1, 1), (0, 0, 0, 0)) &= \sqrt{1 + 1 + 1 + 1} = 2.\end{aligned}$$

The last line cannot be drawn, but it is computed by the same rule as the others.

**1.4 Functions****Definition 1.4.1: Function**

Let  $A$  and  $B$  be sets. A *function*  $f$  from  $A$  to  $B$ , written

$$f : A \rightarrow B,$$

is a rule that assigns to each element  $x \in A$  exactly one element  $f(x) \in B$ .

- $A$  is called the *domain* of  $f$ , written  $\text{dom}(f)$ . It is the set of all possible inputs.
- $B$  is called the *codomain* of  $f$ .

- The *range* of  $f$  is the set of values actually attained,

$$\text{ran}(f) = \{ f(x) \mid x \in A \} \subseteq B.$$

The element  $f(x)$  is called the *image* of  $x$  under  $f$ .

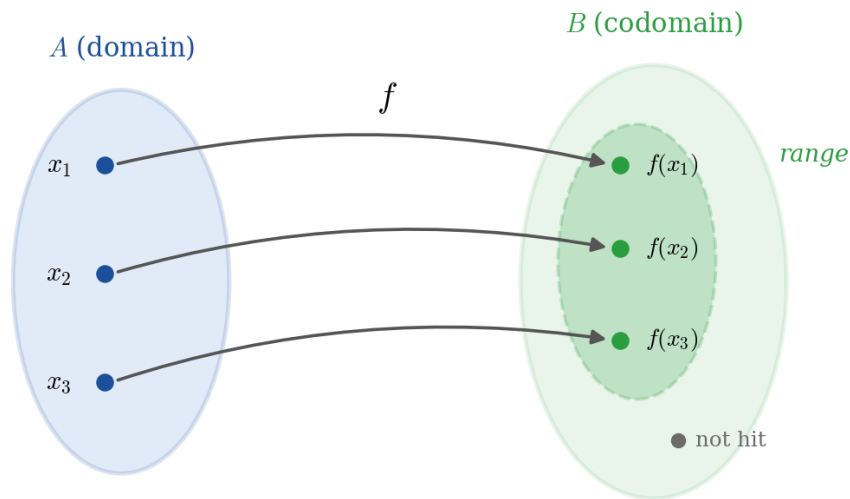


Figure 1.4.1: A function  $f : A \rightarrow B$ . Every element of  $A$  has exactly one arrow leaving it. The range is the part of  $B$  actually hit, and may be smaller than  $B$ .

#### Note:

Two conditions are hidden in the phrase “exactly one”:

- at least one* every  $x \in A$  must be assigned a value, so the rule must work on all of  $A$ ;
- at most one* no  $x$  may be assigned two different values.

The second is what the vertical line test checks graphically.

#### Note:

**Codomain and range are not the same thing.** For  $f : \mathbb{R} \rightarrow \mathbb{R}$  with  $f(x) = x^2$ , the codomain is  $\mathbb{R}$  but the range is  $[0, \infty)$ : no negative number is ever an output. The codomain is declared when the function is written down; the range must be worked out.

### 1.4.1 Notation and variables

When  $A, B \subseteq \mathbb{R}$  it is usual to write  $y = f(x)$ , and then

- $x$  is the *independent variable*, a symbol representing an arbitrary number in the domain;
- $y$  is the *dependent variable*, a symbol representing a number in the range of  $f$ .

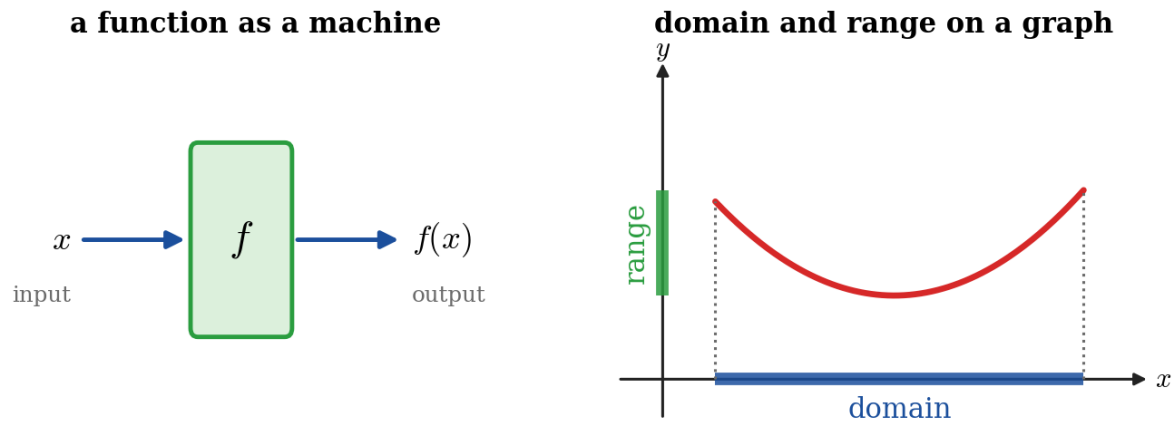


Figure 1.4.2: Left:  $f$  as a machine turning an input into an output. Right: the domain read off the horizontal axis and the range off the vertical axis.

### Definition 1.4.2: Graph of a function

The *graph* of  $f : A \rightarrow B$  with  $A, B \subseteq \mathbb{R}$  is the subset of  $\mathbb{R}^2$  given by

$$\{ (x, f(x)) \mid x \in A \}.$$

#### Note:

The graph is a set of ordered pairs, so it is a subset of  $A \times B \subseteq \mathbb{R} \times \mathbb{R}$ . This is the Cartesian product of Definition 1.1.5 doing its work.

## 1.4.2 Finding the natural domain

When a function is given only by a formula, its *natural domain* is the largest subset of  $\mathbb{R}$  on which the formula makes sense.

### Example 1.4.3: Domains from a formula

Find the domain of each function.

$$(a) f(x) = \sqrt{x+2} \qquad (b) g(x) = \frac{1}{x^2 - x}$$

**Solution:** (a) A square root requires a non-negative argument, so the domain of  $f$  consists of all values of  $x$  such that

$$x + 2 \geq 0 \implies x \geq -2.$$

The domain is  $[-2, \infty)$ , and we write  $f : [-2, \infty) \rightarrow \mathbb{R}$ .

(b) A quotient requires a non-zero denominator. Since  $x^2 - x = x(x - 1)$ , the denominator vanishes at  $x = 0$  and  $x = 1$ , so the domain of  $g$  is

$$\{ x \in \mathbb{R} \mid x \neq 0, x \neq 1 \},$$

which in interval notation is

$$(-\infty, 0) \cup (0, 1) \cup (1, \infty).$$

**Note:**

Two restrictions account for almost every domain question at this level: the argument of an even root must be  $\geq 0$ , and a denominator must be  $\neq 0$ . A third — the argument of a logarithm must be  $> 0$  — appears with the logarithm in [Section 1.5](#).

**1.4.3 The vertical line test****Theorem 1.4.4: Vertical line test**

A curve in the  $xy$ -plane is the graph of a function of  $x$  if and only if no vertical line intersects the curve more than once.

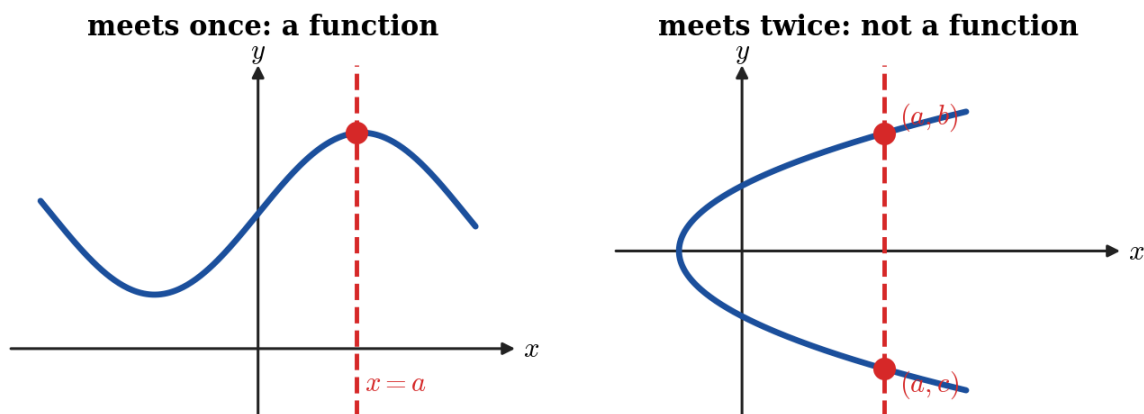


Figure 1.4.3: The vertical line  $x = a$  meets the first curve once, so each input has a single output. It meets the second twice, so  $a$  would be assigned two different values.

**Note:**

The test is exactly the “at most one” half of [Definition 1.4.1](#) drawn as a picture. If the vertical line  $x = a$  met the curve at both  $(a, b)$  and  $(a, c)$  with  $b \neq c$ , the rule would assign two different values to the single input  $a$ , which no function may do.

**Example 1.4.5: A circle is not a graph**

The circle  $x^2 + y^2 = 1$  fails the test: the line  $x = 0$  meets it at  $(0, 1)$  and  $(0, -1)$ . It is therefore not the graph of any function of  $x$ . Splitting it into the upper and lower halves,

$$y = \sqrt{1 - x^2} \quad \text{and} \quad y = -\sqrt{1 - x^2},$$

gives two functions  $[-1, 1] \rightarrow \mathbb{R}$ , each of which does pass the test.

**1.4.4 Piecewise defined functions****Definition 1.4.6: Piecewise defined function**

A *piecewise defined function* is one given by different formulas on different parts of its domain.

**Example 1.4.7: Three piecewise functions**

(a)  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by

$$f(x) = \begin{cases} 1 - x^2 & \text{if } x \leq 2, \\ x^3 & \text{if } x > 2. \end{cases}$$

(b)  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by

$$f(x) = \begin{cases} 1 & \text{if } x \in (-\infty, 0), \\ 2 & \text{if } x \in [0, \infty). \end{cases}$$

(c) The *absolute value function*  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0, \end{cases}$$

that is,  $f(x) = -x$  on  $(-\infty, 0)$  and  $f(x) = x$  on  $[0, \infty)$ .

(d) The function  $f : [0, \infty) \rightarrow \mathbb{R}$  defined by

$$f(x) = \begin{cases} x & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x \geq 1, \end{cases}$$

which rises along the line  $y = x$  and then levels off.

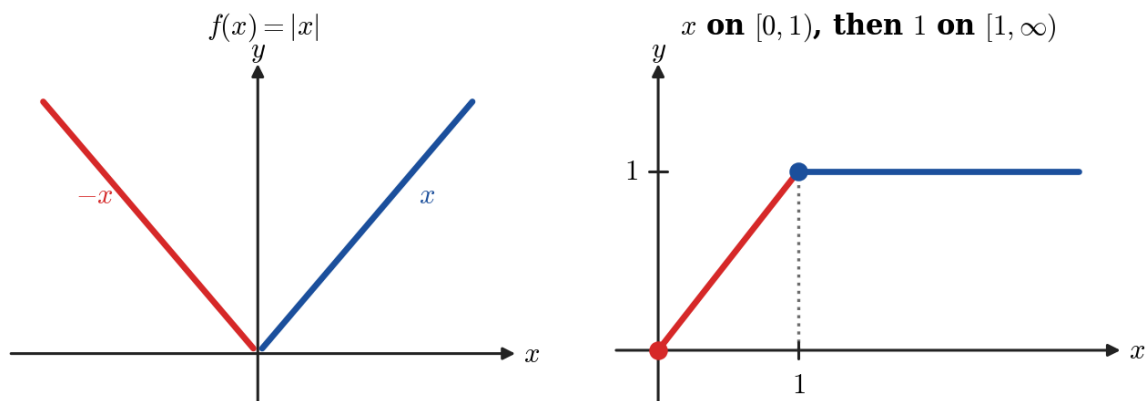


Figure 1.4.4: Two of the examples. The pieces must agree at the joins if the graph is to be unbroken there; both of these do.

**Note:**

The conditions defining the pieces must cover the whole domain and must not overlap otherwise the rule would leave some input without a value, or give one input two values, and [Definition 1.4.1](#) would fail. In part (a) the conditions  $x \leq 2$  and  $x > 2$  do exactly this: together they are all of  $\mathbb{R}$ , and no  $x$  satisfies both.

**Example 1.4.8: Evaluating a piecewise function**

For  $f$  of part (a) above,

$$f(0) = 1 - 0^2 = 1, \quad f(2) = 1 - 2^2 = -3, \quad f(3) = 3^3 = 27.$$

At  $x = 2$  the first formula applies, because the condition there is  $x \leq 2$ . Note that the second formula would give  $2^3 = 8$ , so the two pieces do *not* agree at the join: the graph has a break at  $x = 2$ .

**1.4.5 Even and odd functions****Definition 1.4.9: Even and odd functions**

Let  $f : A \rightarrow \mathbb{R}$ , where  $A \subseteq \mathbb{R}$  is symmetric about the origin, so that  $x \in A$  implies  $-x \in A$ .

- $f$  is *even* if  $f(-x) = f(x)$  for all  $x \in A$ .
- $f$  is *odd* if  $f(-x) = -f(x)$  for all  $x \in A$ .

**Note:**

The domain must be symmetric for the definition to make sense:  $f(-x)$  has to exist before it can be compared with  $f(x)$ . This is why the condition is stated as part of the definition rather than assumed.

**Example 1.4.10: A quadratic and a cubic**

(a)  $f(x) = x^2$  is an *even* function, since

$$f(-x) = (-x)^2 = x^2 = f(x).$$

(b)  $f(x) = x^3$  is an *odd* function, since

$$f(-x) = (-x)^3 = -x^3 = -f(x).$$

**1.4.6 What the symmetry looks like**

- If the graph of  $f$  is symmetric with respect to the  $y$ -axis, then  $f$  is an even function.
- If the graph of  $f$  is symmetric with respect to the origin unchanged by a half turn about it then  $f$  is an odd function.

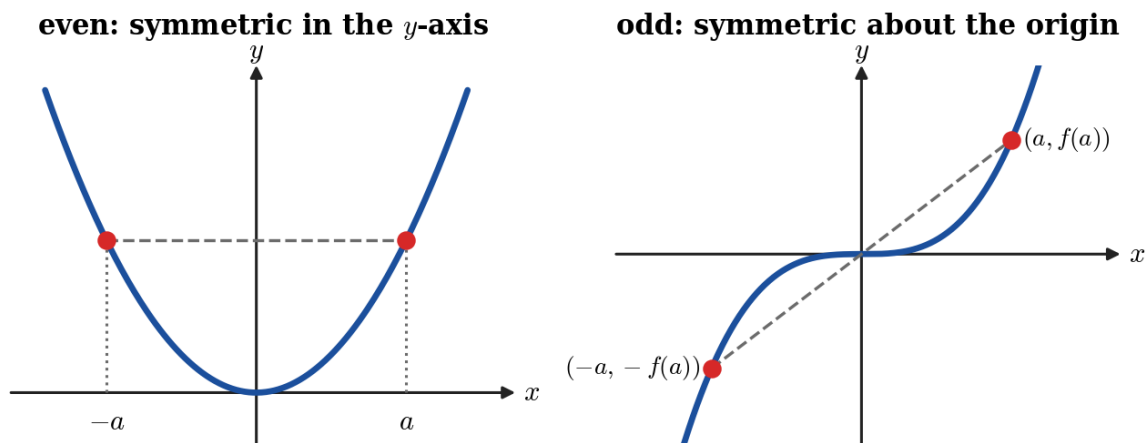


Figure 1.4.5: Even symmetry reflects the graph in the  $y$ -axis; odd symmetry turns it through half a revolution about the origin.

**Note:**

If  $f$  is odd and 0 is in its domain then  $f(0) = -f(0)$ , so  $f(0) = 0$ : the graph of an odd function must pass through the origin. Both examples above confirm this:  $x^3$  passes through the origin. The converse fails, though —  $x^2$  also passes through the origin but is even, not odd.

**Example 1.4.11: Even, odd, or neither**

Determine whether each of the following functions is even, odd, or neither even nor odd.

$$(a) f(x) = x^5 + x \quad (b) g(x) = 1 - x^4 \quad (c) h(x) = 2x - x^2$$

**Solution:** (a) Replacing  $x$  by  $-x$ ,

$$f(-x) = (-x)^5 + (-x) = -x^5 - x = -(x^5 + x) = -f(x),$$

so  $f(-x) = -f(x)$  and  $f$  is an *odd* function.

(b) Replacing  $x$  by  $-x$ ,

$$g(-x) = 1 - (-x)^4 = 1 - x^4 = g(x),$$

so  $g(-x) = g(x)$  and  $g$  is an *even* function.

(c) Replacing  $x$  by  $-x$ ,

$$h(-x) = 2(-x) - (-x)^2 = -2x - x^2 = -(2x + x^2).$$

Comparing with  $h(x) = 2x - x^2$  and  $-h(x) = -2x + x^2$ , we see that

$$h(-x) \neq h(x) \quad \text{and} \quad h(-x) \neq -h(x),$$

so  $h$  is *neither* even nor odd. Most functions are like this, and a single value of  $x$  is enough to show it:  $h(1) = 1$  and  $h(-1) = -3$ , so  $h(-1) \neq h(1)$  and  $h(-1) \neq -h(1)$ .

**Note:**

The pattern in (a) and (b) is general: a polynomial in which every exponent is odd is an odd function, and one in which every exponent is even is an even function. A polynomial mixing the two, as in (c), is neither. This is where the names come from.

**1.4.7 Increasing and decreasing functions****Definition 1.4.12: Increasing and decreasing**

Let  $f : A \rightarrow \mathbb{R}$  and let  $I \subseteq A$  be an interval.

- $f$  is *increasing* on  $I$  if

$$f(x_1) < f(x_2) \quad \text{whenever} \quad x_1 < x_2 \quad \text{in } I.$$

- $f$  is *decreasing* on  $I$  if

$$f(x_1) > f(x_2) \quad \text{whenever} \quad x_1 < x_2 \quad \text{in } I.$$

**Example 1.4.13: A function that does both**

The function  $f(x) = x^2$  is decreasing on the interval  $(-\infty, 0]$  and increasing on  $[0, \infty)$ .

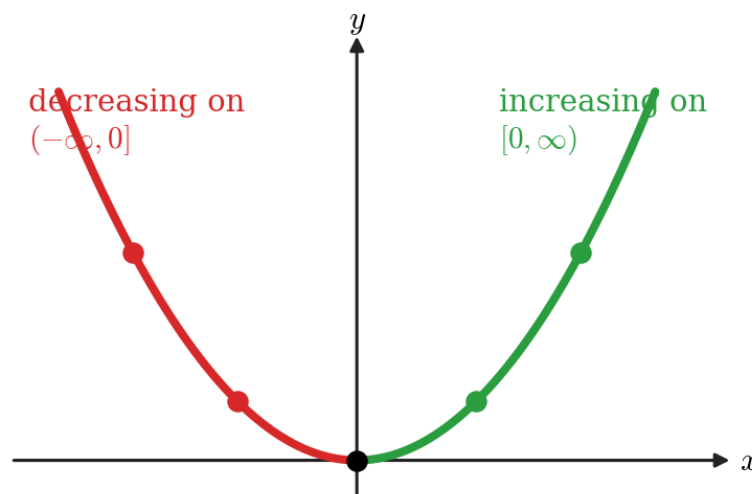


Figure 1.4.6: On the left branch, moving right lowers the height; on the right branch it raises it. A function need not be increasing or decreasing on the whole of its domain.

**Note:**

The definition is about *all* pairs  $x_1 < x_2$  in  $I$ , not just nearby ones, and it must be stated on an interval. Saying “ $x^2$  is increasing” without naming an interval is false:  $-2 < 1$  but  $f(-2) = 4 > 1 = f(1)$ .

## 1.5 Some Standard Families of Functions

### 1.5.1 Polynomials

#### Definition 1.5.1: Polynomial

A function  $p : \mathbb{R} \rightarrow \mathbb{R}$  is called a *polynomial* if

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where  $n$  is a non-negative integer and the numbers  $a_0, a_1, \dots, a_n$  are constants called the *coefficients* of  $p$ .

- For any polynomial the domain is  $\mathbb{R} = (-\infty, \infty)$ .
- If the leading coefficient  $a_n \neq 0$ , then the *degree* of the polynomial is  $n$ .

#### Example 1.5.2: Degree of a polynomial

For

$$p(x) = 3x^5 - 2x^4 + 6x^3 + 2,$$

the leading coefficient is  $a_5 = 3 \neq 0$ , so this is a polynomial of degree 5.

- A polynomial of degree 1 has the form  $p(x) = ax + b$  and is called a *linear function*.
- A polynomial of degree 2 has the form  $p(x) = ax^2 + bx + c$  and is called a *quadratic function*.

#### Note:

The domain of a polynomial is always all of  $\mathbb{R}$  because a polynomial involves only addition and multiplication never a division or a root, which are the two operations that create domain restrictions.

### 1.5.2 Power functions

#### Definition 1.5.3: Power function

A function of the form

$$f(x) = x^a,$$

where  $a$  is a constant, is called a *power function*.

The shape depends on  $a$ . For  $a$  a positive integer, two patterns appear.

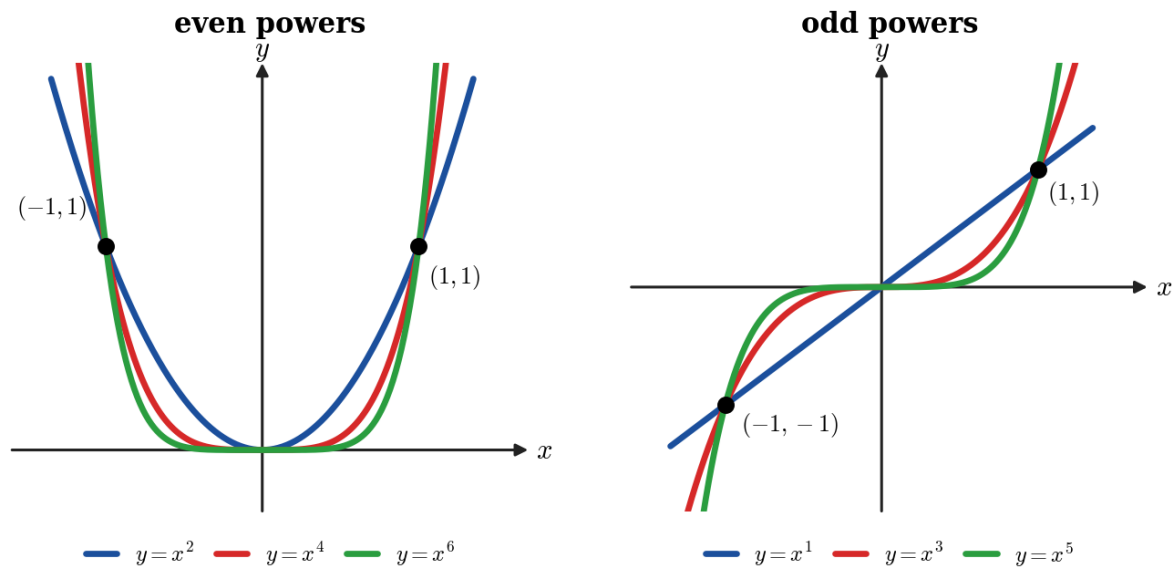


Figure 1.5.1: All these curves pass through  $(1, 1)$ , and through  $(-1, 1)$  or  $(-1, -1)$  according to whether the power is even or odd. The larger the exponent, the flatter the curve near the origin and the steeper beyond  $|x| = 1$ .

- If  $a = n$  is a positive *even* integer,  $f(x) = x^n$  is an even function; its graph is symmetric in the  $y$ -axis, resembles a parabola, and passes through  $(-1, 1)$  and  $(1, 1)$ .
- If  $a = n$  is a positive *odd* integer,  $f(x) = x^n$  is an odd function; its graph is symmetric about the origin and passes through  $(-1, -1)$  and  $(1, 1)$ .

**Note:**

Three cases of  $a$  are worth naming separately:  $a = 1$  gives the line  $y = x$ ;  $a = \frac{1}{2}$  gives the square root  $f(x) = \sqrt{x}$ , whose domain is  $[0, \infty)$ ; and  $a = -1$  gives the reciprocal  $f(x) = \frac{1}{x}$ , whose domain excludes 0. Only for  $a$  a non-negative integer is the domain all of  $\mathbb{R}$ .

### 1.5.3 Rational functions

**Definition 1.5.4: Rational function**

A *rational function* is a quotient of two polynomials,

$$f(x) = \frac{P(x)}{Q(x)},$$

and its domain is the set of all  $x$  with  $Q(x) \neq 0$ .

For example  $f(x) = \frac{2x}{x-3}$  has domain  $\{x \in \mathbb{R} \mid x \neq 3\}$ , while  $g(x) = \frac{x^2-1}{x^2+1}$  has domain all of  $\mathbb{R}$ , because  $x^2+1 \geq 1$  is never zero. Both functions reappear in the section on limits.

### 1.5.4 Exponential, logarithmic and trigonometric functions

These are known from school; the table collects the facts used in this course. Angles are always measured in *radians*.

| Function                         | Domain                       | Range         | Key facts                                                     |
|----------------------------------|------------------------------|---------------|---------------------------------------------------------------|
| $e^x$                            | $\mathbb{R}$                 | $(0, \infty)$ | $e^0 = 1$ , $e^{a+b} = e^a e^b$ , increasing                  |
| $\ln x$                          | $(0, \infty)$                | $\mathbb{R}$  | $\ln 1 = 0$ , $\ln(ab) = \ln a + \ln b$ , $\ln a^r = r \ln a$ |
| $\sin x$ , $\cos x$              | $\mathbb{R}$                 | $[-1, 1]$     | period $2\pi$ , $\sin^2 x + \cos^2 x = 1$                     |
| $\tan x = \frac{\sin x}{\cos x}$ | $x \neq \frac{(2k+1)\pi}{2}$ | $\mathbb{R}$  | period $\pi$                                                  |

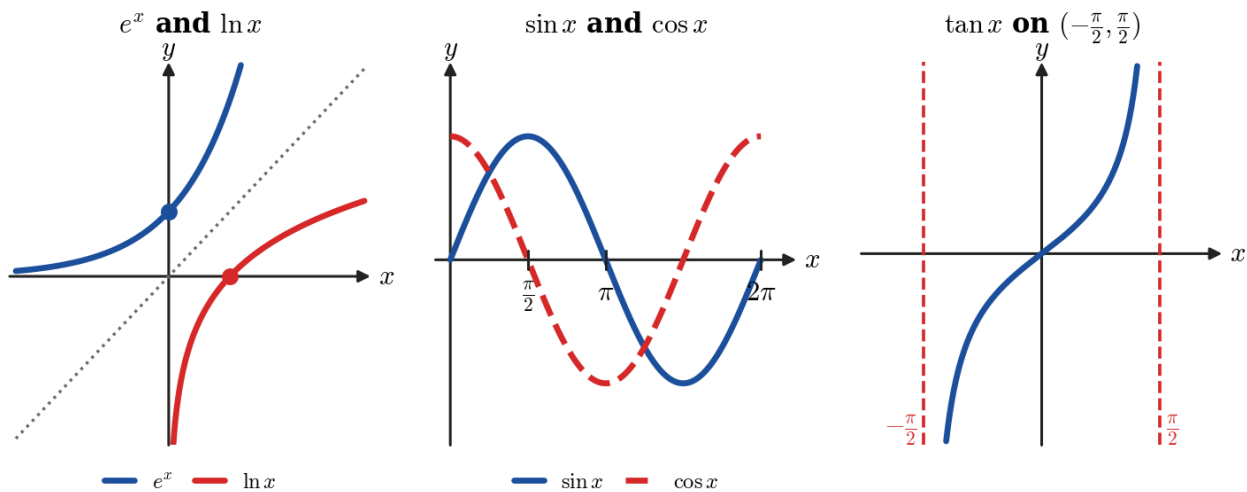


Figure 1.5.2: Left:  $e^x$  and  $\ln x$  are mirror images in the line  $y = x$ , because each undoes the other. Centre:  $\sin x$  is odd and  $\cos x$  is even. Right:  $\tan x$  has vertical asymptotes wherever  $\cos x = 0$ .

**Note:**

The natural logarithm  $\ln x$  is also written  $\log x$ . In Units 2 and 3 of these notes,  $\log x$  *always* means the natural logarithm  $\ln x$  (base  $e$ ). Its domain is  $(0, \infty)$ , which gives the third domain rule: the argument of a logarithm must be positive.

## 1.6 Transformations of Functions

Throughout this section  $c > 0$  is a constant, and  $y = f(x)$  is a given function whose graph is known.

### 1.6.1 Vertical and horizontal shifts

| Equation       | Effect on the graph of $y = f(x)$                         |
|----------------|-----------------------------------------------------------|
| $y = f(x) + c$ | shifts the graph a distance $c$ units <i>upward</i>       |
| $y = f(x) - c$ | shifts the graph a distance $c$ units <i>downward</i>     |
| $y = f(x - c)$ | shifts the graph a distance $c$ units <i>to the right</i> |
| $y = f(x + c)$ | shifts the graph a distance $c$ units <i>to the left</i>  |

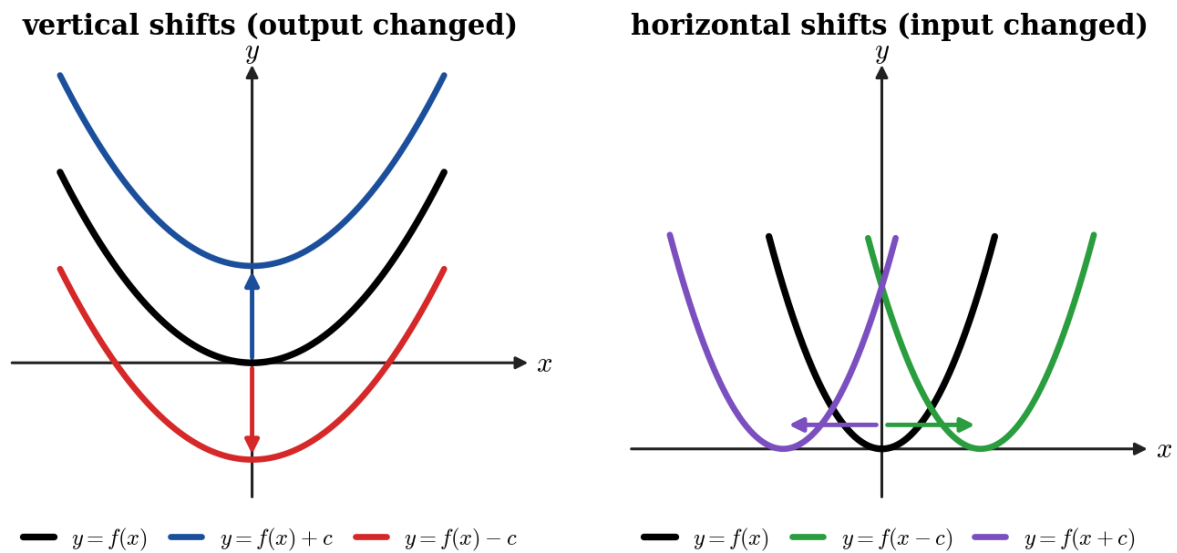


Figure 1.6.1: Left: adding  $c$  to the output moves the graph up or down. Right: replacing  $x$  by  $x \mp c$  moves it sideways — right for  $x - c$ , left for  $x + c$ .

### Common mistake:

The horizontal shifts run opposite to the sign, which is the usual source of error. In  $y = f(x - c)$  the graph moves *right* by  $c$ , not left. The reason is that the new graph reaches at  $x = a + c$  the value the old one had at  $x = a$ , since  $f((a + c) - c) = f(a)$ .

## 1.6.2 Stretching, compressing and reflecting

| Equation                        | Effect on the graph of $y = f(x)$                           |
|---------------------------------|-------------------------------------------------------------|
| $y = cf(x)$                     | stretches the graph <i>vertically</i> by a factor of $c$    |
| $y = \frac{1}{c}f(x)$           | compresses the graph <i>vertically</i> by a factor of $c$   |
| $y = f(cx)$                     | compresses the graph <i>horizontally</i> by a factor of $c$ |
| $y = f\left(\frac{x}{c}\right)$ | stretches the graph <i>horizontally</i> by a factor of $c$  |
| $y = -f(x)$                     | reflects the graph about the $x$ -axis                      |
| $y = f(-x)$                     | reflects the graph about the $y$ -axis                      |

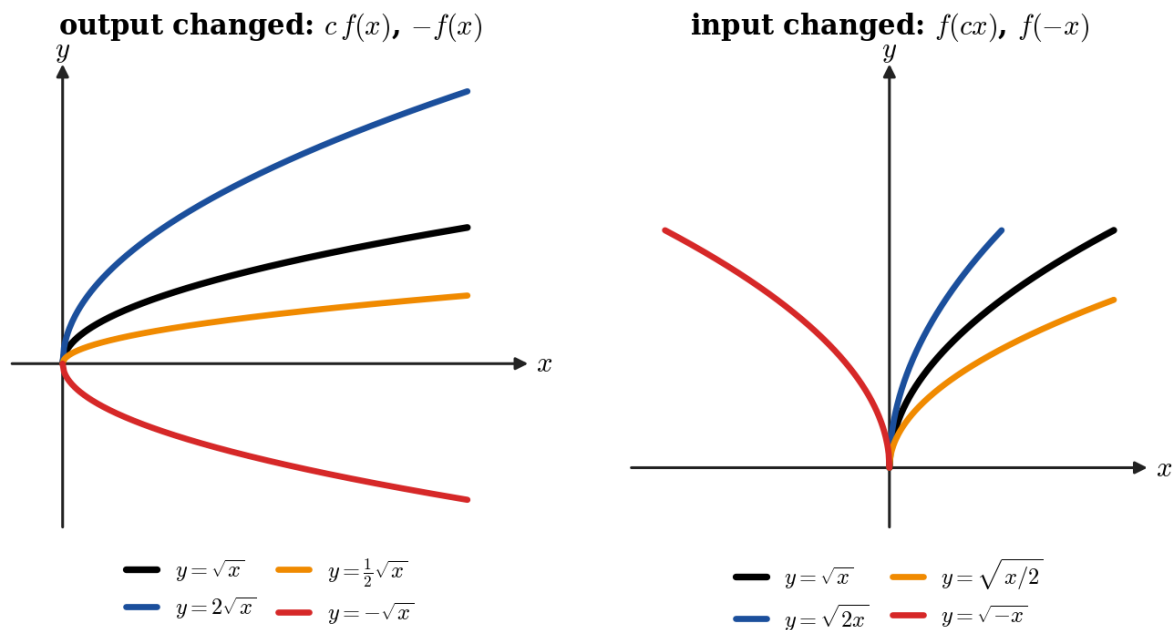


Figure 1.6.2: Shown for  $f(x) = \sqrt{x}$  with  $c = 2$ . Left: multiplying the output scales the graph vertically, and  $-f(x)$  flips it in the  $x$ -axis. Right:  $f(2x)$  is compressed horizontally,  $f(x/2)$  is stretched, and  $f(-x)$  is flipped in the  $y$ -axis.

#### Note:

The same rule governs both tables: **anything done to the output acts on the graph as expected, and anything done to the input acts in the opposite sense.** Adding  $c$  to the output raises the graph; adding  $c$  to the input moves it left. Multiplying the output by  $c$  stretches vertically; multiplying the input by  $c$  compresses horizontally.

#### Example 1.6.1: Reading a transformed function

The graph of  $y = 2(x - 3)^2 + 1$  is obtained from the parabola  $y = x^2$  in three steps:

1. replace  $x$  by  $x - 3$ : shift 3 units to the right;
2. multiply by 2: stretch vertically by a factor of 2;
3. add 1: shift 1 unit upward.

The vertex therefore moves from  $(0, 0)$  to  $(3, 1)$ .

#### Note:

The order matters. Carrying out step 3 before step 2 would give  $2[(x - 3)^2 + 1] = 2(x - 3)^2 + 2$ , a different function. Work from the inside of the formula outwards.

## 1.7 Combining Functions

### 1.7.1 Algebraic combinations

Let  $f$  and  $g$  be two functions with domains  $[a, b]$  and  $[c, d]$  respectively. Their sum, difference, product and quotient are defined pointwise, and each has the domain shown.

| Combination                                       | Domain                                          |
|---------------------------------------------------|-------------------------------------------------|
| $(f + g)(x) = f(x) + g(x)$                        | $[a, b] \cap [c, d]$                            |
| $(f - g)(x) = f(x) - g(x)$                        | $[a, b] \cap [c, d]$                            |
| $(fg)(x) = f(x)g(x)$                              | $[a, b] \cap [c, d]$                            |
| $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ | $\{x \in [a, b] \cap [c, d] \mid g(x) \neq 0\}$ |

**Note:**

The intersection appears because both  $f(x)$  and  $g(x)$  must exist before they can be combined. The quotient carries the extra condition  $g(x) \neq 0$ , so its domain is generally smaller than that of the other three.

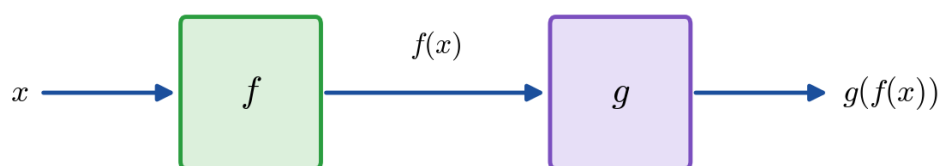
### 1.7.2 Composite functions

**Definition 1.7.1: Composition**

Let  $f$  and  $g$  be functions. The *composite* functions are defined by

$$(f \circ g)(x) = f(g(x)), \quad (g \circ f)(x) = g(f(x)).$$

For  $g \circ f$  to make sense at  $x$ , the value  $f(x)$  must lie in the domain of  $g$ .



$$(g \circ f)(x) = g(f(x)): \text{ } f \text{ acts first, } g \text{ second}$$

Figure 1.7.1: Composition feeds the output of one machine into the next. The right-hand machine acts second, even though  $g$  is written on the left in  $g \circ f$ .

**Example 1.7.2: Composition is not commutative**

Let  $f(x) = x^2$  and  $g(x) = x - 2$ . Then

$$(f \circ g)(x) = f(g(x)) = f(x - 2) = (x - 2)^2,$$

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 - 2.$$

These are different functions: at  $x = 0$  the first gives 4 and the second gives  $-2$ . In general  $f \circ g \neq g \circ f$ .

## 1.8 One-to-One Functions and Inverses

### 1.8.1 One-to-one functions

#### Definition 1.8.1: One-to-one function

A function  $f : A \rightarrow B$  is said to be *one-to-one* (or *injective*) if

$$f(x_1) \neq f(x_2) \quad \text{whenever} \quad x_1 \neq x_2.$$

Equivalently,  $f(x_1) = f(x_2)$  forces  $x_1 = x_2$ : different inputs never share an output.

#### Theorem 1.8.2: Horizontal line test

A function is one-to-one if and only if no horizontal line intersects its graph more than once.

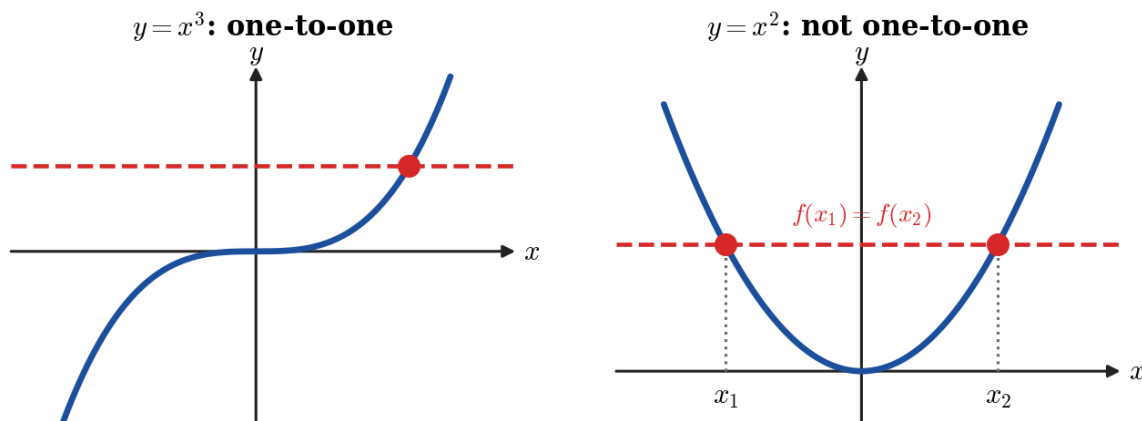


Figure 1.8.1: If a horizontal line meets the graph twice, at  $x_1$  and  $x_2$ , then  $x_1 \neq x_2$  but  $f(x_1) = f(x_2)$ , and the function is not one-to-one.

#### Note:

Compare the two tests. The *vertical* line test decides whether a curve is a function at all; the *horizontal* line test decides whether that function is one-to-one. The first is about inputs having one output, the second about outputs coming from one input.

#### Example 1.8.3: Two power functions

- $f(x) = x^3$  is one-to-one. If  $x_1 \neq x_2$  then  $x_1^3 \neq x_2^3$ , since two different numbers cannot have the same cube. Drawing a horizontal line at any height, it meets the graph at only one point.
- $f(x) = x^2$  on  $\mathbb{R}$  is *not* one-to-one, since  $f(-1) = f(1) = 1$  although  $-1 \neq 1$ . Restricting the domain to  $[0, \infty)$  does make it one-to-one.

### 1.8.2 Inverse functions

#### Definition 1.8.4: Inverse function

Let  $f : A \rightarrow B$  be a one-to-one function with domain  $A$  and range  $B$ . Then its *inverse function*  $f^{-1} : B \rightarrow A$  has domain  $B$  and range  $A$ , and is defined by

$$f^{-1}(y) = x \iff f(x) = y \quad \text{for any } y \in B.$$

Consequently

$$\text{dom}(f^{-1}) = \text{ran}(f), \quad \text{ran}(f^{-1}) = \text{dom}(f).$$

#### Note:

The domain and range trade places. This is why  $f$  must be one-to-one before an inverse can exist: if  $f(x_1) = f(x_2)$  with  $x_1 \neq x_2$ , the rule  $f^{-1}$  would have to send that one output to two different inputs, and [Definition 1.4.1](#) forbids it.

#### Note:

The codomain must be taken to be the *range* of  $f$ , not something larger. For  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = x^3$ , the range is all of  $\mathbb{R}$  and no adjustment is needed. But  $f(x) = x^2$  on  $[0, \infty)$  has range  $[0, \infty)$ , so its inverse  $\sqrt{x}$  is defined on  $[0, \infty)$  only.

### 1.8.3 Finding an inverse

To obtain  $f^{-1}$  from  $y = f(x)$ :

1. write  $y = f(x)$ ;
2. solve for  $x$  in terms of  $y$ ;
3. interchange  $x$  and  $y$ , so that the answer is written as a function of  $x$ .

#### Example 1.8.5: Two inverses

Obtain the inverse function of

$$(a) f(x) = x^3 \qquad (b) f(x) = x^3 + 2$$

**Solution:** (a) Put  $y = x^3$  and solve for  $x$ :

$$x = \sqrt[3]{y}.$$

Interchanging  $x$  and  $y$  gives  $y = \sqrt[3]{x}$ , so

$$f^{-1}(x) = \sqrt[3]{x}.$$

(b) Put  $y = x^3 + 2$  and solve for  $x$ :

$$x^3 = y - 2 \implies x = \sqrt[3]{y - 2}.$$

Interchanging  $x$  and  $y$  gives  $y = \sqrt[3]{x - 2}$ , so

$$f^{-1}(x) = \sqrt[3]{x - 2}.$$

**Note:**

Check an inverse by composing: for part (b),

$$f(f^{-1}(x)) = (\sqrt[3]{x-2})^3 + 2 = x - 2 + 2 = x,$$

$$f^{-1}(f(x)) = \sqrt[3]{(x^3 + 2) - 2} = \sqrt[3]{x^3} = x.$$

Both compositions returning  $x$  is exactly what it means for the two functions to undo each other.

### 1.8.4 The graph of an inverse

Since  $f^{-1}(b) = a$  precisely when  $f(a) = b$ , the point  $(b, a)$  lies on the graph of  $f^{-1}$  precisely when  $(a, b)$  lies on the graph of  $f$ . Interchanging the coordinates of a point reflects it in the line  $y = x$ , so:

**Note:**

The graph of  $f^{-1}$  is the reflection of the graph of  $f$  in the line  $y = x$ .

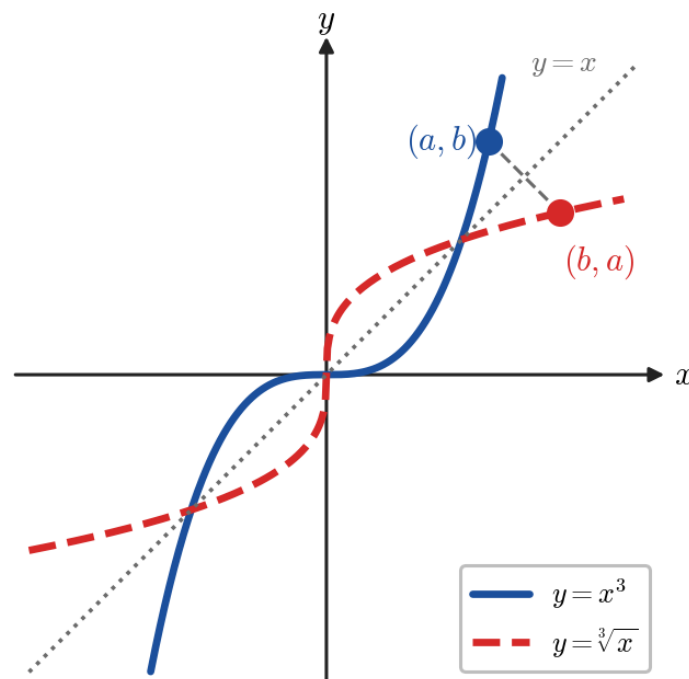


Figure 1.8.2:  $y = x^3$  and  $y = \sqrt[3]{x}$  are mirror images in the dotted line  $y = x$ . The point  $(a, b)$  on one corresponds to  $(b, a)$  on the other.

### 1.8.5 Inverse trigonometric functions

The trigonometric functions are periodic, so they fail the horizontal line test badly. Restricting each one to an interval on which it is one-to-one produces an inverse.

| Restricted function                                      | Inverse                         | Domain $\rightarrow$ range of the inverse                           |
|----------------------------------------------------------|---------------------------------|---------------------------------------------------------------------|
| $\sin x$ on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ | $\sin^{-1} x$ (or $\arcsin x$ ) | $[-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$    |
| $\cos x$ on $[0, \pi]$                                   | $\cos^{-1} x$ (or $\arccos x$ ) | $[-1, 1] \rightarrow [0, \pi]$                                      |
| $\tan x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ | $\tan^{-1} x$ (or $\arctan x$ ) | $\mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ |

So  $\sin^{-1} x$  is *the angle in  $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$  whose sine is  $x$* ; for instance  $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$ ,  $\tan^{-1} 1 = \frac{\pi}{4}$  and  $\cos^{-1} 0 = \frac{\pi}{2}$ . In the same way  $\ln x$  is the inverse of  $e^x$ :  $\ln(e^x) = x$  for all  $x$  and  $e^{\ln x} = x$  for  $x > 0$ .

### Common mistake:

$\sin^{-1} x$  is the *inverse function*, not the reciprocal: in general  $\sin^{-1} x \neq \frac{1}{\sin x}$ . The reciprocal has its own name,  $\operatorname{cosec} x$ . The  $-1$  here means “undo  $\sin$ ”, exactly as in  $f^{-1}$ .

## 1.9 Limits

### 1.9.1 One-sided limits

#### Definition 1.9.1: Left hand limit

We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the *left hand limit* of  $f(x)$  as  $x$  approaches  $a$  is equal to  $L$ , if we can make the values of  $f(x)$  arbitrarily close to  $L$  by taking  $x$  to be sufficiently close to  $a$  and  $x$  less than  $a$ .

#### Definition 1.9.2: Right hand limit

We write

$$\lim_{x \rightarrow a^+} f(x) = L$$

and say the *right hand limit* of  $f(x)$  as  $x$  approaches  $a$  is equal to  $L$ , if we can make the values of  $f(x)$  arbitrarily close to  $L$  by taking  $x$  to be sufficiently close to  $a$  and  $x$  greater than  $a$ .

### Note:

Neither definition says anything about  $f(a)$  itself. The value of the function *at*  $a$  is irrelevant to both one-sided limits; only the values at nearby points on the appropriate side matter. The function need not even be defined at  $a$ .

### Example 1.9.3: A piecewise function at a join

Let

$$f(x) = \begin{cases} x - 3 & \text{if } 1 \leq x \leq 2, \\ 3 & \text{if } 2 < x \leq 4. \end{cases}$$

Find  $\lim_{x \rightarrow 2^+} f(x)$  and  $\lim_{x \rightarrow 2^-} f(x)$ .

**Solution:** For the right hand limit,  $x > 2$  means the second formula applies, so

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2} 3 = 3.$$

For the left hand limit,  $x < 2$  means the first formula applies, so

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2} (x - 3) = 2 - 3 = -1.$$

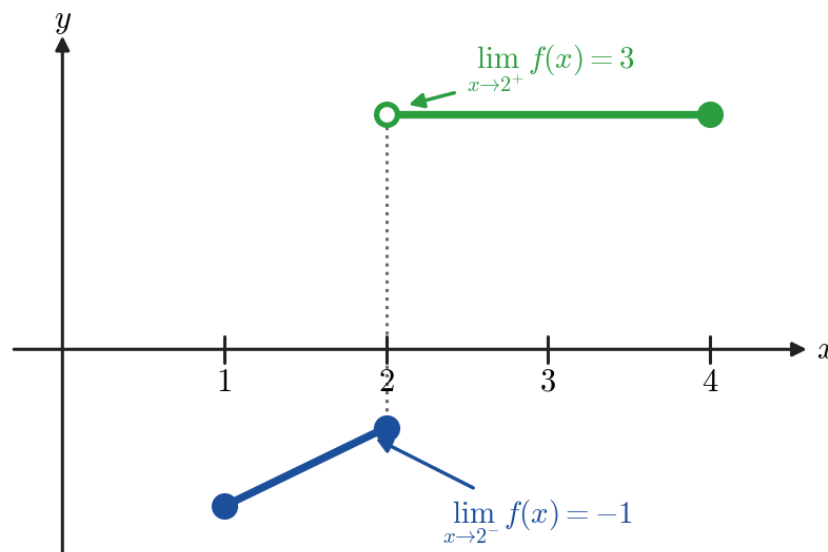


Figure 1.9.1: Approaching  $x = 2$  from the left, the values approach  $-1$ ; approaching from the right, they approach  $3$ . The filled dot marks the value  $f(2) = -1$  that the function actually takes; the open dot marks a height that is approached but never attained.

**Note:**

Here the two one-sided limits are different, so the ordinary two-sided limit  $\lim_{x \rightarrow 2} f(x)$  does *not* exist. Note also that  $\lim_{x \rightarrow 2^-} f(x) = -1$  happens to equal  $f(2)$ , while  $\lim_{x \rightarrow 2^+} f(x) = 3$  does not – further evidence that the limit and the value are separate questions.

## 1.9.2 The two-sided limit

### Definition 1.9.4: Limit of a function

If

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L,$$

then  $L$  is the limit of the function  $f(x)$  as  $x$  tends to  $a$ , and it is denoted by

$$\lim_{x \rightarrow a} f(x) = L.$$

**Note:**

The two-sided limit exists exactly when both one-sided limits exist *and* agree. If either fails, so does the two-sided limit. This is the test used in every example below.

**1.9.3 The  $\varepsilon$ - $\delta$  definition****Definition 1.9.5:  $\varepsilon$ - $\delta$  definition of a limit**

$L$  is the limit of the function  $f(x)$  as  $x \rightarrow a$  if, for every  $\varepsilon > 0$ , there exists  $\delta > 0$  such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - a| < \delta.$$

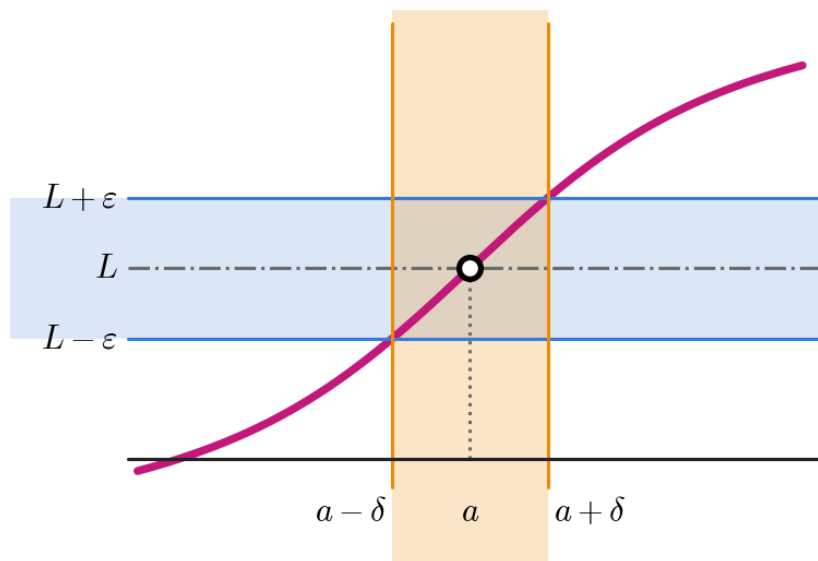


Figure 1.9.2: Given the horizontal band of half-width  $\varepsilon$  about  $L$ , a vertical band of half-width  $\delta$  about  $a$  can be found narrow enough that the graph stays inside the horizontal band.

**Note:**

The order of the quantifiers is the whole content:  $\varepsilon$  is given first and  $\delta$  is found afterwards, so  $\delta$  is allowed to depend on  $\varepsilon$ . The condition  $0 < |x - a|$  excludes  $x = a$  itself, which is why the value  $f(a)$  never affects the limit.

**1.9.4 Limit laws****Theorem 1.9.6: Limit laws**

Suppose that  $c$  is a constant and that

$$\lim_{x \rightarrow a} f(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = L_2$$

both exist. Then:

- (i)  $\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L_1 + L_2,$
- (ii)  $\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L_1 - L_2,$
- (iii)  $\lim_{x \rightarrow a} (c f(x)) = c \lim_{x \rightarrow a} f(x) = cL_1,$
- (iv)  $\lim_{x \rightarrow a} (f(x) g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = L_1 L_2,$
- (v)  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2},$  provided  $L_2 \neq 0.$

### Common mistake:

Every law begins “suppose both limits exist”. The laws cannot manufacture a limit that is not there. In [Problem 1.9.4](#) below,  $\lim_{x \rightarrow 1} f(x)$  does not exist, so no law may be applied to it; and a quotient whose denominator tends to 0 is *not* covered by law (v) at all — it must first be simplified, as in [Problems 1.9.1](#) and [1.9.5](#).

### Theorem 1.9.7: Direct substitution

Let  $f$  be a polynomial, a rational function, a root, a trigonometric, exponential or logarithmic function, or any sum, difference, product, quotient or composite of these. If  $a$  is in the domain of  $f$ , then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

### Note:

For polynomials and rational functions this follows directly from the limit laws; for the other functions it is taken as known. It justifies the last line of most limit calculations: once a troublesome factor has been cancelled, what remains is simply evaluated at  $a$ .

## 1.9.5 The squeeze theorem and a standard trigonometric limit

### Theorem 1.9.8: Squeeze theorem

Suppose  $g(x) \leq f(x) \leq h(x)$  for all  $x$  near  $a$  (except possibly at  $a$  itself), and

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L.$$

Then  $\lim_{x \rightarrow a} f(x) = L$ . The same holds for one-sided limits and for limits as  $x \rightarrow \pm\infty$ .

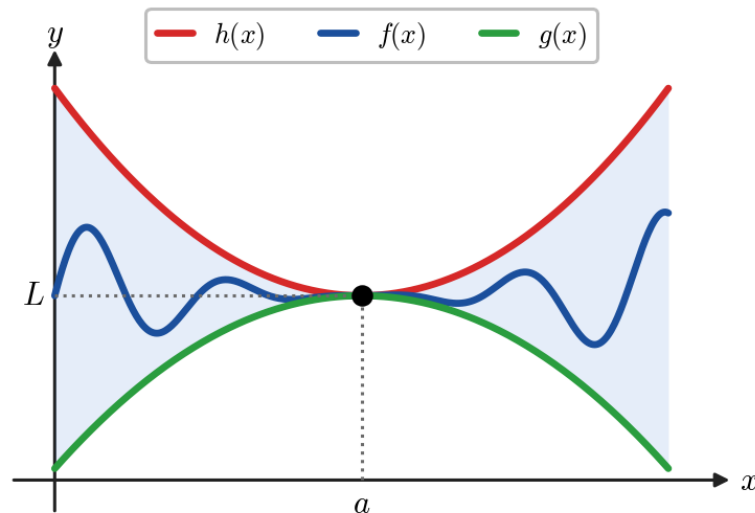


Figure 1.9.3:  $f$  is trapped between  $g$  and  $h$ , and both of those are forced to  $L$  at  $a$ ; so  $f$  has nowhere else to go.

### Corollary 1.9.9: Bounded times vanishing

If  $|B(x)| \leq M$  for all  $x$  near  $a$  and  $\lim_{x \rightarrow a} g(x) = 0$ , then  $\lim_{x \rightarrow a} B(x)g(x) = 0$ .

*Proof.*  $-M|g(x)| \leq B(x)g(x) \leq M|g(x)|$ , and both outer functions tend to 0. Apply [Theorem 1.9.8](#). ■

### Theorem 1.9.10: A standard trigonometric limit

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \quad (\theta \text{ in radians}).$$

*Sketch of proof.* For  $0 < \theta < \frac{\pi}{2}$ , comparing the areas of a triangle, a sector and a larger triangle in the unit circle gives  $\sin \theta < \theta < \tan \theta$ . Dividing by  $\sin \theta > 0$  and inverting,

$$\cos \theta < \frac{\sin \theta}{\theta} < 1.$$

As  $\theta \rightarrow 0^+$ ,  $\cos \theta \rightarrow 1$ , so the squeeze theorem gives the right-hand limit 1. Since  $\frac{\sin \theta}{\theta}$  is an even function, the left-hand limit is the same. ■

#### Note:

Two standard consequences:  $\lim_{x \rightarrow 0} \frac{\sin kx}{x} = k$  (write it as  $k \cdot \frac{\sin kx}{kx}$ ), and  $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$  (write it as  $\frac{\sin x}{x} \cdot \frac{1}{\cos x}$ ).

## 1.9.6 Worked problems on limits

**Problem 1.9.1.** Find  $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ .

**Solution:** Substituting  $x = 2$  gives  $\frac{0}{0}$ , so factor first. Using  $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$ ,

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{x - 2} = \lim_{x \rightarrow 2} (x^2 + 2x + 4) = 4 + 4 + 4 = 12.$$

Cancelling  $(x - 2)$  is legitimate because  $x \neq 2$  throughout the limiting process; only the values near 2 matter, never the value at 2.

**Problem 1.9.2.** Find the limit of  $f(x) = \sin\left(\frac{\pi}{x}\right)$  as  $x \rightarrow 0$ .

**Solution:**

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right) \quad \text{does not exist.}$$

As  $x \rightarrow 0$  the quantity  $\frac{\pi}{x}$  runs through arbitrarily large values, so  $\sin\left(\frac{\pi}{x}\right)$  oscillates between  $-1$  and  $1$  infinitely often and never settles near any single number.

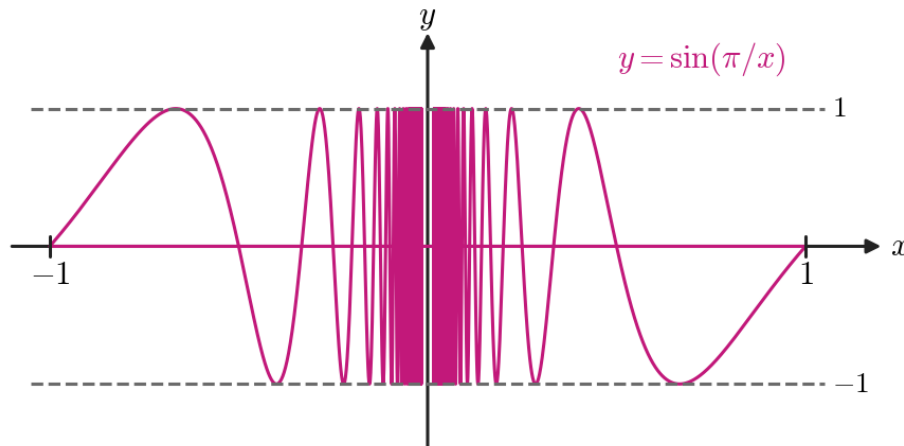


Figure 1.9.4: The oscillations crowd together without diminishing as  $x \rightarrow 0$ , so no value of  $L$  can be approached.

**Problem 1.9.3.** The *Heaviside function*  $H$ , named after the electrical engineer Oliver Heaviside, is defined by

$$H(t) = \begin{cases} 0 & \text{if } t < 0, \\ 1 & \text{if } t \geq 0. \end{cases}$$

Find  $\lim_{t \rightarrow 0} H(t)$ .

**Solution:**

$$\lim_{t \rightarrow 0^-} H(t) = \lim_{t \rightarrow 0^-} 0 = 0, \quad \lim_{t \rightarrow 0^+} H(t) = \lim_{t \rightarrow 0^+} 1 = 1.$$

Since  $\lim_{t \rightarrow 0^-} H(t) \neq \lim_{t \rightarrow 0^+} H(t)$ , the limit of  $H(t)$  does not exist at  $t = 0$ .

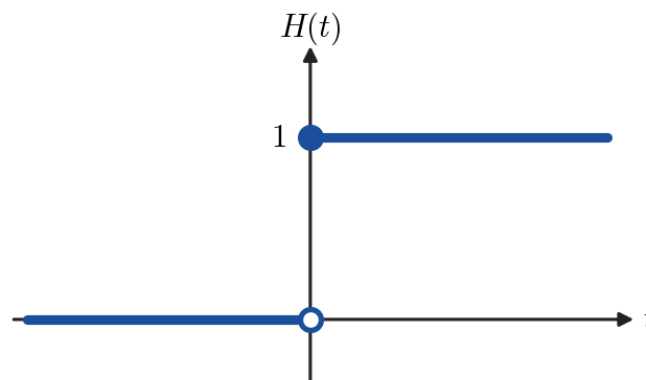


Figure 1.9.5: A jump of size 1 at the origin. The one-sided limits both exist but differ.

**Problem 1.9.4.** Let

$$f(x) = \begin{cases} 2 - x & \text{if } x < -1, \\ x & \text{if } -1 \leq x < 1, \\ (x - 1)^2 & \text{if } x \geq 1. \end{cases}$$

Find  $\lim_{x \rightarrow -1} f(x)$  and  $\lim_{x \rightarrow 1} f(x)$ .

**Solution:** At  $x = -1$ :

$$\text{LHL} = \lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1} (2 - x) = 2 - (-1) = 3,$$

$$\text{RHL} = \lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1} x = -1.$$

Since  $\text{LHL} \neq \text{RHL}$ ,  $\lim_{x \rightarrow -1} f(x)$  does not exist.

At  $x = 1$ , similarly,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1} x = 1, \quad \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1} (x - 1)^2 = 0.$$

Again the two differ, so  $\lim_{x \rightarrow 1} f(x)$  does not exist.

**Problem 1.9.5.** Find  $\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$ .

**Solution:** Substituting  $t = 0$  gives  $\frac{0}{0}$ . Multiply above and below by the conjugate  $\sqrt{t^2 + 9} + 3$ :

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} = \lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} \cdot \frac{\sqrt{t^2 + 9} + 3}{\sqrt{t^2 + 9} + 3} = \lim_{t \rightarrow 0} \frac{t^2 + 9 - 9}{t^2 (\sqrt{t^2 + 9} + 3)},$$

using  $(a - b)(a + b) = a^2 - b^2$  in the numerator. Cancelling  $t^2$ ,

$$= \lim_{t \rightarrow 0} \frac{t^2}{t^2 (\sqrt{t^2 + 9} + 3)} = \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2 + 9} + 3} = \frac{1}{\sqrt{9} + 3} = \frac{1}{6}.$$

**Problem 1.9.6.** Find  $\lim_{x \rightarrow 1} f(x)$ , where

$$f(x) = \begin{cases} x + 1 & \text{if } x \neq 1, \\ \pi & \text{if } x = 1. \end{cases}$$

**Solution:** Here  $f$  is defined at  $x = 1$ , with  $f(1) = \pi$ , but the value of the limit as  $x$  approaches 1 does not depend on  $f(1)$ . For every  $x \neq 1$  the first formula applies, so

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x + 1) = 2.$$

**Note:**

Problems 1.9.2 and 1.9.3 fail for different reasons. In Problem 1.9.3 both one-sided limits exist and are merely unequal a *jump*. In Problem 1.9.2 neither one-sided limit exists at all, because the values oscillate without approaching anything.

**Note:**

[Problem 1.9.6](#) is the clearest illustration of [Definition 1.9.5](#): the condition  $0 < |x - a|$  excludes the point itself, so the limit is 2 no matter what value is assigned at  $x = 1$ . Changing  $\pi$  to any other number would not change the answer.

**1.9.7 Infinite limits****Definition 1.9.11: Infinite limit**

Let  $f$  be defined on both sides of  $a$ , except possibly at  $a$  itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty \quad \text{or} \quad \lim_{x \rightarrow a} f(x) = -\infty$$

means that the values of  $f(x)$  can be made arbitrarily large (or arbitrarily large negative) by taking  $x$  sufficiently close to  $a$ , but not equal to  $a$ .

**Note:**

Writing  $\lim f(x) = \infty$  does *not* mean the limit exists. The symbol  $\infty$  is not a number; the equation is shorthand for a particular way in which the limit fails to exist, namely by growing without bound rather than by oscillating or jumping.

**Definition 1.9.12: Vertical asymptote**

The line  $x = a$  is called a *vertical asymptote* of the curve  $y = f(x)$  if at least one of the following is true:

$$\begin{array}{lll} \lim_{x \rightarrow a^-} f(x) = \infty, & \lim_{x \rightarrow a^+} f(x) = \infty, & \lim_{x \rightarrow a} f(x) = \infty, \\ \lim_{x \rightarrow a^-} f(x) = -\infty, & \lim_{x \rightarrow a^+} f(x) = -\infty, & \lim_{x \rightarrow a} f(x) = -\infty. \end{array}$$

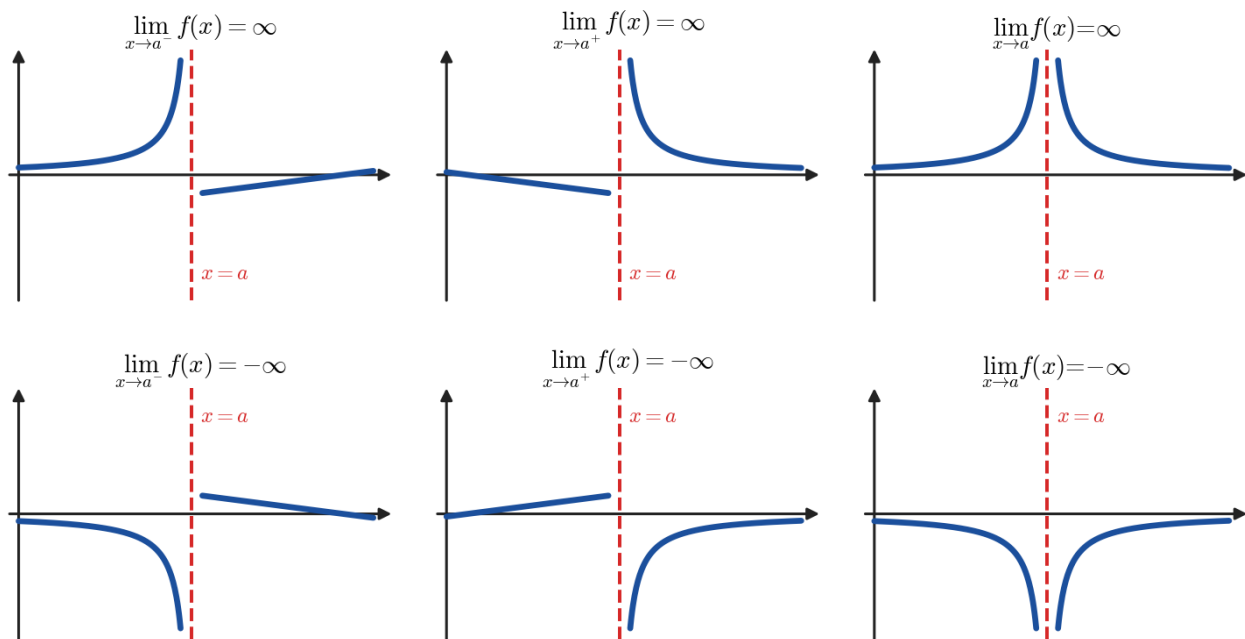


Figure 1.9.6: The six cases. Only one of them need hold for  $x = a$  to be a vertical asymptote the curve may behave quite ordinarily on the other side.

### Example 1.9.13: A vertical asymptote

Consider  $\lim_{x \rightarrow 3} \frac{2x}{x-3}$ .

**Solution:** As  $x \rightarrow 3$  the numerator tends to 6 while the denominator tends to 0, so the quotient grows without bound. Examining the two sides separately:

$$\lim_{x \rightarrow 3^-} \frac{2x}{x-3} = -\infty \quad (\text{denominator negative and small}),$$

$$\lim_{x \rightarrow 3^+} \frac{2x}{x-3} = +\infty \quad (\text{denominator positive and small}).$$

Since at least one of these is infinite, the vertical line  $x = 3$  is an asymptote to the curve  $f(x) = \frac{2x}{x-3}$ .

### Common mistake:

The two one-sided limits here have *opposite* signs, so the two-sided statement  $\lim_{x \rightarrow 3} \frac{2x}{x-3} = \infty$  would not be correct: the function does not grow large in one consistent direction. [Definition 1.9.12](#) requires only *one* of its six conditions, and the two one-sided ones are what hold here. Checking the sign of the denominator on each side is the way to tell the cases apart.

### 1.9.8 Limits at infinity and horizontal asymptotes

#### Definition 1.9.14: Limit at infinity

Let  $f$  be a function defined on some interval  $(a, \infty)$ . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that the values of  $f(x)$  can be made arbitrarily close to  $L$  by taking  $x$  sufficiently large.

Similarly, if  $f$  is defined on some interval  $(-\infty, a)$ , then

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that the values of  $f(x)$  can be made arbitrarily close to  $L$  by taking  $x$  sufficiently large and negative.

#### Definition 1.9.15: Horizontal asymptote

In either case above, the horizontal line  $y = L$  is called a *horizontal asymptote* of the curve  $y = f(x)$ .

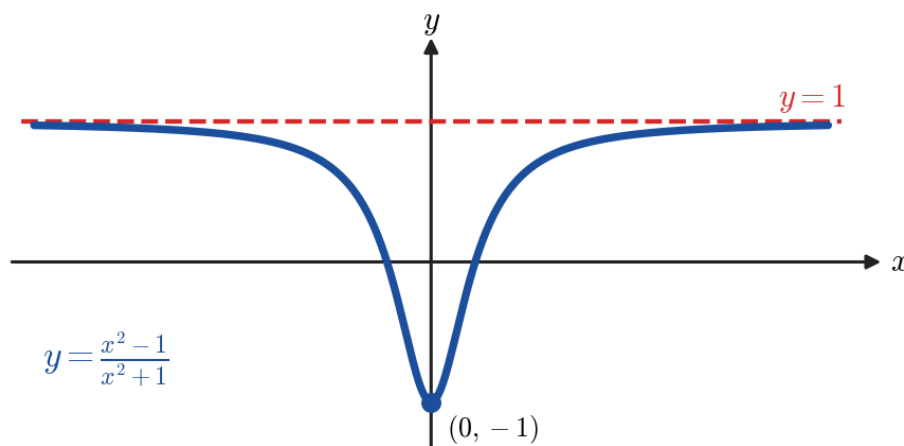


Figure 1.9.7: The curve settles towards the line  $y = 1$  in both directions without ever reaching it.

#### Note:

The basic limit at infinity is  $\lim_{x \rightarrow \pm\infty} \frac{1}{x^p} = 0$  for every  $p > 0$  (for  $x \rightarrow -\infty$ , whenever  $x^p$  is defined), since  $x^p$  grows without bound. Dividing the numerator and denominator of a rational function by the highest power of  $x$  present turns every term but the leading ones into this form.

#### Example 1.9.16: Three limits at infinity

(a)  $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^2 + 1}.$

**Solution:** Divide numerator and denominator by the highest power present,  $x^2$ :

$$\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^2 + 1} = \lim_{x \rightarrow \infty} \frac{x^2 \left(1 - \frac{1}{x^2}\right)}{x^2 \left(1 + \frac{1}{x^2}\right)} = \lim_{x \rightarrow \infty} \frac{1 - \frac{1}{x^2}}{1 + \frac{1}{x^2}} = \frac{1 - 0}{1 + 0} = 1.$$

Hence  $y = 1$  is a horizontal asymptote to the curve  $f(x) = \frac{x^2 - 1}{x^2 + 1}$ .

(b)  $\lim_{x \rightarrow \infty} x^2 \sin\left(\frac{1}{x}\right).$

**Solution:** Put  $u = \frac{1}{x}$ , so that  $u \rightarrow 0^+$  as  $x \rightarrow \infty$  and

$$x^2 \sin\left(\frac{1}{x}\right) = \frac{\sin u}{u^2} = \frac{\sin u}{u} \cdot \frac{1}{u}.$$

As  $u \rightarrow 0^+$  the first factor tends to 1 ([Theorem 1.9.10](#)) while the second grows without bound, so

$$\lim_{x \rightarrow \infty} x^2 \sin\left(\frac{1}{x}\right) = \infty.$$

The limit therefore does not exist as a finite number, and there is no horizontal asymptote.

(c)  $\lim_{x \rightarrow \infty} \frac{\sin x}{x}.$

**Solution:** Since  $-1 \leq \sin x \leq 1$  for all  $x$ , dividing by  $x > 0$  gives

$$-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}.$$

Both outer expressions tend to 0 as  $x \rightarrow \infty$ , so by the squeeze theorem ([Theorem 1.9.8](#))

$$\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0,$$

and  $y = 0$  is a horizontal asymptote to the curve  $f(x) = \frac{\sin x}{x}$ .

### Note:

Parts (b) and (c) both involve a bounded oscillating factor, but they behave oppositely. In (c) the oscillation is divided by something growing, so it is crushed to zero; in (b) it is multiplied by something growing, so it is amplified without bound. The size of the other factor decides the outcome.

## 1.10 Continuity

### Definition 1.10.1: Continuity at a point

If  $\lim_{x \rightarrow a^-} f(x)$  and  $\lim_{x \rightarrow a^+} f(x)$  exist and are equal to the value of the function at  $x = a$ , then  $f$  is *continuous* at  $x = a$ ; that is,

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a).$$

Equivalently, if

$$\lim_{x \rightarrow a} f(x) = f(a),$$

then  $f$  is continuous at  $x = a$ .

### Note:

Three separate things must hold:  $f(a)$  must be defined,  $\lim_{x \rightarrow a} f(x)$  must exist, and the two must be equal. Failing any one of them makes  $f$  discontinuous at  $a$ , and the three failures look different.

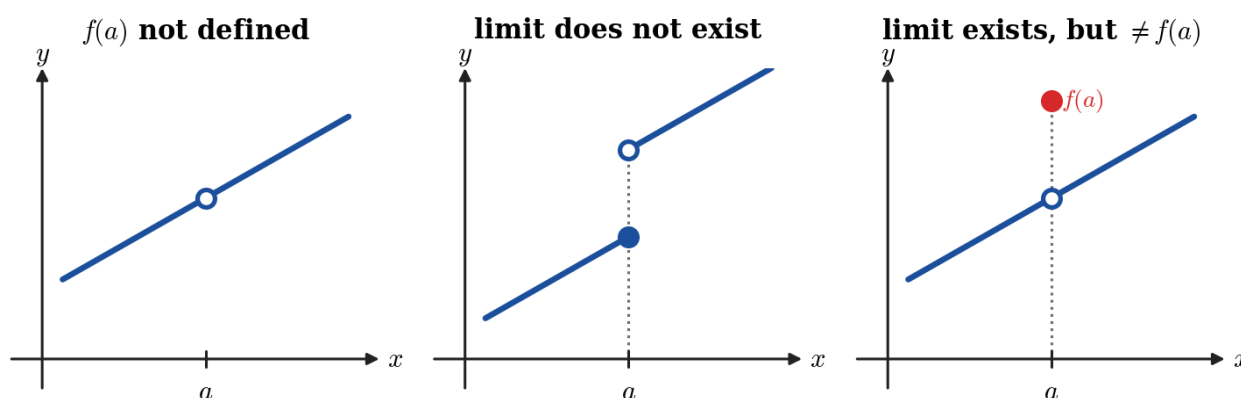


Figure 1.10.1: The three ways continuity can fail at  $a$ . Only in the middle case does the limit itself not exist.

### Definition 1.10.2: Continuity on an interval

$f$  is *continuous on an open interval*  $(a, b)$  if it is continuous at every point of  $(a, b)$ . It is *continuous on the closed interval*  $[a, b]$  if it is continuous on  $(a, b)$  and, at the endpoints,

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow b^-} f(x) = f(b).$$

### Note:

[Theorem 1.9.7](#) says exactly that every polynomial, rational, root, trigonometric, exponential and logarithmic function — and every sum, product, quotient and composite of them — is continuous at each point of its domain. Continuity has to be checked by hand only where a formula changes (piecewise functions) or where a value has been assigned separately. Continuity on a closed interval is the hypothesis of the theorems of Unit 2.

### 1.10.1 Worked problems

**Problem 1.10.1.** Check the continuity of  $f(x) = 1 - \sqrt{1 - x^2}$  on the interval  $[-1, 1]$ .

**Solution:** Let  $-1 \leq a \leq 1$ . By direct substitution ([Theorem 1.9.7](#)),

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} (1 - \sqrt{1 - x^2}) = 1 - \sqrt{1 - a^2} = f(a).$$

Hence  $f(x)$  is continuous on  $[-1, 1]$ .

**Note:**

At the two endpoints  $a = \pm 1$  only a one-sided limit is available, since  $f$  is not defined outside  $[-1, 1]$ . Continuity at an endpoint of a closed interval means one-sided continuity, which is what the calculation above establishes there.

**Problem 1.10.2.** Check the continuity of  $f(x) = |x|$  at  $x = 0$ .

**Solution:** Writing the absolute value piecewise,

$$|x| = \begin{cases} x & \text{if } x > 0, \\ -x & \text{if } x \leq 0, \end{cases}$$

the one-sided limits are

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-x) = 0, \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x) = 0.$$

Since

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = 0 = f(0),$$

the function  $|x|$  is continuous at  $x = 0$ .

**Problem 1.10.3.** Check the continuity of  $f$  at  $x = 0$ , where

$$f(x) = \begin{cases} x^3 + 3 & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$$

**Solution:** Both one-sided limits use the first formula:

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^3 + 3) = 3, \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^3 + 3) = 3.$$

So the limit exists and equals 3, but

$$\lim_{x \rightarrow 0} f(x) = 3 \neq 1 = f(0).$$

Therefore  $f(x)$  is *not* continuous at  $x = 0$ .

**Note:**

This is the third failure in [Fig. 1.10.1](#): the limit exists and the value exists, but they disagree. Redefining  $f(0) = 3$  would repair it, which is why such a point is called a *removable discontinuity*.

**Problem 1.10.4.** Check the continuity of  $f$  at  $x = 0$ , where

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

**Solution:** The factor  $\sin\left(\frac{1}{x}\right)$  oscillates but stays between  $-1$  and  $1$ , so for all  $x \neq 0$

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2.$$

Both outer expressions tend to 0 as  $x \rightarrow 0$ , so by the squeeze theorem ([Theorem 1.9.8](#))

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$

Since this common value equals  $f(0) = 0$ , the function is continuous at  $x = 0$ .

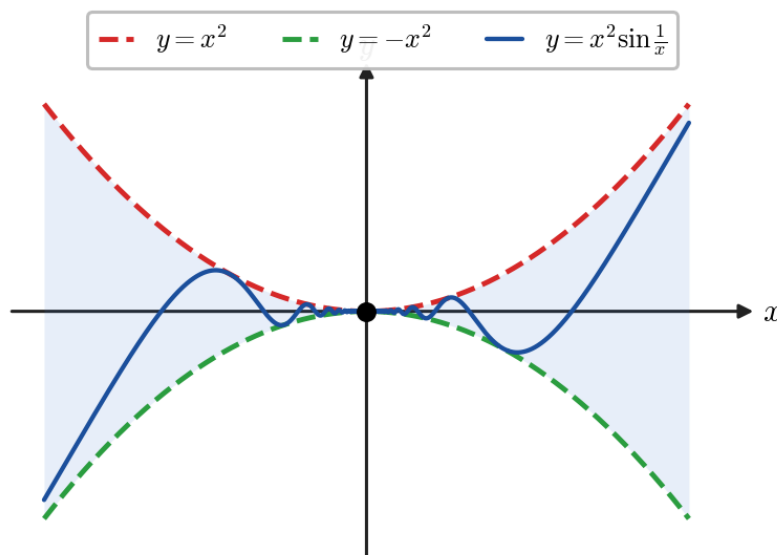


Figure 1.10.2: The oscillation is trapped between  $-x^2$  and  $x^2$ , which pinch together at the origin and force the limit to be 0.

**Note:**

Compare this with [Fig. 1.9.4](#), where  $\sin\left(\frac{\pi}{x}\right)$  alone has no limit at 0. Multiplying by  $x^2$  shrinks the amplitude to nothing, and the limit appears. A bounded factor times something tending to zero tends to zero ([Corollary 1.9.9](#)).

## 1.10.2 The $\varepsilon$ - $\delta$ definition of continuity

### Definition 1.10.3: $\varepsilon$ - $\delta$ continuity

The function  $f(x)$  is continuous at the point  $x = a$  if and only if, for every  $\varepsilon > 0$ , there exists  $\delta > 0$  such that

$$|f(x) - f(a)| < \varepsilon \quad \text{whenever} \quad |x - a| < \delta.$$

**Note:**

This is [Definition 1.9.5](#) with  $L$  replaced by  $f(a)$ , and with the restriction  $0 < |x - a|$  dropped. Allowing  $x = a$  costs nothing here, since  $|f(a) - f(a)| = 0 < \varepsilon$  always holds. That single change is exactly the difference between having a limit and being continuous.

**Example 1.10.4: A continuity proof from the definition**

Show that  $f(x) = 2x + 3$  is continuous at  $x = -5$ .

**Solution:** Since  $f(-5) = 2(-5) + 3 = -10 + 3 = -7$ , we need to show that for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that

$$|f(x) + 7| < \varepsilon \quad \text{whenever} \quad |x + 5| < \delta.$$

Consider the left-hand side:

$$|f(x) + 7| = |2x + 3 + 7| = |2x + 10| = 2|x + 5|.$$

Therefore

$$|f(x) + 7| < \varepsilon \iff 2|x + 5| < \varepsilon \iff |x + 5| < \frac{\varepsilon}{2}.$$

Choosing  $\delta = \frac{\varepsilon}{2}$ , we get

$$|f(x) + 7| < \varepsilon \quad \text{whenever} \quad |x + 5| < \delta,$$

which is what was required. Hence  $f$  is continuous at  $x = -5$ .

**Note:**

The working runs backwards from the goal to find  $\delta$ , then the final sentence states it forwards. That is the standard shape of such a proof: the algebra discovers  $\delta = \varepsilon/2$ , and the last line verifies that this choice does what was asked.

**Example 1.10.5: A continuity proof needing a bound**

Show that  $f(x) = x^2 + 8$  is continuous at  $x = 2$ .

**Solution:** Since  $f(2) = 4 + 8 = 12$ , we need to show that for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that

$$|f(x) - 12| < \varepsilon \quad \text{whenever} \quad |x - 2| < \delta.$$

Consider the left-hand side:

$$|f(x) - 12| = |x^2 + 8 - 12| = |x^2 - 4| = |x + 2||x - 2|.$$

The factor  $|x - 2|$  can be made small by choosing  $\delta$ , but  $|x + 2|$  cannot – it depends on where  $x$  is. So we first bound it. Restrict attention to  $|x - 2| < 1$ , which gives  $1 < x < 3$  and hence

$$3 < x + 2 < 5 \implies |x + 2| < 5.$$

With this restriction in force,

$$|f(x) - 12| = |x + 2| |x - 2| < 5 |x - 2|,$$

and  $5 |x - 2| < \varepsilon$  precisely when  $|x - 2| < \frac{\varepsilon}{5}$ . Choosing

$$\delta = \min \left\{ 1, \frac{\varepsilon}{5} \right\}$$

makes both requirements hold at once, and then

$$|f(x) - 12| < \varepsilon \quad \text{whenever} \quad |x - 2| < \delta,$$

so  $f$  is continuous at  $x = 2$ .

#### Note:

The min is not decoration. The bound  $|x + 2| < 5$  is only valid while  $|x - 2| < 1$ , so  $\delta$  must be at most 1 as well as at most  $\varepsilon/5$ ; taking the smaller of the two secures both. Compare [Example 1.10.4](#), where  $f$  was linear,  $|f(x) - f(a)|$  was exactly  $2|x - a|$ , and no preliminary bound was needed.

## 1.11 Types of Discontinuity

### Definition 1.11.1: Removable, jump and infinite discontinuity

Let  $f$  fail to be continuous at  $x = a$ .

- $f$  has a *removable discontinuity* at  $x = a$  if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) \neq f(a),$$

including the case where  $f(a)$  is not defined at all.

- $f$  has a *jump discontinuity* at  $x = a$  if  $\lim_{x \rightarrow a^-} f(x)$  and  $\lim_{x \rightarrow a^+} f(x)$  both exist but are not equal:

$$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x).$$

- $f$  has an *infinite discontinuity* at  $x = a$  if  $\lim_{x \rightarrow a^-} f(x) = \pm\infty$  or  $\lim_{x \rightarrow a^+} f(x) = \pm\infty$ .

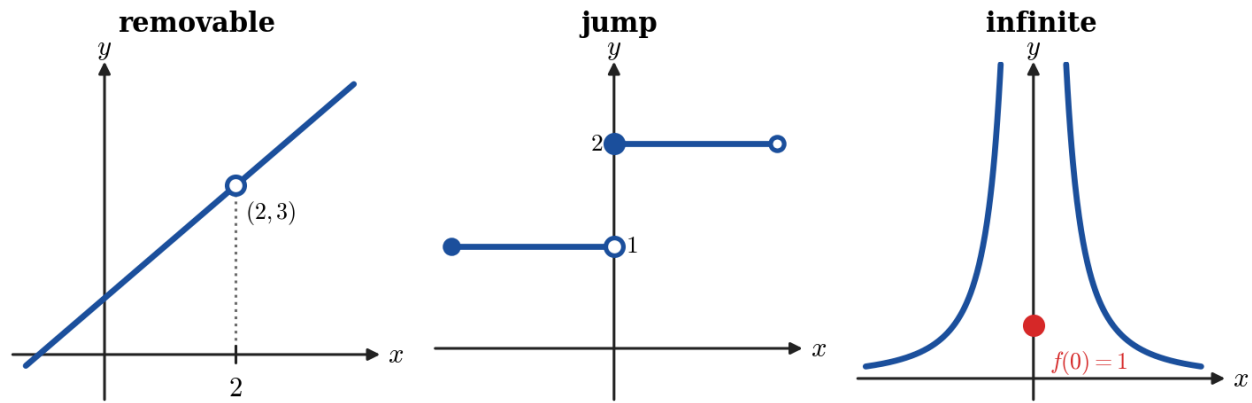


Figure 1.11.1: The three types, drawn on the functions of the worked problems below. Only the removable one can be repaired by changing the function at a single point.

### Note:

The name *removable* is literal: the limit already exists, so redefining  $f(a)$  to equal that limit produces a continuous function. In the other two cases no single value assigned at  $a$  can help in the jump case the two sides disagree, and in the infinite case there is nothing finite to assign.

## 1.11.1 Worked problems

**Problem 1.11.1.** Determine the type of discontinuity of each of the following functions.

(a)  $f(x) = \frac{x^2 - x - 2}{x - 2}.$

**Solution:** At  $x = 2$  the formula gives  $\frac{0}{0}$ , so  $f(2)$  is not defined. Factoring the numerator as  $x^2 - x - 2 = (x - 2)(x + 1)$ ,

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3.$$

Both one-sided limits equal 3, but  $f(2)$  does not exist, so at  $x = 2$  the function  $f$  has a *removable discontinuity*. Redefining

$$f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2, \\ 3 & \text{if } x = 2, \end{cases}$$

makes the function continuous.

(b)  $f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$

**Solution:**

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1}{x^2} = \infty, \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{x^2} = \infty.$$

Since the one-sided limits are infinite,  $f$  has an *infinite discontinuity* at  $x = 0$ .

**Note:**

Both sides give  $+\infty$  here because  $x^2 > 0$  whichever side  $x$  approaches from. Contrast  $\frac{1}{x}$ , whose two sides give  $-\infty$  and  $+\infty$ ; both are infinite discontinuities, but they look different.

$$(c) f(x) = \begin{cases} 1 & \text{if } -5 \leq x < 0, \\ 2 & \text{if } 0 \leq x < 5. \end{cases}$$

**Solution:**

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0} (1) = 1, \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} (2) = 2.$$

Both limits exist but  $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$ , so  $f$  has a *jump discontinuity* at  $x = 0$ .

**Note:**

In part (a) no value is assigned at  $x = 2$ ; in [Problem 1.10.3](#) a value *is* assigned but it is the wrong one. Both discontinuities are removable, which is why [Definition 1.11.1](#) covers the undefined case explicitly: the limit existing is what matters, not whether  $f(a)$  happens to be written down.

## 1.12 Differentiation

### Definition 1.12.1: Derivative

A function  $f(x)$  is said to be *differentiable* at  $x$  if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists. The limit is then denoted by  $f'(x)$ , so that

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

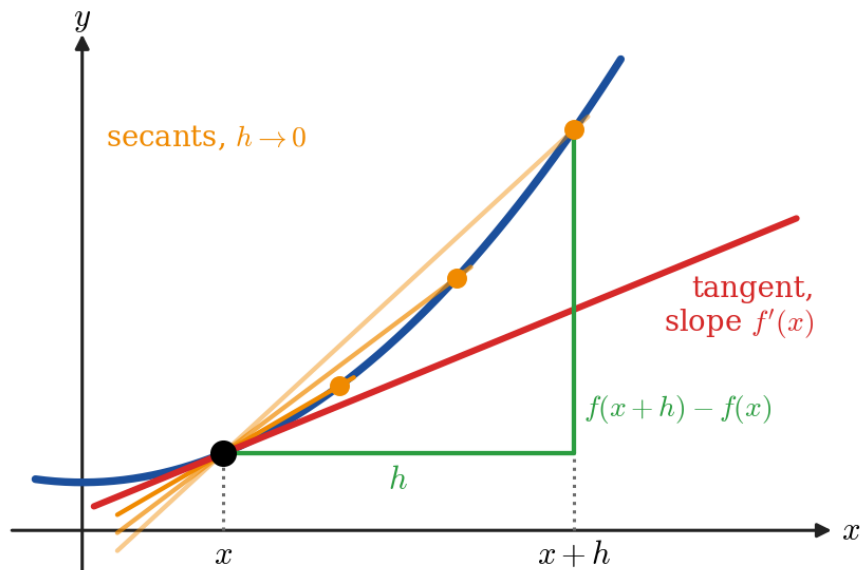


Figure 1.12.1: The quotient  $\frac{f(x+h) - f(x)}{h}$  is the slope of the secant through  $(x, f(x))$  and  $(x+h, f(x+h))$ . As  $h \rightarrow 0$  the second point slides towards the first and the secants turn into the tangent, whose slope is  $f'(x)$ .

**Note:**

The quotient is  $\frac{0}{0}$  at  $h = 0$ , which is why the derivative has to be defined as a *limit* rather than by substitution. Everything about limits in this unit exists to make this one definition work.

**Example 1.12.2: A derivative from the definition**

Find  $f'(x)$  for  $f(x) = x^2$ .

**Solution:**

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \quad (h \neq 0).$$

Letting  $h \rightarrow 0$  gives  $f'(x) = 2x$ .

### 1.12.1 Differentiability and continuity

**Theorem 1.12.3: Differentiable functions are continuous**

If  $f$  is differentiable at  $a$ , then  $f$  is continuous at  $a$ .

*Proof.* For  $x \neq a$ ,

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a).$$

As  $x \rightarrow a$  the first factor tends to  $f'(a)$  and the second to 0, so by limit law (iv) the product tends to  $f'(a) \cdot 0 = 0$ . Hence  $\lim_{x \rightarrow a} f(x) = f(a)$ . ■

**Common mistake:**

The converse of [Theorem 1.12.3](#) is false: a continuous function need not be differentiable.  $f(x) = |x|$  is continuous at 0 — shown in [Problem 1.10.2](#) — yet the quotient  $\frac{|h|}{h}$  equals  $-1$  for  $h < 0$  and  $+1$  for  $h > 0$ , so the limit does not exist and  $|x|$  is not differentiable at 0. A corner is enough to destroy differentiability while leaving continuity intact.

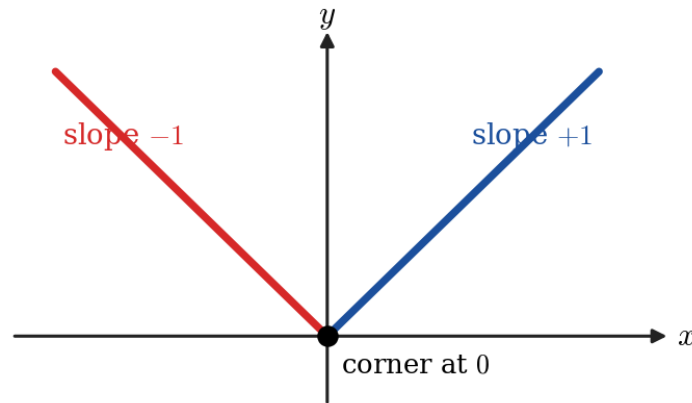


Figure 1.12.2: The graph of  $|x|$  has slope  $-1$  on the left and  $+1$  on the right. At the corner no single tangent line fits, so  $f'(0)$  does not exist.

### 1.12.2 The tangent line and linear approximation

Since  $f'(a)$  is the slope of the tangent at  $(a, f(a))$ , the tangent line there is

$$y = f(a) + f'(a)(x - a).$$

**Definition 1.12.4: Linearisation (one variable)**

The function  $L(x) = f(a) + f'(a)(x - a)$  is the *linearisation* of  $f$  at  $a$ , and the approximation

$$f(x) \approx f(a) + f'(a)(x - a) \quad \text{for } x \text{ near } a$$

is the *tangent line* (or *linear*) *approximation*.

**Example 1.12.5: Estimating a square root**

Estimate  $\sqrt{4.1}$ .

**Solution:** Take  $f(x) = \sqrt{x}$  and  $a = 4$ , a nearby point where everything is known exactly:  $f(4) = 2$  and  $f'(x) = \frac{1}{2\sqrt{x}}$ , so  $f'(4) = \frac{1}{4}$ . Then

$$\sqrt{4.1} \approx f(4) + f'(4)(4.1 - 4) = 2 + \frac{0.1}{4} = 2.025.$$

The true value is  $2.02485\dots$ , so the estimate is correct to three decimal places.

**Note:**

The approximation is good only near  $a$ : the graph bends away from its tangent line as  $x$  moves off. Unit 2 improves on it with Taylor polynomials, which match more

derivatives at  $a$ , and Unit 3 extends it to the tangent *plane* of a surface.

### 1.12.3 Rules and standard derivatives (for reference)

The rules and derivatives below are used without comment in Units 2 and 3. Higher derivatives are obtained by repetition:  $f'' = (f')'$ ,  $f''' = (f'')'$ , and  $f^{(n)}$  denotes the  $n$ th derivative.

| Rules             |                                         |          |                 |                                                     |                           |
|-------------------|-----------------------------------------|----------|-----------------|-----------------------------------------------------|---------------------------|
| Sum               | $(f \pm g)' = f' \pm g'$                |          | Product         | $(fg)' = f'g + fg'$                                 |                           |
| Constant multiple | $(cf)' = cf'$                           |          | Quotient        | $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ |                           |
| Chain rule:       | $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$ |          |                 |                                                     |                           |
| <hr/>             |                                         |          |                 |                                                     |                           |
| $f(x)$            | $f'(x)$                                 | $f(x)$   | $f'(x)$         | $f(x)$                                              | $f'(x)$                   |
| $x^n$             | $nx^{n-1}$                              | $\sin x$ | $\cos x$        | $\sin^{-1} x$                                       | $\frac{1}{\sqrt{1-x^2}}$  |
| $e^x$             | $e^x$                                   | $\cos x$ | $-\sin x$       | $\cos^{-1} x$                                       | $-\frac{1}{\sqrt{1-x^2}}$ |
| $a^x$             | $a^x \ln a$                             | $\tan x$ | $\sec^2 x$      | $\tan^{-1} x$                                       | $\frac{1}{1+x^2}$         |
| $\ln x$           | $\frac{1}{x}$                           | $\sec x$ | $\sec x \tan x$ |                                                     |                           |

#### Note:

**The sign of the derivative.** If  $f'(x) > 0$  at every point of an interval  $I$ , then  $f$  is increasing on  $I$ ; if  $f'(x) < 0$  throughout  $I$ , then  $f$  is decreasing on  $I$  (Definition 1.4.12). This is how “decreasing” is checked in practice, and it is a consequence of Lagrange’s Mean Value Theorem in Unit 2.

## 1.13 Unit 1 Summary

### Sets, numbers and space

- De Morgan:  $(A \cup B)^c = A^c \cap B^c$ ,  $(A \cap B)^c = A^c \cup B^c$ .  $|A \times B| = |A| |B|$ .
- $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ , every inclusion proper;  $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'$ .
- Distance in  $\mathbb{R}^n$ :  $d(\mathbf{a}, \mathbf{b}) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$ .

### Functions

- Natural domain: even root  $\Rightarrow$  argument  $\geq 0$ ; denominator  $\neq 0$ ; logarithm  $\Rightarrow$  argument  $> 0$ .
- Vertical line test: is it a function? Horizontal line test: is it one-to-one?
- Even:  $f(-x) = f(x)$  (symmetric in the  $y$ -axis). Odd:  $f(-x) = -f(x)$  (symmetric about the origin).

- Transformations: changes to the *output* act as expected; changes to the *input* act in the opposite sense.
- Inverse: write  $y = f(x)$ , solve for  $x$ , swap  $x$  and  $y$ .  $\text{dom}(f^{-1}) = \text{ran}(f)$ ; the graph is reflected in  $y = x$ .
- $\sin^{-1} : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$ ,  $\cos^{-1} : [-1, 1] \rightarrow [0, \pi]$ ,  $\tan^{-1} : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$ .

### Limits

- $\lim_{x \rightarrow a} f(x) = L \iff$  both one-sided limits exist and equal  $L$ . The value  $f(a)$  plays no part.
- Limit laws need both limits to exist; direct substitution works for standard functions at points of their domain.
- $\frac{0}{0}$ : factor and cancel, or multiply by the conjugate. At  $\pm\infty$ : divide by the highest power, using  $\frac{1}{x^p} \rightarrow 0$  ( $p > 0$ ).
- Squeeze:  $g \leq f \leq h$ ,  $g, h \rightarrow L \Rightarrow f \rightarrow L$ . Bounded  $\times$  (something  $\rightarrow 0$ )  $\rightarrow 0$ .
- $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ ,  $\lim_{x \rightarrow 0} \frac{\sin kx}{x} = k$ .
- Vertical asymptote  $x = a$ : some one-sided limit is  $\pm\infty$ . Horizontal asymptote  $y = L$ :  $\lim_{x \rightarrow \infty} f = L$  or  $\lim_{x \rightarrow -\infty} f = L$ .

### Continuity and the derivative

- Continuous at  $a$ :  $f(a)$  defined,  $\lim_{x \rightarrow a} f(x)$  exists, and the two are equal.
- Discontinuities: *removable* (limit exists,  $\neq f(a)$  or  $f(a)$  undefined); *jump* (one-sided limits differ); *infinite* (a one-sided limit is  $\pm\infty$ ).
- $\varepsilon$ - $\delta$ : find  $\delta$  in terms of  $\varepsilon$ ; bound any extra factor first and take  $\delta = \min\{1, \cdot\}$ .
- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ . Differentiable  $\Rightarrow$  continuous, not conversely ( $|x|$  at 0).
- Tangent line / linear approximation:  $f(x) \approx f(a) + f'(a)(x - a)$  for  $x$  near  $a$ .

## 1.14 Practice Problems

The problems are arranged in three levels. Level A checks definitions and understanding, level B practises the standard methods, and level C asks you to analyse, justify or interpret a result, including in applied settings. Answers are given in brackets; full solutions are not.

### A. Check your understanding

**P1.** Classify as even, odd or neither: (a)  $x^4 - 3x^2$ , (b)  $x^3 - \frac{1}{x}$ , (c)  $x + \cos x$ , (d)  $x \sin x$ .

[Ans. (a) even (b) odd (c) neither (d) even]

**P2.** Evaluate  $\sin^{-1}(-\frac{1}{2})$ ,  $\tan^{-1} \sqrt{3}$  and  $\cos^{-1} 0$ .

[Ans.  $-\frac{\pi}{6}$ ,  $\frac{\pi}{3}$ ,  $\frac{\pi}{2}$ ]

**P3.** Describe how the graph of  $y = -(x + 2)^2 + 3$  is obtained from  $y = x^2$ .

[**Ans.** shift left 2, reflect in the  $x$ -axis, shift up 3; vertex  $(-2, 3)$ ]

**P4.** Classify the discontinuity of (a)  $\frac{x^2 - 1}{x - 1}$  at  $x = 1$ , (b)  $\frac{|x|}{x}$  at  $x = 0$ , (c)  $\frac{1}{x + 1}$  at  $x = -1$ .

[**Ans.** (a) removable (limit 2) (b) jump (c) infinite]

**P5.** True or false, with a reason: (a) if  $\lim_{x \rightarrow a} f(x)$  exists then  $f(a)$  is defined; (b) every differentiable function is continuous; (c) every continuous function is differentiable.

[**Ans.** (a) F (b) T (c) F]

## B. Standard problems

**P6.** Let  $U = \{1, 2, \dots, 10\}$ ,  $A$  the even numbers in  $U$  and  $B$  the multiples of 3 in  $U$ . Find  $A \cap B$ ,  $A \cup B$  and  $(A \cup B)^c$ .

[**Ans.**  $\{6\}$ ;  $\{2, 3, 4, 6, 8, 9, 10\}$ ;  $\{1, 5, 7\}$ ]

**P7.** Find the natural domain of (a)  $\sqrt{4 - x^2}$ , (b)  $\frac{1}{x^2 - 5x + 6}$ , (c)  $\ln(x - 1) + \sqrt{3 - x}$ .

[**Ans.** (a)  $[-2, 2]$  (b)  $\mathbb{R} \setminus \{2, 3\}$  (c)  $(1, 3]$ ]

**P8.** Find the inverse of  $f(x) = \frac{2x + 1}{x - 3}$ ,  $x \neq 3$ .

[**Ans.**  $f^{-1}(x) = \frac{3x + 1}{x - 2}$ ,  $x \neq 2$ ]

**P9.** For  $f(x) = \sqrt{x}$  and  $g(x) = x^2 - 4$ , find  $f \circ g$  and  $g \circ f$  with their domains.

[**Ans.**  $\sqrt{x^2 - 4}$ ,  $|x| \geq 2$ ;  $x - 4$ ,  $x \geq 0$ ]

**P10.**  $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

[**Ans.** 6]

**P11.**  $\lim_{x \rightarrow 0} \frac{\sqrt{x + 4} - 2}{x}$

[**Ans.**  $\frac{1}{4}$ ]

**P12.**  $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$

[**Ans.** 5]

**P13.**  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$  (Hint:  $1 - \cos x = 2 \sin^2 \frac{x}{2}$ .)

[**Ans.**  $\frac{1}{2}$ ]

**P14.**  $\lim_{x \rightarrow \infty} \frac{3x^2 - x + 2}{5x^2 + 4}$

[**Ans.**  $\frac{3}{5}$ ]

**P15.**  $\lim_{x \rightarrow 0} x \cos \frac{1}{x}$

[**Ans.** 0, by Corollary 1.9.9]

**P16.** Find  $\lim_{x \rightarrow 1} f(x)$  for  $f(x) = x^2$  ( $x < 1$ ),  $f(x) = 2 - x$  ( $x \geq 1$ ).

[**Ans.** LHL = RHL = 1, so the limit is 1]

**P17.** Find the vertical and horizontal asymptotes of  $y = \frac{2x + 1}{x - 2}$ , and the one-sided limits at  $x = 2$ .

[**Ans.**  $x = 2$ ,  $y = 2$ ;  $-\infty$  from the left,  $+\infty$  from the right]

**P18.** Find  $k$  so that  $f(x) = kx + 1$  ( $x \leq 2$ ),  $f(x) = x^2 - 1$  ( $x > 2$ ) is continuous at  $x = 2$ .

[**Ans.**  $k = 1$ ]

**P19.** From the definition, find  $f'(x)$  for (a)  $f(x) = \frac{1}{x}$ , (b)  $f(x) = \sqrt{x}$ .

[ **Ans.** (a)  $-\frac{1}{x^2}$  (b)  $\frac{1}{2\sqrt{x}}$  ]

**P20.** Find the tangent line to  $y = x^3 - 2x$  at  $x = 1$ .

[ **Ans.**  $y = x - 2$  ]

**P21.** Use linear approximation to estimate (a)  $\sqrt[3]{8.1}$ , (b)  $e^{0.02}$ .

[ **Ans.** (a)  $2 + \frac{0.1}{12} \approx 2.0083$  (b)  $1.02$  ]

### C. Analyse, justify and apply

**P22.** Prove from the  $\varepsilon$ - $\delta$  definition that  $f(x) = 3x - 2$  is continuous at  $x = 1$ .

[ **Ans.**  $\delta = \varepsilon/3$  works ]

**P23.** Is  $f(x) = x^2 \sin \frac{1}{x}$  ( $x \neq 0$ ),  $f(0) = 0$  differentiable at 0?

[ **Ans.** yes,  $f'(0) = \lim_{h \rightarrow 0} h \sin \frac{1}{h} = 0$  ]

**P24.** Show that  $|x - 1|$  is continuous but not differentiable at  $x = 1$ .

[ **Ans.** one-sided difference quotients are  $-1$  and  $+1$  ]

**P25.** Construct a function that is continuous everywhere but not differentiable at  $x = 1$  and at  $x = 2$ . Justify your choice.

[ **Ans.** e.g.  $|x - 1| + |x - 2|$  ]

**P26.** A heater is switched on at  $t = 0$ , so its power is  $P(t) = 0$  for  $t < 0$  and  $P(t) = P_0$  for  $t \geq 0$ . Is  $P$  continuous at  $t = 0$ ? Classify the discontinuity and say what it means physically.

[ **Ans.** no; a jump discontinuity (instantaneous switching) ]

**P27.** The side of a square plate is measured as 10 cm with a possible error of 0.05 cm. Use linear approximation to estimate the possible error in the computed area.

[ **Ans.**  $\approx 1 \text{ cm}^2$ , i.e. 1% ]

**P28.** The approximation  $\sin x \approx x$  is justified by  $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ . Use the inequality  $\cos x < \frac{\sin x}{x} < 1$  to bound the relative error at  $x = 0.1$ , and compare with the actual error.

[ **Ans.** below  $1 - \cos 0.1 \approx 0.5\%$ ; actual  $\approx 0.17\%$  ]

# Unit 2      Single Variable Calculus

## 2.1 Rolle's Theorem

### **Theorem 2.1.1: Rolle's Theorem**

Let  $f$  be a function that satisfies the following three conditions:

- (i)  $f$  is continuous on the closed interval  $[a, b]$ ;
- (ii)  $f$  is differentiable on the open interval  $(a, b)$ ;
- (iii)  $f(a) = f(b)$ .

Then there exists a number  $c$  in  $(a, b)$  such that  $f'(c) = 0$ .

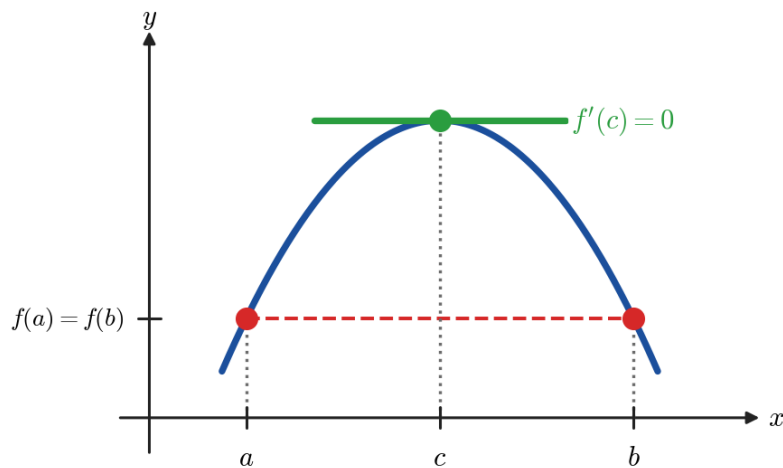


Figure 2.1.1: If the curve has equal ordinates at the ends of  $[a, b]$ , the tangent is horizontal at some interior point  $c$ .

#### **Note:**

There can be more than one such  $c$ . The theorem asserts existence, not uniqueness.

### 2.1.1 Geometrical meaning

Each hypothesis contributes one geometric fact, and together they force the conclusion.

- $f$  is continuous on  $[a, b]$ , so the curve  $y = f(x)$  has no gaps between the points  $(a, f(a))$  and  $(b, f(b))$  — it can be drawn without lifting the pen.
- $f$  is differentiable on  $(a, b)$ , so at every point strictly between  $a$  and  $b$  the curve has a unique tangent line. No corners, no cusps, no vertical tangents.
- $f(a) = f(b)$ , so the two endpoints have the same ordinate and the chord joining them is horizontal.

A curve that starts and finishes at the same height, and is smooth in between, must turn round somewhere. At a turning point the tangent is horizontal. This is the whole content of the

theorem: there is at least one point  $c \in (a, b)$  at which the tangent to the curve is parallel to the  $x$ -axis.

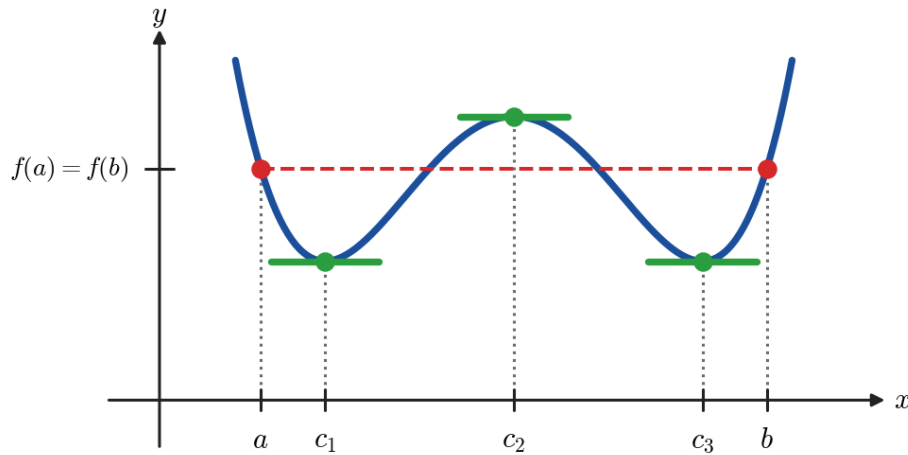


Figure 2.1.2: Rolle's theorem guarantees at least one horizontal tangent, but there may be several. Here  $f'(c_1) = f'(c_2) = f'(c_3) = 0$ , and the theorem names none of them in particular.

### Example 2.1.2: Verifying the hypotheses

Take  $f(x) = x^2 + 4$  on  $[-2, 2]$ , so  $a = -2$  and  $b = 2$ .

$$f(2) = 8, \quad f(-2) = 8 \implies f(2) = f(-2).$$

Being a polynomial,  $f$  is continuous on  $[-2, 2]$  and differentiable on  $(-2, 2)$ , so all three conditions hold. Now  $f'(x) = 2x$  and  $f'(0) = 0$ , and  $c = 0 \in (-2, 2)$  as the theorem predicts.

## 2.1.2 Worked problems

**Problem 2.1.1.** Check whether  $f(x) = x^2 - 6x + 8$  on  $[2, 4]$  satisfies Rolle's theorem.

**Solution:**  $f$  is a polynomial, hence continuous on  $[2, 4]$  and differentiable on  $(2, 4)$ .

$$f(2) = 4 - 12 + 8 = 0, \quad f(4) = 16 - 24 + 8 = 0 \implies f(2) = f(4).$$

All three conditions hold. Now  $f'(x) = 2x - 6$ , so

$$f'(x) = 0 \implies 2x - 6 = 0 \implies x = \frac{6}{2} = 3.$$

Since  $c = 3 \in (2, 4)$ , the function satisfies Rolle's theorem.

**Problem 2.1.2.** Check whether  $f(x) = e^x(\sin x - \cos x)$  on  $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$  satisfies Rolle's theorem.

**Solution:**  $f$  is a product of an exponential and trigonometric functions, hence continuous on  $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$  and differentiable on  $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ . At the endpoints,

$$f\left(\frac{\pi}{4}\right) = e^{\pi/4} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) = 0, \quad f\left(\frac{5\pi}{4}\right) = e^{5\pi/4} \left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = 0,$$

so  $f\left(\frac{\pi}{4}\right) = f\left(\frac{5\pi}{4}\right)$ . Differentiating,

$$f'(x) = e^x \cos x + e^x \sin x + e^x \sin x - e^x \cos x = 2e^x \sin x.$$

Setting  $f'(x) = 0$  gives  $2e^x \sin x = 0$ . Since  $e^x \neq 0$  for all  $x$ ,

$$\sin x = 0 \implies x = n\pi, \quad n = 0, 1, 2, \dots$$

The only such value in  $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$  is  $c = \pi$ . Hence the function satisfies Rolle's theorem.

**Problem 2.1.3.** Check whether  $f(x) = x^2 - 1$  on  $[1, 2]$  satisfies Rolle's theorem.

**Solution:** Here  $f(1) = 0$  and  $f(2) = 3$ , so  $f(1) \neq f(2)$ . The third condition fails, and therefore the function does not satisfy Rolle's theorem on  $[1, 2]$ .

**Problem 2.1.4.** Verify Rolle's theorem for  $f(x) = \sin x$  on  $[0, 2\pi]$ .

**Solution:** The sine function is continuous on  $[0, 2\pi]$  and differentiable on  $(0, 2\pi)$ , and

$$f(0) = \sin 0 = 0 = \sin 2\pi = f(2\pi),$$

so all three hypotheses hold. By Rolle's theorem there is a  $c \in (0, 2\pi)$  with  $f'(c) = 0$ . Since  $f'(x) = \cos x$ ,

$$\cos c = 0 \implies c = (2n + 1)\frac{\pi}{2}, \quad n = 0, 1, 2, \dots$$

Of these, the values lying in the open interval  $(0, 2\pi)$  are

$$c = \frac{\pi}{2} \quad \text{and} \quad c = \frac{3\pi}{2},$$

and  $\cos c = 0$  at both. Rolle's theorem is verified. This is a concrete instance of the non-uniqueness noted above: two admissible values of  $c$ , and the theorem distinguishes neither.

**Problem 2.1.5.** Verify Rolle's theorem for  $f(x) = \log\left(\frac{x^2 + ab}{x(a+b)}\right)$  on  $[a, b]$ , where  $0 < a < b$ .

**Solution:** For  $x \in [a, b]$  we have  $x > 0$  and  $x^2 + ab > 0$ , so the argument of the logarithm is positive throughout and  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . At the endpoints,

$$f(a) = \log\left(\frac{a^2 + ab}{a(a+b)}\right) = \log\left(\frac{a(a+b)}{a(a+b)}\right) = \log 1 = 0,$$

$$f(b) = \log\left(\frac{b^2 + ab}{b(a+b)}\right) = \log\left(\frac{b(a+b)}{b(a+b)}\right) = \log 1 = 0,$$

so  $f(a) = f(b)$  and Rolle's theorem applies: there is a  $c \in (a, b)$  with  $f'(c) = 0$ . Splitting the logarithm before differentiating,

$$f(x) = \log(x^2 + ab) - \log x - \log(a + b),$$

and since  $\log(a + b)$  is a constant,

$$f'(x) = \frac{2x}{x^2 + ab} - \frac{1}{x}.$$

Setting  $f'(c) = 0$ ,

$$\frac{2c}{c^2 + ab} = \frac{1}{c} \implies 2c^2 = c^2 + ab \implies c^2 = ab \implies c = \pm\sqrt{ab}.$$

Only the positive root can lie in  $(a, b)$ , so  $c = \sqrt{ab}$ , the geometric mean of the endpoints. That it really does lie in the interval is the arithmetic–geometric mean inequality in its simplest form: for  $0 < a < b$ ,

$$a = \sqrt{a \cdot a} < \sqrt{ab} < \sqrt{b \cdot b} = b.$$

Rolle's theorem is verified, with  $c = \sqrt{ab}$ .

**Problem 2.1.6.** Verify Rolle's theorem for  $f(x) = 1 - \sqrt[3]{x^2}$  on  $[-1, 1]$ .

**Solution:** Write  $f(x) = 1 - x^{2/3}$ . The function is continuous on  $[-1, 1]$ , and

$$f(-1) = 1 - 1 = 0 = 1 - 1 = f(1),$$

so hypotheses (i) and (iii) hold. Differentiating,

$$f'(x) = -\frac{2}{3}x^{\frac{2}{3}-1} = -\frac{2}{3}x^{-1/3} = -\frac{2}{3\sqrt[3]{x}}.$$

This expression is undefined at  $x = 0$ , and  $x = 0$  lies inside  $(-1, 1)$ . Hypothesis (ii) therefore fails and Rolle's theorem is not applicable.

The conclusion fails too, which is the point of the example. For  $x \neq 0$  we have  $f'(x) \neq 0$ , since  $-\frac{2}{3}x^{-1/3}$  is never zero; so there is no  $c \in (-1, 1)$  with  $f'(c) = 0$ . Geometrically the graph has a *cusp* at the origin: the curve does turn round there, but it does so with a vertical tangent rather than a horizontal one.

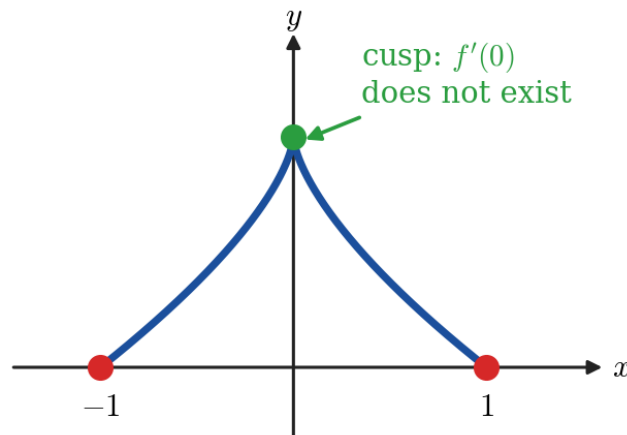


Figure 2.1.3:  $f(x) = 1 - x^{2/3}$  on  $[-1, 1]$ . The endpoint ordinates agree and the curve is unbroken, but the peak is a cusp, not a smooth maximum, so no horizontal tangent exists.

## 2.2 Lagrange's Mean Value Theorem

Lagrange's mean value theorem generalises Rolle's theorem: the requirement  $f(a) = f(b)$  is dropped, and the conclusion becomes a statement about the slope of the chord joining the endpoints.

### **Theorem 2.2.1: Lagrange's Mean Value Theorem**

Let  $f$  be a function that satisfies the following conditions:

- (i)  $f$  is continuous on the closed interval  $[a, b]$ ;
- (ii)  $f$  is differentiable on the open interval  $(a, b)$ .

Then there exists a number  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

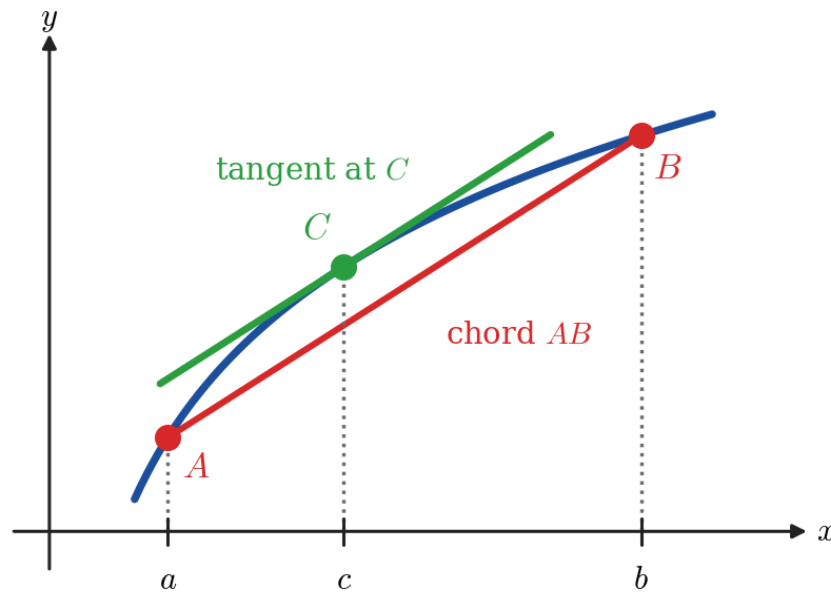


Figure 2.2.1: The tangent at  $c$  is parallel to the chord  $AB$ , whose slope is  $m_{AB} = \frac{f(b) - f(a)}{b - a}$ .

**Note:**

Taking  $f(a) = f(b)$  makes the right-hand side zero, so [Theorem 2.1.1](#) is the special case  $m_{AB} = 0$  of [Theorem 2.2.1](#).

The conclusion is often more convenient in the equivalent *increment* form, obtained by multiplying through by  $b - a$ :

$$f(b) - f(a) = f'(c)(b - a), \quad c \in (a, b).$$

### 2.2.1 Geometrical meaning

- There is at least one point  $(c, f(c))$  on the graph of  $y = f(x)$ , strictly between  $(a, f(a))$  and  $(b, f(b))$ , at which the tangent to the curve is parallel to the chord joining those two endpoints.
- In the language of rates: at some interior point the *instantaneous* rate of change  $f'(c)$  equals the *average* rate of change  $\frac{f(b) - f(a)}{b - a}$  over the whole interval.
- In physical terms, if  $f(t)$  is the position of a moving object at time  $t$ , then its average velocity over the time interval  $[a, b]$  is attained as its instantaneous velocity at some moment inside that interval.

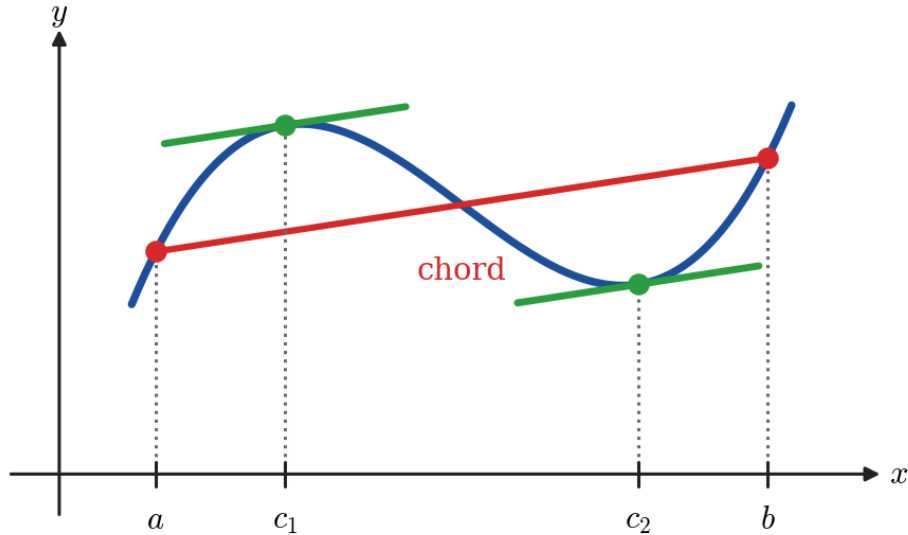


Figure 2.2.2: Two interior points at which the tangent is parallel to the chord. As with Rolle's theorem, the point  $c$  where  $f'(c) = \frac{f(b)-f(a)}{b-a}$  need not be unique.

### 2.2.2 Worked problems

**Problem 2.2.1.** Verify Lagrange's mean value theorem for  $f(x) = x(x-1)(x-2)$  on  $[0, \frac{1}{2}]$ .

**Solution:** Expanding,  $f(x) = (x^2 - x)(x - 2) = x^3 - 3x^2 + 2x$ . Since  $f$  is a polynomial, it is continuous on  $[0, \frac{1}{2}]$  and differentiable on  $(0, \frac{1}{2})$ . At the endpoints,

$$f(0) = 0, \quad f\left(\frac{1}{2}\right) = \frac{1}{8} - \frac{3}{4} + 1 = \frac{1 - 6 + 8}{8} = \frac{3}{8}.$$

Differentiating,  $f'(x) = 3x^2 - 6x + 2$ . By the mean value theorem there is a  $c$  with

$$f'(c) = \frac{f\left(\frac{1}{2}\right) - f(0)}{\frac{1}{2} - 0} = \frac{\frac{3}{8} - 0}{\frac{1}{2}} = \frac{3}{4},$$

so that

$$3c^2 - 6c + 2 = \frac{3}{4} \implies 3c^2 - 6c + \frac{5}{4} = 0 \implies 12c^2 - 24c + 5 = 0.$$

Solving the quadratic,

$$c = \frac{6 \pm \sqrt{(-6)^2 - 4(3)\left(\frac{5}{4}\right)}}{2(3)} = \frac{6 \pm \sqrt{36 - 15}}{6} = \frac{6 \pm \sqrt{21}}{6} = 1 \pm 0.7638,$$

giving  $c = 1.7638$  or  $c = 0.2362$ . Only the second lies in the interval, so  $c = 0.2362 \in (0, \frac{1}{2})$  satisfies the conclusion of the mean value theorem.

**Problem 2.2.2.** Verify Lagrange's mean value theorem for  $f(x) = \log x$  on  $[1, e]$ .

**Solution:**  $f(x) = \log x$  is a logarithmic function, hence continuous on  $[1, e]$  and differentiable on  $(1, e)$ . At the endpoints,

$$f(a) = f(1) = \log_e 1 = 0, \quad f(b) = f(e) = \log_e e = 1, \quad f'(x) = \frac{1}{x}.$$

By the mean value theorem,

$$f'(c) = \frac{f(b) - f(a)}{b - a} \implies \frac{1}{c} = \frac{1 - 0}{e - 1} \implies c = e - 1.$$

Since  $c = e - 1 \approx 1.718 \in (1, e)$ , the conclusion of the mean value theorem is satisfied.

**Problem 2.2.3.** Verify Lagrange's mean value theorem for  $f(x) = \arcsin x = \sin^{-1} x$  on  $[0, 1]$ .

**Solution:**  $f(x) = \sin^{-1} x$  is continuous on  $[0, 1]$  and differentiable on  $(0, 1)$ . At the endpoints,

$$f(a) = \sin^{-1} 0 = 0, \quad f(b) = \sin^{-1} 1 = \frac{\pi}{2}, \quad f'(x) = \frac{1}{\sqrt{1 - x^2}}.$$

By the mean value theorem there is a  $c \in (0, 1)$  with

$$\frac{1}{\sqrt{1 - c^2}} = \frac{\frac{\pi}{2} - 0}{1 - 0} = \frac{\pi}{2} \implies \sqrt{1 - c^2} = \frac{2}{\pi} \implies 1 - c^2 = \frac{4}{\pi^2}.$$

Hence

$$c^2 = 1 - \frac{4}{\pi^2} \implies c = \pm \sqrt{\frac{\pi^2 - 4}{\pi^2}} = \pm \frac{\sqrt{\pi^2 - 4}}{\pi} \approx \pm 0.771.$$

Taking the positive root,  $c \approx 0.771 \in (0, 1)$  satisfies the conclusion of the mean value theorem.

**Note:**

Note that  $f'(x) = \frac{1}{\sqrt{1 - x^2}}$  is undefined at  $x = 1$ , so  $\sin^{-1} x$  is *not* differentiable at the right-hand endpoint. This costs nothing: hypothesis (ii) asks only for differentiability on the *open* interval  $(0, 1)$ , and hypothesis (i) — continuity on the closed interval — does hold at  $x = 1$ . The theorem is stated with the two different intervals precisely so that functions with a vertical tangent at an endpoint, such as this one, are still covered.

**Problem 2.2.4. A speeding ticket by mean value theorem.** You are driving at 55 mph when you pass a police car with radar. Five minutes later, 6 miles down the road, you pass a second police car with radar and you are still going 55 mph. The officer pulls you over and issues a ticket for exceeding the 60 mph speed limit, citing the mean value theorem as proof. Is the ticket justified?

**Solution:** Let  $s(t)$  be the distance travelled, in miles, at time  $t$  hours after passing the first patrol car. Position is a continuous and differentiable function of time, so the mean value theorem applies on  $[0, t_1]$ , where

$$t_1 = 5 \text{ minutes} = \frac{5}{60} = \frac{1}{12} \text{ hour}, \quad s(0) = 0, \quad s(t_1) = 6 \text{ miles}.$$

The average velocity over the interval is

$$\frac{s(t_1) - s(0)}{t_1 - 0} = \frac{6 - 0}{\frac{1}{12}} = 72 \text{ mph}.$$

By the mean value theorem there is a time  $c \in (0, \frac{1}{12})$  with

$$s'(c) = 72 \text{ mph}.$$

Since  $s'(c)$  is the instantaneous velocity at time  $c$ , the car was travelling at exactly 72 mph at some instant between the two patrol cars — well above the 60 mph limit. The ticket is justified.

**Note:**

The readings of 55 mph at the two ends are irrelevant to the argument, and are there to mislead. The mean value theorem says nothing about the endpoints; it converts a measurement of *average* speed, which the two patrol cars together provide, into the existence of a moment of that *instantaneous* speed. Note also what the theorem does not give: it locates no particular instant  $c$ , only guarantees that one exists.

**2.2.3 An inequality from the mean value theorem****Proposition 2.2.2: Bounds for a difference of arctangents**

If  $0 \leq a < b$ , then

$$\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}.$$

*Proof.* Let  $f(x) = \tan^{-1} x$ , so that  $f'(x) = \frac{1}{1+x^2}$ . Being an inverse trigonometric function,  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . By the mean value theorem there exists  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b-a} \quad \text{that is} \quad \frac{1}{1+c^2} = \frac{\tan^{-1} b - \tan^{-1} a}{b-a}. \quad (2.2.1)$$

Since  $0 \leq a < c < b$  and  $x \mapsto x^2$  is increasing on  $[0, \infty)$ ,

$$a^2 < c^2 < b^2 \implies 1+a^2 < 1+c^2 < 1+b^2 \implies \frac{1}{1+b^2} < \frac{1}{1+c^2} < \frac{1}{1+a^2},$$

the last inequality reversing because a larger denominator gives a smaller fraction. Substituting from (2.2.1),

$$\frac{1}{1+b^2} < \frac{\tan^{-1} b - \tan^{-1} a}{b-a} < \frac{1}{1+a^2},$$

and multiplying throughout by  $b-a > 0$  leaves the inequalities unchanged in direction, which gives the stated result. ■

**Note:**

The squaring step needs  $0 \leq a < b$ ; for negative  $a$  the inequality  $a^2 < c^2$  can fail. Both deductions below have  $a = 1 > 0$ , so the hypothesis holds.

**Problem 2.2.5.** Using Proposition 2.2.2, deduce the following.

$$(a) \quad \frac{5\pi+4}{20} < \tan^{-1} 2 < \frac{\pi+2}{4}.$$

**Solution:** Put  $a = 1$  and  $b = 2$  in Proposition 2.2.2:

$$\frac{2-1}{1+2^2} < \tan^{-1} 2 - \tan^{-1} 1 < \frac{2-1}{1+1^2} \implies \frac{1}{5} < \tan^{-1} 2 - \frac{\pi}{4} < \frac{1}{2}.$$

Adding  $\frac{\pi}{4}$  throughout,

$$\frac{1}{5} + \frac{\pi}{4} < \tan^{-1} 2 < \frac{1}{2} + \frac{\pi}{4} \implies \frac{5\pi+4}{20} < \tan^{-1} 2 < \frac{\pi+2}{4}.$$

$$(b) \quad \frac{\pi}{4} + \frac{3}{25} < \tan^{-1}\left(\frac{4}{3}\right) < \frac{\pi}{4} + \frac{1}{6}.$$

**Solution:** Put  $a = 1$  and  $b = \frac{4}{3}$  in [Proposition 2.2.2](#):

$$\frac{\frac{4}{3} - 1}{1 + \left(\frac{4}{3}\right)^2} < \tan^{-1}\left(\frac{4}{3}\right) - \tan^{-1} 1 < \frac{\frac{4}{3} - 1}{1 + 1^2}.$$

The left-hand fraction is  $\frac{1/3}{25/9} = \frac{1}{3} \cdot \frac{9}{25} = \frac{3}{25}$  and the right-hand fraction is  $\frac{1/3}{2} = \frac{1}{6}$ , so

$$\frac{3}{25} < \tan^{-1}\left(\frac{4}{3}\right) - \frac{\pi}{4} < \frac{1}{6} \implies \frac{\pi}{4} + \frac{3}{25} < \tan^{-1}\left(\frac{4}{3}\right) < \frac{\pi}{4} + \frac{1}{6}.$$

The same argument applied to  $\sin^{-1} x$  instead of  $\tan^{-1} x$  gives a companion result. Only the direction of one step changes, and it is worth seeing why.

**Proposition 2.2.3: Bounds for a difference of arcsines**

If  $0 \leq a < b < 1$ , then

$$\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1} b - \sin^{-1} a < \frac{b-a}{\sqrt{1-b^2}}.$$

*Proof.* Let  $f(x) = \sin^{-1} x$ , so that

$$f'(x) = \frac{1}{\sqrt{1-x^2}}, \quad x \neq \pm 1.$$

Since  $b < 1$ , the derivative exists throughout  $(a, b)$ , and  $f$  is continuous on  $[a, b]$ . By the mean value theorem there is a  $c \in (a, b)$  with

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \text{that is} \quad \frac{1}{\sqrt{1-c^2}} = \frac{\sin^{-1} b - \sin^{-1} a}{b - a}, \quad a < c < b < 1. \quad (2.2.2)$$

Now compare  $c$  with each endpoint in turn. From  $a < c$ , and since squaring is increasing on  $[0, \infty)$ ,

$$a^2 < c^2 \implies -a^2 > -c^2 \implies 1 - a^2 > 1 - c^2 \implies \sqrt{1-a^2} > \sqrt{1-c^2},$$

and therefore

$$\frac{1}{\sqrt{1-a^2}} < \frac{1}{\sqrt{1-c^2}}. \quad (2.2.3)$$

From  $c < b$  the same chain runs the other way:

$$c^2 < b^2 \implies 1 - c^2 > 1 - b^2 \implies \sqrt{1-c^2} > \sqrt{1-b^2},$$

so that

$$\frac{1}{\sqrt{1-c^2}} < \frac{1}{\sqrt{1-b^2}}. \quad (2.2.4)$$

Combining (2.2.3), (2.2.2) and (2.2.4),

$$\frac{1}{\sqrt{1-a^2}} < \frac{\sin^{-1} b - \sin^{-1} a}{b - a} < \frac{1}{\sqrt{1-b^2}},$$

and multiplying throughout by  $b - a > 0$  gives the result. ■

**Note:**

Compare this with [Proposition 2.2.2](#). There the derivative  $\frac{1}{1+x^2}$  is *decreasing*, so the bound at  $b$  came out on the left; here  $\frac{1}{\sqrt{1-x^2}}$  is *increasing*, so the bound at  $a$  is on the left instead. The method is identical; only the monotonicity of  $f'$  decides which endpoint supplies which bound. Note also the hypothesis  $b < 1$ , which is needed for  $f'$  to exist on  $(a, b)$  — at  $x = 1$  the derivative blows up and the argument fails.

**Problem 2.2.6.** Using [Proposition 2.2.3](#), deduce that

$$\frac{\pi}{6} + \frac{1}{5\sqrt{3}} < \sin^{-1}\left(\frac{3}{5}\right) < \frac{\pi}{6} + \frac{1}{8}.$$

**Solution:** Put  $a = \frac{1}{2}$  and  $b = \frac{3}{5}$ , which satisfy  $0 \leq a < b < 1$ . Then  $b - a = \frac{3}{5} - \frac{1}{2} = \frac{1}{10}$ , and [Proposition 2.2.3](#) gives

$$\frac{\frac{3}{5} - \frac{1}{2}}{\sqrt{1 - \frac{1}{4}}} < \sin^{-1}\left(\frac{3}{5}\right) - \sin^{-1}\left(\frac{1}{2}\right) < \frac{\frac{3}{5} - \frac{1}{2}}{\sqrt{1 - \left(\frac{3}{5}\right)^2}}.$$

Taking the two outer terms separately,

$$\frac{\frac{1}{10}}{\sqrt{\frac{3}{4}}} = \frac{1}{10} \cdot \frac{2}{\sqrt{3}} = \frac{1}{5\sqrt{3}}, \quad \frac{\frac{1}{10}}{\sqrt{1 - \frac{9}{25}}} = \frac{\frac{1}{10}}{\sqrt{\frac{16}{25}}} = \frac{1}{10} \cdot \frac{5}{4} = \frac{1}{8}.$$

Since  $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$ , the inequality reads

$$\frac{1}{5\sqrt{3}} < \sin^{-1}\left(\frac{3}{5}\right) - \frac{\pi}{6} < \frac{1}{8},$$

and adding  $\frac{\pi}{6}$  throughout gives

$$\boxed{\frac{\pi}{6} + \frac{1}{5\sqrt{3}} < \sin^{-1}\left(\frac{3}{5}\right) < \frac{\pi}{6} + \frac{1}{8}}$$

**Note:**

Numerically the bounds are 0.6391 and 0.6486, and  $\sin^{-1}(0.6) = 0.6435$ , comfortably between them. The two bounds differ by less than 0.01, so the mean value theorem has pinned down  $\sin^{-1}(3/5)$  to about two decimal places using nothing beyond the value of  $\sin^{-1}$  at a single convenient point.

## 2.3 L'Hôpital's Rule

The mean value theorem has one more consequence that is used constantly: a rule for limits of quotients in which the numerator and denominator both tend to 0, or both become infinite.

**Theorem 2.3.1: L'Hôpital's rule**

Let  $f$  and  $g$  be differentiable near  $a$  (except possibly at  $a$ ), with  $g'(x) \neq 0$  there. Suppose that either

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0, \quad \text{or} \quad \lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty.$$

If  $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$  exists (or is  $\pm\infty$ ), then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

The same holds for one-sided limits and for limits as  $x \rightarrow \pm\infty$ .

**Note:**

The rule is proved from a two-function version of Lagrange's theorem (Cauchy's mean value theorem); it is used here without proof. The other indeterminate forms, such as  $0 \cdot \infty$  and  $\infty - \infty$ , are handled by first rewriting the expression as a quotient.

**Common mistake:**

Differentiate the numerator and the denominator *separately* — this is not the quotient rule. And check the form first: the rule applies only to  $\frac{0}{0}$  and  $\frac{\infty}{\infty}$ . For example  $\lim_{x \rightarrow 0} \frac{x+1}{x+2} = \frac{1}{2}$  by direct substitution, whereas a careless application of the rule would give  $\frac{1}{1} = 1$ .

**Example 2.3.2: Using L'Hôpital's rule**

(a)  $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}.$

**Solution:** The form is  $\frac{0}{0}$ . One application gives  $\lim_{x \rightarrow 0} \frac{e^x - 1}{2x}$ , which is again  $\frac{0}{0}$ ; a second gives

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}.$$

(b)  $\lim_{x \rightarrow \infty} \frac{\ln x}{x}.$

**Solution:** The form is  $\frac{\infty}{\infty}$ , and  $\lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$ . The logarithm grows more slowly than any positive power of  $x$ .

(c)  $\lim_{x \rightarrow \infty} \frac{x^k}{e^x}$  for a positive integer  $k$ .

**Solution:** Each application lowers the power of  $x$  by one and leaves  $e^x$  unchanged:

$$\lim_{x \rightarrow \infty} \frac{x^k}{e^x} = \lim_{x \rightarrow \infty} \frac{kx^{k-1}}{e^x} = \cdots = \lim_{x \rightarrow \infty} \frac{k!}{e^x} = 0.$$

So the exponential eventually outgrows every power of  $x$ . This fact is used for the gamma function later in this unit.

(d)  $\lim_{x \rightarrow 0^+} x \ln x$ .

**Solution:** The form is  $0 \cdot (-\infty)$ . Rewriting it as a quotient,

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0.$$

## 2.4 The Definite Integral

Improper integrals are defined as limits of ordinary definite integrals, so it is worth recalling precisely what a definite integral is.

### Definition 2.4.1: Definite integral as a limit of Riemann sums

Let  $f$  be a function defined on  $a \leq x \leq b$ . Divide  $[a, b]$  into  $n$  subintervals of equal width

$$\Delta x = \frac{b - a}{n},$$

and let

$$x_0 = a, x_1, x_2, \dots, x_{n-1}, x_n = b$$

be the  $(n + 1)$  endpoints of these subintervals. Let  $x_1^*, x_2^*, \dots, x_n^*$  be any *sample points*, with  $x_i^*$  lying in the  $i$ th subinterval  $[x_{i-1}, x_i]$ . The *definite integral* of  $f$  from  $a$  to  $b$  is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x,$$

provided this limit exists and takes the same value for every possible choice of sample points. When the limit exists we say  $f$  is *integrable* on  $[a, b]$ .

The sum  $\sum_{i=1}^n f(x_i^*) \Delta x$  is called a *Riemann sum*. Each term  $f(x_i^*) \Delta x$  is the area of a rectangle of width  $\Delta x$  whose height is the value of  $f$  at the sample point, so the whole sum approximates the area under the curve by a staircase of rectangles.

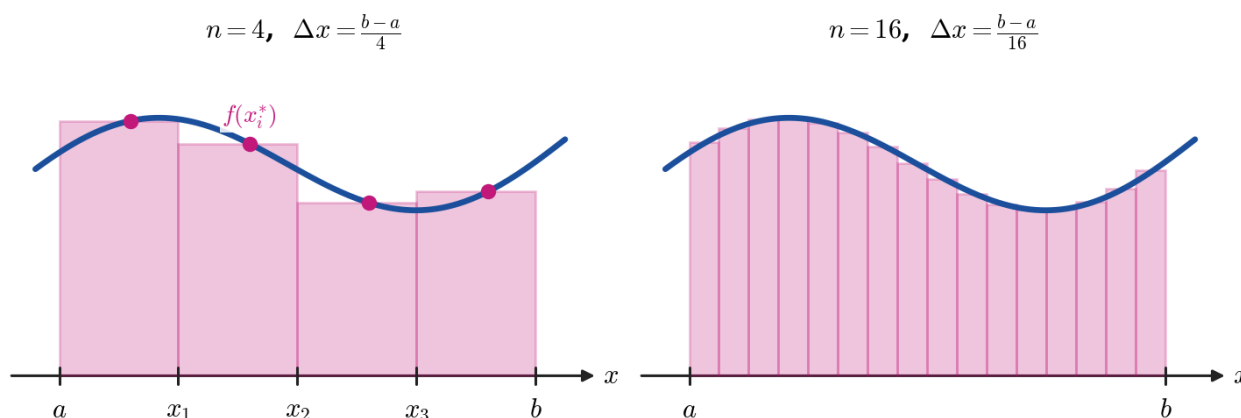


Figure 2.4.1: Riemann sums for the same function with  $n = 4$  and  $n = 16$  rectangles. As  $n$  increases the rectangles narrow, the staircase settles onto the curve, and the sums approach  $\int_a^b f(x) dx$ .

### Note:

Two conditions in the definition do real work and are easy to read past. The limit must exist *and* it must be independent of how the sample points  $x_i^*$  are chosen — left endpoints, right endpoints, midpoints, or anything else must all give the same answer. A function for which different choices give different limits is not integrable. Every function met in this unit is continuous except at finitely many points, and such functions are always integrable, so the condition is automatic here; it matters only for badly behaved functions.

### Note:

Note the two limits at work in this unit and keep them apart. The definite integral is already a limit, taken as  $n \rightarrow \infty$  with the interval  $[a, b]$  fixed and finite. An improper integral is a *second* limit taken on top of that one, as an endpoint runs off to infinity or towards a point where  $f$  blows up. This is why an improper integral is never evaluated directly: the inner definite integral is computed first, as a function of the moving endpoint, and only then is the outer limit taken.

## 2.5 Improper Integrals

### Definition 2.5.1: Improper integral

An integral  $\int_a^b f(x) dx$  is called an *improper integral* if either the interval of integration  $[a, b]$  is infinite, or the integrand  $f$  has an infinite discontinuity in  $[a, b]$ .

- If  $[a, b]$  is infinite,  $\int_a^b f(x) dx$  is an improper integral of *Type 1*.
- If  $f$  has an infinite discontinuity,  $\int_a^b f(x) dx$  is an improper integral of *Type 2*.

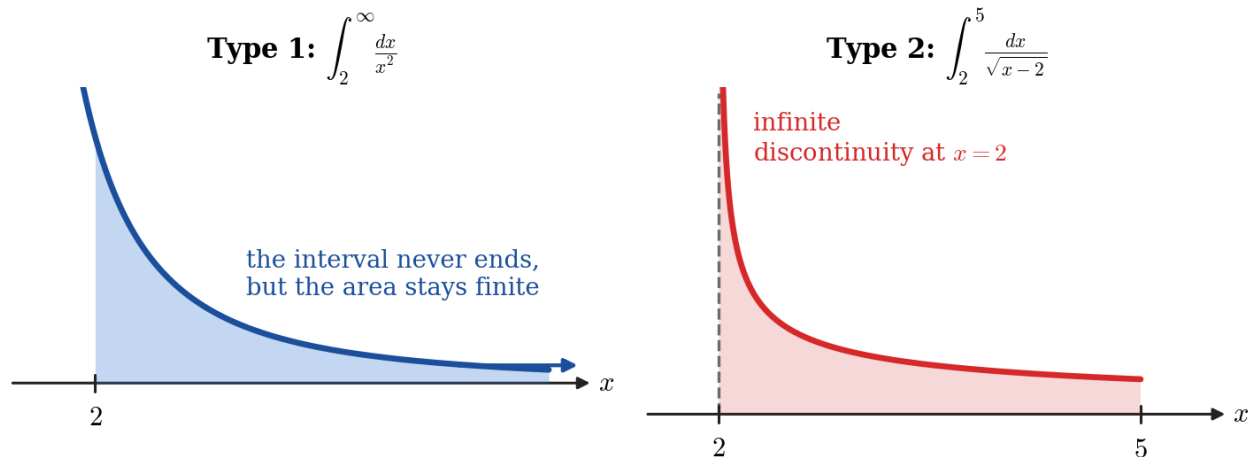


Figure 2.5.1: Two ways an integral can be improper: the region runs off to the right without ending (Type 1), or it runs upwards without ending (Type 2).

### Example 2.5.2: Classifying improper integrals

- (a)  $\int_2^{\infty} \frac{1}{x^2} dx$  Type 1: the interval  $[2, \infty)$  is infinite.
- (b)  $\int_{-\infty}^0 e^x dx$  Type 1: the interval  $(-\infty, 0]$  is infinite.
- (c)  $\int_1^5 \frac{1}{x-1} dx$  Type 2: the integrand  $f(x) = \frac{1}{x-1}$  has  $f(1) = \frac{1}{0} = \infty$ , an infinite discontinuity at the lower limit.
- (d)  $\int_0^{\pi/2} \tan x dx$  Type 2: the integrand  $f(x) = \tan x$  has  $f(\frac{\pi}{2}) = \infty$ , an infinite discontinuity at the upper limit.

### Note:

An integral can be improper in both ways at once. For instance  $\int_0^{\infty} \frac{dx}{x^2}$  has an infinite interval *and* an infinite discontinuity at  $x=0$ ; it must be split, say at  $x=1$ , and each piece examined separately.

## 2.5.1 Improper integrals of Type 1

- (i) If  $\int_a^t f(x) dx$  exists for every number  $t \geq a$ , then

$$\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx,$$

provided this limit exists as a finite number.

- (ii) If  $\int_t^b f(x) dx$  exists for every number  $t \leq b$ , then

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx,$$

provided this limit exists as a finite number.

The improper integrals  $\int_a^\infty f(x) dx$  and  $\int_{-\infty}^b f(x) dx$  are called *convergent* if the limit exists, and *divergent* if the limit does not exist.

(iii) If both  $\int_a^\infty f(x) dx$  and  $\int_{-\infty}^a f(x) dx$  are convergent, then we define

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx, \quad a \in \mathbb{R}.$$

The next two integrals differ only in an exponent, and the regions they measure look much the same, yet one has infinite area and the other has area exactly 1. They are worth doing side by side.

**Problem 2.5.1.** Determine whether  $\int_1^\infty \frac{1}{x} dx$  is convergent or divergent.

**Solution:** The interval  $[1, \infty)$  is infinite, so this is an improper integral of Type 1. By the definition,

$$\int_1^\infty \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} \left[ \log |x| \right]_1^t = \lim_{t \rightarrow \infty} (\log t - \log 1).$$

Since  $\log 1 = 0$ ,

$$\int_1^\infty \frac{1}{x} dx = \lim_{t \rightarrow \infty} \log t = \infty.$$

The limit does not exist as a finite number, so the improper integral  $\int_1^\infty \frac{1}{x} dx$  is **divergent**.

**Problem 2.5.2.** Determine whether  $\int_1^\infty \frac{1}{x^2} dx$  is convergent or divergent.

**Solution:** Again the interval is infinite, so this is Type 1:

$$\int_1^\infty \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \int_1^t x^{-2} dx = \lim_{t \rightarrow \infty} \left[ -\frac{1}{x} \right]_1^t = -\lim_{t \rightarrow \infty} \left( \frac{1}{t} - 1 \right).$$

As  $t \rightarrow \infty$  we have  $\frac{1}{t} \rightarrow 0$ , so

$$\int_1^\infty \frac{1}{x^2} dx = -(0 - 1) = 1.$$

The limit exists finitely, so the improper integral is **convergent**, with value 1.

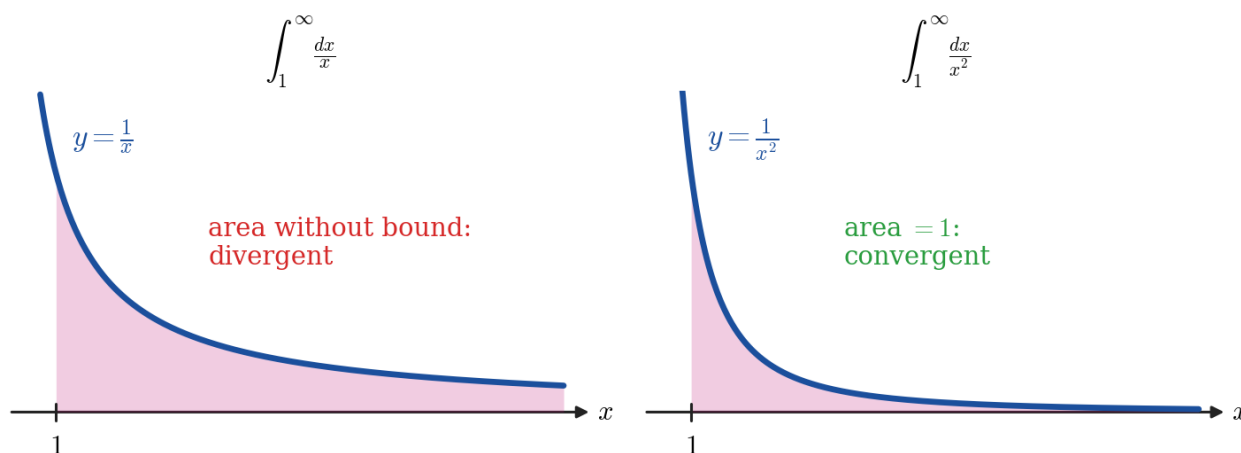


Figure 2.5.2: The two regions of the previous problems. Both are unbounded to the right and both narrow towards the axis, but only the second has finite area.

**Note:**

The contrast is the point. Having the integrand tend to 0 as  $x \rightarrow \infty$  is *not* enough to make a Type 1 integral converge —  $\frac{1}{x} \rightarrow 0$  and its integral still diverges. What matters is how fast the decay happens. The dividing line is settled in general by the  $p$ -integral test below: for  $\int_1^\infty x^{-p} dx$  the answer is convergence exactly when  $p > 1$ , and  $\frac{1}{x}$  sits on the losing side of that line at  $p = 1$  while  $\frac{1}{x^2}$  sits on the winning side at  $p = 2$ .

### 2.5.2 Improper integrals of Type 2

- (i) If  $f$  is continuous on  $[a, b)$  and discontinuous at  $b$ , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx,$$

if this limit exists as a finite number.

- (ii) If  $f$  is continuous on  $(a, b]$  and discontinuous at  $a$ , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx,$$

if this limit exists as a finite number.

The improper integral  $\int_a^b f(x) dx$  is called *convergent* if the corresponding limit exists, and *divergent* if the limit does not exist.

- (iii) If  $f$  has a discontinuity at  $c$ , where  $a < c < b$ , and both  $\int_a^c f(x) dx$  and  $\int_c^b f(x) dx$  are convergent, then we define

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

### 2.5.3 Worked problems

**Problem 2.5.3.** Classify the improper integral and determine whether each integral is convergent or divergent. Evaluate those that are convergent.

(a)  $\int_{-\infty}^0 xe^{-x} dx.$

**Solution:** Here the interval of integration is  $(-\infty, 0]$ , which is infinite, so the given integral is of Type 1. Using  $\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$ , we first evaluate the definite integral by parts, with  $u = x$  and  $dv = e^{-x} dx$ , so that  $v = -e^{-x}$ :

$$I = \int_t^0 xe^{-x} dx = \left[ -xe^{-x} \right]_t^0 - \int_t^0 (-e^{-x}) dx = \left[ -xe^{-x} \right]_t^0 + \left[ -e^{-x} \right]_t^0.$$

Evaluating,

$$I = (0 + te^{-t}) + (-1 + e^{-t}) = te^{-t} + e^{-t} - 1.$$

As  $t \rightarrow -\infty$  we have  $e^{-t} \rightarrow \infty$  and  $te^{-t} \rightarrow -\infty$ , so

$$\lim_{t \rightarrow -\infty} I = -\infty,$$

and the given integral diverges.

(b)  $\int_{-\infty}^0 x e^x dx.$

**Solution:** Again the interval  $(-\infty, 0]$  is infinite, so this is of Type 1. Integrating by parts with  $u = x$  and  $dv = e^x dx$ , so that  $v = e^x$ ,

$$I = \int_t^0 x e^x dx = \left[ x e^x \right]_t^0 - \int_t^0 e^x dx = \left[ x e^x \right]_t^0 - \left[ e^x \right]_t^0 = (0 - t e^t) - (1 - e^t),$$

that is

$$I = -t e^t + e^t - 1.$$

Taking the limit term by term,

$$\lim_{t \rightarrow -\infty} (-t e^t + e^t - 1) = - \lim_{t \rightarrow -\infty} t e^t + \lim_{t \rightarrow -\infty} e^t - 1 = - \lim_{t \rightarrow -\infty} t e^t + 0 - 1.$$

The remaining limit is of the indeterminate form  $-\infty \cdot 0$ ; rewriting it as a quotient,

$$\lim_{t \rightarrow -\infty} t e^t = \lim_{t \rightarrow -\infty} \frac{t}{e^{-t}} = \frac{-\infty}{\infty},$$

and applying L'Hôpital's rule,

$$\lim_{t \rightarrow -\infty} \frac{t}{e^{-t}} = \lim_{t \rightarrow -\infty} \frac{1}{-e^{-t}} = 0.$$

Therefore

$$\lim_{t \rightarrow -\infty} \int_t^0 x e^x dx = 0 + 0 - 1 = -1,$$

so the given integral converges, to  $-1$ .

(c)  $\int_1^{\infty} \frac{\log x}{x} dx.$

**Solution:** Here the interval of integration is  $[1, \infty)$ , which is infinite, so the given integral is an improper integral of Type 1. Thus

$$\int_1^{\infty} \frac{\log x}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{\log x}{x} dx.$$

For the definite integral put  $u = \log x$ , so that  $du = \frac{1}{x} dx$ ; when  $x = 1$ ,  $u = 0$ , and when  $x = t$ ,  $u = \log t$ . Then

$$I = \int_1^t \frac{\log x}{x} dx = \int_0^{\log t} u du = \left[ \frac{u^2}{2} \right]_0^{\log t} = \frac{(\log t)^2}{2}.$$

Hence

$$\lim_{t \rightarrow \infty} \frac{(\log t)^2}{2} = \infty,$$

and the given integral diverges.

(d)  $\int_2^5 \frac{dx}{\sqrt{x-2}}.$

**Solution:** The integrand  $f(x) = \frac{1}{\sqrt{x-2}}$  has an infinite discontinuity at  $x = 2 \in [2, 5]$ , so the given integral is an improper integral of Type 2. The discontinuity is at the lower limit, so

$$\int_2^5 \frac{dx}{\sqrt{x-2}} = \lim_{t \rightarrow 2^+} \int_t^5 \frac{dx}{\sqrt{x-2}} = \lim_{t \rightarrow 2^+} \left[ 2\sqrt{x-2} \right]_{x=t}^{x=5}.$$

Evaluating,

$$= 2 \lim_{t \rightarrow 2^+} (\sqrt{5-2} - \sqrt{t-2}) = 2\sqrt{3} - 2 \lim_{t \rightarrow 2^+} \sqrt{t-2} = 2\sqrt{3} - 0 = 2\sqrt{3}.$$

Hence the given integral converges, to  $2\sqrt{3}$ .

(e)  $\int_0^3 \frac{dx}{\sqrt{9-x^2}}.$

**Solution:** The integrand  $f(x) = \frac{1}{\sqrt{9-x^2}}$  has an infinite discontinuity at  $x = 3 \in [0, 3]$ , so the given integral is an improper integral of Type 2. The discontinuity is at the upper limit, so

$$\int_0^3 \frac{dx}{\sqrt{9-x^2}} = \lim_{t \rightarrow 3^-} \int_0^t \frac{dx}{\sqrt{9-x^2}},$$

and the evaluation uses  $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a} + C$ .

Evaluating,

$$\lim_{t \rightarrow 3^-} \left[ \sin^{-1} \frac{x}{3} \right]_0^t = \lim_{t \rightarrow 3^-} \sin^{-1} \frac{t}{3} - 0 = \sin^{-1} \frac{3}{3} = \sin^{-1} 1 = \frac{\pi}{2}.$$

Hence the given integral converges, to  $\frac{\pi}{2}$ .

(f)  $\int_{-3}^3 \frac{dx}{x^2}.$

**Solution:** The integrand  $f(x) = \frac{1}{x^2}$  has an infinite discontinuity at  $x = 0 \in [-3, 3]$ , so the given integral is an improper integral of Type 2 with the discontinuity *inside* the interval. Using

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx,$$

we split at  $c = 0$ :

$$\int_{-3}^3 \frac{dx}{x^2} = \int_{-3}^0 \frac{dx}{x^2} + \int_0^3 \frac{dx}{x^2}. \quad (2.5.1)$$

Taking the first piece,

$$I_1 = \lim_{t \rightarrow 0^-} \int_{-3}^t \frac{dx}{x^2} = \lim_{t \rightarrow 0^-} \left[ \frac{-1}{x} \right]_{x=-3}^{x=t} = \lim_{t \rightarrow 0^-} \left( \frac{-1}{t} \right) - \frac{1}{3} = \infty,$$

since  $\frac{-1}{t} \rightarrow +\infty$  as  $t \rightarrow 0^-$ . From (2.5.1) it is clear that the given improper integral diverges: it is enough for one piece to diverge.

**Common mistake:**

Ignoring the discontinuity and writing  $\left[\frac{-1}{x}\right]_{-3}^3 = -\frac{1}{3} - \frac{1}{3} = -\frac{2}{3}$  gives a negative answer for the integral of a strictly positive function. That is the standard warning against applying the fundamental theorem across an infinite discontinuity.

(g)  $\int_0^\pi \tan x \, dx.$

**Solution:** As  $x \rightarrow \frac{\pi}{2}$  we have  $\tan x \rightarrow \infty$ , so the integrand has an infinite discontinuity at  $x = \frac{\pi}{2}$ , which lies *inside*  $[0, \pi]$ . This is an improper integral of Type 2 and must be split at the discontinuity:

$$\int_0^\pi \tan x \, dx = \underbrace{\int_0^{\pi/2} \tan x \, dx}_{I_1} + \underbrace{\int_{\pi/2}^\pi \tan x \, dx}_{I_2},$$

each piece being an improper integral of Type 2. Using  $\int \tan x \, dx = -\log |\cos x| + C$ , the first piece is

$$I_1 = \lim_{t \rightarrow \frac{\pi}{2}^-} \left[ -\log |\cos x| \right]_0^t = \lim_{t \rightarrow \frac{\pi}{2}^-} \left( -\log |\cos t| + \underbrace{\log |\cos 0|}_{\log 1 = 0} \right) = - \lim_{t \rightarrow \frac{\pi}{2}^-} \log |\cos t|.$$

As  $t \rightarrow \frac{\pi}{2}^-$  we have  $|\cos t| \rightarrow 0^+$ , so  $\log |\cos t| \rightarrow -\infty$  and therefore

$$I_1 = +\infty.$$

The first piece already diverges, and by the rule used in part (f) that is enough: the given improper integral  $\int_0^\pi \tan x \, dx$  is **divergent**.

**Note:**

It is worth seeing what the second piece does, because it is a trap. By the same antiderivative,

$$\begin{aligned} I_2 &= \lim_{t \rightarrow \frac{\pi}{2}^+} \left[ -\log |\cos x| \right]_t^\pi = \lim_{t \rightarrow \frac{\pi}{2}^+} \left( -\log |\cos \pi| + \underbrace{\log |\cos t|}_{-\log 1 = 0} \right) \\ &= \lim_{t \rightarrow \frac{\pi}{2}^+} \log |\cos t| = -\infty. \end{aligned}$$

So  $I_1 = +\infty$  while  $I_2 = -\infty$  — the two pieces run off in *opposite* directions. They must not be added. The expression  $\infty + (-\infty)$  has no value, and there is no legitimate way to cancel the two halves against each other to obtain the finite answer 0 that the symmetry of  $\tan x$  about  $x = \frac{\pi}{2}$  might suggest. This is exactly why the definition insists that *each* piece converge separately before the sum is formed. The absolute values matter too: on  $(\frac{\pi}{2}, \pi)$  the cosine is negative, and  $\log(\cos x)$  would be undefined there.

(h)  $\int_0^\infty \frac{e^x}{e^{2x} + 3} \, dx.$

**Solution:** The interval is infinite, so this is an improper integral of Type 1:

$$\int_0^{\infty} \frac{e^x}{e^{2x} + 3} dx = \lim_{t \rightarrow \infty} \int_0^t \frac{e^x}{e^{2x} + 3} dx.$$

Put  $u = e^x$ , so that  $du = e^x dx$  and  $e^{2x} = u^2$ . The limits transform as  $x = 0 \Rightarrow u = 1$  and  $x = t \Rightarrow u = e^t$ , so

$$\int_0^t \frac{e^x}{e^{2x} + 3} dx = \int_1^{e^t} \frac{du}{u^2 + 3} = \int_1^{e^t} \frac{du}{u^2 + (\sqrt{3})^2}.$$

Using  $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$  with  $a = \sqrt{3}$ ,

$$= \frac{1}{\sqrt{3}} \left[ \tan^{-1} \frac{u}{\sqrt{3}} \right]_1^{e^t} = \frac{1}{\sqrt{3}} \left( \tan^{-1} \frac{e^t}{\sqrt{3}} - \tan^{-1} \frac{1}{\sqrt{3}} \right).$$

Now take  $t \rightarrow \infty$ . Since  $e^t \rightarrow \infty$  we have  $\tan^{-1} \frac{e^t}{\sqrt{3}} \rightarrow \frac{\pi}{2}$ , while  $\tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$ . Hence

$$\int_0^{\infty} \frac{e^x}{e^{2x} + 3} dx = \frac{1}{\sqrt{3}} \left( \frac{\pi}{2} - \frac{\pi}{6} \right) = \frac{1}{\sqrt{3}} \cdot \frac{\pi}{3} = \frac{\pi}{3\sqrt{3}} \approx 0.6046.$$

The limit exists finitely, so the integral is **convergent**, with value  $\frac{\pi}{3\sqrt{3}}$ .

**Note:**

Note the order of operations. The substitution is carried out on the *proper* integral  $\int_0^t$ , where  $u = e^x$  is a genuine change of variable between finite limits; only afterwards is  $t \rightarrow \infty$  taken. It is never legitimate to substitute directly into a symbol such as  $\int_0^{\infty}$ , because  $\infty$  is not a value that  $u$  can take.

(i)  $\int_{-1}^0 \frac{e^{1/x}}{x^2} dx.$

**Solution:** As  $x \rightarrow 0^-$  the exponent  $\frac{1}{x} \rightarrow -\infty$ , and the integrand is undefined at  $x = 0$ ; the discontinuity sits at the upper limit, so this is an improper integral of Type 2:

$$\int_{-1}^0 \frac{e^{1/x}}{x^2} dx = \lim_{t \rightarrow 0^-} \int_{-1}^t \frac{e^{1/x}}{x^2} dx.$$

Put  $u = \frac{1}{x}$ , so that  $-\frac{1}{x^2} dx = du$ , that is  $\frac{1}{x^2} dx = -du$ . The limits become  $x = -1 \Rightarrow u = -1$  and  $x = t \Rightarrow u = \frac{1}{t}$ , so

$$\int_{-1}^t \frac{e^{1/x}}{x^2} dx = - \int_{-1}^{1/t} e^u du = - \left[ e^u \right]_{-1}^{1/t} = - \left( e^{1/t} - e^{-1} \right).$$

As  $t \rightarrow 0^-$  the quantity  $\frac{1}{t} \rightarrow -\infty$ , so  $e^{1/t} \rightarrow e^{-\infty} = 0$ . Therefore

$$\int_{-1}^0 \frac{e^{1/x}}{x^2} dx = - \left( 0 - e^{-1} \right) = \frac{1}{e} \approx 0.3679.$$

The integral is **convergent**, with value  $\frac{1}{e}$ .

**Note:**

The one-sided approach is essential here. Had  $t \rightarrow 0^+$  instead, then  $\frac{1}{t} \rightarrow +\infty$  and  $e^{1/t} \rightarrow \infty$ , and the integral  $\int_0^1 \frac{e^{1/x}}{x^2} dx$  would diverge. The same integrand gives a finite area on one side of the origin and an infinite one on the other, so the sign of the approach cannot be dropped.

**2.5.4 The two  $p$ -integrals**

**Question.** For what values of  $p$  is  $\int_1^\infty \frac{1}{x^p} dx$  convergent?

**Solution: Case (i):**  $p = 1$ .

$$I = \int_1^\infty \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} [\log x]_1^t = \lim_{t \rightarrow \infty} \left( \log t - \underbrace{\log 1}_{=0} \right) = \infty,$$

so the integral diverges.

**Case (ii):**  $p < 1$ .

$$I = \lim_{t \rightarrow \infty} \int_1^t x^{-p} dx = \lim_{t \rightarrow \infty} \left[ \frac{x^{-p+1}}{-p+1} \right]_1^t = \frac{1}{1-p} \lim_{t \rightarrow \infty} (t^{1-p} - 1).$$

Since  $p < 1$  we have  $1 - p > 0$ , so  $t^{1-p} \rightarrow \infty$  and therefore  $I = \infty$ . The integral diverges.

**Case (iii):**  $p > 1$ .

$$I = \lim_{t \rightarrow \infty} \left[ \frac{x^{-p+1}}{1-p} \right]_1^t = \frac{1}{1-p} \lim_{t \rightarrow \infty} (t^{1-p} - 1).$$

Now  $p > 1$  gives  $1 - p < 0$ , so

$$\lim_{t \rightarrow \infty} t^{1-p} = \lim_{t \rightarrow \infty} \frac{1}{t^{p-1}} = 0,$$

and hence  $I = \frac{-1}{1-p} = \frac{1}{p-1}$ , a finite number. The integral converges.

**Note:**

$$\int_1^\infty \frac{1}{x^p} dx = \begin{cases} \text{converges to } \frac{1}{p-1}, & p > 1, \\ \text{diverges,} & p \leq 1. \end{cases}$$

**Question.** For what values of  $p$  is  $\int_0^1 \frac{1}{x^p} dx$  convergent?

Here the integrand has an infinite discontinuity at the lower limit  $x = 0$  whenever  $p > 0$ , so this is an improper integral of Type 2.

**Case (i):**  $p = 1$ .

$$I = \int_0^1 \frac{1}{x} dx = \lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{x} dx = \lim_{t \rightarrow 0^+} [\log x]_t^1 = \lim_{t \rightarrow 0^+} (-\log t) = \lim_{t \rightarrow 0^+} \log \frac{1}{t} = \infty,$$

so the integral diverges.

**Case (ii):**  $p > 1$ , so that  $1 - p < 0$ .

$$I = \lim_{t \rightarrow 0^+} \int_t^1 x^{-p} dx = \lim_{t \rightarrow 0^+} \left[ \frac{x^{1-p}}{1-p} \right]_t^1 = \frac{1}{1-p} \lim_{t \rightarrow 0^+} (1 - t^{1-p}).$$

Since  $1 - p < 0$ , the quantity  $t^{1-p} = \frac{1}{t^{p-1}} \rightarrow \infty$  as  $t \rightarrow 0^+$  a small positive number raised to a negative power. Hence  $I = \infty$  and the integral diverges. **Case (iii):**  $p < 1$ , so that  $1 - p > 0$ .

$$I = \lim_{t \rightarrow 0^+} \int_t^1 x^{-p} dx = \lim_{t \rightarrow 0^+} \left[ \frac{x^{1-p}}{1-p} \right]_t^1 = \frac{1}{1-p} - \frac{1}{1-p} \underbrace{\lim_{t \rightarrow 0^+} t^{1-p}}_{=0} = \frac{1}{1-p},$$

a finite number, so the integral converges.

**Note:**

$$\int_0^1 \frac{1}{x^p} dx = \begin{cases} \text{converges to } \frac{1}{1-p}, & p < 1, \\ \text{diverges,} & p \geq 1. \end{cases}$$

The two conditions are opposite. Near infinity  $\frac{1}{x^p}$  must decay *fast* to be integrable, so  $p$  has to be large; near zero it must blow up *slowly*, so  $p$  has to be small. Neither integral converges at  $p = 1$ .

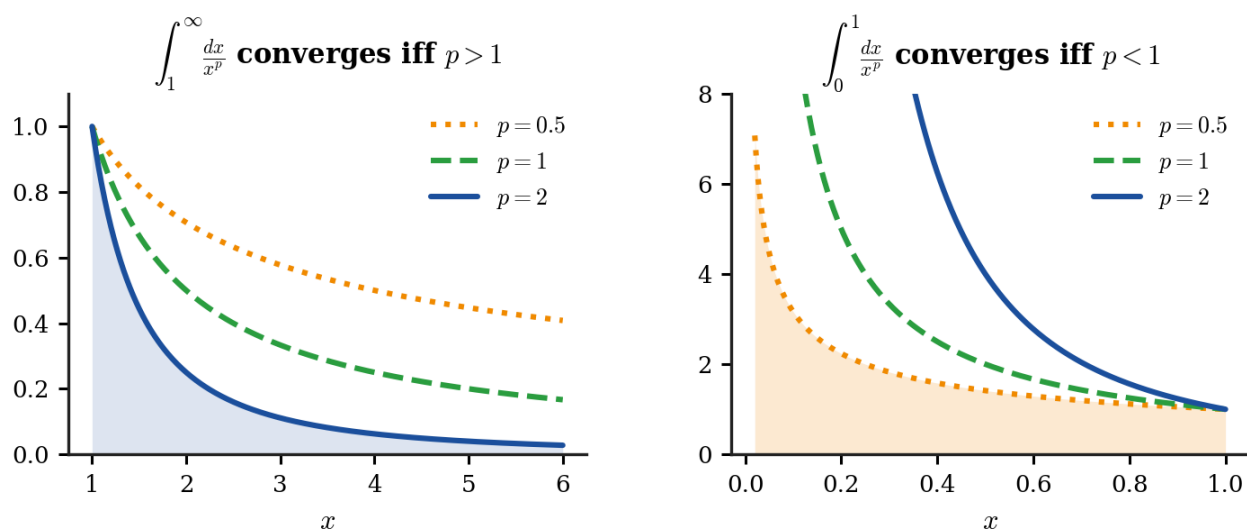


Figure 2.5.3: The same family of curves on the two intervals. The shaded region is finite in each panel:  $p = 2$  on the left,  $p = \frac{1}{2}$  on the right.

### 2.5.5 Further worked problems

**Problem 2.5.4.** Evaluate the following improper integrals, or show that they diverge.

(a)  $\int_{-\infty}^0 x e^{-x^2} dx.$

**Solution:** Here the interval of integration is  $(-\infty, 0]$ , which is infinite, so the given integral is an improper integral of Type 1. Put  $x^2 = u$ , so that  $2x dx = du$ ; when  $x \rightarrow -\infty$ ,  $u \rightarrow \infty$ ,

and when  $x \rightarrow 0$ ,  $u \rightarrow 0$ . For the definite integral from  $t$  to 0 this gives

$$I = \int_t^0 x e^{-x^2} dx = \frac{1}{2} \int_{t^2}^0 e^{-u} du = -\frac{1}{2} \int_0^{t^2} e^{-u} du = -\frac{1}{2} \left[ -e^{-u} \right]_0^{t^2} = \frac{1}{2} (e^{-t^2} - 1).$$

As  $t \rightarrow -\infty$  we have  $t^2 \rightarrow \infty$ , so  $e^{-t^2} \rightarrow 0$  and

$$\lim_{t \rightarrow -\infty} I = \frac{1}{2}(0 - 1) = -\frac{1}{2}.$$

Hence the given integral converges, to  $-\frac{1}{2}$ .

**Note:**

The answer must be negative: on  $(-\infty, 0)$  the factor  $x$  is negative while  $e^{-x^2} > 0$ , so the integrand is negative throughout.

(b)  $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx.$

**Solution:** The interval  $(-\infty, \infty)$  is infinite, so this is of Type 1, and it must be split at a convenient point, here  $x = 0$ :

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \int_{-\infty}^0 \frac{dx}{1+x^2} + \int_0^{\infty} \frac{dx}{1+x^2} = \lim_{t \rightarrow -\infty} \left[ \tan^{-1} x \right]_t^0 + \lim_{t \rightarrow \infty} \left[ \tan^{-1} x \right]_0^t.$$

Evaluating each piece,

$$= \lim_{t \rightarrow -\infty} (\tan^{-1} 0 - \tan^{-1} t) + \lim_{t \rightarrow \infty} (\tan^{-1} t - \tan^{-1} 0) = -\left(-\frac{\pi}{2}\right) + \frac{\pi}{2} = \pi.$$

Hence the given integral converges, to  $\pi$ .

(c)  $\int_3^{\infty} \frac{dx}{(x-2)^{3/2}}.$

**Solution:** The interval  $[3, \infty)$  is infinite, so this is an improper integral of Type 1. Note that the integrand is continuous on  $[3, \infty)$  the trouble spot  $x = 2$  lies outside the interval. Thus

$$I = \lim_{t \rightarrow \infty} \int_3^t (x-2)^{-3/2} dx = \lim_{t \rightarrow \infty} \left[ \frac{(x-2)^{-1/2}}{-\frac{1}{2}} \right]_3^t = -2 \lim_{t \rightarrow \infty} \left( \frac{1}{\sqrt{t-2}} - \frac{1}{\sqrt{3-2}} \right).$$

Since  $\frac{1}{\sqrt{t-2}} \rightarrow 0$ ,

$$I = -2(0 - 1) = 2,$$

and the given integral converges, to 2.

### 2.5.6 The comparison test

Sometimes an improper integral cannot be evaluated in closed form, and yet its convergence or divergence can still be settled by comparing it with an integral that can. The  $p$ -integrals of the previous subsection are the usual yardsticks.

**Theorem 2.5.3: Comparison test for improper integrals**

Suppose  $f$  and  $g$  are continuous functions with

$$f(x) \geq g(x) \geq 0 \quad \text{for every } x \geq a.$$

Then:

- (i) If  $\int_a^\infty f(x) dx$  is convergent, then  $\int_a^\infty g(x) dx$  is convergent.
- (ii) If  $\int_a^\infty g(x) dx$  is divergent, then  $\int_a^\infty f(x) dx$  is divergent.

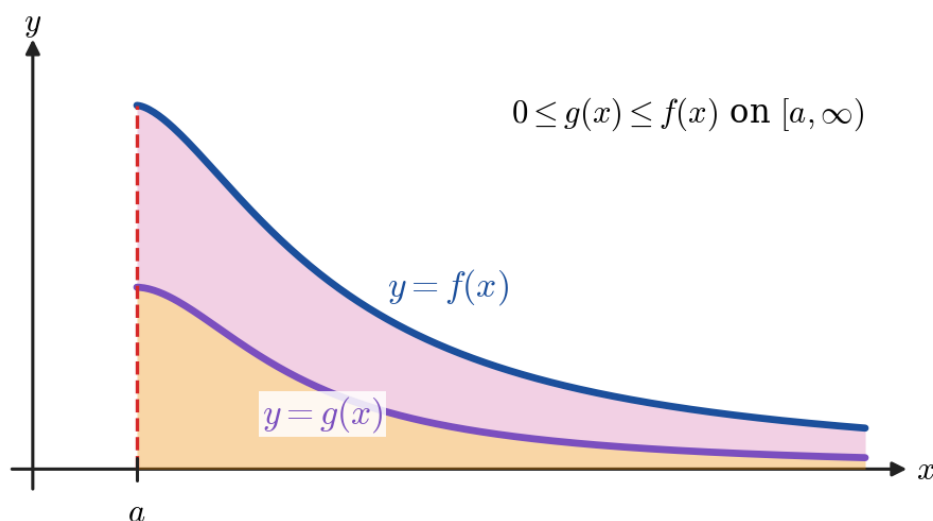


Figure 2.5.4: The graph of  $g$  lies between the  $x$ -axis and the graph of  $f$  throughout  $[a, \infty)$ , so the area under  $g$  is contained in the area under  $f$ . A finite outer area forces a finite inner one; an infinite inner area forces an infinite outer one.

The geometric reading is the whole proof in miniature. The area under  $g$  sits inside the area under  $f$ , so:

- if the area under the larger curve  $f$  is finite, then the area under the smaller curve  $g$ , being part of it, is finite too;
- if the area under the smaller curve  $g$  is infinite, then the area under the larger curve  $f$ , which contains it, is infinite too.

**Note:**

The two useless directions are worth stating explicitly, because they are where the test is most often misapplied.

- If  $\int_a^\infty g$  converges,  $\int_a^\infty f$  may or may not converge. A small function being integrable says nothing about a larger one.
- If  $\int_a^\infty f$  diverges,  $\int_a^\infty g$  may or may not diverge. A large function failing says nothing about a smaller one.

In short, the test only ever transfers good news downwards and bad news upwards. To

prove convergence, find a *larger* convergent integral; to prove divergence, find a *smaller* divergent one. Choosing the inequality the wrong way round is the commonest error in this topic, and it produces an argument that proves nothing while looking complete.

**Note:**

The hypothesis  $g(x) \geq 0$  is not decoration. The test compares *areas*, and the argument that one area is contained in another needs both regions to lie above the axis. For integrands that change sign, convergence must be settled by other means — typically by applying the test to  $|f|$  instead.

### 2.5.7 Worked problems on the comparison test

In each problem the strategy is the same: bound the given integrand above or below by a function whose integral is already known — almost always one of the  $p$ -integrals or an exponential — and read off the conclusion from [Theorem 2.5.3](#). Which direction to bound in is decided by what is to be proved.

**Problem 2.5.5.** Show that  $\int_0^\infty e^{-x^2} dx$  is convergent.

**Solution:** The useful inequality  $x^2 \geq x$  holds only for  $x \geq 1$ , so split the interval there:

$$\int_0^\infty e^{-x^2} dx = \underbrace{\int_0^1 e^{-x^2} dx}_{\text{proper}} + \underbrace{\int_1^\infty e^{-x^2} dx}_{\text{improper, Type 1}}.$$

The first integral is a definite integral of a continuous function over a finite interval, so it is a proper integral and is certainly finite; nothing needs to be proved about it.

For the second, note that for  $x \geq 1$ ,

$$x^2 \geq x \implies -x^2 \leq -x \implies e^{-x^2} \leq e^{-x},$$

the last step because  $t \mapsto e^t$  is increasing. Both functions are positive, so [Theorem 2.5.3](#) applies with  $f(x) = e^{-x}$  and  $g(x) = e^{-x^2}$ . Now

$$\int_1^\infty e^{-x} dx = \lim_{t \rightarrow \infty} \left[ -e^{-x} \right]_1^t = -\lim_{t \rightarrow \infty} (e^{-t} - e^{-1}) = -(0 - e^{-1}) = \frac{1}{e},$$

which is finite, so  $\int_1^\infty e^{-x} dx$  converges. By part (i) of the comparison test,  $\int_1^\infty e^{-x^2} dx$  converges too. Being the sum of a proper integral and a convergent improper one,  $\int_0^\infty e^{-x^2} dx$  is **convergent**.

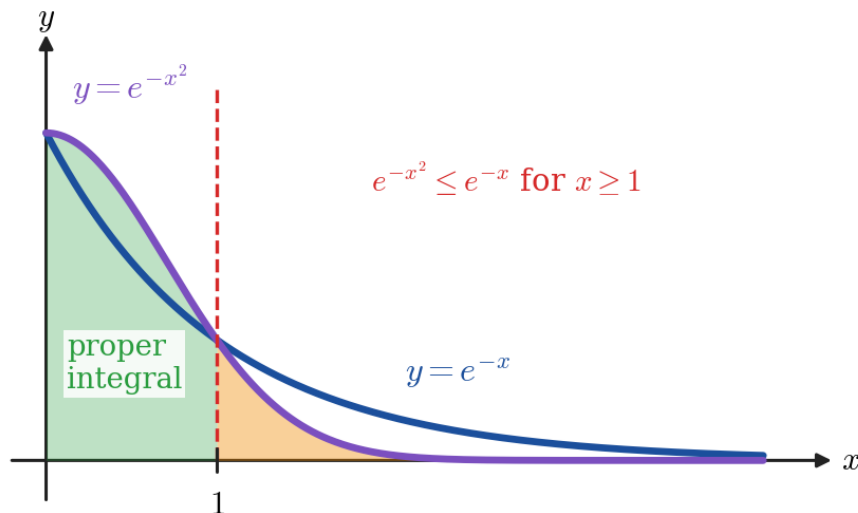


Figure 2.5.5: On  $[1, \infty)$  the curve  $e^{-x^2}$  lies below  $e^{-x}$ , so its tail area is finite. On  $[0, 1]$  the inequality runs the other way, which is exactly why that part is split off and handled as a proper integral instead.

**Note:**

The splitting point is not arbitrary and it is not cosmetic. Below  $x = 1$  we have  $x^2 \leq x$ , so  $e^{-x^2} \geq e^{-x}$  and the comparison reverses — applying the test on  $[0, \infty)$  in one go would be invalid. Convergence of a Type 1 integral is decided entirely by the tail, so any finite piece may be detached in this way whenever it makes the inequality work.

**Note:**

The test settles *whether* the integral converges without producing its value, and that is all it can do. The value here is  $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \approx 0.8862$ , the Gaussian integral, which no elementary antiderivative will give —  $e^{-x^2}$  has none. That is precisely the situation the comparison test exists for.

**Problem 2.5.6.** Test  $\int_1^\infty \frac{1 + e^{-x}}{x} dx$  for convergence.

**Solution:** For each  $x \geq 1$  we have  $e^{-x} > 0$ , so  $1 + e^{-x} > 1$  and hence, dividing by  $x > 0$ ,

$$f(x) = \frac{1 + e^{-x}}{x} > \frac{1}{x} = g(x) \geq 0.$$

The smaller function is the one whose integral is known:

$$\int_1^\infty \frac{1}{x} dx = \lim_{t \rightarrow \infty} \left[ \log |x| \right]_1^t = \lim_{t \rightarrow \infty} (\log t - 0) = \infty,$$

so  $\int_1^\infty \frac{dx}{x}$  diverges. By part (ii) of the comparison test, the larger integral diverges as well:  $\int_1^\infty \frac{1 + e^{-x}}{x} dx$  is **divergent**.

**Note:**

The term  $e^{-x}$  decays to 0 and contributes only a finite amount to the total, so it cannot rescue the integral; the divergence is driven entirely by the  $\frac{1}{x}$  part, which is the

$p$ -integral with  $p = 1$ .

**Problem 2.5.7.** Test  $\int_1^\infty \frac{\sin^2 x}{x^2} dx$  for convergence.

**Solution:** For every  $x$  we have  $0 \leq \sin^2 x \leq 1$ , so for  $x \geq 1$ ,

$$g(x) = \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2} = f(x),$$

and both sides are non-negative, as the test requires. The larger integral is the  $p$ -integral with  $p = 2$ :

$$\int_1^\infty \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left[ -\frac{1}{x} \right]_1^t = -\lim_{t \rightarrow \infty} \left( \frac{1}{t} - 1 \right) = 1,$$

which converges. By part (i) of the comparison test,  $\int_1^\infty \frac{\sin^2 x}{x^2} dx$  is **convergent**.

**Note:**

Squaring the sine is what makes this work. The test needs a non-negative integrand, and  $\frac{\sin x}{x^2}$  oscillates in sign, so the comparison test could not be applied to it directly — one would bound  $\left| \frac{\sin x}{x^2} \right| \leq \frac{1}{x^2}$  instead and argue via absolute convergence.

**Problem 2.5.8.** Test  $\int_1^\infty \frac{dx}{x^2(1+e^x)}$  for convergence.

**Solution:** For every  $x \geq 1$  we have  $e^x > 0$ , so  $1 + e^x > 1$  and therefore  $x^2(1 + e^x) > x^2$ . Taking reciprocals reverses the inequality:

$$g(x) = \frac{1}{x^2(1+e^x)} < \frac{1}{x^2} = f(x),$$

with both sides positive. As computed in the previous problem,  $\int_1^\infty \frac{dx}{x^2} = 1$  converges, so by part (i) of the comparison test the given integral is **convergent**.

**Problem 2.5.9.** Test  $\int_2^\infty \frac{2 + \sin x}{x - 1} dx$  for convergence.

**Solution:** For every  $x$  we have  $\sin x \geq -1$ , so

$$2 + \sin x \geq 2 - 1 = 1.$$

Since  $x - 1 > 0$  for  $x \geq 2$ , dividing through gives

$$f(x) = \frac{2 + \sin x}{x - 1} \geq \frac{1}{x - 1} = g(x) \geq 0.$$

For the smaller function, substituting  $u = x - 1$  or integrating directly,

$$\int_2^\infty \frac{dx}{x - 1} = \lim_{t \rightarrow \infty} \left[ \log(x - 1) \right]_2^t = \lim_{t \rightarrow \infty} \left( \log(t - 1) - \underbrace{\log 1}_{=0} \right) = \infty,$$

so  $\int_2^\infty \frac{dx}{x - 1}$  diverges. By part (ii) of the comparison test,  $\int_2^\infty \frac{2 + \sin x}{x - 1} dx$  is **divergent**.

**Note:**

The oscillating numerator never gets a chance to cause trouble because it is bounded away from zero:  $2 + \sin x$  stays between 1 and 3 for every  $x$ . The integrand is therefore squeezed between  $\frac{1}{x-1}$  and  $\frac{3}{x-1}$ , both of which diverge, so the shift from  $x$  to  $x-1$  and the wobble from  $\sin x$  are equally irrelevant — this is the  $p = 1$  case in disguise.

**Note:**

**Choosing the comparison function.** In every problem above the comparison was made against  $\frac{1}{x^p}$  or an exponential. The practical recipe is to look at the integrand's behaviour for large  $x$ , discard bounded factors that neither vanish nor blow up, and compare with whatever power of  $x$  remains. In [Problem 2.5.6](#) the integrand behaves like  $\frac{1}{x}$ ; in [Problems 2.5.7](#) and [2.5.8](#) it is dominated by  $\frac{1}{x^2}$ ; in [Problem 2.5.9](#) it behaves like  $\frac{1}{x}$  once the bounded numerator is discarded. Then bound in the direction the conclusion needs — above for convergence, below for divergence.

## 2.5.8 The limit comparison test

The comparison test needs a genuine inequality between the two integrands, and finding one can be fiddly — in [Problem 2.5.9](#) above the inequality  $\frac{2+\sin x}{x-1} \geq \frac{1}{x-1}$  had to be manufactured by bounding the numerator. Often it is clear that two integrands *behave alike* for large  $x$  without either dominating the other. The following test uses only that comparison of behaviour.

**Theorem 2.5.4: Limit comparison test**

Let  $f$  and  $g$  be positive continuous functions on  $[a, \infty)$ , and suppose

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L, \quad \text{where } 0 < L < \infty.$$

Then  $\int_a^\infty f(x) dx$  and  $\int_a^\infty g(x) dx$  either both converge or both diverge.

**Note:**

The conclusion is symmetric in  $f$  and  $g$ : the two integrals share a fate, and the test does not care which is which. This is a real convenience over [Theorem 2.5.3](#), where getting the inequality the wrong way round gives an argument that proves nothing.

**Note:**

Both restrictions on  $L$  matter, and the test says nothing when either fails. If  $L = 0$  then  $f$  is eventually much smaller than  $g$ , and convergence of  $\int g$  carries down to  $\int f$  but divergence of  $\int g$  says nothing. If  $L = \infty$  the roles are reversed. And if the limit does not exist at all — as happens whenever the ratio oscillates without settling — the test is simply unavailable, and [Theorem 2.5.3](#) must be used instead.

**Example 2.5.5: Comparing with a  $p$ -integral**

Show that  $\int_1^\infty \frac{dx}{1+x^2}$  is convergent, by comparison with  $\int_1^\infty \frac{dx}{x^2}$ .

**Solution:** Take

$$f(x) = \frac{1}{1+x^2}, \quad g(x) = \frac{1}{x^2},$$

both positive and continuous on  $[1, \infty)$ . Forming the ratio,

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{\frac{1}{1+x^2}}{\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{x^2}{1+x^2}.$$

Dividing numerator and denominator by  $x^2$ ,

$$= \lim_{x \rightarrow \infty} \frac{x^2}{x^2 \left( \frac{1}{x^2} + 1 \right)} = \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x^2} + 1} = \frac{1}{0+1} = 1.$$

So  $L = 1$ , which satisfies  $0 < L < \infty$ . Since  $\int_1^\infty \frac{dx}{x^2} = 1$  is convergent — the  $p$ -integral with  $p = 2$  — [Theorem 2.5.4](#) gives that  $\int_1^\infty \frac{dx}{1+x^2}$  is **convergent** as well.

**Note:**

Here the ordinary comparison test would also have worked, since  $\frac{1}{1+x^2} < \frac{1}{x^2}$  outright. The limit test earns its keep when no such clean inequality is available. Note too that, as with the ordinary test, convergence is established without the value being produced — though in this case it happens to be computable:

$$\int_1^\infty \frac{dx}{1+x^2} = \lim_{t \rightarrow \infty} \left[ \tan^{-1} x \right]_1^t = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \approx 0.7854,$$

comfortably less than the value 1 of the comparison integral, as it must be.

**Note:**

**Which test to reach for.** Both tests reduce an unknown integral to a known one, almost always a  $p$ -integral. Use [Theorem 2.5.3](#) when a clean inequality is visible; use [Theorem 2.5.4](#) when the integrands merely behave alike for large  $x$ , for instance when the integrand is a ratio of polynomials. For  $\frac{P(x)}{Q(x)}$  with  $\deg Q - \deg P = k$ , comparing against  $\frac{1}{x^k}$  always produces a finite non-zero  $L$ , so the question reduces at once to whether  $k > 1$ .

## 2.6 Gamma and Beta Functions

### 2.6.1 The gamma function

**Definition 2.6.1: Gamma function**

The gamma function is defined by

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, \quad n \in \mathbb{R}, \quad n > 0, \quad (2.6.1)$$

where  $x$  is a dummy variable.

**Note:**

This is an improper integral of Type 1 because of the infinite upper limit. It is *also* of Type 2 when  $0 < n < 1$ : then  $n - 1 < 0$ , so  $x^{n-1} = \frac{1}{x^{1-n}} \rightarrow \infty$  as  $x \rightarrow 0^+$ , an infinite discontinuity at the lower limit.

**2.6.2 Does the defining integral converge?**

Being improper in two ways at once, the integral in (2.6.1) has to be split before anything can be said about it. Take any  $c > 0$  — the choice  $c = 1$  is convenient — and write

$$\Gamma(n) = \underbrace{\int_0^c e^{-x} x^{n-1} dx}_{\text{Type 2}} + \underbrace{\int_c^{\infty} e^{-x} x^{n-1} dx}_{\text{Type 1}}.$$

Both pieces must converge for  $\Gamma(n)$  to be defined, and each is settled by the comparison test.

**The piece near 0.** On  $(0, c]$  we have  $0 < e^{-x} \leq 1$ , so

$$0 < e^{-x} x^{n-1} \leq x^{n-1}.$$

The comparison integral is a  $p$ -integral of the second kind, with  $p = 1 - n$ :

$$\int_0^c x^{n-1} dx = \int_0^c \frac{dx}{x^{1-n}},$$

which converges precisely when  $1 - n < 1$ , that is when  $n > 0$ . So the first piece converges for every  $n > 0$ , and diverges for  $n \leq 0$ .

**The piece near  $\infty$ .** Here the exponential decides everything. Since  $x^{n-1}e^{-x/2} \rightarrow 0$  as  $x \rightarrow \infty$  for any fixed  $n$ , there is a constant  $M$  with  $x^{n-1} \leq Me^{x/2}$  for all large  $x$ , and hence

$$0 < e^{-x} x^{n-1} \leq Me^{-x/2}.$$

Since  $\int_c^{\infty} Me^{-x/2} dx = 2Me^{-c/2}$  is finite, the comparison test gives convergence of the second piece — for *every* real  $n$ .

Both pieces converge exactly when  $n > 0$ , which is why the definition carries that restriction.

**Note:**

The two ends fail for opposite reasons, and it is worth keeping them apart. At the upper end nothing can go wrong:  $e^{-x}$  decays faster than any power of  $x$  grows, so the tail converges whatever  $n$  is. The only genuine constraint comes from the lower end, where  $x^{n-1}$  blows up too fast to be integrable as soon as  $n \leq 0$ . This is the same  $p$ -integral dividing line met earlier, seen from the other side.

### 2.6.3 Properties

(i)  $\Gamma(1) = 1$ .

**Solution:** Putting  $n = 1$  in (2.6.1),

$$\Gamma(1) = \int_0^{\infty} e^{-x} x^{1-1} dx = \int_0^{\infty} e^{-x} dx = \lim_{t \rightarrow \infty} \int_0^t e^{-x} dx = \lim_{t \rightarrow \infty} \left[ -e^{-x} \right]_0^t.$$

Evaluating,

$$\Gamma(1) = \lim_{t \rightarrow \infty} \left[ -e^{-t} - (-e^0) \right] = \lim_{t \rightarrow \infty} [1 - e^{-t}] = 1 - 0 = 1.$$

(ii)  $\Gamma(n+1) = n\Gamma(n)$  for  $n \in \mathbb{R}$ ,  $n > 0$ .

**Solution:** From the definition,

$$\Gamma(n+1) = \int_0^{\infty} e^{-x} x^{n+1-1} dx = \int_0^{\infty} e^{-x} x^n dx = \lim_{t \rightarrow \infty} \int_0^t x^n e^{-x} dx.$$

Integrating by parts with  $u = x^n$  and  $dv = e^{-x} dx$ , so that  $v = -e^{-x}$ ,

$$\Gamma(n+1) = \lim_{t \rightarrow \infty} \left\{ \left[ -x^n e^{-x} \right]_0^t - \int_0^t n x^{n-1} (-e^{-x}) dx \right\}.$$

The boundary term vanishes at both ends, and the remaining integral is  $\Gamma(n)$  again:

$$\begin{aligned} \Gamma(n+1) &= \lim_{t \rightarrow \infty} \left( -t^n e^{-t} - 0 \right) + n \lim_{t \rightarrow \infty} \int_0^t e^{-x} x^{n-1} dx \\ &= -\lim_{t \rightarrow \infty} \frac{t^n}{e^t} + n \int_0^{\infty} e^{-x} x^{n-1} dx. \end{aligned}$$

Since an exponential grows faster than any power (Example 2.3.2(c)),  $\lim_{t \rightarrow \infty} \frac{t^n}{e^t} = 0$ , and therefore

$$\Gamma(n+1) = n\Gamma(n). \quad (2.6.2)$$

### 2.6.4 Consequences of the recurrence

If  $n$  is a positive integer, repeated application of (2.6.2) gives

$$\begin{aligned} \Gamma(n+1) &= n\Gamma(n) = n(n-1)\Gamma(n-1) = n(n-1)\Gamma(n-2+1) \\ &= n(n-1)(n-2)(n-3) \cdots 3 \cdot 2 \cdot \Gamma(1) = n(n-1) \cdots 3 \cdot 2 \cdot 1 = n!. \end{aligned}$$

Hence

$$\boxed{\Gamma(n+1) = n!} \quad \text{or} \quad \boxed{\Gamma(n) = (n-1)!} \quad n \in \mathbb{Z}^+.$$

Rearranging (2.6.2) gives

$$\Gamma(n) = \frac{\Gamma(n+1)}{n}, \quad (2.6.3)$$

which extends the definition to negative  $n$ . Repeated application of (2.6.3) yields

$$\Gamma(n) = \frac{\Gamma(n+1)}{n} = \frac{\Gamma(n+2)}{n(n+1)} = \frac{\Gamma(n+3)}{n(n+1)(n+2)} = \cdots = \frac{\Gamma(n+k+1)}{n(n+1) \cdots (n+k)}, \quad k \in \mathbb{Z}.$$

Each of these has a zero in the denominator when  $n$  is zero or a negative integer, so

the gamma function is not defined at  $n = 0, -1, -2, -3, \dots$

The value  $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ , on which every half-integer value rests, needs the beta function and is derived in Proposition 2.6.3.

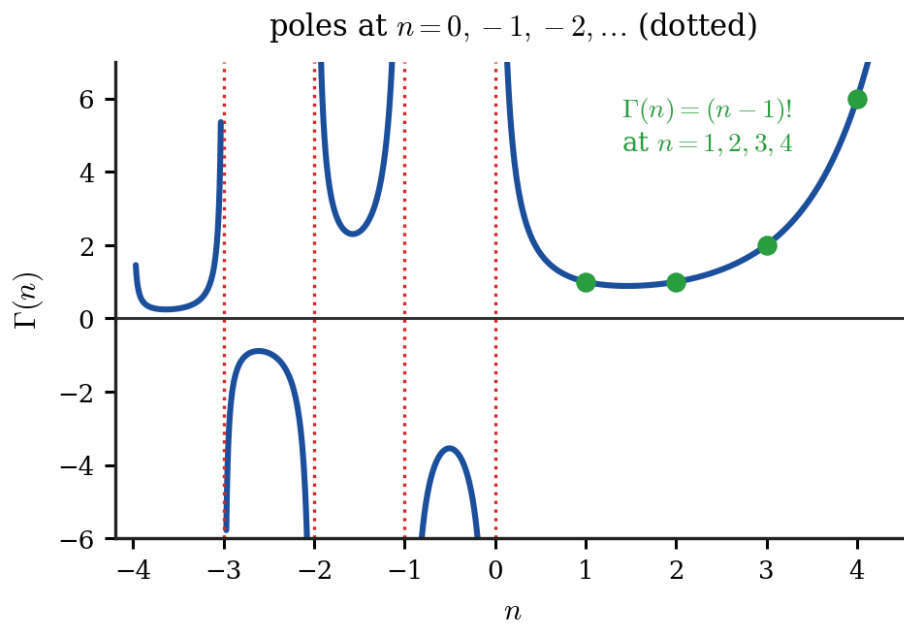


Figure 2.6.1: The gamma function. It passes through the factorials at the positive integers and has a vertical asymptote at every non-positive integer, which is where (2.6.3) breaks down.

## 2.6.5 The beta function

### Definition 2.6.2: Beta function

The beta function is given by

$$\beta(m, n) = B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx, \quad m, n \in \mathbb{R}, \quad m, n > 0.$$

#### Note:

This is an improper integral of Type 2 whenever  $m < 1$  or  $n < 1$ . If  $m < 1$  then  $x^{m-1} = \frac{1}{x^{1-m}} \rightarrow \infty$  as  $x \rightarrow 0^+$ , and similarly  $n < 1$  gives a blow-up at  $x = 1$ . For  $m, n \geq 1$  the integrand is bounded on  $[0, 1]$  and the integral is an ordinary one.

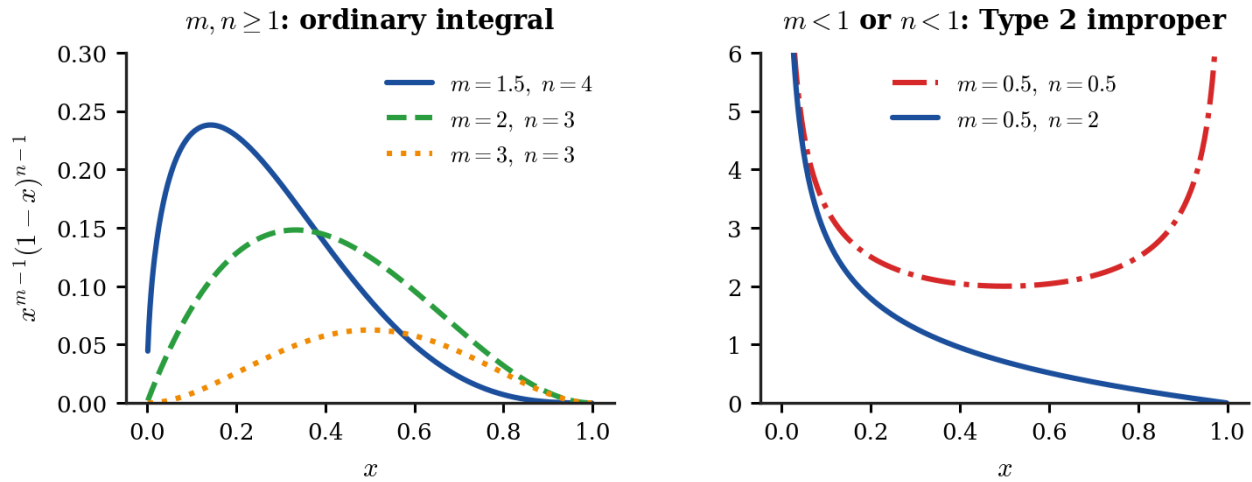


Figure 2.6.2: The beta integrand on  $[0, 1]$ . Exponents below 1 send the curve to infinity at the corresponding endpoint.

### 2.6.6 Properties

- (i)  **$B$  is symmetric:**  $B(m, n) = B(n, m)$ .

**Solution:** Consider

$$B(n, m) = \int_0^1 x^{n-1}(1-x)^{m-1} dx.$$

Put  $1-x = t$ , so that  $x = 1-t$  and  $-dx = dt$ . When  $x = 0$ ,  $t = 1$ ; when  $x = 1$ ,  $t = 0$ . Hence

$$B(n, m) = \int_1^0 (1-t)^{n-1} t^{m-1} (-dt) = \int_0^1 (1-t)^{n-1} t^{m-1} dt = \int_0^1 t^{m-1} (1-t)^{n-1} dt,$$

and since  $t$  is a dummy variable this is exactly  $B(m, n)$ . Therefore  $B(n, m) = B(m, n)$ .

- (ii) **Trigonometric form:**  $B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$ .

**Solution:** Put  $x = \sin^2 \theta$ , so that  $dx = 2 \sin \theta \cos \theta d\theta$  and  $1-x = \cos^2 \theta$ . When  $x = 0$ ,  $\sin^2 \theta = 0$  so  $\theta = 0$ ; when  $x = 1$ ,  $\sin^2 \theta = 1$  so  $\theta = \frac{\pi}{2}$ . Then

$$\begin{aligned} B(m, n) &= \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-2} \theta \cos^{2n-2} \theta \sin \theta \cos \theta d\theta, \end{aligned}$$

that is,

$$B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

Now let  $2m-1 = p$  and  $2n-1 = q$ , so that  $m = \frac{p+1}{2}$  and  $n = \frac{q+1}{2}$ . Substituting,

$$B\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = 2 \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta$$

and hence the reduction rule for trigonometric integrals,

$$\boxed{\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)} \quad (2.6.4)$$

(iii) **Relation between the beta and gamma functions:**

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}. \quad (2.6.5)$$

Its proof uses a double integral and is omitted here.

(iv) **Splitting identity:**  $B(m, n) = B(m+1, n) + B(m, n+1)$ .

**Solution:** Work from the right-hand side. Replacing  $m$  by  $m+1$  in the definition,

$$B(m+1, n) = \int_0^1 x^m (1-x)^{n-1} dx,$$

and replacing  $n$  by  $n+1$ ,

$$B(m, n+1) = \int_0^1 x^{m-1} (1-x)^n dx.$$

Adding these and taking out the common factor  $x^{m-1}(1-x)^{n-1}$ ,

$$\begin{aligned} B(m+1, n) + B(m, n+1) &= \int_0^1 \left[ x^m (1-x)^{n-1} + x^{m-1} (1-x)^n \right] dx \\ &= \int_0^1 x^{m-1} (1-x)^{n-1} \left[ x + (1-x) \right] dx. \end{aligned}$$

The bracket collapses, since  $x + (1-x) = 1$ , leaving

$$= \int_0^1 x^{m-1} (1-x)^{n-1} dx = B(m, n),$$

which is the required identity.

#### Note:

The whole proof is the observation that  $x + (1-x) = 1$ , and the two terms were built to make that factorisation available: raising the first index by one contributes the factor  $x$ , raising the second contributes  $(1-x)$ . The identity is often used in the rearranged form

$$B(m+1, n) = B(m, n) - B(m, n+1),$$

which lowers a beta function's index by one and pairs with the gamma recurrence  $\Gamma(n+1) = n\Gamma(n)$  as a way of stepping between neighbouring values.

#### Note:

As a check, take  $m = \frac{5}{2}$  and  $n = \frac{3}{2}$ . Then  $B\left(\frac{5}{2}, \frac{3}{2}\right) = \frac{\pi}{16} \approx 0.19635$ , while  $B\left(\frac{7}{2}, \frac{3}{2}\right) + B\left(\frac{5}{2}, \frac{5}{2}\right) \approx 0.12272 + 0.07363 = 0.19635$ , as the identity requires.

**Note:**

**Reflection property of  $\Gamma$ :**  $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi}$  for  $0 < n < 1$ . Putting  $n = \frac{1}{2}$  recovers  $\Gamma\left(\frac{1}{2}\right)^2 = \pi$ , that is,  $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ .

**Note:**

**Alternate form of the beta function:**

$$\beta(m, n) = \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}} dt, \quad m > 0, n > 0.$$

**Proposition 2.6.3: The value of  $\Gamma\left(\frac{1}{2}\right)$** 

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

*Proof.* This one cannot be got from the recurrence, which only ever relates  $\Gamma$  at points differing by an integer. The shortest route uses the beta function, and in particular the two expressions for  $B$  established above — the gamma relation (2.6.5) and the trigonometric form. Evaluate  $B\left(\frac{1}{2}, \frac{1}{2}\right)$  both ways.

First, by (2.6.5) with  $m = n = \frac{1}{2}$ ,

$$B\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2} + \frac{1}{2}\right)} = \frac{\left[\Gamma\left(\frac{1}{2}\right)\right]^2}{\Gamma(1)} = \left[\Gamma\left(\frac{1}{2}\right)\right]^2, \quad (2.6.6)$$

using  $\Gamma(1) = 1$ .

Second, by the trigonometric form  $B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$  with  $m = n = \frac{1}{2}$ , both exponents become  $2 \cdot \frac{1}{2} - 1 = 0$ , so  $\sin^0 \theta = \cos^0 \theta = 1$  and the integrand collapses to 1:

$$B\left(\frac{1}{2}, \frac{1}{2}\right) = 2 \int_0^{\pi/2} d\theta = 2 \left[\theta\right]_0^{\pi/2} = 2 \left(\frac{\pi}{2} - 0\right) = \pi. \quad (2.6.7)$$

Comparing (2.6.6) and (2.6.7),

$$\left[\Gamma\left(\frac{1}{2}\right)\right]^2 = \pi \quad \Rightarrow \quad \boxed{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$$

The positive square root is the right one because  $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty e^{-x} x^{-1/2} dx$  is an integral of a strictly positive function, hence positive. ■

**Note:**

Every half-integer value of  $\Gamma$  rests on this one. The recurrence  $\Gamma(n+1) = n\Gamma(n)$  moves in steps of 1, so it connects  $\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$  to each other but never to an integer, and the whole half-integer ladder has to be anchored by an independent calculation. That is what this property supplies;  $\Gamma(1) = 1$  anchors the integer ladder in the same way.

### 2.6.7 A useful special case

Putting  $q = 0$  in (2.6.4) and using (2.6.5),

$$\int_0^{\pi/2} \sin^m \theta \, d\theta = \int_0^{\pi/2} \cos^m \theta \, d\theta = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{m+2}{2}\right)},$$

and since  $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ ,

$$\boxed{\int_0^{\pi/2} \sin^m \theta \, d\theta = \int_0^{\pi/2} \cos^m \theta \, d\theta = \frac{\sqrt{\pi}}{2} \cdot \frac{\Gamma\left(\frac{m+1}{2}\right)}{\Gamma\left(\frac{m+2}{2}\right)}}$$

The two integrals are equal because  $B$  is symmetric, by property (i).

### 2.6.8 Worked problems

**Problem 2.6.1.** Evaluate the following gamma values.

(a)  $\Gamma\left(\frac{3}{2}\right)$ .

**Solution:** Using  $\Gamma(n+1) = n \Gamma(n)$  with  $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ ,

$$\Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2} + 1\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

(b)  $\Gamma\left(\frac{7}{2}\right)$ .

**Solution:**

$$\Gamma\left(\frac{7}{2}\right) = \Gamma\left(\frac{5}{2} + 1\right) = \frac{5}{2} \Gamma\left(\frac{5}{2}\right) = \frac{5}{2} \cdot \frac{3}{2} \Gamma\left(\frac{3}{2}\right) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{15}{8} \sqrt{\pi}.$$

**Note:**

In general, for a positive integer  $n$ ,

$$\Gamma\left(\frac{2n+1}{2}\right) = \frac{2n-1}{2} \cdot \frac{2n-3}{2} \cdots \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}.$$

(c)  $\Gamma\left(-\frac{1}{2}\right)$ .

**Solution:** Here  $n = -\frac{1}{2}$ , so  $n+1 = \frac{1}{2}$ . Using  $\Gamma(n) = \frac{\Gamma(n+1)}{n}$ ,

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}} = -2\sqrt{\pi}.$$

(d)  $\Gamma\left(-\frac{7}{2}\right)$ .

**Solution:** Applying (2.6.3) down the chain,

$$\Gamma\left(-\frac{7}{2}\right) = \frac{\Gamma\left(-\frac{5}{2}\right)}{-\frac{7}{2}} = -\frac{2}{7} \cdot \left(-\frac{2}{5}\right) \Gamma\left(-\frac{3}{2}\right) = \frac{4}{35} \Gamma\left(-\frac{3}{2}\right) = \frac{4}{35} \cdot \left(-\frac{2}{3}\right) \Gamma\left(-\frac{1}{2}\right),$$

so that

$$\Gamma\left(-\frac{7}{2}\right) = -\frac{8}{105} \cdot (-2\sqrt{\pi}) = \frac{16}{105} \sqrt{\pi}.$$

**Problem 2.6.2.** Evaluate  $\beta\left(\frac{9}{2}, \frac{5}{2}\right)$ .

Using  $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$  with  $m = \frac{9}{2}$  and  $n = \frac{5}{2}$ , so that  $m+n = 7$ ,

$$\beta\left(\frac{9}{2}, \frac{5}{2}\right) = \frac{\Gamma\left(\frac{9}{2}\right)\Gamma\left(\frac{5}{2}\right)}{\Gamma(7)} = \frac{\left(\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}\right) \left(\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}\right)}{6!}.$$

The numerator is  $\frac{105}{16} \sqrt{\pi} \cdot \frac{3}{4} \sqrt{\pi} = \frac{315}{64} \pi$ , and  $6! = 720$ , so

$$\beta\left(\frac{9}{2}, \frac{5}{2}\right) = \frac{315\pi}{64 \cdot 720} = \frac{7\pi}{1024}.$$

**Problem 2.6.3.** Evaluate the following integrals.

(a)  $\int_0^1 x^{1/2}(1-x)^3 dx$ .

**Solution:** The given integral is a beta function with  $m-1 = \frac{1}{2}$  and  $n-1 = 3$ , so  $m = \frac{3}{2}$  and  $n = 4$ :

$$\int_0^1 x^{\frac{3}{2}-1}(1-x)^{4-1} dx = \beta\left(\frac{3}{2}, 4\right) = \frac{\Gamma\left(\frac{3}{2}\right)\Gamma(4)}{\Gamma\left(\frac{3}{2}+4\right)} = \frac{\Gamma\left(\frac{3}{2}\right) \cdot 3!}{\Gamma\left(\frac{11}{2}\right)}.$$

Expanding the denominator with  $\Gamma(n+1) = n\Gamma(n)$ ,

$$\Gamma\left(\frac{11}{2}\right) = \frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \Gamma\left(\frac{3}{2}\right),$$

so the factor  $\Gamma\left(\frac{3}{2}\right)$  cancels and

$$\beta\left(\frac{3}{2}, 4\right) = \frac{6}{\frac{9 \cdot 7 \cdot 5 \cdot 3}{16}} = \frac{6 \cdot 16}{945} = \frac{32}{315}.$$

Hence  $\int_0^1 x^{1/2}(1-x)^3 dx = \frac{32}{315}$ .

(b)  $\int_0^{\pi/2} \sin^6 \theta d\theta$ .

**Solution:** Using (2.6.4) with  $p = 6$  and  $q = 0$ ,

$$\int_0^{\pi/2} \sin^6 \theta d\theta = \frac{1}{2} \beta\left(\frac{6+1}{2}, \frac{0+1}{2}\right) = \frac{1}{2} \beta\left(\frac{7}{2}, \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(4)}.$$

Now  $\Gamma\left(\frac{7}{2}\right) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi} = \frac{15}{8}\sqrt{\pi}$ ,  $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$  and  $\Gamma(4) = 3! = 6$ , so

$$\int_0^{\pi/2} \sin^6 \theta \, d\theta = \frac{1}{2} \cdot \frac{\frac{15}{8}\pi}{6} = \frac{15\pi}{96} = \frac{5\pi}{32}.$$

(c)  $\int_0^{\pi/2} \sin^3 \theta \, d\theta.$

**Solution:** Using (2.6.4) with  $p = 3$  and  $q = 0$ , so that  $\frac{p+1}{2} = \frac{4}{2} = 2$  and  $\frac{q+1}{2} = \frac{1}{2}$ ,

$$\int_0^{\pi/2} \sin^3 \theta \, d\theta = \frac{1}{2} \beta\left(2, \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{\Gamma(2) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{5}{2}\right)}.$$

Now  $\Gamma(2) = 1! = 1$ ,  $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$  and  $\Gamma\left(\frac{5}{2}\right) = \frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi}$ , so the factor  $\sqrt{\pi}$  cancels:

$$\int_0^{\pi/2} \sin^3 \theta \, d\theta = \frac{1}{2} \cdot \frac{1 \cdot \sqrt{\pi}}{\frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi}} = \frac{1}{2} \cdot \frac{4}{3} = \frac{2}{3}.$$

(d)  $\int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta.$

**Solution:** Write the integrand in terms of sine and cosine:

$$\int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta = \int_0^{\pi/2} \left( \frac{\sin \theta}{\cos \theta} \right)^{1/2} d\theta = \int_0^{\pi/2} \sin^{1/2} \theta \cos^{-1/2} \theta \, d\theta,$$

which is of the form (2.6.4) with  $p = \frac{1}{2}$  and  $q = -\frac{1}{2}$ . Then

$$\frac{p+1}{2} = \frac{\frac{1}{2}+1}{2} = \frac{3}{4}, \quad \frac{q+1}{2} = \frac{-\frac{1}{2}+1}{2} = \frac{1}{4},$$

so that

$$\int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta = \frac{1}{2} \beta\left(\frac{3}{4}, \frac{1}{4}\right) = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)}{\Gamma(1)} = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(1 - \frac{1}{4}\right),$$

since  $\Gamma(1) = 1$  and  $\frac{3}{4} = 1 - \frac{1}{4}$ . Applying the reflection property  $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi}$  with  $n = \frac{1}{4}$ ,

$$\int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta = \frac{1}{2} \cdot \frac{\pi}{\sin \frac{\pi}{4}} = \frac{1}{2} \cdot \frac{\pi}{\frac{1}{\sqrt{2}}} = \frac{\pi}{\sqrt{2}}.$$

#### Note:

This integral has no elementary antiderivative. The beta function converts it into two gamma values, and the reflection property then evaluates them together without either being known separately.

**Problem 2.6.4.** Find the area bounded by the curve  $x^3 + y^3 = 8$  in the first quadrant.

**Solution:** In the first quadrant the curve meets the axes at  $(2, 0)$  and  $(0, 2)$ , since  $x = 0$  gives  $y^3 = 8$  and  $y = 0$  gives  $x^3 = 8$ . Solving for  $y$ ,

$$y^3 = 8 - x^3 \quad \implies \quad y = (8 - x^3)^{1/3},$$

so the required area is

$$A = \int_0^2 y \, dx = \int_0^2 (8 - x^3)^{1/3} \, dx.$$

Take the factor 8 outside the cube root, writing  $8 - x^3 = 8 \left[ 1 - \left(\frac{x}{2}\right)^3 \right]$  and  $8^{1/3} = 2$ :

$$A = 2 \int_0^2 \left[ 1 - \left(\frac{x}{2}\right)^3 \right]^{1/3} \, dx.$$

Now substitute  $\left(\frac{x}{2}\right)^3 = t$ , so that  $x = 2t^{1/3}$  and

$$dx = \frac{2}{3} t^{-2/3} \, dt.$$

The limits transform as  $x = 0 \Rightarrow t = 0$  and  $x = 2 \Rightarrow t = 1$ , giving

$$A = 2 \int_0^1 (1 - t)^{1/3} \cdot \frac{2}{3} t^{-2/3} \, dt = \frac{4}{3} \int_0^1 t^{-2/3} (1 - t)^{1/3} \, dt.$$

Comparing the integral with  $B(m, n) = \int_0^1 t^{m-1} (1 - t)^{n-1} \, dt$ ,

$$m - 1 = -\frac{2}{3} \implies m = \frac{1}{3}, \quad n - 1 = \frac{1}{3} \implies n = \frac{4}{3},$$

so that

$$A = \frac{4}{3} B\left(\frac{1}{3}, \frac{4}{3}\right) = \frac{4}{3} \cdot \frac{\Gamma\left(\frac{1}{3}\right) \Gamma\left(\frac{4}{3}\right)}{\Gamma\left(\frac{1}{3} + \frac{4}{3}\right)}.$$

Both remaining gamma values reduce by the recurrence  $\Gamma(x + 1) = x \Gamma(x)$ :

$$\Gamma\left(\frac{4}{3}\right) = \Gamma\left(\frac{1}{3} + 1\right) = \frac{1}{3} \Gamma\left(\frac{1}{3}\right), \quad \Gamma\left(\frac{5}{3}\right) = \Gamma\left(\frac{2}{3} + 1\right) = \frac{2}{3} \Gamma\left(\frac{2}{3}\right).$$

Substituting both,

$$A = \frac{4}{3} \cdot \frac{\Gamma\left(\frac{1}{3}\right) \cdot \frac{1}{3} \Gamma\left(\frac{1}{3}\right)}{\frac{2}{3} \Gamma\left(\frac{2}{3}\right)} = \frac{4}{3} \cdot \frac{1}{2} \cdot \frac{\left[\Gamma\left(\frac{1}{3}\right)\right]^2}{\Gamma\left(\frac{2}{3}\right)},$$

that is

$$A = \frac{2}{3} \cdot \frac{\left[\Gamma\left(\frac{1}{3}\right)\right]^2}{\Gamma\left(\frac{2}{3}\right)} \approx 3.5333$$

#### Note:

The numerical value is worth having, since the answer is otherwise an unevaluated

expression. With  $\Gamma\left(\frac{1}{3}\right) \approx 2.6789$  and  $\Gamma\left(\frac{2}{3}\right) \approx 1.3541$ ,

$$A \approx \frac{2}{3} \cdot \frac{(2.6789)^2}{1.3541} \approx 3.5333.$$

As a sanity check, the region sits inside the square  $[0, 2] \times [0, 2]$  of area 4 and contains the triangle with vertices  $(0, 0)$ ,  $(2, 0)$ ,  $(0, 2)$  of area 2, so the answer had to lie between 2 and 4. The curve  $x^3 + y^3 = 8$  bulges outwards from the straight line  $x + y = 2$ , which is why the area exceeds the triangle's.

### Note:

This is the pattern for turning an area into a beta function: scale the variable so the region runs from 0 to 1, then substitute whatever power of the variable appears in the equation of the curve. Here  $\left(\frac{x}{2}\right)^3 = t$  did both jobs at once. The integral  $\int_0^2 (8 - x^3)^{1/3} dx$  has no elementary antiderivative, so without the beta function there is no closed form at all.

## 2.7 Sequences and Series

### 2.7.1 Sequences

#### Definition 2.7.1: Sequence

A *sequence* of real numbers is a function  $f : \mathbb{N} \rightarrow \mathbb{R}$ . The value of  $f$  at  $n$  is denoted by  $f(n) = a_n$  and is called the  $n$ th term of the sequence. The sequence itself is written  $\{a_n\}$  or  $\{a_n\}_{n=1}^\infty$ .

#### Example 2.7.2: Some standard sequences

- (a)  $a_n = k$  (constant):  $\{a_n\} = \{k, k, k, \dots\}$ ; as  $n \rightarrow \infty$ ,  $f(n) = k$ .
- (b)  $a_n = n$ :  $\{a_n\} = \{1, 2, 3, \dots, n, \dots\}$ ; as  $n \rightarrow \infty$ ,  $f(n) \rightarrow \infty$ .
- (c)  $a_n = n^2$ :  $\{a_n\} = \{1^2, 2^2, 3^2, \dots\} = \{1, 4, 9, \dots\}$ ; as  $n \rightarrow \infty$ ,  $f(n) \rightarrow \infty$ .
- (d)  $a_n = \frac{1}{n}$ :  $\{a_n\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$ ; as  $n \rightarrow \infty$ ,  $f(n) \rightarrow 0$ .
- (e)  $a_n = (-1)^n = \cos n\pi$ :  $\{a_n\} = \{-1, 1, -1, 1, \dots\}$ ; the terms oscillate and have no limit.
- (f)  $a_n = 2^{1/n}$ :  $\{a_n\} = \{2, 2^{1/2}, 2^{1/3}, \dots\}$ ; as  $n \rightarrow \infty$ ,  $f(n) \rightarrow 1$ , since  $2^{1/n} \rightarrow 2^0 = 1$ .
- (g) Fibonacci sequence (recursive definition):  $f_1 = f_2 = 1$  and  $f_n = f_{n-1} + f_{n-2}$ , giving  $\{f_n\} = \{1, 1, 2, 3, 5, \dots\}$ .

### Note:

A sequence *converges* if its terms approach a single finite number, and *diverges* otherwise — for instance when the terms grow without bound or oscillate.

## 2.7.2 Limit of a sequence

### Definition 2.7.3: Limit of a sequence

A real number  $L$  is said to be the limit of the sequence  $\{a_n\}_{n=1}^{\infty}$ , denoted

$$\lim_{n \rightarrow \infty} a_n = L,$$

if  $a_n$  approaches  $L$  as  $n$  approaches infinity. In that case the sequence is called *convergent*; if no such  $L$  exists, the sequence is called *divergent*.

Referring to [Example 2.7.2](#):

#### Convergent

(a)  $\lim_{n \rightarrow \infty} a_n = k$

(d)  $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

(f)  $\lim_{n \rightarrow \infty} 2^{1/n} = 1$

#### Divergent

(b)  $\lim_{n \rightarrow \infty} n = \infty$

(c)  $\lim_{n \rightarrow \infty} n^2 = \infty$

(e)  $\lim_{n \rightarrow \infty} (-1)^n$  does not exist

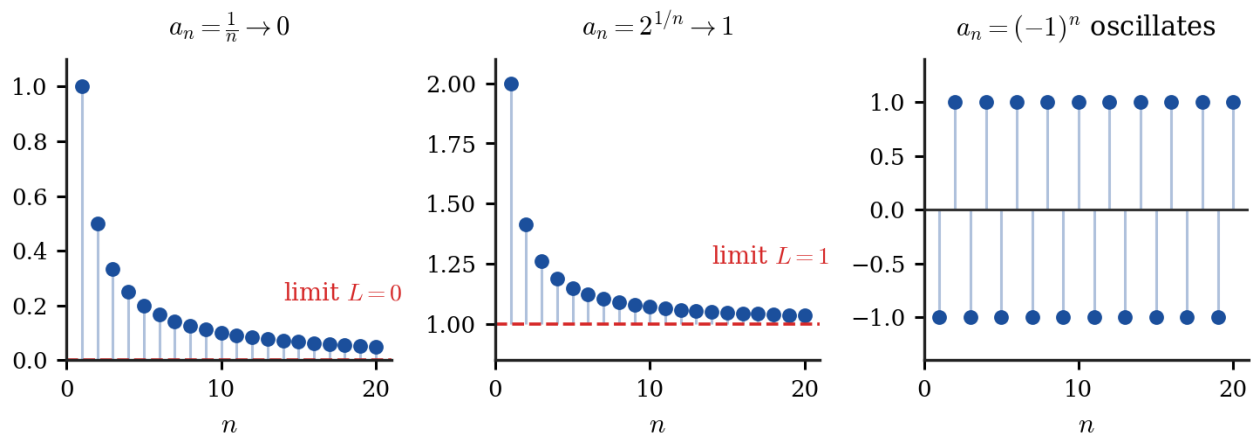


Figure 2.7.1: A sequence is defined only at integer values of  $n$ . The first two settle down to a limit; the third oscillates between  $-1$  and  $1$  and so has no limit.

## 2.7.3 Standard limits of sequences

- (i) If  $p > 0$ , then  $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ .
- (ii) If  $p > 0$ , then  $\lim_{n \rightarrow \infty} p^{1/n} = 1$ .
- (iii)  $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$ .
- (iv)  $\lim_{n \rightarrow \infty} n^{1/n} = 1$ .
- (v) If  $|x| < 1$ , then  $\lim_{n \rightarrow \infty} x^n = 0$ .
- (vi) If  $\alpha, \beta > 0$ , then  $\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1 + \beta)^n} = 0$ .

(vii)  $\lim_{n \rightarrow \infty} (n!)^{1/n} = \infty$ , where  $n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$ .

**Note:**

In (vi) the numerator  $n^\alpha$  is a power function and the denominator  $(1 + \beta)^n$  is an exponential function. The exponential grows faster, and so approaches infinity before the power function does; hence the ratio tends to 0.

## 2.7.4 Two useful results

### *Theorem 2.7.4: Continuous functions preserve sequence limits*

If  $\lim_{n \rightarrow \infty} a_n = L$  and the function  $f$  is continuous at  $L$ , then

$$\lim_{n \rightarrow \infty} f(a_n) = f(L) = f\left(\lim_{n \rightarrow \infty} a_n\right).$$

In words, a continuous function may be taken inside the limit. This is what licenses steps such as  $\lim \sqrt{a_n} = \sqrt{\lim a_n}$  or  $\lim e^{a_n} = e^{\lim a_n}$ , which are otherwise unjustified.

**Note:**

Continuity at  $L$  is the whole hypothesis, and it cannot be dropped. Take  $a_n = \frac{1}{n} \rightarrow 0$  and let  $f$  be the function that is 1 at 0 and 0 elsewhere. Then  $f(a_n) = 0$  for every  $n$ , so  $\lim f(a_n) = 0$ , whereas  $f(0) = 1$ . The limit and the function cannot be interchanged because  $f$  is discontinuous at the limit point.

### *Theorem 2.7.5: The absolute value test for sequences*

If  $\lim_{n \rightarrow \infty} |a_n| = 0$ , then  $\lim_{n \rightarrow \infty} a_n = 0$ .

**Note:**

This is the one situation in which the sign of the terms may be ignored, and it works only for the limit 0. It is false for any other limit:  $a_n = (-1)^n$  has  $|a_n| = 1 \rightarrow 1$ , yet  $\{a_n\}$  has no limit at all. The reason is that  $|a_n| \rightarrow 0$  forces  $a_n$  into a shrinking interval around 0, while  $|a_n| \rightarrow L \neq 0$  leaves the terms free to alternate between  $L$  and  $-L$ .

## 2.7.5 Worked problems on sequences

**Problem 2.7.1.** Discuss whether each of the following sequences converges or diverges.

(a)  $\{x_n\} = \left\{ \frac{n}{n+1} \right\} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots \right\}.$

**Solution:** Dividing numerator and denominator by  $n$ ,

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{n}{n\left(1 + \frac{1}{n}\right)} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = \frac{1}{1+0} = 1.$$

The limit exists finitely, so the sequence is **convergent**, with limit 1.

$$(b) \{x_n\} = \left\{ \frac{n}{\sqrt{10+n^2}} \right\}.$$

**Solution:** Take  $n$  out of the square root, using  $\sqrt{n^2} = n$  for  $n > 0$ :

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{10+n^2}} = \lim_{n \rightarrow \infty} \frac{n}{n\sqrt{1+\frac{10}{n^2}}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+\frac{10}{n^2}}} = \frac{1}{\sqrt{1+0}} = 1.$$

The sequence is **convergent**, with limit 1. Note that moving the limit inside the square root is exactly [Theorem 2.7.4](#), with  $f(x) = \sqrt{x}$  continuous at 1.

$$(c) \{x_n\} = \left\{ \frac{\ln n}{n} \right\}.$$

**Solution:** Both  $\ln n \rightarrow \infty$  and  $n \rightarrow \infty$ , so the quotient is of the indeterminate form  $\frac{\infty}{\infty}$ . Applying L'Hôpital's rule ([Theorem 2.3.1](#)) to the corresponding function of a real variable,

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

The sequence is **convergent**, with limit 0.

**Note:**

L'Hôpital's rule is a statement about functions of a continuous variable, not about sequences, so strictly it is applied to  $f(x) = \frac{\ln x}{x}$  and the result transferred to the sequence by restricting  $x$  to integer values. The transfer is legitimate in this direction: if  $f(x) \rightarrow L$  as  $x \rightarrow \infty$  then  $f(n) \rightarrow L$  as well. It is not legitimate in reverse —  $\sin(n\pi) = 0$  for every integer  $n$ , but  $\sin(\pi x)$  has no limit as  $x \rightarrow \infty$ .

$$(d) \{x_n\} = \{(-1)^n\}.$$

**Solution:** The terms are  $-1, 1, -1, 1, \dots$ , oscillating between  $-1$  and  $1$  and never settling. No single value is approached, so  $\lim_{n \rightarrow \infty} (-1)^n$  does not exist and the sequence is **divergent**.

$$(e) \{x_n\} = \left\{ \frac{(-1)^n}{n} \right\} = \left\{ -1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, \dots \right\}.$$

**Solution:** The terms alternate in sign, so the limit is not immediate. Take absolute values:

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{n} \right| = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

By [Theorem 2.7.5](#),  $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$  and the sequence is **convergent**.

**Note:**

Compare this with (d). Both sequences alternate in sign, but here the oscillation is damped: the terms are squeezed between  $-\frac{1}{n}$  and  $\frac{1}{n}$ , and both bounds close in on 0. In (d) the oscillation has constant amplitude and nothing closes in, so alternation by itself decides nothing.

(f)  $\{x_n\} = \{\sqrt{n+1} - \sqrt{n}\}.$

**Solution:** As written this is of the indeterminate form  $\infty - \infty$ . Rationalise by multiplying and dividing by the conjugate:

$$\sqrt{n+1} - \sqrt{n} = (\sqrt{n+1} - \sqrt{n}) \cdot \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}.$$

The denominator increases without bound, so

$$\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1} + \sqrt{n}} = 0,$$

and the sequence is **convergent**, with limit 0.

**Note:**

The two terms each tend to infinity, so the difference cannot be evaluated term by term —  $\infty - \infty$  is not a number. Rationalising converts the difference into a quotient, where the competition between the two parts is visible: the numerator is the constant 1 while the denominator grows, so the gap between consecutive square roots closes.

## 2.7.6 Series

### Definition 2.7.6: Infinite series

The infinite series associated with a sequence  $\{a_n\}_{n=1}^{\infty}$  is

$$a_1 + a_2 + \cdots + a_n + \cdots \quad \text{or} \quad \sum_{n=1}^{\infty} a_n.$$

### Definition 2.7.7: Sequence of partial sums

The *sequence of partial sums*  $\{S_n\}$  of the series  $\sum a_n$  is defined by

$$S_1 = a_1, \quad S_2 = a_1 + a_2 = S_1 + a_2, \quad S_3 = a_1 + a_2 + a_3 = S_2 + a_3, \quad \dots$$

and in general

$$S_n = \sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n = S_{n-1} + a_n. \quad (2.7.1)$$

The series  $\sum_{n=1}^{\infty} a_n$  is said to *converge* if the sequence of partial sums  $\{S_n\}$  converges. In that case, if  $L = \lim_{n \rightarrow \infty} S_n$ , we call  $L$  the sum of the series and write  $\sum_{n=1}^{\infty} a_n = L$ . If  $\sum a_n$  does not converge, the series is said to *diverge*.

## 2.7.7 A necessary condition for convergence

From (2.7.1) we have  $a_n = S_n - S_{n-1}$ . If  $\lim_{n \rightarrow \infty} S_n = L$ , that is, the series converges to  $L$ , then

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} [S_n - S_{n-1}] = \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} S_{n-1} = L - L = 0.$$

**Theorem 2.7.8:  $n$ th term test for divergence**

If  $\lim_{n \rightarrow \infty} a_n \neq 0$ , then  $\sum_{n=1}^{\infty} a_n$  diverges.

**Common mistake:**

The converse is false:  $\lim a_n = 0$  does *not* guarantee convergence. The harmonic series  $\sum \frac{1}{n}$  has  $a_n \rightarrow 0$  and still diverges (Section 2.7.10). The  $n$ th term test can only ever prove divergence.

**Example 2.7.9: Divergent series**

- (a)  $\sum 2^{1/n}$  diverges, since  $2^{1/n} \rightarrow 1 \neq 0$ .
- (b)  $\sum n$  and  $\sum n^2$  diverge, since the terms tend to  $\infty$ .
- (c)  $\sum_{n=1}^{\infty} (-1)^n$  diverges, since  $(-1)^n$  has no limit.

**Example 2.7.10: The  $n$ th term test at work**

Test  $\sum_{n=1}^{\infty} \frac{n^2}{n^2 - 5n + 2}$  for convergence.

**Solution:** By the  $n$ th term test,

$$\lim_{n \rightarrow \infty} \frac{n^2}{n^2 - 5n + 2} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2 \left(1 - \frac{5}{n} + \frac{2}{n^2}\right)} = \frac{1}{1 - 0 + 0} = 1 \neq 0,$$

so the series is **divergent**.

**Note:**

The first few terms of this series are negative, since  $n^2 - 5n + 2 < 0$  for  $n = 1, 2, 3, 4$ . That does not affect the conclusion — the  $n$ th term test needs no sign condition, and in any case convergence is unchanged by altering finitely many terms. It would, however, block the integral test, which requires positive terms.

**2.7.8 Series whose partial sums can be found**

The definition of convergence is a statement about the sequence  $\{S_n\}$  of partial sums, so in principle a series can be settled by finding  $S_n$  in closed form and taking its limit. In practice a closed form is rarely available; the following series is one of the few where it is, and it is worth doing once in full.

**Problem 2.7.2.** Check the convergence of  $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ , and find its sum if it converges.

**Solution:** Let  $a_n = \frac{1}{n(n+1)}$ , so that the  $n$ th partial sum is

$$S_n = \sum_{i=1}^n \frac{1}{i(i+1)}.$$

Split the general term by partial fractions. Writing

$$\frac{1}{i(i+1)} = \frac{A}{i} + \frac{B}{i+1} \quad \Rightarrow \quad 1 = A(i+1) + Bi,$$

and comparing coefficients: the constant terms give  $A = 1$ , and the coefficients of  $i$  give  $0 = A + B$ , so  $B = -1$ . Hence

$$\frac{1}{i(i+1)} = \frac{1}{i} - \frac{1}{i+1}.$$

Substituting into the partial sum and writing the terms out,

$$\begin{aligned} S_n &= \sum_{i=1}^n \left( \frac{1}{i} - \frac{1}{i+1} \right) \\ &= \left( \frac{1}{1} - \frac{1}{2} \right) + \left( \frac{1}{2} - \frac{1}{3} \right) + \left( \frac{1}{3} - \frac{1}{4} \right) + \cdots + \left( \frac{1}{n} - \frac{1}{n+1} \right). \end{aligned}$$

Every interior term appears twice with opposite signs and cancels: the  $-\frac{1}{2}$  against the  $+\frac{1}{2}$ , the  $-\frac{1}{3}$  against the  $+\frac{1}{3}$ , and so on. Only the first and last survive,

$$S_n = 1 - \frac{1}{n+1}.$$

Now take the limit:

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left( 1 - \frac{1}{n+1} \right) = 1 - 0 = 1.$$

The sequence of partial sums converges, so the series is **convergent**, and

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1}$$

#### Note:

A series whose partial sums collapse in this way is called *telescoping*, after the way the terms slide into one another. The pattern to look for is a general term expressible as  $b_i - b_{i+1}$ : then  $S_n = b_1 - b_{n+1}$  and the whole question reduces to  $\lim b_{n+1}$ . Partial fractions is the usual way of producing that form.

#### Note:

Note what this example does that the convergence tests do not. The convergence tests that follow decide *whether* a series converges but never produce its sum. Here the sum is obtained exactly, because  $S_n$  was available in closed form. That is the exception rather than the rule — for  $\sum \frac{1}{n^2}$ , for instance, convergence is easy but the value  $\frac{\pi^2}{6}$  is not obtainable by any elementary manipulation of the partial sums.

### 2.7.9 The geometric series

Written with the index starting at 1, the geometric series is

$$\sum_{n=1}^{\infty} a r^{n-1} = a + ar + ar^2 + ar^3 + \cdots,$$

where  $a \neq 0$  is the first term and  $r$  the common ratio. Its partial sums can be found exactly. Writing

$$S_n = a + ar + \cdots + ar^{n-1}, \quad rS_n = ar + ar^2 + \cdots + ar^n$$

and subtracting, every term but two cancels:  $(1 - r)S_n = a(1 - r^n)$ , so

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad (r \neq 1).$$

If  $|r| < 1$  then  $r^n \rightarrow 0$  (standard limit (v)) and  $S_n \rightarrow \frac{a}{1 - r}$ . If  $|r| \geq 1$  the terms  $ar^{n-1}$  do not tend to 0, so the series diverges by [Theorem 2.7.8](#). Hence the series converges if  $|r| < 1$ , with

$$\boxed{\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1 - r}, \quad |r| < 1} \quad (2.7.2)$$

and diverges if  $|r| \geq 1$ .

#### Note:

Starting the index at 0 gives the equivalent form  $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ ; only the indexing differs, and both have first term  $a$ . The  $ar^{n-1}$  form is usually the more convenient one in problems, because the first term of the series is read off directly as  $a$  rather than having to be computed.

**Problem 2.7.3.** Check the convergence of  $\sum_{n=1}^{\infty} \frac{e^{2n}}{6^{n-1}}$ .

**Solution:** Rewrite the general term to expose a geometric ratio. Splitting off one factor of  $e^2$ ,

$$\frac{e^{2n}}{6^{n-1}} = e^2 \cdot \frac{e^{2n-2}}{6^{n-1}} = e^2 \cdot \frac{e^{2(n-1)}}{6^{n-1}} = e^2 \left( \frac{e^2}{6} \right)^{n-1}.$$

This is a geometric series with

$$a = e^2, \quad r = \frac{e^2}{6} \approx \frac{7.389}{6} \approx 1.2315.$$

Since  $|r| > 1$ , the series is **divergent**.

#### Note:

The ratio is close enough to 1 that it cannot be judged by eye, and getting it the wrong side of the boundary reverses the answer. Since  $e^2 = 7.389$  exceeds 6, the ratio exceeds 1 and the terms grow; had the denominator been  $8^{n-1}$  instead, the ratio would have been  $\frac{7.389}{8} < 1$  and the series would have converged, to  $\frac{a}{1-r} = \frac{e^2}{1-e^2/8}$ .

## 2.7.10 The integral test

A series of positive decreasing terms and an improper integral of the corresponding function measure almost the same thing, and either one settles the other.

**Theorem 2.7.11: Integral test**

Suppose  $f$  is continuous, positive and decreasing on  $[1, \infty)$ , and let  $a_n = f(n)$ . Then

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \int_1^{\infty} f(x) dx \text{ converges.}$$

Equivalently:

- if  $\int_1^{\infty} f(x) dx$  is convergent, then  $\sum_{n=1}^{\infty} a_n$  is convergent;
- if  $\int_1^{\infty} f(x) dx$  is divergent, then  $\sum_{n=1}^{\infty} a_n$  is divergent.

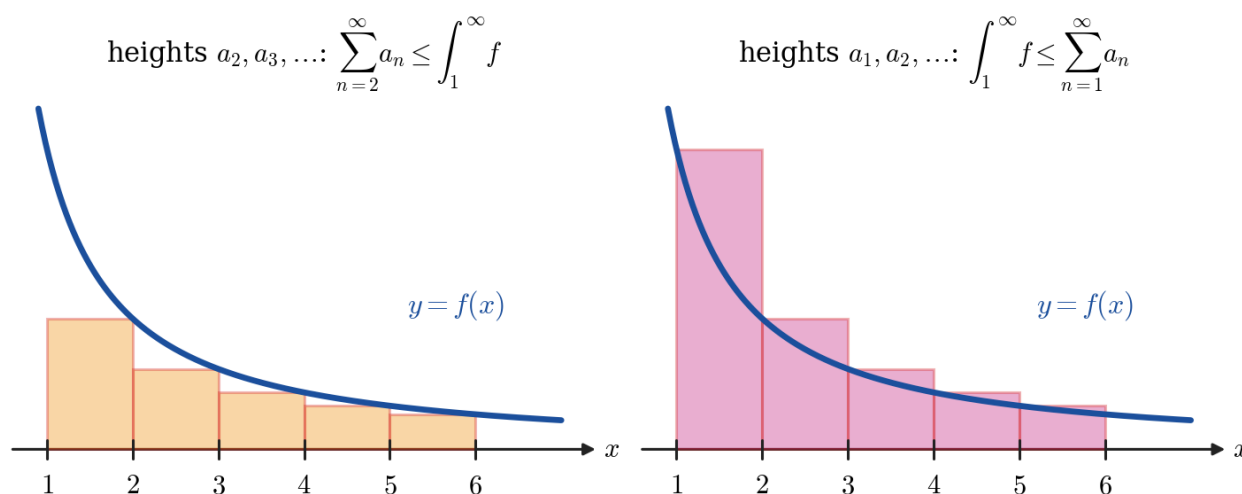


Figure 2.7.2: Rectangles of unit width and height  $a_n$  against the curve  $y = f(x)$ . Inscribed on the left, circumscribed on the right. The series and the integral trap each other between the two pictures, so neither can be finite without the other.

The two pictures are the proof. Taking the rectangle on  $[n, n+1]$  to have height  $f(n+1)$  puts it under the curve, giving  $\sum_{n=2}^{\infty} a_n \leq \int_1^{\infty} f(x) dx$ ; taking it to have height  $f(n)$  puts it over the curve, giving  $\int_1^{\infty} f(x) dx \leq \sum_{n=1}^{\infty} a_n$ . A finite integral therefore forces a finite sum, and an infinite integral forces an infinite sum.

**Note:**

All three hypotheses are needed. *Positive* and *decreasing* are what make the rectangles nest against the curve; for a series whose terms oscillate in sign, or one whose terms rise before they fall, the picture collapses and the test does not apply. In practice “decreasing” is checked by showing  $f'(x) < 0$ . If  $f$  only becomes decreasing beyond some  $x = N$ , the test may still be applied on  $[N, \infty)$ , since altering finitely many terms cannot change whether a series converges.

**Common mistake:**

**The test does not give the sum.** It compares the series with the integral closely enough to decide convergence and no more; the two values are different, and the inequalities above only bracket one by the other. This is the single commonest error

made with the integral test — see part (c) of [Problem 2.7.4](#) below, where the integral equals 2 while the series does not.

### 2.7.11 Worked problems on the integral test

**Problem 2.7.4.** Test the following series for convergence using the integral test.

- (a) The  $p$ -series  $\sum_{n=1}^{\infty} \frac{1}{n^p}$ , where  $p \in \mathbb{R}$ .

**Solution:** Take  $a_n = \frac{1}{n^p} = f(n)$ , so that  $f(x) = \frac{1}{x^p}$  on  $[1, \infty)$ . For  $p > 0$  this function is continuous, positive and decreasing there, so the integral test applies. From the  $p$ -integral ([Section 2.5.4](#)),

$$\int_1^{\infty} \frac{1}{x^p} dx \quad \text{converges if } p > 1 \quad \text{and diverges if } p \leq 1.$$

Hence, by the integral test,

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges if } p > 1, \text{ diverges if } p \leq 1$$

#### Note:

For  $p \leq 0$  the function is not decreasing and the test does not apply — but nothing is lost, because then  $\frac{1}{n^p} = n^{|p|} \not\rightarrow 0$  and the  $n$ th term test already gives divergence. The harmonic series is the case  $p = 1$ , sitting exactly on the boundary and diverging.

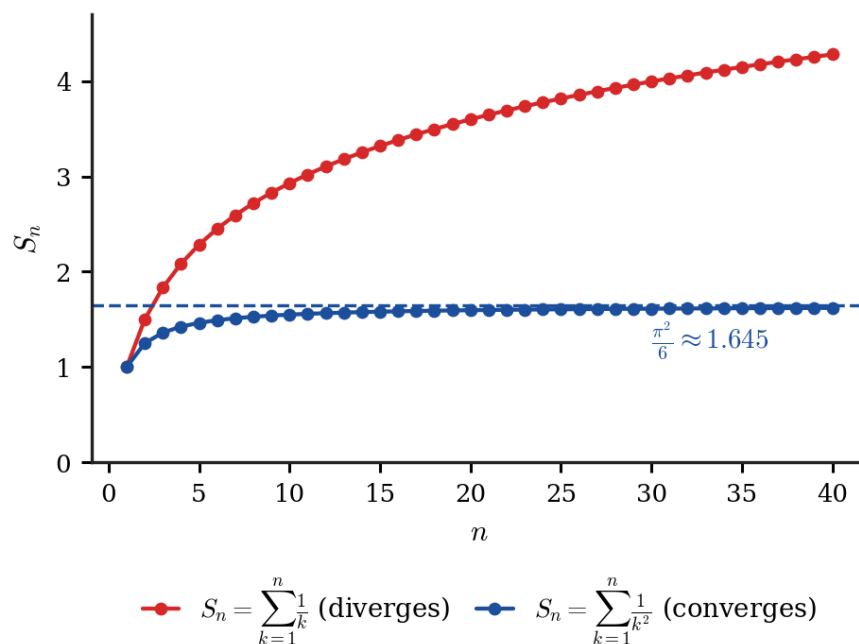


Figure 2.7.3: Partial sums of  $\sum \frac{1}{k}$  keep climbing without bound, while those of  $\sum \frac{1}{k^2}$  level off at  $\pi^2/6$ .

(b)  $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}.$

**Solution:** Here  $a_n = f(n) = \frac{1}{n^2 + 1}$ , so take  $f(x) = \frac{1}{x^2 + 1}$  on  $[1, \infty)$ . This is continuous and positive there, and differentiating,

$$f'(x) = -\frac{2x}{(x^2 + 1)^2} < 0 \quad \text{for } x \geq 1,$$

so  $f$  is decreasing. The integral test applies. Now

$$\int_1^{\infty} \frac{dx}{x^2 + 1} = \lim_{t \rightarrow \infty} \left[ \tan^{-1} x \right]_1^t = \lim_{t \rightarrow \infty} (\tan^{-1} t - \tan^{-1} 1) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4},$$

a finite number, so the integral converges. By the integral test,  $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$  is **convergent**.

(c)  $1 + \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \cdots$

**Solution:** The  $n$ th term is  $\frac{1}{n\sqrt{n}}$ , and since  $n\sqrt{n} = n \cdot n^{1/2} = n^{3/2}$ , the series is

$$\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}.$$

Take  $f(x) = \frac{1}{x^{3/2}}$ , which is continuous, positive and decreasing on  $[1, \infty)$ . Then

$$\int_1^{\infty} \frac{dx}{x^{3/2}} = \lim_{t \rightarrow \infty} \int_1^t x^{-3/2} dx = \lim_{t \rightarrow \infty} \left[ \frac{x^{-1/2}}{-\frac{1}{2}} \right]_1^t = -2 \lim_{t \rightarrow \infty} \left( \frac{1}{\sqrt{t}} - 1 \right) = -2(0 - 1) = 2.$$

The integral converges, so by the integral test the series is **convergent**.

**Note:**

**The series does not sum to 2.** It is tempting to carry the value of the integral across to the series, but the integral test transfers only the *fact* of convergence. Here the integral equals 2 while

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}} = \zeta\left(\frac{3}{2}\right) \approx 2.6124,$$

which is not 2 and has no elementary closed form at all. The relation between them is only the bracketing from Fig. 2.7.2: with  $a_1 = 1$ , the right-hand picture gives  $2 \leq \sum a_n$  and the left-hand picture gives  $\sum a_n \leq 1 + 2 = 3$ , and indeed  $2 \leq 2.6124 \leq 3$ . That is as much as the test can say about the value.

(d)  $1 + \frac{1}{3} + \frac{1}{7} + \frac{1}{11} + \frac{1}{15} + \cdots$

**Solution:** After the leading 1, the denominators run 3, 7, 11, 15, ..., an arithmetic progression with  $n$ th term  $4n - 1$ . The series is therefore

$$1 + \sum_{n=1}^{\infty} \frac{1}{4n - 1}.$$

Take  $f(x) = \frac{1}{4x-1}$ , which on  $[1, \infty)$  is continuous — the only singularity is at  $x = \frac{1}{4}$ , outside the interval — positive, and decreasing, since the denominator increases. Then

$$\int_1^\infty \frac{dx}{4x-1} = \frac{1}{4} \lim_{t \rightarrow \infty} \int_1^t \frac{4 dx}{4x-1} = \frac{1}{4} \lim_{t \rightarrow \infty} \left[ \log(4x-1) \right]_1^t,$$

where the factor 4 was introduced to make the numerator the derivative of the denominator, using  $\int \frac{f'(x)}{f(x)} dx = \log |f(x)| + C$ . Hence

$$= \frac{1}{4} \lim_{t \rightarrow \infty} (\log(4t-1) - \log 3) = \infty,$$

so the integral diverges. By the integral test  $\sum_{n=1}^\infty \frac{1}{4n-1}$  diverges, and adding the single term 1 cannot repair that, so the given series is **divergent**.

(e)  $\sum_{n=1}^\infty \frac{\log n}{n}$ .

**Solution:** Take  $f(x) = \frac{\log x}{x}$  on  $[1, \infty)$ . Before applying the test, check the hypotheses honestly. Differentiating by the quotient rule,

$$f'(x) = \frac{x \cdot \frac{1}{x} - \log x}{x^2} = \frac{1 - \log x}{x^2},$$

which is *positive* for  $x < e$  and negative for  $x > e$ . So  $f$  increases on  $[1, e]$  and only decreases on  $[e, \infty)$ ; it is also zero, not positive, at  $x = 1$ . The hypotheses therefore fail on  $[1, \infty)$  and hold on  $[3, \infty)$ , so apply the test there instead:

$$\int_3^\infty \frac{\log x}{x} dx = \lim_{t \rightarrow \infty} \left[ \frac{(\log x)^2}{2} \right]_3^t = \lim_{t \rightarrow \infty} \frac{(\log t)^2 - (\log 3)^2}{2} = \infty,$$

substituting  $u = \log x$ ,  $du = \frac{dx}{x}$ . The integral diverges, so  $\sum_{n=3}^\infty \frac{\log n}{n}$  diverges, and since the first two terms are finite the full series  $\sum_{n=1}^\infty \frac{\log n}{n}$  is **divergent**.

**Note:**

It is worth being careful about the starting point rather than asserting that  $\frac{\log x}{x}$  is decreasing on all of  $[1, \infty)$ , which is false: the function rises from 0 at  $x = 1$  to its maximum  $\frac{1}{e}$  at  $x = e$  before turning down. Shifting the lower limit to any point past  $e$  costs nothing, because convergence of a series is never affected by finitely many terms. The habit of verifying the hypotheses on the interval actually used is what makes the test safe to apply.

## 2.7.12 Absolute and conditional convergence

### Definition 2.7.12: Absolute and conditional convergence

The series  $\sum a_n$  is said to be *absolutely convergent* if  $\sum |a_n|$  is convergent. If  $\sum a_n$  converges but  $\sum |a_n|$  does not, the series is said to be *conditionally convergent*.

**Example 2.7.13: A conditionally convergent series**

The series  $\sum \frac{(-1)^n}{n}$  converges, whereas

$$\sum |a_n| = \sum \left| \frac{(-1)^n}{n} \right| = \sum \frac{1}{n}$$

diverges. Hence the series is conditionally, but not absolutely, convergent.

The convergence claimed in [Example 2.7.13](#) can be proved with the integral test. The test cannot be applied directly, since the terms alternate in sign, but it can be applied after the terms are paired.

Grouping the terms of  $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$  two at a time,

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots = \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} - \frac{1}{2n}\right),$$

which now has *positive* terms, since  $\frac{1}{2n-1} > \frac{1}{2n}$ . Taking  $f(x) = \frac{1}{2x-1} - \frac{1}{2x}$ , continuous, positive and decreasing on  $[1, \infty)$ ,

$$\int_1^{\infty} \left(\frac{1}{2x-1} - \frac{1}{2x}\right) dx = \frac{1}{2} \lim_{t \rightarrow \infty} \left[ \log(2x-1) - \log(2x) \right]_1^t = \frac{1}{2} \lim_{t \rightarrow \infty} \left[ \log\left(\frac{2x-1}{2x}\right) \right]_1^t.$$

Evaluating,

$$= \frac{1}{2} \lim_{t \rightarrow \infty} \left( \log\left(\frac{2t-1}{2t}\right) - \log\left(\frac{1}{2}\right) \right) = \frac{1}{2} (\log 1 + \log 2) = \frac{\log 2}{2} \approx 0.3466,$$

using  $\frac{2t-1}{2t} = 1 - \frac{1}{2t} \rightarrow 1$ . The integral converges, so by the integral test the paired series converges, and therefore  $\sum \frac{(-1)^{n+1}}{n}$  converges.

On the other hand,

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$$

is the harmonic series, which diverges. The series therefore converges but not absolutely: it is **conditionally convergent**, confirming [Example 2.7.13](#).

**Note:**

The pairing step deserves a word, because regrouping the terms of an infinite series is not free — it is exactly what fails for  $\sum (-1)^n$ , where  $(1-1) + (1-1) + \dots = 0$  but  $1 - (1-1) - (1-1) - \dots = 1$ . What makes it legitimate here is that  $a_n = \frac{(-1)^{n+1}}{n} \rightarrow 0$ . Under that condition the partial sums of the paired series are a subsequence of the original partial sums, namely  $S_2, S_4, S_6, \dots$ , and the odd ones differ from their neighbours by  $a_n \rightarrow 0$ ; so if the even-indexed partial sums converge, all of them converge to the same limit. For  $\sum (-1)^n$  the terms do not tend to 0 and the argument is unavailable.

**Note:**

Note that the integral's value  $\frac{\log 2}{2} \approx 0.3466$  is not the sum of the series, which is  $\log 2 \approx 0.6931$  — the same trap as in part (c) of [Problem 2.7.4](#). The integral test has again decided convergence and nothing more.

**Note:**

$\sum |a_n|$  convergent implies  $\sum a_n$  convergent; that is, absolute convergence  $\Rightarrow$  convergence. The converse is not true, as [Example 2.7.13](#) shows.

**2.7.13 Tests for absolute convergence****Theorem 2.7.14: Ratio test**

Given a series  $\sum a_n$ , let  $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ .

- (i) If  $L < 1$ , the series  $\sum a_n$  converges absolutely, and hence converges.
- (ii) If  $L > 1$ , or if the limit does not exist, the series  $\sum a_n$  diverges.
- (iii) If  $L = 1$ , the ratio test is inconclusive: the series can converge or diverge.

**Theorem 2.7.15: Root test**

Given a series  $\sum a_n$ , let  $L = \lim_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ .

- (i) If  $L < 1$ , the series  $\sum a_n$  converges absolutely, and hence converges.
- (ii) If  $L > 1$ , or if the limit does not exist, the series  $\sum a_n$  diverges.
- (iii) If  $L = 1$ , the root test is inconclusive.

**2.7.14 Worked problems**

**Problem 2.7.5.** Test the convergence of the given series using the root test.

(a)  $\sum_{n=1}^{\infty} \left( \frac{2n+3}{3n+2} \right)^n$ .

**Solution:** Here  $a_n = \left( \frac{2n+3}{3n+2} \right)^n$ , which is positive for all  $n \geq 1$ . Consider

$$L = \lim_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} \left[ \left( \frac{2n+3}{3n+2} \right)^n \right]^{1/n} = \lim_{n \rightarrow \infty} \frac{2n+3}{3n+2}.$$

Dividing numerator and denominator by  $n$ ,

$$L = \lim_{n \rightarrow \infty} \frac{n \left( 2 + \frac{3}{n} \right)}{n \left( 3 + \frac{2}{n} \right)} = \frac{2 + \lim_{n \rightarrow \infty} \frac{3}{n}}{3 + \lim_{n \rightarrow \infty} \frac{2}{n}} = \frac{2+0}{3+0} = \frac{2}{3},$$

using  $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ . Since  $L = \frac{2}{3} < 1$ , the given series converges by the root test.

$$(b) \sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}.$$

**Solution:** Here  $a_n = \left(1 + \frac{1}{n}\right)^{-n^2}$ , which is positive for all  $n$ . Consider

$$L = \lim_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} \left[ \left(1 + \frac{1}{n}\right)^{-n^2} \right]^{1/n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{-n} = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n}.$$

Since  $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$ ,

$$L = \frac{1}{e} < 1,$$

and the series is convergent by the root test.

$$(c) \sum_{n=1}^{\infty} \frac{7}{(2n+5)^n}.$$

**Solution:** Here  $a_n = \frac{7}{(2n+5)^n}$ . Consider

$$L = \lim_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} \left[ \frac{7}{(2n+5)^n} \right]^{1/n} = \lim_{n \rightarrow \infty} \frac{7^{1/n}}{2n+5} = \left( \lim_{n \rightarrow \infty} 7^{1/n} \right) \left( \lim_{n \rightarrow \infty} \frac{1}{2n+5} \right).$$

Now  $\lim_{n \rightarrow \infty} 7^{1/n} = 1$  by standard limit (ii), and  $2n+5 \rightarrow \infty$ , so  $\frac{1}{2n+5} \rightarrow 0$ . Hence

$$L = 1 \cdot 0 = 0 < 1,$$

and the series is convergent by the root test.

**Problem 2.7.6.** Examine the convergence of the series  $\sum_{n=1}^{\infty} \left(\frac{nx}{n+1}\right)^n$ , where  $x \in \mathbb{R}$ .

Here  $a_n = \left(\frac{nx}{n+1}\right)^n$ , and  $x$  may be positive or negative, so the absolute value must be kept. By the root test,

$$L = \lim_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} \left| \left(\frac{nx}{n+1}\right)^n \right|^{1/n} = \lim_{n \rightarrow \infty} \left| \left(\frac{n}{n+1}\right)^n x^n \right|^{1/n},$$

using  $|ab| = |a||b|$  and  $|a|^2 = (|a|)^2$ . Therefore

$$L = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right) \cdot \lim_{n \rightarrow \infty} (|x|^n)^{1/n} = |x| \lim_{n \rightarrow \infty} \frac{n}{n\left(1 + \frac{1}{n}\right)} = |x| \cdot \frac{1}{1+0} = |x|.$$

By the root test:

- (i) the series is convergent if  $|x| < 1$ ;
- (ii) the series is divergent if  $|x| > 1$ ;
- (iii) the test is inconclusive if  $|x| = 1$ .

When  $|x| = 1$  the root test says nothing, so use the  $n$ th term test. With  $|x| = 1$ ,

$$|a_n| = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1 + \frac{1}{n}\right)^n} \implies \lim_{n \rightarrow \infty} |a_n| = \frac{1}{e} \neq 0.$$

Since  $a_n \not\rightarrow 0$ , the series diverges at  $x = 1$  and at  $x = -1$  by [Theorem 2.7.8](#). Collecting the three cases:

convergent for  $x \in (-1, 1)$ ,      divergent for  $x \in (-\infty, -1] \cup [1, \infty)$ .

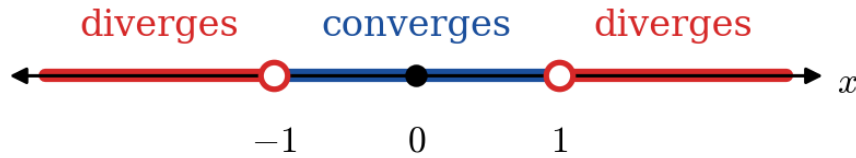


Figure 2.7.4: The series converges exactly on the open interval  $(-1, 1)$ . The endpoints are excluded: there the root test is inconclusive, but the  $n$ th term test gives divergence.

**Problem 2.7.7.** Test the convergence of the given series using the ratio test.

(a)  $\sum_{n=1}^{\infty} \frac{2(n-1)}{3^n + 1}.$

**Solution:** Given  $a_n = \frac{2(n-1)}{3^n + 1}$ , replacing  $n$  by  $n+1$ ,

$$a_{n+1} = \frac{2((n+1)-1)}{3^{n+1} + 1} = \frac{2n}{3^{n+1} + 1}.$$

Hence

$$\frac{a_{n+1}}{a_n} = \left(\frac{2n}{3^{n+1} + 1}\right) \left(\frac{3^n + 1}{2(n-1)}\right) = \left(\frac{2n}{2(n-1)}\right) \left(\frac{3^n + 1}{3^{n+1} + 1}\right),$$

and factoring  $3^n$  out of the numerator and  $3^{n+1}$  out of the denominator,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n-1}\right) \left(\frac{3^n \left(1 + \frac{1}{3^n}\right)}{3^{n+1} \left(1 + \frac{1}{3^{n+1}}\right)}\right).$$

Dividing numerator and denominator of the first bracket by  $n$ ,

$$\frac{a_{n+1}}{a_n} = \left(\frac{1}{1 - \frac{1}{n}}\right) \left(\frac{1 + \left(\frac{1}{3}\right)^n}{3 \left(1 + \left(\frac{1}{3}\right)^{n+1}\right)}\right).$$

Taking the limit, and using  $\lim_{n \rightarrow \infty} x^n = 0$  for  $|x| < 1$ ,

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{3} \cdot \frac{1}{\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)} \cdot \frac{1 + \lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^n}{1 + \lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^{n+1}} = \frac{1}{3} \cdot \frac{1}{1-0} \cdot \frac{1+0}{1+0} = \frac{1}{3}.$$

Since  $L = \frac{1}{3} < 1$ , the given series converges by the ratio test.

$$(b) \sum_{n=2}^{\infty} \frac{3^{n+2}}{\ln n}.$$

**Solution:** Given  $a_n = \frac{3^{n+2}}{\ln n}$ , replacing  $n$  by  $n+1$ ,

$$a_{n+1} = \frac{3^{(n+1)+2}}{\ln(n+1)} = \frac{3^{n+3}}{\ln(n+1)},$$

so that

$$\frac{a_{n+1}}{a_n} = \left( \frac{3^{n+3}}{\ln(n+1)} \right) \left( \frac{\ln n}{3^{n+2}} \right) = 3 \frac{\ln n}{\ln(n+1)}.$$

Hence

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 3 \lim_{n \rightarrow \infty} \frac{\ln n}{\ln(n+1)} = 3 \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{n+1}} = 3 \lim_{n \rightarrow \infty} \frac{n+1}{n} = 3 \lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right) = 3,$$

where L'Hôpital's rule was applied to the indeterminate form  $\frac{\infty}{\infty}$ . Since  $L = 3 > 1$ , the given series diverges by the ratio test.

**Alternative working.** Without L'Hôpital's rule, write  $\ln(n+1) = \ln\left[n\left(1 + \frac{1}{n}\right)\right] = \ln n + \ln\left(1 + \frac{1}{n}\right)$ , so that

$$\frac{a_{n+1}}{a_n} = 3 \frac{\ln n}{\ln n + \ln\left(1 + \frac{1}{n}\right)} = \frac{3}{1 + \frac{\ln\left(1 + \frac{1}{n}\right)}{\ln n}}.$$

As  $n \rightarrow \infty$  the numerator  $\ln\left(1 + \frac{1}{n}\right) \rightarrow \ln 1 = 0$  while  $\ln n \rightarrow \infty$ , so the second term in the denominator tends to 0 and

$$L = \frac{3}{1+0} = 3 > 1,$$

giving divergence, as before.

**Problem 2.7.8.** Discuss the convergence of the series

$$\frac{x}{1+x} + \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} + \cdots + \frac{x^n}{1+x^n} + \cdots \quad (x \in \mathbb{R}).$$

Here

$$a_n = \frac{x^n}{1+x^n} = \frac{1+x^n-1}{1+x^n} = 1 - \frac{1}{1+x^n},$$

a form that will be convenient for the  $n$ th term test. Recall that

$$\lim_{n \rightarrow \infty} x^n = \infty \text{ if } |x| > 1, \quad \lim_{n \rightarrow \infty} x^n = 0 \text{ if } |x| < 1,$$

while for  $x = 1$  we have  $x^n = 1$  and for  $x = -1$  we have  $x^n = \pm 1$  alternately. The behaviour therefore splits into three cases.

**Case (i):**  $|x| < 1$ .

Here  $\lim_{n \rightarrow \infty} x^n = 0$ . Using the ratio test with

$$a_n = \frac{x^n}{1+x^n}, \quad a_{n+1} = \frac{x^{n+1}}{1+x^{n+1}},$$

we compute  $\frac{a_{n+1}}{a_n}$  as follows. Forming the ratio,

$$\frac{a_{n+1}}{a_n} = \left( \frac{x^{n+1}}{1+x^{n+1}} \right) \left( \frac{1+x^n}{x^n} \right) = x \cdot \frac{x^n \left(1 + \frac{1}{x^n}\right)}{x^{n+1} \left(1 + \frac{1}{x^{n+1}}\right)} \cdot x = x \frac{1+x^n}{1+x^{n+1}}.$$

Hence

$$L = \lim_{n \rightarrow \infty} \left| x \cdot \frac{1+x^n}{1+x^{n+1}} \right| = |x| \left| \frac{1 + \lim_{n \rightarrow \infty} x^n}{1 + \lim_{n \rightarrow \infty} x^{n+1}} \right| = |x| \left| \frac{1+0}{1+0} \right| = |x|,$$

where the limit could be taken inside because  $t \mapsto \frac{1+t}{1+t}$  is continuous at  $t = 0$ ; for a continuous function,  $\lim_{x \rightarrow a} f(x) = f(a)$ , that is,  $f\left(\lim_{x \rightarrow a} x\right) = f(a)$ . By the case assumption  $|x| < 1$ , so  $L < 1$  and the given series *converges* by the ratio test.

**Case (ii):**  $|x| > 1$ .

Here  $\lim_{n \rightarrow \infty} x^n = \infty$  in magnitude, so  $\frac{1}{1+x^n} \rightarrow 0$  and

$$L = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left( 1 - \frac{1}{1+x^n} \right) = 1 - \frac{1}{1 + \lim_{n \rightarrow \infty} x^n} = 1 - 0 = 1.$$

Since  $L = 1 \neq 0$ , the given series *diverges* by the  $n$ th term test ([Theorem 2.7.8](#)).

**Case (iii):**  $|x| = 1$ .

Here  $x = 1$  or  $x = -1$ , and the two must be treated separately.

**If  $x = 1$ :** every term equals

$$a_n = \frac{1^n}{1+1^n} = \frac{1}{2}, \quad \text{so} \quad \lim_{n \rightarrow \infty} a_n = \frac{1}{2} \neq 0,$$

and the series *diverges* by the  $n$ th term test.

**If  $x = -1$ :** then  $x^n = -1$  whenever  $n$  is odd, so the denominator  $1+x^n = 0$  and those terms are not defined. The series therefore does not exist at  $x = -1$ .

#### Note:

Collecting the cases: the series converges for  $|x| < 1$ , diverges for  $|x| > 1$  and for  $x = 1$ , and is undefined at  $x = -1$ .

### 2.7.15 Worked problems

**Problem 2.7.9.** Determine whether the following series converge or diverge, applying either the ratio test or the root test.

$$(a) \sum_{n=1}^{\infty} \frac{n}{5^n}.$$

**Solution:** Here  $a_n = \frac{n}{5^n}$  and  $a_{n+1} = \frac{n+1}{5^{n+1}}$ , so

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+1}{5^{n+1}} \cdot \frac{5^n}{n} \right| = \frac{1}{5} \lim_{n \rightarrow \infty} \frac{n+1}{n} = \frac{1}{5} \lim_{n \rightarrow \infty} \frac{n \left(1 + \frac{1}{n}\right)}{n} = \frac{1}{5}.$$

Since  $L = \frac{1}{5} < 1$ , by [Theorem 2.7.14](#) the series converges absolutely, and hence **converges**.

$$(b) \sum_{n=1}^{\infty} \frac{10^n}{(n+1)4^{2n+1}}.$$

**Solution:** With  $a_n = \frac{10^n}{(n+1)4^{2n+1}}$  we have  $a_{n+1} = \frac{10^{n+1}}{(n+2)4^{2n+3}}$ , so

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{10^{n+1}}{(n+2)4^{2n+3}} \cdot \frac{(n+1)4^{2n+1}}{10^n} = \frac{10}{4^2} \cdot \frac{n+1}{n+2}.$$

Dividing the last fraction through by  $n$ ,

$$L = \lim_{n \rightarrow \infty} \frac{10}{16} \cdot \frac{n \left(1 + \frac{1}{n}\right)}{n \left(1 + \frac{2}{n}\right)} = \frac{10}{16} = \frac{5}{8}.$$

Since  $L = \frac{5}{8} < 1$ , the series converges absolutely and hence **converges**.

$$(c) \frac{2}{3} + \frac{2 \cdot 5}{3 \cdot 5} + \frac{2 \cdot 5 \cdot 8}{3 \cdot 5 \cdot 7} + \frac{2 \cdot 5 \cdot 8 \cdot 11}{3 \cdot 5 \cdot 7 \cdot 9} + \dots$$

**Solution:** Read off the two patterns. In the numerator the factors run  $2, 5, 8, 11, \dots$ , an arithmetic progression with  $n$ th term  $3n - 1$ ; in the denominator they run  $3, 5, 7, 9, \dots$ , with  $n$ th term  $2n + 1$ . Hence

$$a_n = \frac{2 \cdot 5 \cdot 8 \cdot 11 \cdots (3n - 1)}{3 \cdot 5 \cdot 7 \cdot 9 \cdots (2n + 1)}, \quad a_{n+1} = \frac{2 \cdot 5 \cdots (3n - 1)(3n + 2)}{3 \cdot 5 \cdots (2n + 1)(2n + 3)}.$$

Forming the ratio, every factor common to the two products cancels and only the newly introduced ones remain:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{3n + 2}{2n + 3}.$$

Therefore

$$L = \lim_{n \rightarrow \infty} \frac{3n + 2}{2n + 3} = \lim_{n \rightarrow \infty} \frac{n \left(3 + \frac{2}{n}\right)}{n \left(2 + \frac{3}{n}\right)} = \frac{3}{2}.$$

Since  $L = \frac{3}{2} > 1$ , by [Theorem 2.7.14](#) the series is **divergent**.

#### Note:

The cancellation is what makes the ratio test the natural tool for series built from products. Neither  $a_n$  nor  $a_{n+1}$  has a usable closed form, but their *ratio* does, because consecutive terms differ by exactly one factor top and bottom. The same applies to any series containing  $n!$ , which is why a factorial in the general term is a reliable signal to reach for the ratio test rather than the root test.

$$(d) \sum_{n=1}^{\infty} \frac{n!}{n^n}.$$

**Solution:** Take  $a_n = \frac{n!}{n^n}$ , so that  $a_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}}$ . Then

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!}.$$

Now  $(n+1)! = (n+1)n!$  and  $(n+1)^{n+1} = (n+1)(n+1)^n$ , so the factors  $(n+1)$  and  $n!$  cancel, leaving

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{n^n}{(n+1)^n} = \left( \frac{n}{n+1} \right)^n = \frac{1}{\left( \frac{n+1}{n} \right)^n} = \frac{1}{\left( 1 + \frac{1}{n} \right)^n}.$$

Using the standard limit  $\left( 1 + \frac{1}{n} \right)^n \rightarrow e$ ,

$$L = \lim_{n \rightarrow \infty} \frac{1}{\left( 1 + \frac{1}{n} \right)^n} = \frac{1}{e} \approx 0.3679.$$

Since  $L = \frac{1}{e} < 1$ , the series **converges**.

**Note:**

The root test also settles this, but less comfortably: it requires  $\lim \frac{(n!)^{1/n}}{n}$ , which is again  $\frac{1}{e}$  but needs Stirling's approximation to evaluate. When both tests apply they always give the same  $L$ ; the ratio test is preferred whenever factorials or products are present, and the root test whenever the general term is an  $n$ th power.

(e)  $\sum_{n=1}^{\infty} \frac{(-2)^n}{n^2}.$

**Solution:** The general term carries an  $n$ th power, so use the root test. Taking absolute values first,

$$|a_n|^{1/n} = \left| \frac{(-2)^n}{n^2} \right|^{1/n} = \left( \frac{2^n}{n^2} \right)^{1/n} = \frac{2}{(n^{1/n})^2}.$$

Using the standard limit  $n^{1/n} \rightarrow 1$ ,

$$L = \lim_{n \rightarrow \infty} \frac{2}{(n^{1/n})^2} = \frac{2}{1^2} = 2.$$

Since  $L = 2 > 1$ , by [Theorem 2.7.15](#) the series is **divergent**.

**Note:**

Divergence is also clear from the  $n$ th term test:  $\frac{2^n}{n^2}$  grows without bound, because an exponential beats any power, so  $a_n \not\rightarrow 0$ . Whenever the root or ratio test returns  $L > 1$  the terms in fact tend to infinity, so the  $n$ th term test always agrees — which is a useful check but never a contradiction.

(f)  $\sum_{n=1}^{\infty} \frac{x^n}{n!},$  where  $x \in \mathbb{R}.$

**Solution:** With  $a_n = \frac{x^n}{n!}$  and  $a_{n+1} = \frac{x^{n+1}}{(n+1)!},$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \left| \frac{x \cdot n!}{(n+1)n!} \right| = \frac{|x|}{n+1}.$$

For any fixed real  $x$  the numerator is constant while the denominator grows without bound, so

$$L = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1.$$

The series therefore converges absolutely for **every** real  $x$ .

**Note:**

This is the series for  $e^x$ , met later in this unit as a Maclaurin series, and the computation just made is the proof that its radius of convergence is infinite. Note how the factorial does the work: it eventually outgrows  $|x|^n$  no matter how large  $x$  is, which is why no restriction on  $x$  appears.

When the general term contains a variable  $x$ , the same tests return a limit  $L$  that depends on  $x$ , and the conditions  $L < 1$  and  $L > 1$  become conditions on  $x$ . This is how the interval on which a power series converges is found.

**Problem 2.7.10.** For each series, find the values of  $x$  for which it converges.

(a)  $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2+1}.$

**Solution:** With  $a_n = \frac{(x-2)^n}{n^2+1}$  and  $a_{n+1} = \frac{(x-2)^{n+1}}{(n+1)^2+1},$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(x-2)^{n+1}}{(n+1)^2+1} \cdot \frac{n^2+1}{(x-2)^n} \right| = |x-2| \cdot \frac{n^2+1}{(n+1)^2+1}.$$

Dividing the fraction through by  $n^2$ ,

$$\frac{n^2+1}{(n+1)^2+1} = \frac{n^2 \left(1 + \frac{1}{n^2}\right)}{n^2 \left[\left(1 + \frac{1}{n}\right)^2 + \frac{1}{n^2}\right]} \rightarrow \frac{1}{1} = 1,$$

so that  $L = |x-2|$ . The ratio test therefore gives absolute convergence when

$$|x-2| < 1 \iff -1 < x-2 < 1 \iff 1 < x < 3,$$

and divergence when  $|x-2| > 1$ , that is when  $x < 1$  or  $x > 3$ .

**Common mistake:**

The ratio test says nothing at the two endpoints  $x = 1$  and  $x = 3$ , where  $L = 1$ . They must be examined separately by substituting each value back into the series. At  $x = 3$  the series becomes  $\sum \frac{1}{n^2+1}$ , shown convergent by the integral test earlier; at  $x = 1$  it becomes  $\sum \frac{(-1)^n}{n^2+1}$ , which converges absolutely for the same reason. So this particular series converges on the closed interval  $[1, 3]$  — but that is an extra piece of work, not something the ratio test delivered.

(b)  $\sum_{n=1}^{\infty} \frac{x^n}{2n-1}.$

**Solution:** With  $a_n = \frac{x^n}{2n-1}$  and  $a_{n+1} = \frac{x^{n+1}}{2n+1}$ ,

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{2n+1} \cdot \frac{2n-1}{x^n} \right| = |x| \cdot \frac{2n-1}{2n+1} = |x| \cdot \frac{n(2 - \frac{1}{n})}{n(2 + \frac{1}{n})}.$$

Letting  $n \rightarrow \infty$ , the fraction tends to  $\frac{2}{2} = 1$ , so

$$L = |x|.$$

By the ratio test the series converges absolutely for  $|x| < 1$  and diverges for  $|x| > 1$ .

**Note:**

At the endpoints the test is again silent. Substituting  $x = 1$  gives  $\sum \frac{1}{2n-1}$ , which diverges — it is essentially the harmonic series with the even terms removed. Substituting  $x = -1$  gives  $\sum \frac{(-1)^n}{2n-1}$ , which converges conditionally. The two endpoints therefore behave differently, which is why neither can be guessed from the other.

## 2.8 Power Series, Taylor and Maclaurin Series

### Definition 2.8.1: Power series

A *power series* centred at  $x = a$  is a series of the form

$$c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots = \sum_{n=0}^{\infty} c_n(x-a)^n,$$

where the constants  $c_0, c_1, c_2, \dots \in \mathbb{R}$  are called the *coefficients* of the power series.

### 2.8.1 Where a power series converges

For any given power series  $\sum_{n=0}^{\infty} c_n(x-a)^n$  there are exactly three possibilities:

- (i) the series converges for all  $x \in \mathbb{R}$ ;
- (ii) the series converges only at  $x = a$ ;
- (iii) there is a number  $R > 0$  such that the series converges for  $|x-a| < R$  and diverges for  $|x-a| > R$ .

In the third case  $R$  is called the *radius of convergence*, centred at  $x = a$ . In the first case the radius of convergence is infinite; in the second it is zero.

**Note:**

Every power series converges at  $x = a$ , where all terms after the first vanish, so case (ii) is the smallest the set of convergence can be. The tests of the previous section are what produce  $R$ : applying the ratio test to  $\sum c_n(x-a)^n$  gives a limit of the form  $L = |x-a| \cdot \ell$ , and the condition  $L < 1$  reads  $|x-a| < \frac{1}{\ell} = R$ . Both worked problems above are instances —  $\sum \frac{(x-2)^n}{n^2+1}$  has  $a = 2$  and  $R = 1$ , while  $\sum \frac{x^n}{n!}$  has  $R = \infty$ .

**Note:**

The behaviour at the two endpoints  $|x - a| = R$  is not covered by the trichotomy and must always be checked separately, by substituting  $x = a - R$  and  $x = a + R$  into the series. As the problems above show, the series may converge at both endpoints, at neither, or at one but not the other.

**2.8.2 Determining the coefficients**

Suppose  $f$  has a power series representation about  $x = a$ , say

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots \quad (2.8.1)$$

The coefficients are then forced — they can be read off one at a time by differentiating and substituting  $x = a$ , which kills every term that still carries a factor of  $(x-a)$ .

Putting  $x = a$  in (2.8.1) directly gives

$$c_0 = f(a).$$

Differentiating (2.8.1) once,

$$f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + 4c_4(x-a)^3 + \cdots \quad (2.8.2)$$

and putting  $x = a$ ,

$$c_1 = f'(a).$$

Differentiating (2.8.2) again,

$$f''(x) = 2c_2 + 3 \cdot 2c_3(x-a) + 4 \cdot 3c_4(x-a)^2 + \cdots$$

and putting  $x = a$  gives  $2c_2 = f''(a)$ , that is

$$c_2 = \frac{f''(a)}{2!}.$$

A third differentiation gives  $3 \cdot 2 \cdot 1 c_3 = f'''(a)$ , so  $c_3 = \frac{f'''(a)}{3!}$ , and continuing in the same way, after  $n$  differentiations the constant term is  $n! c_n$ , whence

$$c_n = \frac{f^{(n)}(a)}{n!}$$

Substituting these coefficients back into (2.8.1) produces the Taylor series.

**Definition 2.8.2: Taylor series**

Suppose  $f$  has a power series representation around  $x = a$ . Then the *Taylor series expansion* of  $f$  near  $a$  is given by

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

**Note:**

The  $n = 0$  term is  $\frac{f^{(0)}(a)}{0!}(x-a)^0 = f(a)$ , with the conventions  $f^{(0)} = f$  and  $0! = 1$ .  
Each coefficient  $\frac{f^{(n)}(a)}{n!}$  is a real number.

**Definition 2.8.3: Maclaurin series**

The *Maclaurin series* of a function is the Taylor series of  $f$  centred at  $a = 0$ :

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

**Note:**

Truncating the Taylor series after the term in  $(x-a)^n$  gives the *Taylor polynomial* of degree  $n$ ,

$$T_n(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

**2.8.3 Worked problems**

**Problem 2.8.1.** Obtain the Taylor series expansion of  $\log_e x$  about  $x = 1$  up to the term containing the fourth degree, and hence determine the approximate value of  $\log_e 1.1$ .

**Solution:** Given  $f(x) = \log_e x$ , with  $a = 1$  and  $n = 4$ . Differentiating repeatedly and evaluating at  $x = 1$ :

$$\begin{aligned} f(x) &= \log_e x, & f(1) &= \log_e 1 = 0, \\ f'(x) &= \frac{1}{x}, & f'(1) &= 1, \\ f''(x) &= -\frac{1}{x^2}, & f''(1) &= -1, \\ f'''(x) &= \frac{2}{x^3}, & f'''(1) &= 2, \\ f^{(4)}(x) &= -\frac{6}{x^4}, & f^{(4)}(1) &= -6. \end{aligned}$$

In general  $f^{(n)}(x) = (-1)^{n-1} \frac{(n-1)!}{x^n}$ .

**Note:**

Note the difference between  $f^4(x)$ , meaning  $(f(x))^4$ , and  $f^{(4)}(x)$ , meaning the fourth derivative of  $f$ .

The Taylor expansion of  $f$  up to the fourth degree term is

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \frac{f^{(4)}(a)}{4!}(x-a)^4,$$

so, substituting  $a = 1$  and the values above,

$$\log_e x \approx 0 + 1(x-1) + \frac{(-1)}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 + \frac{(-6)}{4!}(x-1)^4,$$

and since  $2! = 2$ ,  $3! = 6$  and  $4! = 24$ ,

$$\log_e x \approx (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4}. \quad (2.8.3)$$

To find  $\log_e 1.1$ , put  $x = 1.1 = 1 + 0.1$ , so that  $x - 1 = 0.1$  in (2.8.3):

$$\log_e 1.1 \approx 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4} = 0.1 - \frac{0.01}{2} + \frac{0.001}{3} - \frac{0.0001}{4},$$

that is

$$\log_e 1.1 \approx 0.1 - 0.005 + 0.000333 - 0.000025 = 0.095308.$$

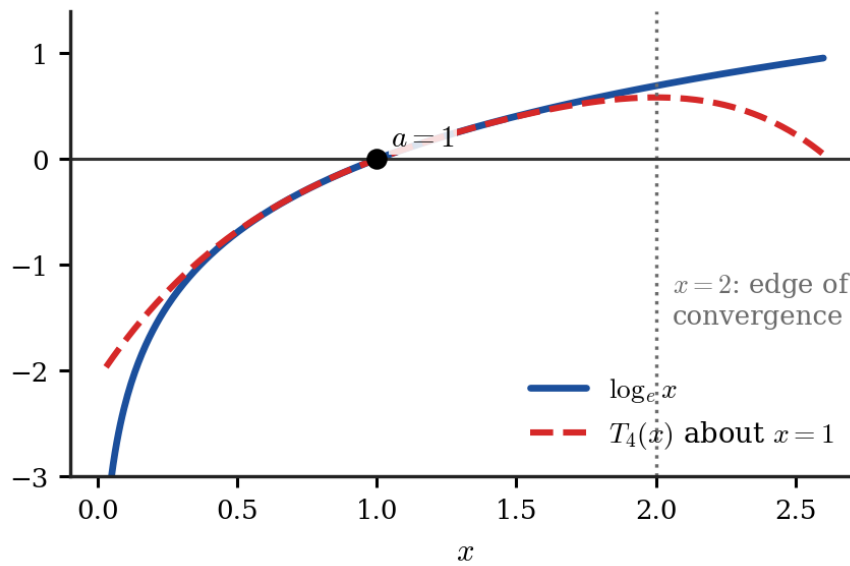


Figure 2.8.1:  $T_4$  tracks  $\log_e x$  closely near the centre  $a = 1$  and pulls away as  $x$  approaches the ends of  $(0, 2]$ , outside which the series does not converge.

**Problem 2.8.2.** Obtain the Taylor series expansion of the function  $\sin x$  at  $a = \frac{\pi}{6}$ .

**Solution:** Given  $f(x) = \sin x$  with  $a = \frac{\pi}{6}$ . The derivatives cycle with period four:

$$\begin{aligned} f(x) &= \sin x, & f\left(\frac{\pi}{6}\right) &= \sin \frac{\pi}{6} = \frac{1}{2}, \\ f'(x) &= \cos x, & f'\left(\frac{\pi}{6}\right) &= \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}, \\ f''(x) &= -\sin x, & f''\left(\frac{\pi}{6}\right) &= -\sin \frac{\pi}{6} = -\frac{1}{2}, \\ f'''(x) &= -\cos x, & f'''\left(\frac{\pi}{6}\right) &= -\cos \frac{\pi}{6} = -\frac{\sqrt{3}}{2}, \\ f^{(4)}(x) &= \sin x, & f^{(4)}\left(\frac{\pi}{6}\right) &= \sin \frac{\pi}{6} = \frac{1}{2}. \end{aligned}$$

The Taylor series expansion of  $f$  near  $x = a$  is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n = f\left(\frac{\pi}{6}\right) + \frac{f'\left(\frac{\pi}{6}\right)}{1!} \left(x - \frac{\pi}{6}\right) + \frac{f''\left(\frac{\pi}{6}\right)}{2!} \left(x - \frac{\pi}{6}\right)^2 + \dots$$

Substituting the values found above,

$$\sin x = \frac{1}{2} + \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{6}\right) + \frac{\left(-\frac{1}{2}\right)}{2!} \left(x - \frac{\pi}{6}\right)^2 + \frac{\left(-\frac{\sqrt{3}}{2}\right)}{3!} \left(x - \frac{\pi}{6}\right)^3 + \frac{\left(\frac{1}{2}\right)}{4!} \left(x - \frac{\pi}{6}\right)^4 + \dots$$

that is,

$$\sin x = \frac{1}{2} + \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{6}\right) - \frac{1}{4} \left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12} \left(x - \frac{\pi}{6}\right)^3 + \frac{1}{48} \left(x - \frac{\pi}{6}\right)^4 + \cdots \quad (2.8.4)$$

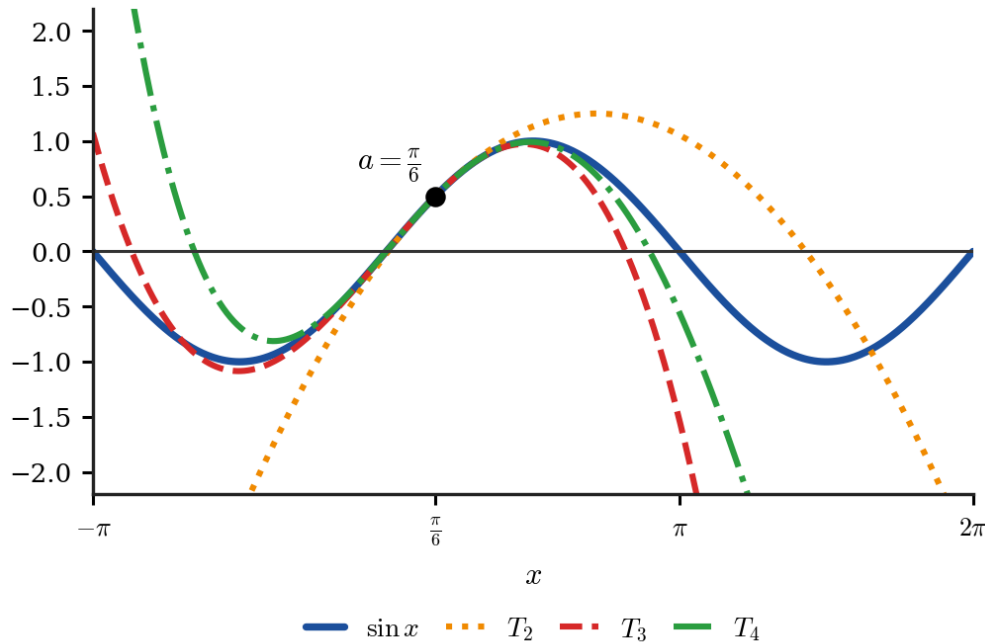


Figure 2.8.2: Successive Taylor polynomials of  $\sin x$  about  $a = \pi/6$ . Each extra term widens the stretch over which the polynomial follows the curve, but every one of them eventually leaves it.

**Problem 2.8.3.** Find the approximate value of  $\sin 29^\circ$  using the third degree Taylor polynomial.

**Solution:** Write  $29^\circ = 30^\circ - 1^\circ$ , so in radians

$$x = \frac{\pi}{6} - \frac{\pi}{180} \implies x - \frac{\pi}{6} = -\frac{\pi}{180}.$$

Substituting into (2.8.4) and keeping terms up to the third degree,

$$\sin 29^\circ \approx \frac{1}{2} + \frac{\sqrt{3}}{2} \left(-\frac{\pi}{180}\right) - \frac{1}{4} \left(-\frac{\pi}{180}\right)^2 - \frac{\sqrt{3}}{12} \left(-\frac{\pi}{180}\right)^3.$$

With  $\frac{\pi}{180} = 0.0174533$ , the terms are

$$\sin 29^\circ \approx 0.5 - 0.0151150 - 0.0000762 + 0.0000008,$$

the third term being negative and the fourth positive because  $\left(-\frac{\pi}{180}\right)^3$  is itself negative. Hence

$$\sin 29^\circ \approx 0.4848096.$$

#### Note:

The true value is  $\sin 29^\circ = 0.48480962\dots$ , so the cubic agrees to seven decimal places. The accuracy comes from the smallness of  $x - a = -\frac{\pi}{180} \approx -0.0175$ : each successive

term carries another factor of it, so the first neglected term is only  $\frac{1}{48} \left(\frac{\pi}{180}\right)^4 \approx 1.9 \times 10^{-9}$ . This is why the centre is taken at  $\frac{\pi}{6} = 30^\circ$  rather than at 0; expanding about 0 would need  $x \approx 0.506$  and several more terms for the same accuracy.

**Problem 2.8.4.** Expand  $e^{\sin x}$  by Maclaurin series up to the term containing  $x^4$ .

**Solution:** Given  $f(x) = e^{\sin x}$ , with  $a = 0$  and  $n = 4$ . Then

$$f(0) = e^{\sin 0} = e^0 = 1.$$

Differentiating,  $f'(x) = \frac{d}{dx} e^{\sin x} = e^{\sin x} \cos x = f(x) \cos x$ , so  $f'(0) = 1 \cdot 1 = 1$ . Differentiating the relation  $f'(x) = f(x) \cos x$  repeatedly keeps the working short:

$$f''(x) = f'(x) \cos x - f(x) \sin x,$$

$$f''(0) = (1)(1) - (1)(0) = 1;$$

$$f'''(x) = f''(x) \cos x - f'(x) \sin x - [f'(x) \sin x + f(x) \cos x]$$

$$= f''(x) \cos x - 2f'(x) \sin x - f(x) \cos x,$$

$$f'''(0) = (1)(1) - 0 - (1)(1) = 0;$$

$$f^{(4)}(x) = f'''(x) \cos x - 3f''(x) \sin x - 3f'(x) \cos x + f(x) \sin x,$$

$$f^{(4)}(0) = 0 - 0 - 3(1)(1) + 0 = -3.$$

The Maclaurin series up to the fourth degree term is

$$f(x) \approx f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4,$$

so that

$$e^{\sin x} \approx 1 + x + \frac{1}{2}x^2 + 0 \cdot x^3 + \frac{(-3)}{4!}x^4 \quad \Rightarrow \quad e^{\sin x} \approx 1 + x + \frac{x^2}{2} - \frac{x^4}{8}.$$

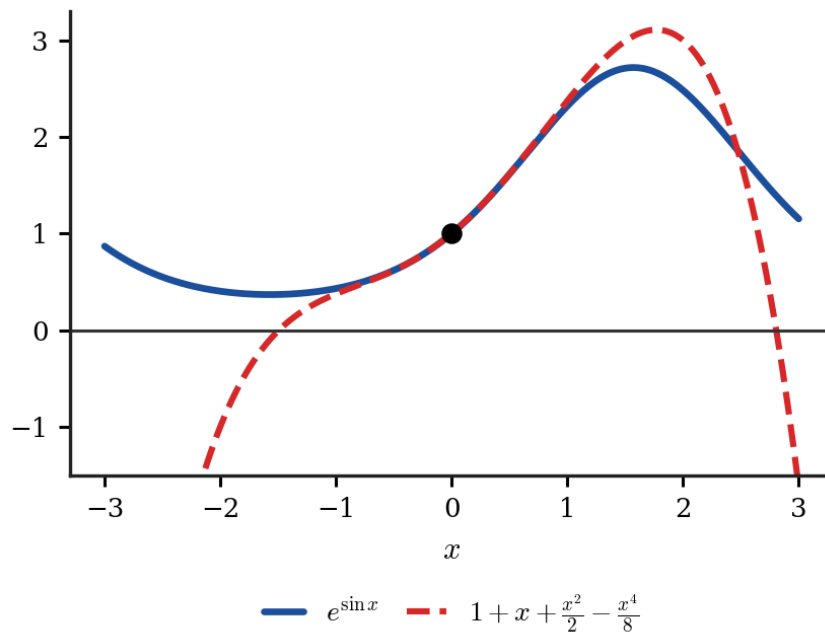


Figure 2.8.3: The fourth degree Maclaurin polynomial of  $e^{\sin x}$  is accurate near the origin and diverges from the function beyond roughly  $|x| > 1.5$ .

**Problem 2.8.5.** Obtain the Maclaurin series expansion of  $\log(1 + \cos x)$  up to the term containing  $x^4$ .

**Solution:** Given  $f(x) = \log(1 + \cos x)$ , with  $a = 0$  and  $n = 4$ . Then

$$f(0) = \log(1 + \cos 0) = \log 2.$$

Differentiating,

$$f'(x) = \frac{-\sin x}{1 + \cos x} \implies f'(0) = \frac{0}{2} = 0.$$

For the second derivative, by the quotient rule,

$$f''(x) = - \left[ \frac{(1 + \cos x) \cos x + \sin x (\sin x)}{(1 + \cos x)^2} \right] = - \left[ \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} \right] = - \frac{1 + \cos x}{(1 + \cos x)^2},$$

using  $\sin^2 x + \cos^2 x = 1$ . Hence

$$f''(x) = - \frac{1}{1 + \cos x} \implies f''(0) = - \frac{1}{1 + 1} = - \frac{1}{2}.$$

Continuing the differentiation, from  $f''(x) = - \frac{1}{1 + \cos x}$ ,

$$f'''(x) = - \frac{\sin x}{(1 + \cos x)^2} \implies f'''(0) = 0,$$

and differentiating the quotient  $f''(x) = \frac{-1}{1 + \cos x}$  once more,

$$f^{(4)}(x) = \frac{f''(x)(1 + \cos x) + f'(x)(-\sin x)}{(1 + \cos x)^2} \implies f^{(4)}(0) = \frac{-\frac{1}{2}(1 + 1) + 0}{(1 + 1)^2} = - \frac{1}{4}.$$

The Maclaurin series up to the fourth degree term is

$$f(x) \approx f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4,$$

so that

$$\log(1 + \cos x) \approx \log 2 + 0 + \frac{\left(-\frac{1}{2}\right)}{2!}x^2 + 0 + \frac{\left(-\frac{1}{4}\right)}{4!}x^4 \implies \log(1 + \cos x) \approx \log 2 - \frac{x^2}{4} - \frac{x^4}{96}.$$

**Problem 2.8.6.** Using the Maclaurin series, evaluate  $\int e^{-x^2} dx$  as an infinite series.

**Solution:** Given  $f(x) = e^{-x^2}$ , so  $f(0) = e^{-0^2} = 1$ . Differentiating,

$$f'(x) = -2x e^{-x^2} = -2x f(x) \implies f'(0) = 0.$$

Differentiating the relation  $f'(x) = -2x f(x)$  repeatedly,

$$\begin{aligned} f''(x) &= -2[xf'(x) + f(x)] = -2xf'(x) - 2f(x), & f''(0) &= 0 - 2 = -2; \\ f'''(x) &= -2xf''(x) - 4f'(x), & f'''(0) &= 0 - 0 = 0; \\ f^{(4)}(x) &= -2xf'''(x) - 6f''(x), & f^{(4)}(0) &= 0 - 6(-2) = 12. \end{aligned}$$

Hence the Maclaurin series is

$$e^{-x^2} = 1 + 0 \cdot x + \frac{(-2)}{2!}x^2 + \frac{0}{3!}x^3 + \frac{12}{4!}x^4 + \dots = 1 - x^2 + \frac{x^4}{2} - \dots$$

Integrating term by term,

$$\int e^{-x^2} dx = \int \left(1 - x^2 + \frac{x^4}{2} - \dots\right) dx = \int 1 dx - \int x^2 dx + \int \frac{x^4}{2} dx - \dots$$

giving

$$\int e^{-x^2} dx = x - \frac{x^3}{3} + \frac{x^5}{10} - \dots + C.$$

**Note:**

Replacing  $x$  by  $-x^2$  in the series for  $e^x$  gives the general term directly:  $e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}$ , so

$$\int e^{-x^2} dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!(2n+1)} + C.$$

There is no elementary antiderivative of  $e^{-x^2}$ , which is why the series is the answer rather than a step towards one.

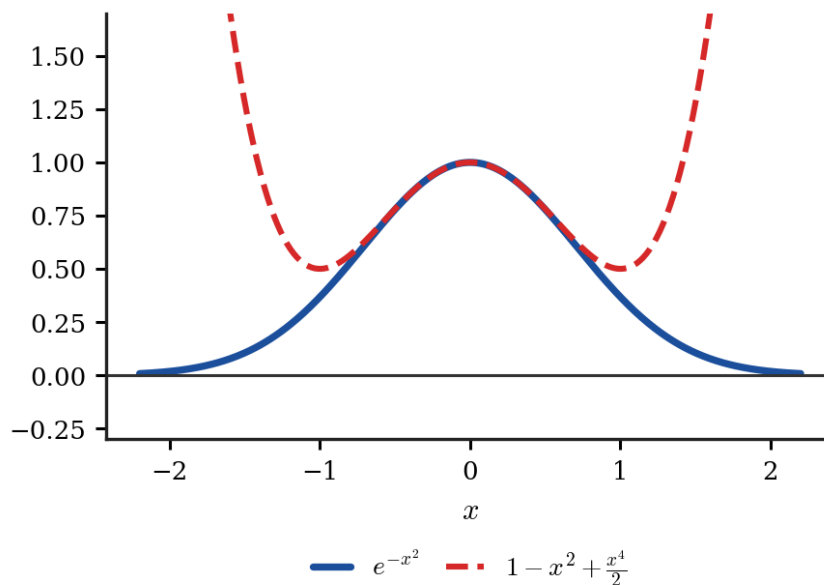


Figure 2.8.4:  $e^{-x^2}$  and its fourth degree Maclaurin polynomial. Term by term integration of the series is accurate over the range where the two curves agree.

## 2.8.4 Standard Maclaurin expansions

Obtain the Maclaurin series of the following functions.

(i)  $e^x$  (ii)  $\sin x$  (iii)  $\cos x$  (iv)  $\log(1+x)$  (v)  $(1+x)^k$ ,  $k \in \mathbb{R}$  (vi)  $\tan^{-1} x$

(i)  $f(x) = e^x$ , with  $a = 0$ .

**Solution:** Since every derivative of  $e^x$  is  $e^x$ , we have  $f^{(n)}(x) = e^x$  and therefore  $f^{(n)}(0) = 1$  for all  $n$ , in particular  $f(0) = 1$ . The Maclaurin series of the function is

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

so that

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

**Note:**

The geometric series will be used repeatedly below:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n, \quad |x| < 1.$$

(ii)  $f(x) = \sin x$ .

**Solution:** The derivatives repeat with period four:

$$\begin{aligned} f(x) &= \sin x, & f(0) &= 0; & f'(x) &= \cos x, & f'(0) &= 1; \\ f''(x) &= -\sin x, & f''(0) &= 0; & f'''(x) &= -\cos x, & f'''(0) &= -1; \\ f^{(4)}(x) &= \sin x, & f^{(4)}(0) &= 0. \end{aligned}$$

Hence every even order derivative vanishes at 0, while the odd order ones alternate in sign:

$$f^{(2n)}(0) = 0, \quad f^{(2n+1)}(0) = (-1)^n \quad \text{for } n = 0, 1, 2, \dots$$

The Maclaurin series is  $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$ , and keeping only the surviving terms,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + \frac{(-1)^n}{(2n+1)!}x^{2n+1} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1}$$

(iii)  $f(x) = \cos x$ .

**Solution:** Similarly,

$$\begin{aligned} f(x) &= \cos x, & f(0) &= 1; & f'(x) &= -\sin x, & f'(0) &= 0; \\ f''(x) &= -\cos x, & f''(0) &= -1; & f'''(x) &= \sin x, & f'''(0) &= 0; \\ f^{(4)}(x) &= \cos x, & f^{(4)}(0) &= 1; & f^{(5)}(x) &= -\sin x, & f^{(5)}(0) &= 0. \end{aligned}$$

Now the odd order derivatives vanish at 0 and the even ones alternate, so

$$\cos x = 1 + 0 \cdot x - \frac{x^2}{2!} + 0 \cdot x^3 + \frac{x^4}{4!} + \cdots$$

that is,

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots + \frac{(-1)^n}{(2n)!}x^{2n} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}x^{2n}$$

(iv)  $f(x) = \log(1+x)$ .

**Solution:** Differentiating,

$$\begin{aligned} f(x) &= \log(1+x), & f(0) &= 0; & f'(x) &= \frac{1}{1+x}, & f'(0) &= 1; \\ f''(x) &= \frac{-1}{(1+x)^2}, & f''(0) &= -1; & f'''(x) &= \frac{2}{(1+x)^3}, & f'''(0) &= 2; \\ f^{(4)}(x) &= \frac{-6}{(1+x)^4}, & f^{(4)}(0) &= -6; & f^{(5)}(x) &= \frac{24}{(1+x)^5}, & f^{(5)}(0) &= 24. \end{aligned}$$

In general

$$f^{(n)}(x) = (-1)^{n-1} \frac{(n-1)!}{(1+x)^n} \implies f^{(n)}(0) = (-1)^{n-1} (n-1)!.$$

Substituting into the Maclaurin series,

$$\log(1+x) = \frac{x}{1} + \frac{(-1)}{2!}x^2 + \frac{2}{3!}x^3 + \frac{(-6)}{4!}x^4 + \frac{24}{5!}x^5 + \dots + \frac{(-1)^{n-1}(n-1)!}{n!}x^n + \dots$$

and since  $\frac{(n-1)!}{n!} = \frac{1}{n}$ ,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots + \frac{(-1)^{n-1}}{n}x^n + \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}x^n$$

valid for  $-1 < x \leq 1$ .

**Note:**

The sum starts at  $n = 1$ , not  $n = 0$ : the term  $\frac{(-1)^{-1}}{0}x^0$  is not defined, and  $f(0) = 0$  contributes nothing anyway.

(v)  $f(x) = (1+x)^k$ , where  $k \in \mathbb{R}$ .

**Solution:** Differentiating,

$$\begin{aligned} f(x) &= (1+x)^k, & f(0) &= 1; \\ f'(x) &= k(1+x)^{k-1}, & f'(0) &= k; \\ f''(x) &= k(k-1)(1+x)^{k-2}, & f''(0) &= k(k-1), \end{aligned}$$

and in general

$$f^{(n)}(x) = k(k-1)(k-2) \cdots (k-(n-1))(1+x)^{k-n} \implies f^{(n)}(0) = k(k-1) \cdots (k-n+1).$$

Substituting into the Maclaurin series gives the *binomial series*

$$\begin{aligned} (1+x)^k &= 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 \\ &\quad + \dots + \frac{k(k-1) \cdots (k-(n-1))}{n!}x^n + \dots \end{aligned}$$

valid for  $|x| < 1$ . Writing the coefficient as  $\binom{k}{n} = \frac{k(k-1)\cdots(k-n+1)}{n!}$ , the series takes the compact form

$$(1+x)^k = \binom{k}{0} + \binom{k}{1}x + \binom{k}{2}x^2 + \cdots + \binom{k}{n}x^n + \cdots = \sum_{n=0}^{\infty} \binom{k}{n}x^n.$$

**Note:**

When  $k$  is a non-negative integer the coefficient can also be written as  $\frac{k!}{(k-n)!n!}$ , every term beyond  $n = k$  vanishes, and the series terminates as the ordinary binomial theorem. For general real  $k$  the factorial form has no meaning and the series is genuinely infinite.

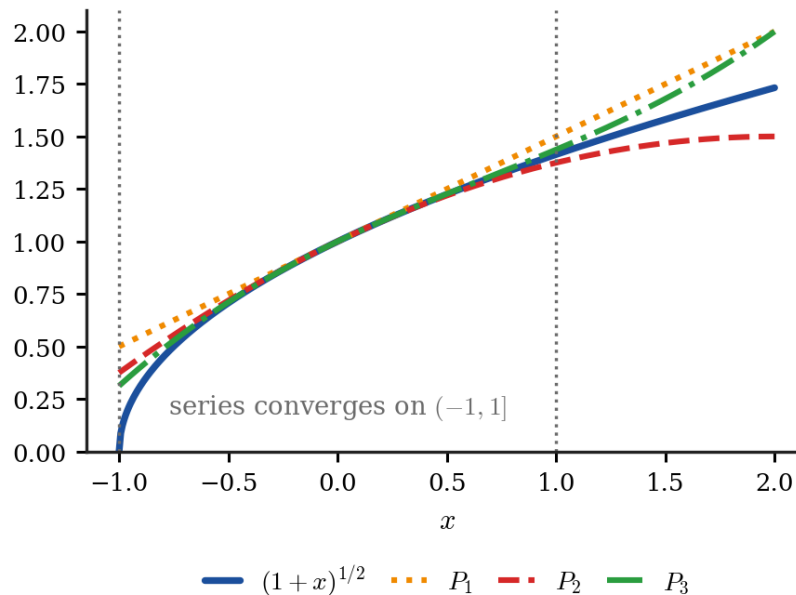


Figure 2.8.5: The binomial series for  $(1+x)^{1/2}$ . Successive partial sums approach the function on  $(-1, 1)$  and separate from it outside.

(vi)  $f(x) = \tan^{-1} x$ .

**Solution:** Rather than differentiating repeatedly, start from

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2} \implies \tan^{-1} x = \int \frac{1}{1+x^2} dx.$$

In the geometric series  $\frac{1}{1-x} = 1 + x + x^2 + \cdots$ , valid for  $|x| < 1$ , replace  $x$  by  $-x^2$ :

$$\frac{1}{1+x^2} = 1 + (-x^2) + (-x^2)^2 + (-x^2)^3 + \cdots = 1 - x^2 + x^4 - x^6 + \cdots$$

Integrating term by term,

$$\tan^{-1} x = \int (1 - x^2 + x^4 - x^6 + \cdots) dx = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + \frac{(-1)^n}{2n+1} x^{2n+1} + \cdots$$

The constant of integration is zero because  $\tan^{-1} 0 = 0$ . Hence

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

valid for  $|x| \leq 1$ .

Truncating a Taylor or Maclaurin series after finitely many terms leaves a polynomial, and evaluating that polynomial gives a numerical estimate of the function. The problem below is typical: the centre  $a$  is chosen at a nearby point where the function and its derivatives are known exactly.

**Problem 2.8.7.** Use an appropriate Taylor polynomial of degree 3 to estimate  $\sqrt[3]{1.3}$ .

**Solution:** Write  $\sqrt[3]{1.3} = (1 + 0.3)^{1/3}$ , so the binomial series applies with  $x = 0.3$  and  $k = \frac{1}{3}$ . Truncating after the cubic term,

$$(1+x)^k \approx 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3.$$

Compute the three coefficients separately. First,

$$k = \frac{1}{3}, \quad \frac{k(k-1)}{2!} = \frac{\frac{1}{3}(\frac{1}{3}-1)}{2} = \frac{\frac{1}{3} \cdot (-\frac{2}{3})}{2} = -\frac{2}{18} = -\frac{1}{9},$$

and

$$\frac{k(k-1)(k-2)}{3!} = \frac{\frac{1}{3}(-\frac{2}{3})(-\frac{5}{3})}{6} = \frac{\frac{10}{27}}{6} = \frac{10}{162} = \frac{5}{81}.$$

Substituting  $x = 0.3$ , so that  $x^2 = 0.09$  and  $x^3 = 0.027$ ,

$$\begin{aligned} (1.3)^{1/3} &\approx 1 + \frac{1}{3}(0.3) - \frac{1}{9}(0.09) + \frac{5}{81}(0.027) \\ &= 1 + 0.1 - 0.01 + 0.001667 \\ &= 1.091667. \end{aligned}$$

Hence

$$\sqrt[3]{1.3} \approx 1.0917$$

#### Note:

**Watch the sign of the quadratic coefficient.** Since  $k = \frac{1}{3} < 1$ , the factor  $(k-1)$  is negative and so is  $\frac{k(k-1)}{2!} = -\frac{1}{9}$ . The quadratic term therefore *subtracts* 0.01; writing it as  $+\frac{1}{9}$  would give 1.1117 instead, an error in the second decimal place. The alternating signs are characteristic of the binomial series for fractional  $k$ , since  $(k-1), (k-2), (k-3), \dots$  are then all negative.

#### Note:

The true value is  $\sqrt[3]{1.3} = 1.0913929$ , so the degree three estimate is accurate to about 0.0003. Note that  $x = 0.3$  satisfies  $|x| < 1$ , which is the interval of validity of the binomial series — for  $\sqrt[3]{3}$ , say, one would have  $x = 2$  and the series would diverge, so the number would first have to be rewritten, for instance as  $\sqrt[3]{3} = 2\sqrt[3]{1 - \frac{5}{8}}$ .

### 2.8.5 Further Maclaurin expansions

For functions built by composition, differentiating repeatedly from scratch quickly becomes unmanageable. The trick used below is to differentiate the *relation* rather than the function: write  $y'$  in terms of  $y$ , then differentiate that identity, so each new derivative is expressed using the ones already computed and evaluated at  $x = 0$ .

**Problem 2.8.8.** Find the first few terms of the Maclaurin series of the following functions.

(a)  $y = e^{x \sin x}$ .

**Solution:** At  $x = 0$ ,  $y(0) = e^0 = 1$ . Differentiating,

$$y' = e^{x \sin x} (\sin x + x \cos x) = y (\sin x + x \cos x),$$

so  $y'(0) = y(0)(\sin 0 + 0) = 0$ . Differentiating this product,

$$y'' = y(\cos x + \cos x - x \sin x) + y'(\sin x + x \cos x) = y(2 \cos x - x \sin x) + y'(\sin x + x \cos x),$$

and at  $x = 0$ , using  $y(0) = 1$  and  $y'(0) = 0$ ,

$$y''(0) = 1 \cdot (2 - 0) + 0 = 2.$$

Differentiating once more,

$$\begin{aligned} y''' &= y(-2 \sin x - x \cos x - \sin x) + y'(2 \cos x - x \sin x) \\ &\quad + y'(\cos x - x \sin x + \cos x) + y''(\sin x + x \cos x), \end{aligned}$$

and at  $x = 0$  every surviving bracket is multiplied by  $y'(0) = 0$  or by  $\sin 0 = 0$ , giving

$$y'''(0) = 1 \cdot 0 + 0 \cdot 2 + 0 \cdot 2 + 2 \cdot 0 = 0.$$

Substituting into the Maclaurin series  $y = y(0) + \frac{y'(0)}{1!}x + \frac{y''(0)}{2!}x^2 + \frac{y'''(0)}{3!}x^3 + \dots$ ,

$$e^{x \sin x} = 1 + \frac{2}{2!}x^2 + \dots = 1 + x^2 + \dots$$

**Note:**

Continuing the calculation gives  $e^{x \sin x} = 1 + x^2 + \frac{x^4}{3} + \frac{x^6}{120} + \dots$ . Only even powers appear, which is forced by symmetry:  $x \sin x$  is an even function, so  $e^{x \sin x}$  is even, and every odd order derivative must vanish at 0. Noticing this in advance halves the work —  $y'(0)$  and  $y'''(0)$  did not really need computing.

(b)  $y = \log(\sec x + \tan x)$ .

**Solution:** At  $x = 0$ ,  $y(0) = \log(\sec 0 + \tan 0) = \log(1 + 0) = \log 1 = 0$ . Differentiating and simplifying,

$$y' = \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} = \frac{\sec x (\tan x + \sec x)}{\sec x + \tan x} = \sec x,$$

the bracket cancelling against the denominator. Hence  $y'(0) = \sec 0 = 1$ . The remaining derivatives are now those of  $\sec x$ :

$$\begin{aligned} y'' &= \sec x \tan x, & y''(0) &= 1 \cdot 0 = 0; \\ y''' &= \sec x \sec^2 x + \tan x \cdot \sec x \tan x = \sec^3 x + \sec x \tan^2 x, & y'''(0) &= 1 + 0 = 1; \\ y^{(4)} &= 3 \sec^2 x \sec x \tan x + \sec x (2 \tan x \sec^2 x) + \tan^2 x \sec x \tan x, & y^{(4)}(0) &= 0. \end{aligned}$$

Substituting into the Maclaurin series,

$$\log(\sec x + \tan x) = x + \frac{x^3}{3!} + \cdots = x + \frac{x^3}{6} + \frac{x^5}{24} + \cdots$$

**Note:**

The simplification  $y' = \sec x$  is the whole problem. Without it one is differentiating a logarithm of a sum of two trigonometric functions repeatedly; with it, the work is the familiar derivatives of  $\sec x$ . Only odd powers survive because  $\log(\sec x + \tan x)$  is an odd function.

Some integrands have no elementary antiderivative. Replacing the integrand by its Maclaurin polynomial and integrating term by term gives a usable numerical estimate, and the accuracy can be judged by comparison with the exact value where one is available.

**Problem 2.8.9.** Use the Maclaurin series to estimate  $\int_0^1 e^{\sin x} dx$ .

**Solution:** Expanding  $e^{\sin x}$  as a Maclaurin series — done earlier in this unit —

$$e^{\sin x} = 1 + x + \frac{x^2}{2!} - \frac{3x^4}{4!} + \cdots$$

Integrating term by term between 0 and 1,

$$\int_0^1 e^{\sin x} dx \approx \left[ x + \frac{x^2}{2} + \frac{x^3}{3 \cdot 2!} - \frac{3}{4!} \cdot \frac{x^5}{5} \right]_0^1 = 1 + \frac{1}{2} + \frac{1}{3!} - \frac{3}{5!}.$$

Evaluating,

$$= 1 + 0.5 + 0.1667 - 0.025 = 1.6417.$$

**Note:**

The exact value is  $\int_0^1 e^{\sin x} dx = 1.6319$ , so the estimate is high by about 0.0098 — roughly 0.6%. The error is larger than in the earlier Taylor estimates because the interval reaches out to  $x = 1$ , where the neglected terms are no longer small; the terms of the series are being evaluated at the far edge of their useful range rather than close to the centre  $x = 0$ . Taking more terms, or splitting  $[0, 1]$  and re-centring, would tighten it.

**Problem 2.8.10.** Using the Maclaurin series for  $\tan^{-1} x$ , estimate  $\int_0^{1/\sqrt{3}} \tan^{-1} x dx$ .

**Solution:** From standard expansion (vi),  $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$  for  $|x| \leq 1$ . Truncating after the fifth degree term,

$$\tan^{-1} x \approx x - \frac{x^3}{3} + \frac{x^5}{5}.$$

Integrating this from 0 to  $\frac{1}{\sqrt{3}}$ ,

$$\int_0^{1/\sqrt{3}} \tan^{-1} x \, dx \approx \left[ \frac{x^2}{2} - \frac{x^4}{4 \cdot 3} + \frac{x^6}{6 \cdot 5} \right]_0^{1/\sqrt{3}} = \frac{1}{3 \cdot 2} - \frac{1}{3^2 \cdot 12} + \frac{1}{3^3 \cdot 30},$$

using  $\left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$ ,  $\left(\frac{1}{\sqrt{3}}\right)^4 = \frac{1}{9}$  and  $\left(\frac{1}{\sqrt{3}}\right)^6 = \frac{1}{27}$ . Hence

$$\approx 0.16667 - 0.00926 + 0.00123 = 0.15864.$$

### Note:

Integrating by parts gives the exact value

$$\int_0^{1/\sqrt{3}} \tan^{-1} x \, dx = \frac{\sqrt{3}\pi}{18} + \frac{\log 3}{2} - \log 2 = 0.158459,$$

against the series estimate 0.158642 — agreement to three decimal places, an error of 0.00018. The accuracy here is far better than in the previous problem because the upper limit  $\frac{1}{\sqrt{3}} \approx 0.577$  is well inside the interval of convergence  $|x| < 1$ , so the neglected terms are still genuinely small. At  $x = 1$  the same series would converge only barely, and at  $x > 1$  not at all.

## 2.9 Unit 2 Summary

### Mean value theorems and L'Hôpital's rule

- Rolle:  $f$  continuous on  $[a, b]$ , differentiable on  $(a, b)$ ,  $f(a) = f(b) \Rightarrow f'(c) = 0$  for some  $c \in (a, b)$ .
- Lagrange: continuity and differentiability alone  $\Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$ ; used to prove inequalities by bounding  $f'(c)$ .
- L'Hôpital: for  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$ ,  $\lim \frac{f}{g} = \lim \frac{f'}{g'}$ ; rewrite  $0 \cdot \infty$  as a quotient first.

### Improper integrals

- Type 1 (infinite interval) and Type 2 (infinite integrand): replace the bad endpoint by  $t$ , integrate, then let  $t$  tend to it. Split at every bad point; the integral converges only if every piece does.
- $\int_1^\infty \frac{dx}{x^p}$  converges iff  $p > 1$ ;  $\int_0^1 \frac{dx}{x^p}$  converges iff  $p < 1$ .
- Comparison ( $0 \leq g \leq f$ ):  $\int f$  finite  $\Rightarrow \int g$  finite;  $\int g$  infinite  $\Rightarrow \int f$  infinite. Limit comparison:  $f/g \rightarrow L \in (0, \infty) \Rightarrow$  both behave alike.

### Gamma and beta functions

- $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$ ;  $\Gamma(n+1) = n\Gamma(n)$ ;  $\Gamma(n+1) = n!$ ;  $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ .

- $B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} = B(n, m).$
- $\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right).$

### Sequences and series

- $\sum a_n$  converges  $\iff$  the partial sums  $S_n$  converge.  $a_n \not\rightarrow 0 \Rightarrow$  divergence (never the converse).
- Geometric:  $\sum_{n \geq 1} ar^{n-1} = \frac{a}{1-r}$  for  $|r| < 1$ , divergent for  $|r| \geq 1$ .  $p$ -series:  $\sum \frac{1}{n^p}$  converges iff  $p > 1$ .
- Integral test (positive, decreasing, continuous  $f$ ):  $\sum f(n)$  and  $\int_1^\infty f$  converge together — but the values differ.
- Ratio / root test:  $L < 1$  absolute convergence,  $L > 1$  divergence,  $L = 1$  no information. Ratio for factorials and products, root for  $n$ th powers.
- Absolute convergence  $\Rightarrow$  convergence;  $\sum \frac{(-1)^{n+1}}{n}$  converges only conditionally.

### Power, Taylor and Maclaurin series

- $\sum c_n(x-a)^n$  converges for  $|x-a| < R$  (ratio test) and diverges for  $|x-a| > R$ ; test the endpoints separately.
- $f(x) = \sum \frac{f^{(n)}(a)}{n!}(x-a)^n$ ; Maclaurin when  $a = 0$ .
- $e^x = \sum \frac{x^n}{n!}$ ,  $\sin x = \sum \frac{(-1)^n x^{2n+1}}{(2n+1)!}$ ,  $\cos x = \sum \frac{(-1)^n x^{2n}}{(2n)!}$  (all  $x$ );  $\log(1+x) = \sum_{n \geq 1} \frac{(-1)^{n+1} x^n}{n}$  ( $-1 < x \leq 1$ );  $(1+x)^k = \sum \binom{k}{n} x^n$  ( $|x| < 1$ );  $\tan^{-1} x = \sum \frac{(-1)^n x^{2n+1}}{2n+1}$  ( $|x| \leq 1$ ).
- Estimates are best when the centre  $a$  is close to the point of evaluation.

## 2.10 Practice Problems

The problems are arranged in three levels. Level A checks definitions and understanding, level B practises the standard methods, and level C asks you to analyse, justify or interpret a result, including in applied settings. Answers are given in brackets; full solutions are not.

### A. Check your understanding

- P1.** State the three hypotheses of Rolle's theorem. Which one fails for  $f(x) = |x|$  on  $[-1, 1]$ ?  
[Ans. differentiability at  $x = 0$ ]
- P2.** Classify each as Type 1, Type 2 or both: (a)  $\int_1^\infty \frac{dx}{x^3}$ , (b)  $\int_0^1 \ln x dx$ , (c)  $\int_0^\infty \frac{dx}{\sqrt{x}}$ .  
[Ans. (a) Type 1 (b) Type 2 (c) both]
- P3.** For which  $p$  does  $\sum \frac{1}{n^p}$  converge? For which  $p$  does  $\int_0^1 \frac{dx}{x^p}$  converge?  
[Ans.  $p > 1$ ;  $p < 1$ ]
- P4.** True or false, with a reason: (a) if  $a_n \rightarrow 0$  then  $\sum a_n$  converges; (b) an absolutely convergent series converges; (c) if the ratio test gives  $L = 1$  the series diverges.

[Ans. (a) F (b) T (c) F]

**P5.** Write the first four non-zero terms of the Maclaurin series of  $\cos x$ , and state where it converges.  
[Ans.  $1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720}$ ; all real  $x$ ]

**P6.** Write  $\Gamma(n+1)$  for a positive integer  $n$ , and the value of  $\Gamma(\frac{1}{2})$ . [Ans.  $n!$ ;  $\sqrt{\pi}$ ]

## B. Standard problems

**P7.** Verify Rolle's theorem for  $x^2 - 5x + 6$  on  $[2, 3]$ . [Ans.  $c = \frac{5}{2}$ ]

**P8.** Verify Lagrange's theorem for  $x^3$  on  $[1, 3]$ . [Ans.  $c = \sqrt{13/3} \approx 2.082$ ]

**P9.** Evaluate (a)  $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$ , (b)  $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ . [Ans. (a)  $\frac{1}{6}$  (b)  $\frac{1}{3}$ ]

**P10.** Evaluate (a)  $\int_0^\infty e^{-2x} dx$ , (b)  $\int_0^1 \frac{dx}{\sqrt{1-x}}$ , (c)  $\int_1^\infty \frac{dx}{x(x+1)}$ . [Ans. (a)  $\frac{1}{2}$  (b) 2 (c)  $\ln 2$ ]

**P11.** Evaluate  $\Gamma(\frac{9}{2})$  and  $\Gamma(-\frac{5}{2})$ . [Ans.  $\frac{105}{16}\sqrt{\pi}$ ;  $-\frac{8}{15}\sqrt{\pi}$ ]

**P12.** Evaluate (a)  $\int_0^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta$ , (b)  $\int_0^1 x^4(1-x)^3 dx$ . [Ans. (a)  $\frac{\pi}{32}$  (b)  $\frac{1}{280}$ ]

**P13.** Test for convergence: (a)  $\sum \frac{n^2}{2^n}$ , (b)  $\sum \left(\frac{n}{2n+1}\right)^n$ , (c)  $\sum_{n \geq 2} \frac{1}{n \ln n}$ .  
[Ans. (a) converges (b) converges (c) diverges]

**P14.** Find the sum of  $\sum_{n=1}^\infty \frac{3^{n+1}}{5^n}$ . [Ans.  $\frac{9}{2}$ ]

**P15.** Find the Maclaurin series of  $e^x \sin x$  up to the term in  $x^3$ . [Ans.  $x + x^2 + \frac{x^3}{3}$ ]

**P16.** Expand  $\sqrt{x}$  about  $x = 4$  up to the second-degree term, and estimate  $\sqrt{4.2}$ .  
[Ans.  $2 + \frac{x-4}{4} - \frac{(x-4)^2}{64}$ ; 2.0494]

## C. Analyse, justify and apply

**P17.** Use Lagrange's theorem to show that  $|\sin a - \sin b| \leq |a - b|$  for all real  $a, b$ .  
[Ans.  $|\cos c| \leq 1$ ]

**P18.** Find the interval of convergence of  $\sum_{n=1}^\infty \frac{(x-1)^n}{n 3^n}$ , examining both endpoints.  
[Ans.  $[-2, 4)$ ]

**P19.** Decide, with a suitable comparison, whether (a)  $\int_1^\infty \frac{dx}{\sqrt{x^3+1}}$  and (b)  $\int_1^\infty \frac{x+1}{x^2} dx$  converge.  
[Ans. (a) converges (b) diverges]

**P20.** A student argues that  $\sum_{n=1}^\infty \frac{1}{n^2} = 1$  because  $\int_1^\infty \frac{dx}{x^2} = 1$ . Evaluate the argument, and use the integral test inequalities to bracket the true sum.  
[Ans. invalid;  $1 \leq \sum \leq 2$  (the sum is  $\frac{\pi^2}{6}$ )]

**P21.** Show that  $\int_0^\infty x^n e^{-ax} dx = \frac{\Gamma(n+1)}{a^{n+1}}$  for  $a > 0$ , and use it to evaluate  $\int_0^\infty x^3 e^{-2x} dx$ .  
[Ans.  $\frac{3}{8}$ ]

- P22.** A discharging capacitor delivers current  $i(t) = I_0 e^{-t/\tau}$ . The total charge delivered is  $Q = \int_0^\infty i(t) dt$ . Find  $Q$ , and explain why an infinite time interval gives a finite charge. [Ans.  $Q = I_0 \tau$ ]
- P23.** The small-angle approximation  $\sin \theta \approx \theta$  is used in pendulum and beam problems. Using the next term of the Maclaurin series, estimate the relative error of this approximation at  $\theta = 0.2$  rad. [Ans.  $\approx \frac{\theta^3}{6} = 0.0013$ , about 0.67%]
- P24.** Construct (a) a series with  $a_n \rightarrow 0$  that converges only conditionally, and (b) a series with  $a_n > 0$ ,  $a_n \rightarrow 0$  that diverges. Justify each. [Ans. e.g. (a)  $\sum \frac{(-1)^{n+1}}{n}$  (b)  $\sum \frac{1}{\sqrt{n}}$ ]

# Unit 3 Two Variable Calculus

## 3.1 Functions of Several Variables

### Definition 3.1.1: Function of two variables

A function  $f$  of two variables is a rule that assigns to each ordered pair of real numbers  $(x, y)$  in a set  $D$  a unique real number denoted by  $f(x, y)$ . The set  $D$  is the *domain* of  $f$ , and its *range* is the set of values that  $f$  takes on, that is

$$\{ f(x, y) \mid (x, y) \in D \}.$$

Here  $D$  is a subset of the real plane, so such a function can be written as

$$z = f(x, y), \quad f : \mathbb{R}^2 \rightarrow \mathbb{R}.$$

We often write  $z = f(x, y)$  to make explicit the value taken on by  $f$  at the general point  $(x, y)$ . The variables  $x$  and  $y$  are the independent variables and  $z$  is the dependent variable.

### Example 3.1.2: Two functions of two variables

$$(a) f(x, y) = \frac{\sqrt{x+y+1}}{x-1} \quad (b) f(x, y) = x \ln(y^2 - x)$$

In (a) the square root requires  $x + y + 1 \geq 0$  and the denominator requires  $x \neq 1$ , so the domain is the closed half plane above the line  $y = -x - 1$ , with the vertical line  $x = 1$  removed. In (b) the logarithm requires  $y^2 - x > 0$ , so the domain is the region  $x < y^2$ , to the left of the parabola  $x = y^2$ .

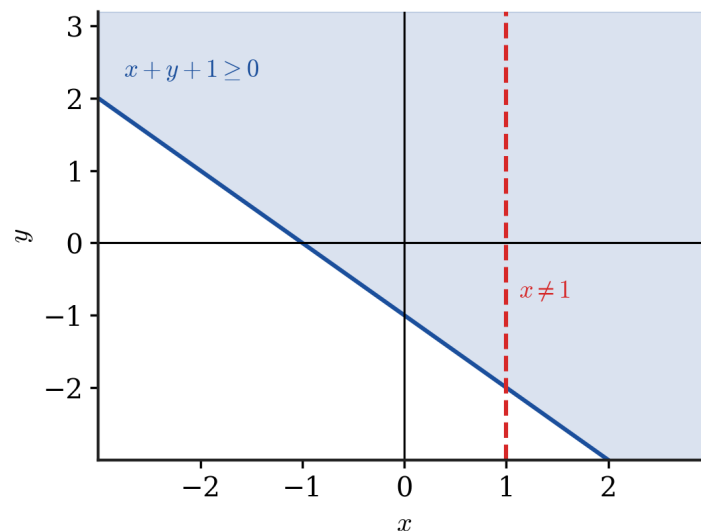


Figure 3.1.1: Domain of  $f(x, y) = \frac{\sqrt{x+y+1}}{x-1}$ . For a function of two variables the domain is a region of the plane, not an interval.

**Definition 3.1.3: Graph of a function of two variables**

If  $f$  is a function of two variables with domain  $D$ , then the graph of  $f$  is the set of all points  $(x, y, z)$  in  $\mathbb{R}^3$  such that  $z = f(x, y)$  and  $(x, y)$  is in  $D$ .

**Note:**

The graph is represented geometrically in three dimensions and looks like a surface sitting above (or below) the region  $D$  in the  $xy$ -plane.

**3.1.1 Level curves****Definition 3.1.4: Level curve**

The *level curves* of a function  $f$  of two variables are the curves with equations

$$f(x, y) = k,$$

where  $k$  is a constant taken from the range of  $f$ .

A level curve is the set of points of the domain at which  $f$  takes one fixed value. Geometrically it is the horizontal slice of the surface  $z = f(x, y)$  at height  $k$ , projected down onto the  $xy$ -plane; a contour line on a map is exactly this, with  $f$  the height of the ground. Drawing several level curves for equally spaced values of  $k$  gives a picture of the whole surface on flat paper: where the curves crowd together the surface is steep, where they spread out it is nearly flat.

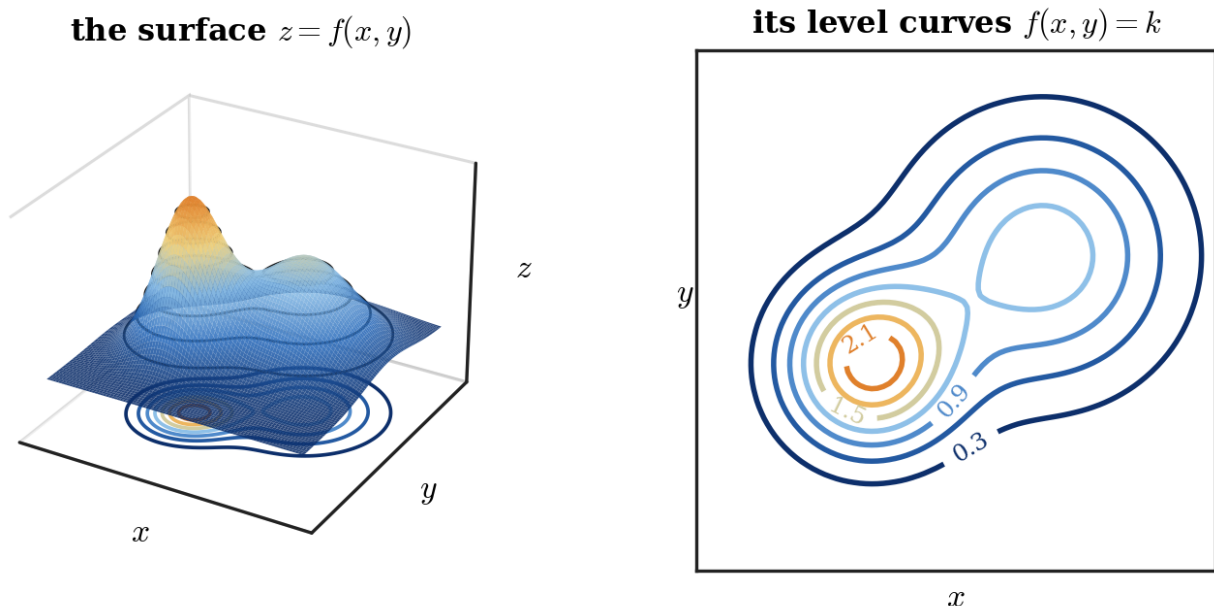


Figure 3.1.2: A surface with two peaks and its level curves. Each curve is one horizontal slice; the tightly packed curves around the taller peak show where the surface is steep.

**3.1.2 Limits**

For a function of one variable there are only two ways to approach a point, from the left and from the right, and the two-sided limit exists when those two agree. In the plane a point can

be approached along infinitely many paths — straight lines at any angle, parabolas, spirals — and the limit is required to be the same along every one of them.

#### Definition 3.1.5: Limit of a function of two variables

If  $f(x, y)$  approaches the same value  $L$  as  $(x, y) \rightarrow (a, b)$  along *every* path in the domain, then  $L$  is the limit of  $f$  at  $(a, b)$ , written

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L.$$

If two paths give different values, the limit does not exist.

#### Common mistake:

The two halves of this definition are not equally usable. To *disprove* the existence of a limit it is enough to exhibit two paths giving different values, which is a finite piece of work. To *prove* a limit exists, checking paths is never enough — there are infinitely many, and a function can approach the same value along every straight line and still fail to have a limit. Existence has to be established by other means, typically by an  $\varepsilon$ - $\delta$  argument or by squeezing  $|f(x, y) - L|$  between 0 and something that tends to 0. The problems below are all of the disproving kind.

### 3.1.3 Worked problems on limits

**Problem 3.1.1.** Does  $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$  exist?

**Solution:** Approach the origin along two different paths.

**Along  $y = 0$**  (the  $x$ -axis). Substituting  $y = 0$  before taking the limit,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x \cdot 0}{x^2 + 0} = \lim_{x \rightarrow 0} \frac{0}{x^2} = 0.$$

**Along  $y = x$ .** Substituting  $y = x$ ,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x \cdot x}{x^2 + x^2} = \lim_{x \rightarrow 0} \frac{x^2}{2x^2} = \frac{1}{2}.$$

The two paths give 0 and  $\frac{1}{2}$ , which differ. Hence the limit **does not exist**.

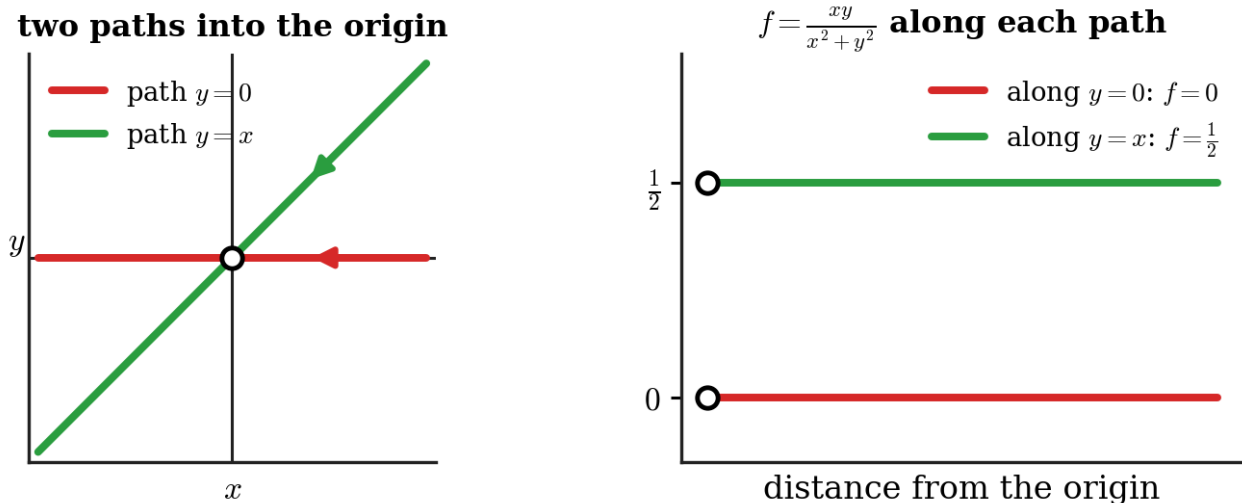


Figure 3.1.3: Approaching the origin along  $y = 0$  and along  $y = x$ . On the left, the two paths in the domain; on the right, the value of  $f$  at each point of the path. Along either path  $f$  is *constant*, at 0 and at  $\frac{1}{2}$  respectively, so the two approaches disagree however close to the origin one gets.

**Note:**

Note what the right-hand panel shows: along each straight line through the origin this function is constant, because numerator and denominator are both homogeneous of degree 2 and the  $x^2$  cancels. Substituting  $y = mx$  gives  $\frac{m}{1+m^2}$ , a different value for every slope  $m$ . So it is not merely that two paths disagree — every line through the origin gives its own answer.

**Problem 3.1.2.** Show that  $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$  does not exist.

**Solution:** Along  $x = 0$  (the  $y$ -axis),

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{y \rightarrow 0} \frac{0 - y^2}{0 + y^2} = \lim_{y \rightarrow 0} \frac{-y^2}{y^2} = -1.$$

Along  $y = 0$  (the  $x$ -axis),

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x^2 - 0}{x^2 + 0} = \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1.$$

The limits along the two axes are  $-1$  and  $1$ , which differ, so the limit **does not exist**.

**Note:**

Both problems have the same shape: a quotient of homogeneous polynomials of equal degree, undefined at the origin because numerator and denominator vanish together. The two coordinate axes are always worth trying first, since substituting  $x = 0$  or  $y = 0$  usually collapses the expression to something immediate. If they happen to agree, try  $y = x$ , then  $y = mx$ , then  $y = x^2$ .

### 3.1.4 Continuity

#### Definition 3.1.6: Continuity of a function of two variables

The function  $f(x, y)$  is *continuous* at  $(a, b)$  if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

The function  $f$  is said to be continuous on a domain  $D$  if it is continuous at every point of  $D$ .

#### Note:

As in one variable, this single equation carries three separate requirements:  $f(a, b)$  must be defined, the limit must exist, and the two must be equal. The functions in the problems above fail at the first two — neither is defined at the origin, and neither has a limit there — so no value assigned at  $(0, 0)$  could make either continuous.

#### Note:

Polynomials in  $x$  and  $y$  are continuous everywhere on  $\mathbb{R}^2$ , and a rational function is continuous wherever its denominator is non-zero. Sums, products, quotients and compositions of continuous functions are continuous wherever they are defined. In practice this means continuity is only ever in question at the isolated points where a denominator vanishes or a piecewise definition changes, which is where the path test above is needed.

## 3.2 Partial Derivatives

#### Definition 3.2.1: Partial derivatives

If  $f$  is a function of two variables, its *partial derivatives* are the functions  $f_x$  and  $f_y$  defined by

$$f_x = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}, \quad f_y = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h},$$

provided these limits exist.

### 3.2.1 Geometric interpretation

Taking  $y$  as a constant, say  $y = b$ , gives a plane. The intersection of that plane with the surface  $z = f(x, y)$  gives a curve, and the partial derivative  $f_x(a, b)$  is the slope of the tangent to that curve at the point  $(a, b, f(a, b))$ . Holding  $x$  constant instead gives  $f_y$  in the same way.

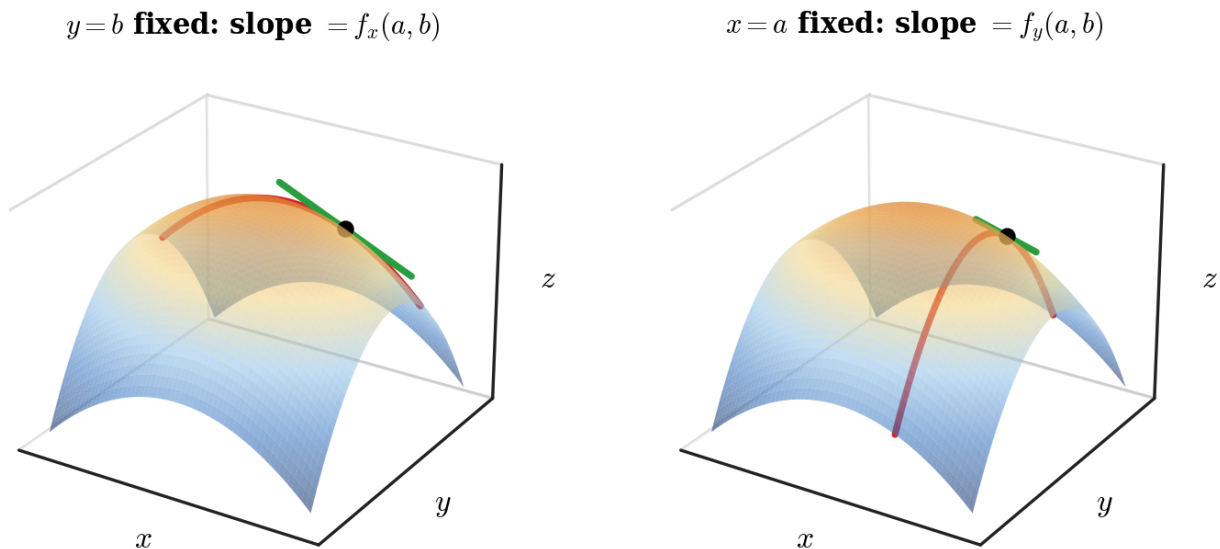


Figure 3.2.1: Each partial derivative is an ordinary derivative along one slice. The plane cuts a curve out of the surface (in red) and the partial derivative is the slope of the tangent to that curve (in green).

### 3.2.2 Notation

If  $z = f(x, y)$ , we write

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x}, \quad f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y}.$$

#### Definition 3.2.2: Gradient

The *gradient* of  $f(x, y)$  is the vector of its first order partial derivatives,

$$\nabla f = (f_x, f_y) = f_x \mathbf{i} + f_y \mathbf{j},$$

and for  $f(x, y, z)$  it is  $\nabla f = (f_x, f_y, f_z)$ .

#### Note:

At each point,  $\nabla f$  points in the direction in which  $f$  increases fastest, its length  $|\nabla f| = \sqrt{f_x^2 + f_y^2}$  is that fastest rate, and it is perpendicular to the level curve (or level surface) through the point. These facts are quoted without proof; they are used in the chain rule and in the method of Lagrange multipliers.

### 3.2.3 Rules for finding the partial derivatives of $z = f(x, y)$

1. To find  $f_x$ , regard  $y$  as a constant and differentiate  $f(x, y)$  with respect to  $x$ .
2. To find  $f_y$ , regard  $x$  as a constant and differentiate  $f(x, y)$  with respect to  $y$ .

### 3.2.4 Higher derivatives

The second order partial derivatives of  $f(x, y)$ , where  $z = f(x, y)$ , are given by

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2},$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x},$$

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y},$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}.$$

#### Common mistake:

Read the subscripts left to right in the order the derivatives are taken:  $f_{xy}$  means differentiate with respect to  $x$  first, then  $y$ . The  $\partial$ -notation reads in the opposite order,  $\frac{\partial^2 f}{\partial y \partial x}$ , which is the same object.

#### Note:

The operators  $D_x$  and  $D_y$  are linear operators, that is

$$D_x [\alpha f(x, y) + \beta g(x, y)] = \alpha D_x f(x, y) + \beta D_x g(x, y),$$

and similarly for  $D_y$ .

### 3.2.5 Worked problems

**Problem 3.2.1.** Find the first order partial derivatives of the following.

(a)  $f(x, y) = x^2 + y^2$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$  and  $y$ . Treating  $y$  as a constant,

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} [x^2 + y^2] = \frac{\partial}{\partial x} (x^2) + \frac{\partial}{\partial x} (y^2) = 2x + 0 = 2x,$$

the term  $y^2$  vanishing because it is treated as a constant. Similarly, treating  $x$  as a constant,

$$f_y = \frac{\partial}{\partial y} [x^2 + y^2] = 0 + 2y = 2y.$$

(b)  $f(x, y) = y^5 - 3xy$ .

**Solution:**

$$f_x = \frac{\partial}{\partial x} [y^5 - 3xy] = \frac{\partial}{\partial x} (y^5) - 3 \frac{\partial}{\partial x} (xy) = 0 - 3y = -3y,$$

$$f_y = \frac{\partial}{\partial y} [y^5 - 3xy] = \frac{\partial}{\partial y} (y^5) - 3 \frac{\partial}{\partial y} (xy) = 5y^4 - 3x.$$

(c)  $f(x, y) = e^{xy} \log y$ .

**Solution:** For  $f_x$ , the factor  $\log y$  is a constant, and the chain rule applies to  $e^{xy}$ :

$$f_x = \frac{\partial}{\partial x} [e^{xy} \log y] = \log y \frac{\partial}{\partial x} e^{xy} = \log y \cdot e^{xy} \frac{\partial}{\partial x} (xy) = y \log y e^{xy}.$$

For  $f_y$ , both factors depend on  $y$ , so the product rule is needed:

$$f_y = \frac{\partial}{\partial y} [e^{xy} \log y] = e^{xy} \frac{\partial}{\partial y} \log y + \log y \frac{\partial}{\partial y} e^{xy} = \frac{e^{xy}}{y} + \log y \cdot e^{xy} \cdot x,$$

that is

$$f_y = x \log y e^{xy} + \frac{e^{xy}}{y}.$$

**Note:**

The product rule is needed whenever both factors depend on the variable being differentiated. In  $f_x$  above,  $\log y$  carries no  $x$ , so it comes out as a constant multiplier instead.

(d)  $f(x, y, z) = x^2y + zy + z^2$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$ ,  $y$  and  $z$  in turn, treating the other two as constants each time:

$$f_x = \frac{\partial}{\partial x} [x^2y + zy + z^2] = 2xy + 0 + 0 = 2xy,$$

$$f_y = \frac{\partial}{\partial y} [x^2y + zy + z^2] = x^2 + z + 0 = x^2 + z,$$

$$f_z = \frac{\partial}{\partial z} [x^2y + zy + z^2] = 0 + y + 2z = y + 2z.$$

(e)  $f(x, y, z) = x + \sin(y + 3z)$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$ ,  $y$  and  $z$ .

$$f_x = \frac{\partial}{\partial x} [x + \sin(y + 3z)] = \frac{\partial}{\partial x} x + \frac{\partial}{\partial x} \sin(y + 3z) = 1 + 0 = 1.$$

For  $f_y$ , the chain rule gives

$$f_y = \frac{\partial}{\partial y} [x + \sin(y + 3z)] = 0 + \cos(y + 3z) \frac{\partial}{\partial y} (y + 3z) = \cos(y + 3z),$$

and likewise

$$f_z = 0 + \cos(y + 3z) \frac{\partial}{\partial z} (y + 3z) = 3 \cos(y + 3z).$$

(f)  $f(x, y, z) = e^{-(x^2+y^2+z^2)}$ .

**Solution:** Each derivative uses the chain rule on the exponent:

$$f_x = \frac{\partial f}{\partial x} = e^{-(x^2+y^2+z^2)} \frac{\partial}{\partial x} [-(x^2 + y^2 + z^2)] = -2x e^{-(x^2+y^2+z^2)}.$$

By the same working in the other two variables,

$$f_y = -2y e^{-(x^2+y^2+z^2)}, \quad f_z = -2z e^{-(x^2+y^2+z^2)}.$$

**Note:**

The function is symmetric in  $x$ ,  $y$  and  $z$ , so once  $f_x$  is known the other two follow by swapping the letters. Checking that a symmetry of the function survives into the answer is a quick way to catch slips.

(g)  $f(x, y) = x \cos y + ye^x$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$  and  $y$ . Treating  $y$  as a constant,

$$f_x = \frac{\partial}{\partial x} [x \cos y + ye^x] = \cos y \frac{\partial}{\partial x} x + y \frac{\partial}{\partial x} e^x = \cos y + ye^x.$$

Treating  $x$  as a constant,

$$f_y = \frac{\partial}{\partial y} [x \cos y + ye^x] = x \frac{\partial}{\partial y} \cos y + e^x \frac{\partial}{\partial y} y = -x \sin y + e^x.$$

(h)  $f(x, y) = \frac{2y}{y + \cos x}$ .

**Solution:** Both derivatives use the quotient rule. Treating  $y$  as a constant,

$$f_x = \frac{\partial}{\partial x} \left[ \frac{2y}{y + \cos x} \right] = \frac{(y + \cos x) \frac{\partial}{\partial x} (2y) - (2y) \frac{\partial}{\partial x} (y + \cos x)}{(y + \cos x)^2},$$

and since  $\frac{\partial}{\partial x} (2y) = 0$  and  $\frac{\partial}{\partial x} (y + \cos x) = -\sin x$ ,

$$f_x = \frac{(y + \cos x) \cdot 0 + 2y \sin x}{(y + \cos x)^2} = \frac{2y \sin x}{(y + \cos x)^2}.$$

Treating  $x$  as a constant,

$$f_y = \frac{(y + \cos x) \frac{\partial}{\partial y} (2y) - (2y) \frac{\partial}{\partial y} (y + \cos x)}{(y + \cos x)^2} = \frac{2y + 2 \cos x - 2y}{(y + \cos x)^2} = \frac{2 \cos x}{(y + \cos x)^2}.$$

(i)  $f(x, y) = x^y$ .

**Solution:** With  $y$  constant,  $x^y$  is an ordinary power of  $x$ :

$$f_x = \frac{\partial}{\partial x} x^y = y x^{y-1}.$$

With  $x$  constant,  $x^y$  is an exponential in  $y$ , and  $\frac{d}{dt} a^t = a^t \log a$  for constant  $a$ , so

$$f_y = \frac{\partial}{\partial y} x^y = x^y \log x.$$

**Note:**

The same expression is differentiated by two different rules depending on which letter is held fixed. Deciding which of the two rules applies before differentiating is the whole content of the problem.

**Problem 3.2.2.** Find all the second order partial derivatives of the following.

(a)  $f(x, y) = x^3y^5 + 2x^4y$ .

**Solution:** First the two first order derivatives:

$$f_x = \frac{\partial}{\partial x} [x^3y^5 + 2x^4y] = 3x^2y^5 + 8x^3y, \quad f_y = \frac{\partial}{\partial y} [x^3y^5 + 2x^4y] = 5x^3y^4 + 2x^4.$$

Now differentiate  $f_x$  and  $f_y$  partially with respect to  $x$  and  $y$ :

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x} (f_x) = \frac{\partial}{\partial x} [3x^2y^5 + 8x^3y] = 6xy^5 + 24x^2y, \\ f_{xy} &= \frac{\partial}{\partial y} (f_x) = \frac{\partial}{\partial y} [3x^2y^5 + 8x^3y] = 15x^2y^4 + 8x^3, \\ f_{yx} &= \frac{\partial}{\partial x} (f_y) = \frac{\partial}{\partial x} [5x^3y^4 + 2x^4] = 15x^2y^4 + 8x^3, \\ f_{yy} &= \frac{\partial}{\partial y} (f_y) = \frac{\partial}{\partial y} [5x^3y^4 + 2x^4] = 20x^3y^3 + 0 = 20x^3y^3. \end{aligned}$$

**Note:**

Here  $f_{xy} = f_{yx}$ , and the same happens in the next problem. This is not an accident: whenever the mixed partials are continuous near a point, the order of differentiation does not matter there. Getting two different answers is a reliable sign of an arithmetic slip.

(b)  $f(x, y) = x^3 + x^2y^3 - 2y^2$ .

**Solution:** The first order derivatives are

$$\begin{aligned} f_x &= \frac{\partial}{\partial x} [x^3 + x^2y^3 - 2y^2] = 3x^2 + 2xy^3 - 0 = 3x^2 + 2xy^3, \\ f_y &= \frac{\partial}{\partial y} [x^3 + x^2y^3 - 2y^2] = 0 + 3x^2y^2 - 4y = 3x^2y^2 - 4y. \end{aligned}$$

Differentiating  $f_x$  and  $f_y$  with respect to  $x$  and  $y$ ,

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x} [3x^2 + 2xy^3] = 6x + 2y^3, \\ f_{xy} &= \frac{\partial}{\partial y} [3x^2 + 2xy^3] = 6xy^2, \\ f_{yx} &= \frac{\partial}{\partial x} [3x^2y^2 - 4y] = 6xy^2 - 0 = 6xy^2, \\ f_{yy} &= \frac{\partial}{\partial y} [3x^2y^2 - 4y] = 6x^2y - 4. \end{aligned}$$

**Problem 3.2.3.** Find the indicated partial derivatives.

- (a)  $f(x, y) = 4 - x^2 - 2y^2$ ; find  $f_x(1, 1)$  and  $f_y(1, 1)$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$  and  $y$ :

$$f_x = \frac{\partial}{\partial x} [4 - x^2 - 2y^2] = -2x \implies f_x(x, y) = -2x,$$

$$f_y = \frac{\partial}{\partial y} [4 - x^2 - 2y^2] = -4y \implies f_y(x, y) = -4y.$$

Putting  $x = y = 1$ ,

$$f_x(1, 1) = -2(1) = -2, \quad f_y(1, 1) = -4(1) = -4.$$

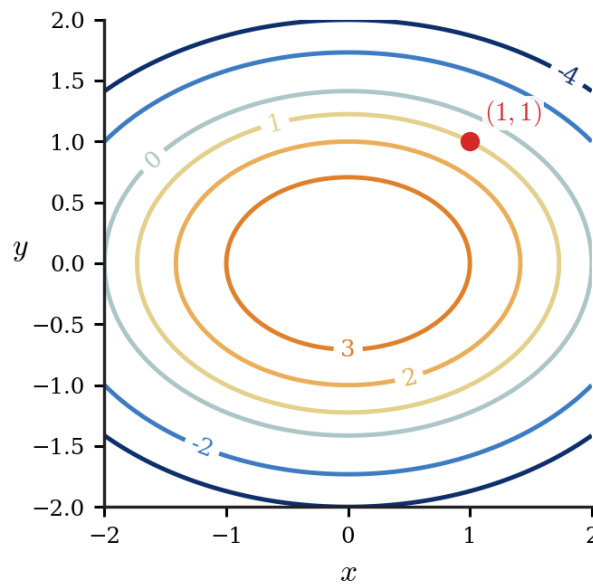


Figure 3.2.2: Level curves of  $f(x, y) = 4 - x^2 - 2y^2$ . Both partial derivatives are negative at  $(1, 1)$ , matching the fact that the level values fall as one moves away from the origin.

- (b)  $f(x, y) = x^3 + 2x^2y^2 + y^3$ ; find  $f_{xx}(1, 2)$ ,  $f_{yy}(1, 2)$  and  $f_{yx}(1, 2)$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$  and  $y$ :

$$f_x = \frac{\partial}{\partial x} [x^3 + 2x^2y^2 + y^3] = 3x^2 + 4xy^2 + 0 = 3x^2 + 4xy^2,$$

$$f_y = \frac{\partial}{\partial y} [x^3 + 2x^2y^2 + y^3] = 0 + 4x^2y + 3y^2 = 3y^2 + 4x^2y.$$

Now differentiate  $f_x$  and  $f_y$  again. First,

$$f_{xx} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} [3x^2 + 4xy^2] = 6x + 4y^2,$$

so at  $x = 1, y = 2$ ,

$$f_{xx}(1, 2) = 6(1) + 4(2)^2 = 6 + 16 = 22.$$

Next,

$$f_{yy} = \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} [3y^2 + 4x^2y] = 6y + 4x^2,$$

so that

$$f_{yy}(1, 2) = 6(2) + 4(1)^2 = 12 + 4 = 16.$$

Finally,

$$f_{yx} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} [3y^2 + 4x^2y] = 8xy,$$

and therefore

$$f_{yx}(1, 2) = 8(1)(2) = 16.$$

**Note:**

That  $f_{yy}(1, 2)$  and  $f_{yx}(1, 2)$  both come out as 16 is a coincidence of this particular point, not an instance of Clairaut's theorem. The theorem relates  $f_{xy}$  to  $f_{yx}$ , not  $f_{yy}$  to anything.

(c)  $f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$ ; find  $f_x(2, 3)$ .

**Solution:** Differentiate  $f$  with respect to  $x$ , treating  $y$  as a constant. Using  $\frac{d}{dt} \tan^{-1} t = \frac{1}{1+t^2}$  together with the chain rule,

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x}\right).$$

The first factor simplifies to  $\frac{1}{\frac{x^2+y^2}{x^2}} = \frac{x^2}{x^2+y^2}$ , and the second is  $\frac{\partial}{\partial x} (yx^{-1}) = -\frac{y}{x^2}$ . Hence

$$f_x = \frac{x^2}{x^2+y^2} \cdot \left(\frac{-y}{x^2}\right) = \frac{-y}{x^2+y^2} \implies f_x(x, y) = \frac{-y}{x^2+y^2}.$$

Taking  $(x, y) \rightarrow (2, 3)$ ,

$$f_x(2, 3) = \frac{-3}{2^2+3^2} = \frac{-3}{4+9} = -\frac{3}{13}.$$

**Note:**

The  $x^2$  introduced by clearing the fraction inside the arctangent cancels against the  $x^2$  from differentiating  $\frac{y}{x}$ , which is why the answer is so much simpler than the intermediate steps.

### 3.3 Clairaut's Theorem

**Theorem 3.3.1: Clairaut's Theorem**

Suppose  $f$  is defined on a disk  $D$  that contains the point  $(a, b)$ . If the functions  $f_{xy}$  and  $f_{yx}$  are both continuous on  $D$ , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

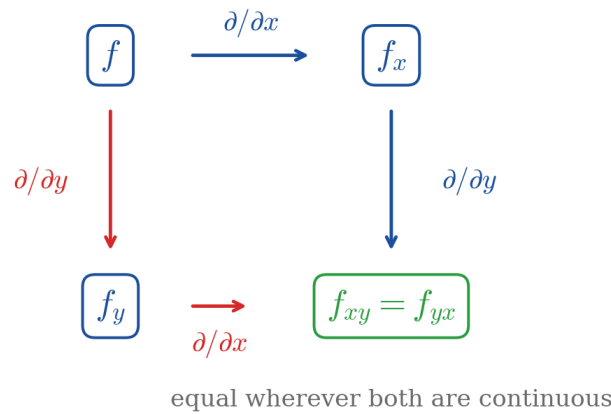


Figure 3.3.1: Either route from  $f$  to the mixed second derivative gives the same function, provided both mixed partials are continuous.

**Note:**

The same conclusion extends to third order derivatives: under the corresponding continuity assumptions,

$$f_{xyy} = f_{yxy} = f_{yyx},$$

and in general only the number of derivatives taken in each variable matters, not their order.

### 3.3.1 Worked problems

**Problem 3.3.1.** Verify the conclusion of Clairaut's theorem for the given functions.

(a)  $u = \cos(x^2y)$ .

**Solution:** Differentiate  $u$  partially with respect to  $x$  and  $y$ :

$$u_x = \frac{\partial u}{\partial x} = -\sin(x^2y) \frac{\partial}{\partial x}(x^2y) = -2xy \sin(x^2y),$$

$$u_y = \frac{\partial u}{\partial y} = -\sin(x^2y) \frac{\partial}{\partial y}(x^2y) = -x^2 \sin(x^2y).$$

Now differentiate  $u_x$  with respect to  $y$ , using the product rule on  $y \sin(x^2y)$ :

$$u_{xy} = \frac{\partial}{\partial y}(u_x) = -2x \left[ \sin(x^2y) + y \frac{\partial}{\partial y} \sin(x^2y) \right] = -2x \left[ \sin(x^2y) + x^2y \cos(x^2y) \right],$$

that is

$$u_{xy} = -2x \sin(x^2y) - 2x^3y \cos(x^2y).$$

Differentiating  $u_y$  with respect to  $x$ , again by the product rule,

$$u_{yx} = \frac{\partial}{\partial x}(u_y) = \sin(x^2y) \frac{\partial}{\partial x}(-x^2) - x^2 \frac{\partial}{\partial x} \sin(x^2y),$$

so that

$$u_{yx} = -2x \sin(x^2y) - 2x^3y \cos(x^2y).$$

Therefore  $u_{xy} = u_{yx}$ , and Clairaut's theorem is verified.

(b)  $u = e^x + x \log y + y \log x$ .

**Solution:** Differentiate  $u$  partially with respect to  $x$  and  $y$ :

$$u_x = \frac{\partial}{\partial x} [e^x + x \log y + y \log x] = e^x + \log y + \frac{y}{x},$$

$$u_y = \frac{\partial}{\partial y} [e^x + x \log y + y \log x] = \frac{x}{y} + \log x.$$

Differentiating  $u_x$  with respect to  $y$  and  $u_y$  with respect to  $x$ ,

$$u_{xy} = \frac{\partial}{\partial y} \left[ e^x + \log y + \frac{y}{x} \right] = \frac{1}{y} + \frac{1}{x}, \quad u_{yx} = \frac{\partial}{\partial x} \left[ \frac{x}{y} + \log x \right] = \frac{1}{y} + \frac{1}{x}.$$

Hence  $u_{xy} = u_{yx}$  and Clairaut's theorem is verified.

**Note:**

The domain matters here:  $\log x$  and  $\log y$  require  $x > 0$  and  $y > 0$ , and the mixed partials are continuous on that open quarter plane. Clairaut's theorem applies there, which is exactly where the function is defined.

(c)  $u = e^{xy} \sin y$ .

**Solution:** Differentiate  $u$  partially with respect to  $x$  and  $y$ . Since  $\sin y$  is constant with respect to  $x$ ,

$$u_x = \frac{\partial}{\partial x} [e^{xy} \sin y] = \sin y e^{xy} \frac{\partial}{\partial x} (xy) = y \sin y e^{xy} = y u,$$

writing the answer in terms of  $u$  itself to keep the next step short. For  $u_y$  the product rule is needed:

$$u_y = \sin y \frac{\partial}{\partial y} e^{xy} + e^{xy} \frac{\partial}{\partial y} \sin y = x \sin y e^{xy} + e^{xy} \cos y = x u + e^{xy} \cos y.$$

Now differentiate  $u_x$  with respect to  $y$ :

$$u_{xy} = \frac{\partial}{\partial y} [y u] = u \frac{\partial y}{\partial y} + y \frac{\partial u}{\partial y} = u + y (x u + e^{xy} \cos y),$$

that is

$$u_{xy} = e^{xy} \sin y + xy e^{xy} \sin y + y e^{xy} \cos y.$$

Differentiating  $u_y$  with respect to  $x$ ,

$$u_{yx} = \frac{\partial}{\partial x} [x u + e^{xy} \cos y] = u \frac{\partial x}{\partial x} + x \frac{\partial u}{\partial x} + \cos y \frac{\partial}{\partial x} e^{xy} = u + x(y u) + y e^{xy} \cos y,$$

that is

$$u_{yx} = e^{xy} \sin y + xy e^{xy} \sin y + y e^{xy} \cos y.$$

Hence  $u_{xy} = u_{yx}$  and Clairaut's theorem is verified.

(d)  $u = \sin(3x + 2y)$ .

**Solution:** Differentiate  $u$  partially with respect to  $x$  and  $y$ :

$$u_x = \frac{\partial}{\partial x} \sin(3x + 2y) = 3 \cos(3x + 2y), \quad u_y = \frac{\partial}{\partial y} \sin(3x + 2y) = 2 \cos(3x + 2y).$$

Differentiating  $u_x$  with respect to  $y$  and  $u_y$  with respect to  $x$ ,

$$u_{xy} = \frac{\partial}{\partial y} [3 \cos(3x + 2y)] = -6 \sin(3x + 2y),$$

$$u_{yx} = \frac{\partial}{\partial x} [2 \cos(3x + 2y)] = -6 \sin(3x + 2y).$$

Hence  $u_{xy} = u_{yx}$  and Clairaut's theorem is verified.

**Note:**

The two first derivatives carry different constants, 3 and 2, yet both mixed derivatives carry the product  $3 \times 2 = 6$ . That is the general pattern for  $\sin(ax + by)$ , and a quick way to check such an answer.

(e)  $u = xy^2 + x^2y^3 + x^3y^4$ .

**Solution:** Differentiate  $u$  partially with respect to  $x$  and  $y$ :

$$u_x = \frac{\partial}{\partial x} [xy^2 + x^2y^3 + x^3y^4] = y^2 + 2xy^3 + 3x^2y^4,$$

$$u_y = \frac{\partial}{\partial y} [xy^2 + x^2y^3 + x^3y^4] = 2xy + 3x^2y^2 + 4x^3y^3.$$

Differentiating  $u_x$  with respect to  $y$  and  $u_y$  with respect to  $x$ ,

$$u_{xy} = \frac{\partial}{\partial y} [y^2 + 2xy^3 + 3x^2y^4] = 2y + 6xy^2 + 12x^2y^3,$$

$$u_{yx} = \frac{\partial}{\partial x} [2xy + 3x^2y^2 + 4x^3y^3] = 2y + 6xy^2 + 12x^2y^3.$$

Hence  $u_{xy} = u_{yx}$  and Clairaut's theorem is verified.

(f)  $u = x^y + y^x$ .

**Solution:** Each term is a power in one letter and an exponential in the other, so the two rules from problem 1(j) are both needed. Differentiating with respect to  $x$ ,

$$u_x = \frac{\partial}{\partial x} [x^y + y^x] = y x^{y-1} + y^x \log y,$$

and with respect to  $y$ ,

$$u_y = \frac{\partial}{\partial y} [x^y + y^x] = x^y \log x + x y^{x-1}.$$

Now differentiate  $u_x$  with respect to  $y$ . The first term needs the product rule, and the second the product rule together with  $\frac{\partial}{\partial y} \log y = \frac{1}{y}$ :

$$u_{xy} = \frac{\partial}{\partial y} [y x^{y-1} + y^x \log y] = [x^{y-1} + y x^{y-1} \log x] + \left[ x y^{x-1} \log y + \frac{y^x}{y} \right],$$

and since  $\frac{y^x}{y} = y^{x-1}$ ,

$$u_{xy} = y x^{y-1} \log x + x^{y-1} + y^{x-1} + x y^{x-1} \log y.$$

Differentiating  $u_y$  with respect to  $x$  in the same way,

$$u_{yx} = \frac{\partial}{\partial x} [x^y \log x + x y^{x-1}] = \left[ y x^{y-1} \log x + \frac{x^y}{x} \right] + [y^{x-1} + x y^{x-1} \log y],$$

that is

$$u_{yx} = y x^{y-1} \log x + x^{y-1} + y^{x-1} + x y^{x-1} \log y.$$

Hence  $u_{xy} = u_{yx}$  and Clairaut's theorem is verified.

### 3.4 Harmonic Functions

#### Definition 3.4.1: Harmonic function

A function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  is called *harmonic* provided that it has second order partial derivatives and satisfies Laplace's equation

$$f_{xx}(x, y) + f_{yy}(x, y) = 0 \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

#### Note:

Laplace's equation says the two pure second derivatives cancel: however much the surface curves upward in the  $x$  direction, it curves downward by exactly as much in the  $y$  direction. A harmonic function therefore has no interior peak or pit every stationary point is a saddle.

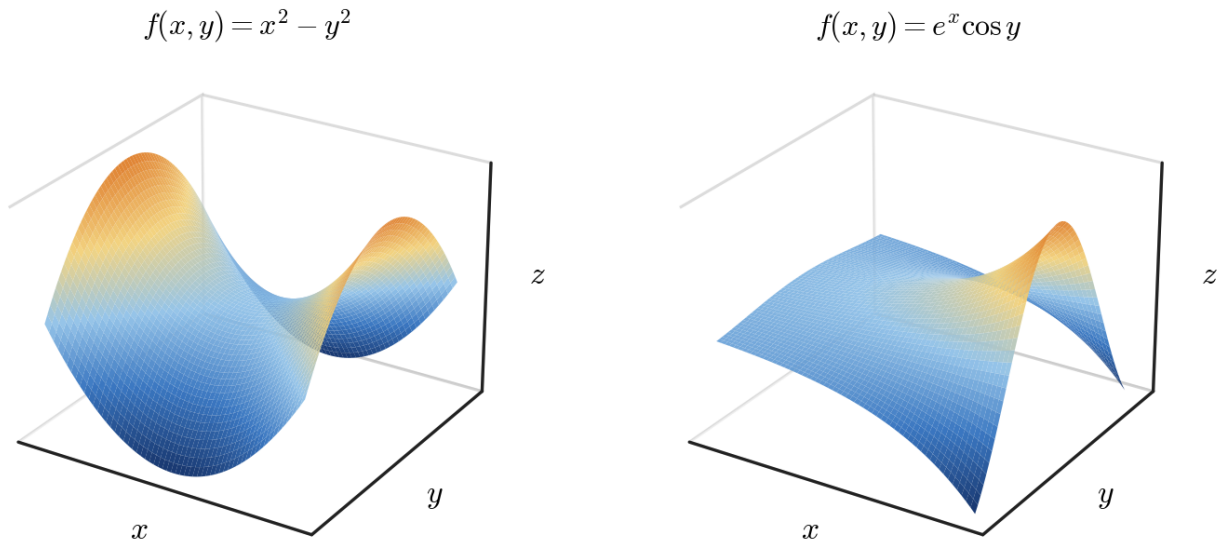


Figure 3.4.1: Two harmonic functions. Neither surface has a summit or a basin; each rises in one direction exactly as fast as it falls in the other.

### 3.4.1 Worked problems

**Problem 3.4.1.** Determine whether the following functions are harmonic.

(a)  $f(x, y) = e^x \cos y$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$  twice:

$$f_x = \frac{\partial}{\partial x} [e^x \cos y] = e^x \cos y \implies f_{xx} = \frac{\partial}{\partial x} (f_x) = e^x \cos y.$$

$$f_{xx} = e^x \cos y \tag{3.4.1}$$

Now differentiate partially with respect to  $y$  twice:

$$f_y = \frac{\partial}{\partial y} [e^x \cos y] = -e^x \sin y \implies f_{yy} = \frac{\partial}{\partial y} (f_y) = -e^x \frac{\partial}{\partial y} \sin y = -e^x \cos y.$$

$$f_{yy} = -e^x \cos y \tag{3.4.2}$$

Adding (3.4.1) and (3.4.2),

$$f_{xx} + f_{yy} = e^x \cos y - e^x \cos y = 0,$$

so  $f$  is a harmonic function.

(b)  $f(x, y) = x^2 - y^2$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$  twice:

$$f_x = \frac{\partial}{\partial x} (x^2 - y^2) = 2x \implies f_{xx} = \frac{\partial}{\partial x} (2x) = 2.$$

$$f_{xx} = 2 \tag{3.4.3}$$

Then with respect to  $y$  twice:

$$f_y = \frac{\partial}{\partial y} (x^2 - y^2) = -2y \implies f_{yy} = \frac{\partial}{\partial y} (-2y) = -2.$$

$$f_{yy} = -2 \quad (3.4.4)$$

Adding (3.4.3) and (3.4.4),

$$f_{xx} + f_{yy} = 2 - 2 = 0,$$

so  $f$  is a harmonic function.

(c)  $f(x, y) = x^2 - y^3$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$  twice:

$$f_x = \frac{\partial}{\partial x} [x^2 - y^3] = 2x \implies f_{xx} = \frac{\partial}{\partial x} (2x) = 2.$$

Then with respect to  $y$  twice:

$$f_y = \frac{\partial}{\partial y} [x^2 - y^3] = -3y^2 \implies f_{yy} = \frac{\partial}{\partial y} (-3y^2) = -6y.$$

Adding,

$$f_{xx} + f_{yy} = 2 - 6y \neq 0,$$

so  $f$  is *not* a harmonic function.

**Note:**

The sum vanishes only on the single line  $y = \frac{1}{3}$ , and Laplace's equation must hold at every point of the domain. Holding at some points is not enough.

**Problem 3.4.2.** If  $y = f(x + at) + g(x - at)$ , show that  $y_{tt} = a^2 y_{xx}$ .

**Solution:** Here  $y$  is a function of the two variables  $x$  and  $t$ , written  $y(t, x) = f(x + at) + g(x - at)$ , where  $f$  and  $g$  are arbitrary twice differentiable functions of one variable.

Differentiating with respect to  $t$ , the chain rule gives  $\frac{\partial}{\partial t}(x + at) = a$  and  $\frac{\partial}{\partial t}(x - at) = -a$ , so

$$y_t = \frac{\partial}{\partial t} [f(x + at) + g(x - at)] = f'(x + at) \frac{\partial}{\partial t}(x + at) + g'(x - at) \frac{\partial}{\partial t}(x - at)$$

that is,

$$y_t = a f'(x + at) - a g'(x - at).$$

Differentiating once more with respect to  $t$ ,

$$y_{tt} = \frac{\partial}{\partial t} [a f'(x + at) - a g'(x - at)] = a f''(x + at) \cdot a - a g''(x - at) \cdot (-a),$$

so that

$$y_{tt} = a^2 f''(x + at) + a^2 g''(x - at). \quad (3.4.5)$$

Now differentiate with respect to  $x$ . Since  $\frac{\partial}{\partial x}(x \pm at) = 1$ ,

$$y_x = \frac{\partial}{\partial x} [f(x + at) + g(x - at)] = f'(x + at) + g'(x - at),$$

and differentiating again,

$$y_{xx} = \frac{\partial}{\partial x} [f'(x + at) + g'(x - at)] = f''(x + at) + g''(x - at). \quad (3.4.6)$$

From (3.4.5),

$$y_{tt} = a^2 f''(x + at) + a^2 g''(x - at) = a^2 [f''(x + at) + g''(x - at)],$$

and substituting (3.4.6) gives

$$y_{tt} = a^2 y_{xx}$$

**Note:**

This is the one dimensional wave equation. The two pieces are shapes travelling without changing form:  $f(x + at)$  moves to the left and  $g(x - at)$  to the right, both at speed  $a$ . That  $f$  and  $g$  are arbitrary is the point every solution of the wave equation has this shape.

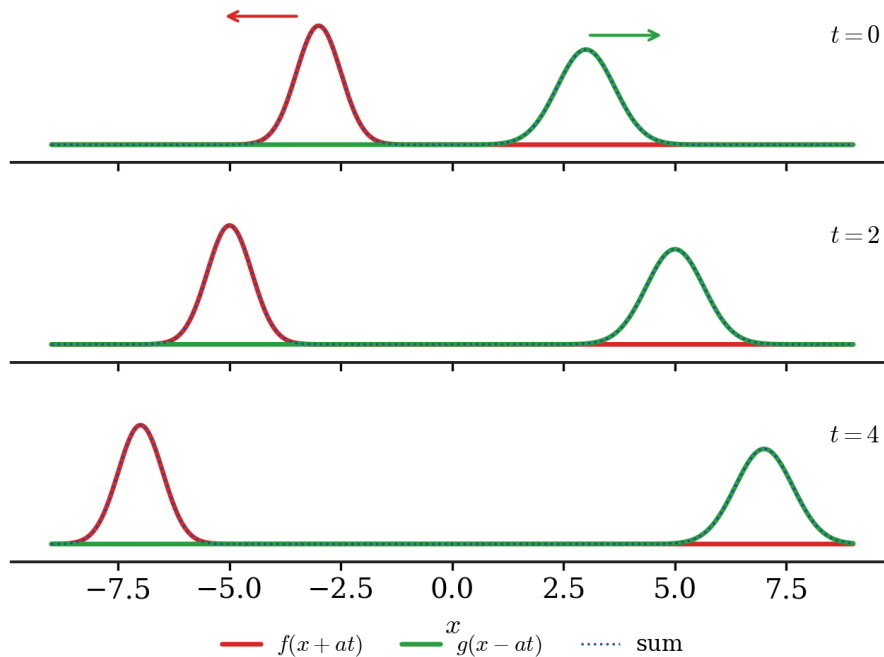


Figure 3.4.2: A solution  $y = f(x + at) + g(x - at)$  at three times. Neither bump changes shape; they simply move in opposite directions at speed  $a$ .

**Problem 3.4.3.** Show that  $u_{xx} - 2u_{xy} + u_{yy} = 0$ , where  $u = x f(x + y) + y g(x + y)$ .

**Solution:** Here  $f$  and  $g$  are arbitrary twice differentiable functions of the single variable  $x + y$ . Since

$$\frac{\partial}{\partial x}(x + y) = \frac{\partial}{\partial y}(x + y) = 1,$$

the chain rule contributes no extra factors, and to keep the working readable the argument  $(x + y)$  is written out in full each time.

Differentiating  $u$  with respect to  $x$ , using the product rule on  $x f(x + y)$ ,

$$u_x = \frac{\partial}{\partial x} [x f(x + y) + y g(x + y)] = x f'(x + y) + f(x + y) + y g'(x + y),$$

and with respect to  $y$ , using the product rule on  $y g(x + y)$ ,

$$u_y = \frac{\partial}{\partial y} [x f(x + y) + y g(x + y)] = x f'(x + y) + y g'(x + y) + g(x + y).$$

### Second order derivatives.

Differentiating  $u_x$  with respect to  $x$ ,

$$u_{xx} = \frac{\partial}{\partial x} [x f'(x + y) + f(x + y) + y g'(x + y)],$$

so that

$$u_{xx} = x f''(x + y) + f'(x + y) + f'(x + y) + y g''(x + y). \quad (3.4.7)$$

Differentiating  $u_x$  with respect to  $y$ ,

$$u_{xy} = \frac{\partial}{\partial y} [x f'(x + y) + f(x + y) + y g'(x + y)],$$

so that

$$u_{xy} = x f''(x + y) + f'(x + y) + y g''(x + y) + g'(x + y). \quad (3.4.8)$$

Differentiating  $u_y$  with respect to  $y$ ,

$$u_{yy} = \frac{\partial}{\partial y} [x f'(x + y) + y g'(x + y) + g(x + y)],$$

so that

$$u_{yy} = x f''(x + y) + y g''(x + y) + g'(x + y) + g'(x + y). \quad (3.4.9)$$

### Combining.

Substituting (3.4.7), (3.4.8) and (3.4.9) into the left-hand side, and abbreviating the argument  $(x + y)$ ,

$$\begin{aligned} u_{xx} - 2u_{xy} + u_{yy} &= [x f'' + 2f' + y g''] \\ &\quad - 2[x f'' + f' + y g'' + g'] \\ &\quad + [x f'' + y g'' + 2g']. \end{aligned}$$

Collecting the four kinds of term separately,

$$\begin{aligned} x f'' : & \quad 1 - 2 + 1 = 0, \\ f' : & \quad 2 - 2 = 0, \\ y g'' : & \quad 1 - 2 + 1 = 0, \\ g' : & \quad -2 + 2 = 0. \end{aligned}$$

Every coefficient vanishes, and therefore

$$u_{xx} - 2u_{xy} + u_{yy} = 0$$

#### Note:

The result holds for *any* twice differentiable  $f$  and  $g$ , because nothing about their form was used only that they depend on  $x$  and  $y$  through the single combination  $x + y$ .

## 3.5 Equation of a Tangent Plane

**Theorem 3.5.1: Tangent plane to a surface**

Suppose  $f$  has continuous partial derivatives. An equation of the tangent plane to the surface  $z = f(x, y)$  at the point  $P(x_0, y_0, z_0)$  is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0). \quad (3.5.1)$$

**Note:**

Here  $z_0 = f(x_0, y_0)$ . If  $z_0$  is not given, substitute  $x_0$  and  $y_0$  into  $z = f(x, y)$  to obtain it.

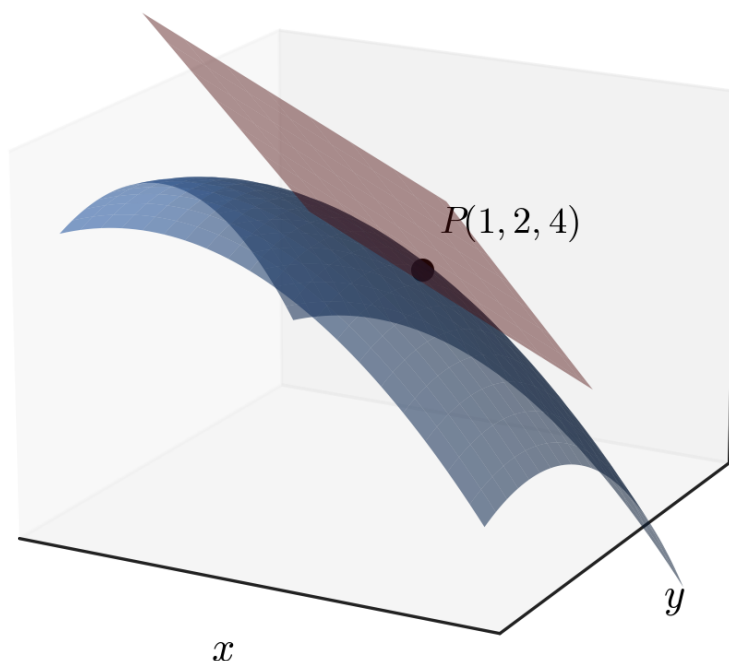


Figure 3.5.1: The tangent plane to  $z = 9 - x^2 - y^2$  at  $P(1, 2, 4)$ . Surface and plane agree at  $P$  and separate steadily away from it.

### 3.5.1 Worked problems

**Problem 3.5.1.** For each of the following surfaces, obtain the equation of the tangent plane at the point indicated.

- (a)  $z = 9 - x^2 - y^2$  at  $(1, 2, 4)$ .

**Solution:** Here  $f(x, y) = z = 9 - x^2 - y^2$  and  $(x_0, y_0, z_0) = (1, 2, 4)$ . The partial derivatives are

$$f_x = \frac{\partial}{\partial x} [9 - x^2 - y^2] = -2x, \quad f_y = \frac{\partial}{\partial y} [9 - x^2 - y^2] = -2y,$$

so at the given point

$$f_x(x_0, y_0) = f_x(1, 2) = -2(1) = -2, \quad f_y(x_0, y_0) = f_y(1, 2) = -2(2) = -4.$$

Substituting into (3.5.1),

$$z - 4 = -2(x - 1) + (-4)(y - 2) \implies 2x + 4y + z = 14 \implies \boxed{z = -2x - 4y + 14}$$

**Verification.** Take a point close to  $P$ , say  $(1.1, 2.1)$ . On the surface,

$$f(1.1, 2.1) = 9 - (1.1)^2 - (2.1)^2 = 9 - 1.21 - 4.41 = 3.38,$$

while on the tangent plane,

$$z(1.1, 2.1) = -2(1.1) - 4(2.1) + 14 = -2.2 - 8.4 + 14 = 3.4.$$

So  $f(1.1, 2.1) \approx z(1.1, 2.1)$ , agreeing to two decimal places.

**Note:**

The approximation is only valid when the point taken is near  $P(x_0, y_0, z_0)$ . At  $(2, 3)$ , for instance, the surface gives  $-4$  and the plane gives  $-2$  an error of 2 rather than 0.02.

(b)  $z = x \cos y - ye^x$  at  $(0, 0, 0)$ .

**Solution:** Here  $f(x, y) = z = x \cos y - ye^x$  and  $(x_0, y_0, z_0) = (0, 0, 0)$ . Then

$$f_x = \frac{\partial}{\partial x} [x \cos y - ye^x] = \cos y - ye^x \implies f_x(0, 0) = \cos 0 - 0 \cdot e^0 = 1.$$

Differentiating with respect to  $y$ ,

$$f_y = \frac{\partial}{\partial y} [x \cos y - ye^x] = -x \sin y - e^x \implies f_y(x_0, y_0) = f_y(0, 0) = -(0) \sin 0 - e^0 = -1.$$

Substituting into (3.5.1),

$$z - 0 = 1(x - 0) + (-1)(y - 0) \implies \boxed{z = x - y} \quad \text{that is} \quad x - y - z = 0.$$

(c)  $z = 3y^2 - 2x^2 + x$  at  $(2, -1, -3)$ .

**Solution:** Here  $f(x, y) = z = 3y^2 - 2x^2 + x$  and  $(x_0, y_0, z_0) = (2, -1, -3)$ . Checking the third coordinate,

$$f(2, -1) = 3(-1)^2 - 2(2)^2 + 2 = 3 - 8 + 2 = -3,$$

as given. The partial derivatives are

$$f_x = \frac{\partial}{\partial x} [3y^2 - 2x^2 + x] = -4x + 1 \implies f_x(2, -1) = -4(2) + 1 = -7,$$

$$f_y = \frac{\partial}{\partial y} [3y^2 - 2x^2 + x] = 6y \implies f_y(2, -1) = 6(-1) = -6.$$

Substituting into (3.5.1),

$$z - (-3) = (-7)(x - 2) + (-6)(y - (-1)) \implies z + 3 = -7x + 14 - 6y - 6,$$

and therefore

$$\boxed{z = -7x - 6y + 5}$$

**Note:**

It is worth substituting  $(x_0, y_0)$  back into  $f$  whenever  $z_0$  is supplied, as was done above. If the two disagree the point is not on the surface and (3.5.1) does not apply.

**Problem 3.5.2.** Obtain the equation of the tangent plane to the sphere  $x^2 + y^2 + z^2 = 14$  at the point  $(2, 1, 3)$ .

**Solution:** The surface is not presented in the form  $z = f(x, y)$ , so solve for  $z$  first. Since the point of contact has  $z = 3 > 0$ , take the upper hemisphere:

$$z^2 = 14 - x^2 - y^2 \quad \implies \quad f(x, y) = z = \sqrt{14 - x^2 - y^2}.$$

As a check, at  $(x_0, y_0) = (2, 1)$ ,

$$z_0 = \sqrt{14 - 4 - 1} = \sqrt{9} = 3,$$

which agrees with the given point. Differentiating,

$$f_x = \frac{-2x}{2\sqrt{14 - x^2 - y^2}} = \frac{-x}{\sqrt{14 - x^2 - y^2}}, \quad f_y = \frac{-y}{\sqrt{14 - x^2 - y^2}},$$

so that at  $(2, 1)$ , where the square root equals 3,

$$f_x(2, 1) = \frac{-2}{3}, \quad f_y(2, 1) = \frac{-1}{3}.$$

Substituting into  $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ ,

$$z - 3 = -\frac{2}{3}(x - 2) - \frac{1}{3}(y - 1).$$

Multiplying through by 3 and collecting terms,

$$3z - 9 = -2x + 4 - y + 1 \quad \implies \quad \boxed{2x + y + 3z = 14}$$

**Note:**

Two checks are available and both are quick. First, the point of contact must satisfy the plane:  $2(2) + 1 + 3(3) = 4 + 1 + 9 = 14$ . Second, for a sphere centred at the origin the tangent plane at  $P$  is perpendicular to the radius  $OP$ , so the plane's normal vector must be parallel to the position vector of  $P$ . Here the normal is  $(2, 1, 3)$ , which is exactly  $P$  — as it must be.

**Note:**

Choosing the positive square root was a real decision, not a formality. The equation  $x^2 + y^2 + z^2 = 14$  defines two functions of  $(x, y)$ , an upper and a lower hemisphere, and only the upper one passes through  $(2, 1, 3)$ . Had the point been  $(2, 1, -3)$  the negative root would have been needed and both partial derivatives would have changed sign, giving the plane  $2x + y - 3z = 14$ .

### 3.6 Linear Approximation

The tangent plane at a point is the plane that best matches the surface there, so near the point of contact its height is a good estimate of the height of the surface. Writing that estimate as a formula gives the two-variable analogue of the tangent line approximation from Unit 1.

**Definition 3.6.1: Linearisation and linear approximation**

Let  $f$  have continuous partial derivatives at  $(x_0, y_0)$ . The linear function whose graph is the tangent plane at that point,

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0),$$

is called the *linearisation* of  $f$  at  $(x_0, y_0)$ , and the approximation

$$f(x, y) \approx f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

is called the *linear approximation*, or the *tangent plane approximation*, of  $f$  at  $(x_0, y_0)$ .

**Note:**

$L$  and the tangent plane are the same object described two ways: the tangent plane is a set of points in space, and  $L$  is the function whose graph that set is. Rearranging the tangent plane equation  $z - z_0 = f_x(x - x_0) + f_y(y - y_0)$  into  $z = \dots$  gives  $L(x, y)$  immediately.

**Note:**

The approximation is only good *near*  $(x_0, y_0)$ , and the error grows with distance from it. This is the exact counterpart of the situation in Unit 2, where a Taylor polynomial about  $a$  was accurate close to  $a$  and drifted further out.  $L$  is in fact the first degree Taylor polynomial of  $f$  in two variables.

**3.6.1 Worked problems**

**Problem 3.6.1.** Obtain the tangent plane approximation of  $f(x, y) = 2x^2 + y^2$  at  $(1, 1)$ .

**Solution:** Here  $x_0 = 1$  and  $y_0 = 1$ , and

$$f(1, 1) = 2(1)^2 + (1)^2 = 2 + 1 = 3.$$

Differentiating partially,

$$f_x = 4x, \quad f_x(1, 1) = 4; \quad f_y = 2y, \quad f_y(1, 1) = 2.$$

Substituting into the linear approximation,

$$f(x, y) \approx 3 + 4(x - 1) + 2(y - 1) = 3 + 4x - 4 + 2y - 2,$$

that is

$$f(x, y) \approx 4x + 2y - 3$$

**Note:**

Testing the approximation at  $(1.05, 0.95)$  gives  $L = 4(1.05) + 2(0.95) - 3 = 3.1$  against the true value  $2(1.05)^2 + (0.95)^2 = 3.1075$ , an error of 0.0075. The error is second order in the displacement: doubling the distance from  $(1, 1)$  quadruples it.

**Problem 3.6.2.** Obtain the linear approximation of  $f(x, y) = e^{2x-y}$  at  $(1, -1)$ .

**Solution:** With  $x_0 = 1$  and  $y_0 = -1$ ,

$$f(1, -1) = e^{2(1)-(-1)} = e^{2+1} = e^3.$$

Differentiating, and using the chain rule on the exponent,

$$f_x = 2e^{2x-y}, \quad f_x(1, -1) = 2e^3; \quad f_y = -e^{2x-y}, \quad f_y(1, -1) = -e^3.$$

Substituting, and noting that  $y - y_0 = y - (-1) = y + 1$ ,

$$f(x, y) \approx e^3 + 2e^3(x - 1) - e^3(y + 1).$$

Taking out the common factor  $e^3$ ,

$$\approx e^3(1 + 2x - 2 - y - 1),$$

and so

$$f(x, y) \approx e^3(2x - y - 2)$$

#### Common mistake:

Watch the sign when  $y_0$  is negative: the factor is  $y - y_0 = y + 1$ , not  $y - 1$ . Getting this wrong is the commonest slip in these problems and shifts the whole plane. As a check, the approximation must be exact at the point itself: putting  $x = 1$ ,  $y = -1$  gives  $e^3(2 + 1 - 2) = e^3$ , which is  $f(1, -1)$ .

## 3.7 The Chain Rule

### Theorem 3.7.1: Chain rule, Case 1

Suppose  $z = f(x, y)$  is a differentiable function of  $x$  and  $y$ , where  $x = g(t)$  and  $y = h(t)$  are both differentiable functions of  $t$ . Then  $z$  is a differentiable function of  $t$  and

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}. \quad (3.7.1)$$

### Theorem 3.7.2: Chain rule, Case 2

Suppose  $z = f(x, y)$  is a differentiable function of  $x$  and  $y$ , where  $x = g(s, t)$  and  $y = h(s, t)$  are differentiable functions of  $s$  and  $t$ . Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}, \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}.$$

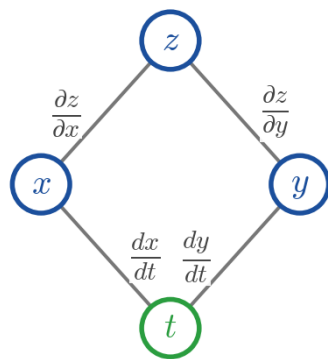
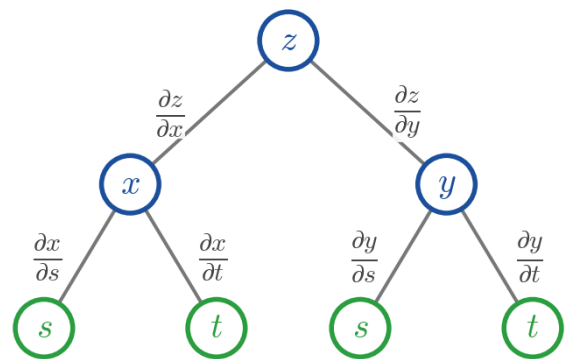
**Case 1:**  $x = g(t)$ ,  $y = h(t)$ **Case 2:**  $x = g(s, t)$ ,  $y = h(s, t)$ 

Figure 3.7.1: Reading the chain rule off a tree. Each path from the top to the bottom contributes one product, and the products are added.

**Note:**

Number of terms = number of intermediate variables.

Number of derivatives to be written = number of independent variables.

### 3.7.1 Worked problems

**Problem 3.7.1.** Use the chain rule to find  $\frac{dz}{dt}$  or  $\frac{dw}{dt}$ . Evaluate the derivative at the given value of  $t$ .

- (a)  $z = xy^3 - x^2y$ , where  $x = t^2 + 1$  and  $y = t^2 - 1$ ; at  $t = 0$ .

**Solution:** The derivatives of the intermediate variables are

$$\frac{dx}{dt} = \frac{d}{dt} [t^2 + 1] = 2t, \quad \frac{dy}{dt} = \frac{d}{dt} [t^2 - 1] = 2t,$$

and the partial derivatives of  $z$  are

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} [xy^3 - x^2y] = y^3 - 2xy, \quad \frac{\partial z}{\partial y} = \frac{\partial}{\partial y} [xy^3 - x^2y] = 3xy^2 - x^2.$$

Substituting into (3.7.1),

$$\frac{dz}{dt} = (y^3 - 2xy)(2t) + (3xy^2 - x^2)(2t). \quad (3.7.2)$$

Putting  $t = 0$  into  $x(t) = t^2 + 1$  and  $y(t) = t^2 - 1$  gives  $x(0) = 1$  and  $y(0) = -1$ . Substituting these and  $t = 0$  into (3.7.2),

$$\left. \frac{dz}{dt} \right|_{t=0} = [(-1)^3 - 2(1)(-1)](0) + [3(1)(-1)^2 - (1)^2](0) = 0.$$

**Note:**

Both terms carry the factor  $2t$ , which vanishes at  $t = 0$ , so the answer is zero whatever the brackets contain. The brackets are still worth evaluating as a check on the working.

- (b)  $z = \sin x \cos y$ , where  $x = \sqrt{t}$  and  $y = \frac{1}{t}$ ; at  $t = \pi$ .

**Solution:** The derivatives of the intermediate variables are

$$\frac{dx}{dt} = \frac{d}{dt}(\sqrt{t}) = \frac{1}{2\sqrt{t}}, \quad \frac{dy}{dt} = \frac{d}{dt}\left(\frac{1}{t}\right) = -\frac{1}{t^2},$$

and the partial derivatives of  $z$  are

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(\sin x \cos y) = \cos x \cos y, \quad \frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(\sin x \cos y) = -\sin x \sin y.$$

Substituting into (3.7.1),

$$\frac{dz}{dt} = (\cos x \cos y) \left(\frac{1}{2\sqrt{t}}\right) + (-\sin x \sin y) \left(\frac{-1}{t^2}\right) = \frac{\cos x \cos y}{2\sqrt{t}} + \frac{\sin x \sin y}{t^2}. \quad (3.7.3)$$

Putting  $t = \pi$  gives  $x(\pi) = \sqrt{\pi}$  and  $y(\pi) = \frac{1}{\pi}$ , so from (3.7.3),

$$\left.\frac{dz}{dt}\right|_{t=\pi} = \frac{\cos \sqrt{\pi} \cos \frac{1}{\pi}}{2\sqrt{\pi}} + \frac{\sin \sqrt{\pi} \sin \frac{1}{\pi}}{\pi^2}.$$

Numerically,  $\cos \sqrt{\pi} \approx -0.2010$ ,  $\cos \frac{1}{\pi} \approx 0.9497$ ,  $\sin \sqrt{\pi} \approx 0.9796$  and  $\sin \frac{1}{\pi} \approx 0.3130$ , giving

$$\left.\frac{dz}{dt}\right|_{t=\pi} \approx \frac{-0.1909}{3.5449} + \frac{0.3066}{9.8696} \approx -0.0539 + 0.0311 \approx -0.023.$$

**Note:**

The angles here are in radians and are not standard values, so the answer cannot be simplified in closed form. Leaving it as the exact expression and giving a decimal alongside is the appropriate finish.

- (c)  $w = z - \sin xy$ , where  $x = t$ ,  $y = \ln t$  and  $z = e^{t-1}$ ; at  $t = 1$ .

**Solution:** With three intermediate variables the chain rule has three terms,

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}. \quad (3.7.4)$$

The partial derivatives of  $w$  are

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x}[z - \sin xy] = -y \cos xy, \quad \frac{\partial w}{\partial y} = -x \cos xy, \quad \frac{\partial w}{\partial z} = 1,$$

and the derivatives of the intermediate variables are

$$\frac{dx}{dt} = \frac{d}{dt}(t) = 1, \quad \frac{dy}{dt} = \frac{d}{dt}(\ln t) = \frac{1}{t}, \quad \frac{dz}{dt} = \frac{d}{dt}e^{t-1} = e^{t-1}.$$

Substituting into (3.7.4),

$$\frac{dw}{dt} = -y \cos xy + (-x \cos xy) \left( \frac{1}{t} \right) + e^{t-1} = -y \cos xy - \frac{x \cos xy}{t} + e^{t-1}.$$

At  $t = 1$  we have  $x(1) = 1$ ,  $y(1) = \ln 1 = 0$  and  $z(1) = e^0 = 1$ , so

$$\left. \frac{dw}{dt} \right|_{t=1} = -(0) \cos 0 - \frac{(1) \cos 0}{1} + e^0 = 0 - 1 + 1 = 0.$$

(d)  $z = x^2 + y^2$ , where  $x = \cos t + \sin t$  and  $y = \cos t - \sin t$ ; at  $t = 0$ .

**Solution:** The partial derivatives of  $z$  are

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} [x^2 + y^2] = 2x, \quad \frac{\partial z}{\partial y} = 2y,$$

and the derivatives of the intermediate variables are

$$\begin{aligned} \frac{dx}{dt} &= \frac{d}{dt} [\cos t + \sin t] = -\sin t + \cos t = \cos t - \sin t, \\ \frac{dy}{dt} &= \frac{d}{dt} [\cos t - \sin t] = -\sin t - \cos t = -(\sin t + \cos t). \end{aligned}$$

Substituting into (3.7.1),

$$\frac{dz}{dt} = 2x (\cos t - \sin t) - 2y (\sin t + \cos t).$$

At  $t = 0$ ,  $x(0) = 0 + 1 = 1$  and  $y(0) = 1 - 0 = 1$ , so

$$\left. \frac{dz}{dt} \right|_{t=0} = 2(1)(1 - 0) - 2(1)(0 + 1) = 2 - 2 = 0.$$

(e)  $w = x e^{y/z}$ , where  $x = t^2$ ,  $y = 1 - t$  and  $z = 1 + 2t$ ; at  $t = 3$ .

**Solution:** The partial derivatives of  $w$  are

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x} [x e^{y/z}] = e^{y/z}, \quad \frac{\partial w}{\partial y} = \frac{x e^{y/z}}{z}, \quad \frac{\partial w}{\partial z} = -\frac{xy e^{y/z}}{z^2},$$

and the derivatives of the intermediate variables are

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = -1, \quad \frac{dz}{dt} = 2.$$

Substituting into (3.7.4),

$$\frac{dw}{dt} = e^{y/z}(2t) + \frac{x e^{y/z}}{z}(-1) - \frac{xy e^{y/z}}{z^2}(2) = e^{y/z} \left[ 2t - \frac{x}{z} - \frac{2xy}{z^2} \right].$$

At  $t = 3$ ,

$$x(3) = 3^2 = 9, \quad y(3) = 1 - 3 = -2, \quad z(3) = 1 + 2(3) = 7,$$

so  $y/z = -\frac{2}{7}$  and

$$\left. \frac{dw}{dt} \right|_{t=3} = e^{-2/7} \left[ 2(3) - \frac{9}{7} - \frac{2(9)(-2)}{7^2} \right] = e^{-2/7} \left[ 6 - \frac{9}{7} + \frac{36}{49} \right].$$

Putting the bracket over a common denominator of 49,

$$\left. \frac{dw}{dt} \right|_{t=3} = e^{-2/7} \left( \frac{294 - 63 + 36}{49} \right) = e^{-2/7} \left( \frac{267}{49} \right).$$

(f)  $w = \ln \sqrt{x^2 + y^2 + z^2}$ , where  $x = \sin t$ ,  $y = \cos t$  and  $z = \tan t$ ; at  $t = 0$ .

**Solution:** The partial derivatives of  $w$  are

$$\frac{\partial w}{\partial x} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \cdot \frac{\partial}{\partial x} \sqrt{x^2 + y^2 + z^2} = \frac{2x}{2(x^2 + y^2 + z^2)} = \frac{x}{x^2 + y^2 + z^2},$$

and by the same working,

$$\frac{\partial w}{\partial y} = \frac{y}{x^2 + y^2 + z^2}, \quad \frac{\partial w}{\partial z} = \frac{z}{x^2 + y^2 + z^2}.$$

The derivatives of the intermediate variables are

$$\frac{dx}{dt} = \cos t, \quad \frac{dy}{dt} = -\sin t, \quad \frac{dz}{dt} = \sec^2 t.$$

Substituting into (3.7.4),

$$\frac{dw}{dt} = \frac{x \cos t}{x^2 + y^2 + z^2} - \frac{y \sin t}{x^2 + y^2 + z^2} + \frac{z \sec^2 t}{x^2 + y^2 + z^2}.$$

At  $t = 0$ ,  $x(0) = \sin 0 = 0$ ,  $y(0) = \cos 0 = 1$  and  $z(0) = \tan 0 = 0$ , so  $x^2 + y^2 + z^2 = 1$  and

$$\left. \frac{dw}{dt} \right|_{t=0} = \frac{(0)(1)}{1} - \frac{(1)(0)}{1} + \frac{(0)(1)}{1} = 0.$$

**Note:**

Writing  $w = \frac{1}{2} \ln(x^2 + y^2 + z^2)$  before differentiating removes the square root and gives the same three derivatives in one line each.

**Problem 3.7.2.** Use the chain rule to find the indicated partial derivatives.

(a)  $z = (x - y)^2$ , where  $x = s^2t$  and  $y = st^2$ . Find  $\frac{\partial z}{\partial s}$  and  $\frac{\partial z}{\partial t}$  when  $s = 4$ ,  $t = 2$ .

**Solution:** By the chain rule,

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}, \quad (3.7.5)$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}. \quad (3.7.6)$$

The partial derivatives of  $z$  are

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} [(x - y)^2] = 2(x - y), \quad \frac{\partial z}{\partial y} = \frac{\partial}{\partial y} [(x - y)^2] = -2(x - y),$$

and those of the intermediate variables are

$$\frac{\partial x}{\partial s} = \frac{\partial}{\partial s} [s^2t] = 2st, \quad \frac{\partial x}{\partial t} = s^2, \quad \frac{\partial y}{\partial s} = t^2, \quad \frac{\partial y}{\partial t} = 2st.$$

Substituting into (3.7.5),

$$\frac{\partial z}{\partial s} = 2(x - y)(2st) - 2(x - y)(t^2) = 2(x - y)[2st - t^2], \quad (3.7.7)$$

and into (3.7.6),

$$\frac{\partial z}{\partial t} = 2(x - y)(s^2) - 2(x - y)(2st) = 2(x - y)[s^2 - 2st]. \quad (3.7.8)$$

At  $(s, t) = (4, 2)$ ,

$$x(4, 2) = 4^2(2) = 32, \quad y(4, 2) = 4(2)^2 = 16, \quad x - y = 16.$$

Putting these into (3.7.7) and (3.7.8),

$$\left. \frac{\partial z}{\partial s} \right|_{(4,2)} = 2(32 - 16)[2(4)(2) - 2^2] = 32(16 - 4) = 384,$$

$$\left. \frac{\partial z}{\partial t} \right|_{(4,2)} = 2(32 - 16)[4^2 - 2(4)(2)] = 32(16 - 16) = 0.$$

- (b)  $z = x^4 + y^2$ , where  $x = s + 2t - u$  and  $y = stu^2$ . Find  $\frac{\partial z}{\partial s}$ ,  $\frac{\partial z}{\partial t}$  and  $\frac{\partial z}{\partial u}$  when  $s = 4$ ,  $t = 2$ ,  $u = 1$ .

**Solution:** There are two intermediate variables and three independent ones, so each of the three derivatives is a sum of two terms:

$$\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}, & \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}, \\ \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}. \end{aligned}$$

The partial derivatives of  $z$  are

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} [x^4 + y^2] = 4x^3, \quad \frac{\partial z}{\partial y} = 2y,$$

and those of the intermediate variables are

$$\begin{aligned} \frac{\partial x}{\partial s} &= 1, & \frac{\partial x}{\partial t} &= 2, & \frac{\partial x}{\partial u} &= -1, \\ \frac{\partial y}{\partial s} &= tu^2, & \frac{\partial y}{\partial t} &= su^2, & \frac{\partial y}{\partial u} &= 2stu. \end{aligned}$$

Therefore

$$\begin{aligned} \frac{\partial z}{\partial s} &= 4x^3(1) + 2y(tu^2) = 4x^3 + 2y tu^2, \\ \frac{\partial z}{\partial t} &= 4x^3(2) + 2y(su^2) = 8x^3 + 2y su^2, \\ \frac{\partial z}{\partial u} &= 4x^3(-1) + 2y(2stu) = -4x^3 + 4y stu. \end{aligned}$$

At  $(s, t, u) = (4, 2, 1)$ ,

$$x(4, 2, 1) = 4 + 2(2) - 1 = 7, \quad y(4, 2, 1) = 4(2)(1)^2 = 8.$$

Substituting,

$$\left. \frac{\partial z}{\partial s} \right|_{(4,2,1)} = 4(7)^3 + 2(8)(2)(1)^2 = 1372 + 32 = 1404,$$

$$\left. \frac{\partial z}{\partial t} \right|_{(4,2,1)} = 8(7)^3 + 2(8)(4)(1)^2 = 2744 + 64 = 2808,$$

$$\left. \frac{\partial z}{\partial u} \right|_{(4,2,1)} = -4(7)^3 + 4(8)(4)(2)(1) = -1372 + 256 = -1116.$$

(c)  $z = \tan^{-1}(x^2 + y^2)$ , where  $x = s \ln t$  and  $y = te^s$ . Find  $\frac{\partial z}{\partial s}$  and  $\frac{\partial z}{\partial t}$  when  $s = 1$ ,  $t = 1$ .

**Solution:** The partial derivatives of  $z$  are

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} \tan^{-1}(x^2 + y^2) = \frac{2x}{1 + (x^2 + y^2)^2}, \quad \frac{\partial z}{\partial y} = \frac{2y}{1 + (x^2 + y^2)^2},$$

and those of the intermediate variables are

$$\frac{\partial x}{\partial s} = \ln t, \quad \frac{\partial x}{\partial t} = \frac{s}{t}, \quad \frac{\partial y}{\partial s} = te^s, \quad \frac{\partial y}{\partial t} = e^s.$$

Substituting into (3.7.5) and (3.7.6),

$$\frac{\partial z}{\partial s} = \frac{2x \ln t}{1 + (x^2 + y^2)^2} + \frac{2y te^s}{1 + (x^2 + y^2)^2}, \quad (3.7.9)$$

$$\frac{\partial z}{\partial t} = \frac{2x \frac{s}{t}}{1 + (x^2 + y^2)^2} + \frac{2y e^s}{1 + (x^2 + y^2)^2}. \quad (3.7.10)$$

At  $(s, t) = (1, 1)$ ,

$$x(1, 1) = (1) \ln 1 = 0, \quad y(1, 1) = (1)e^1 = e,$$

so that

$$x^2 + y^2 = 0 + e^2 = e^2 \quad \text{and} \quad 1 + (x^2 + y^2)^2 = 1 + e^4.$$

Putting these into (3.7.9) and (3.7.10),

$$\left. \frac{\partial z}{\partial s} \right|_{(1,1)} = \frac{2(0) \ln 1}{1 + e^4} + \frac{2(e)(1)e}{1 + e^4} = \frac{2e^2}{1 + e^4} \approx 0.266,$$

$$\left. \frac{\partial z}{\partial t} \right|_{(1,1)} = \frac{2(0) \left(\frac{1}{1}\right)}{1 + e^4} + \frac{2(e)e}{1 + e^4} = \frac{2e^2}{1 + e^4} \approx 0.266.$$

#### Note:

Both derivatives come out equal here because  $x(1, 1) = 0$  kills the first term in each, and the surviving second terms happen to agree. It is a coincidence of this point, not a general fact.

(d)  $z = e^r \cos \theta$ , where  $r = st$  and  $\theta = st^2$ . Find  $\frac{\partial z}{\partial s}$  and  $\frac{\partial z}{\partial t}$  when  $s = 1$ ,  $t = 1$ .

**Solution:** The partial derivatives of  $z$  are

$$\frac{\partial z}{\partial r} = \frac{\partial}{\partial r} [e^r \cos \theta] = e^r \cos \theta, \quad \frac{\partial z}{\partial \theta} = -e^r \sin \theta,$$

and those of the intermediate variables are

$$\frac{\partial r}{\partial s} = t, \quad \frac{\partial r}{\partial t} = s, \quad \frac{\partial \theta}{\partial s} = t^2, \quad \frac{\partial \theta}{\partial t} = 2st.$$

Therefore

$$\frac{\partial z}{\partial s} = t e^r \cos \theta - t^2 e^r \sin \theta, \quad (3.7.11)$$

$$\frac{\partial z}{\partial t} = s e^r \cos \theta - 2st e^r \sin \theta. \quad (3.7.12)$$

At  $(s, t) = (1, 1)$  we have  $r(1, 1) = 1$  and  $\theta(1, 1) = 1$ , so from (3.7.11) and (3.7.12),

$$\left. \frac{\partial z}{\partial s} \right|_{(1,1)} = e (\cos 1 - \sin 1) \approx -0.819,$$

$$\left. \frac{\partial z}{\partial t} \right|_{(1,1)} = e (\cos 1 - 2 \sin 1) \approx -3.106.$$

**Problem 3.7.3.** If  $z = f(x, y)$  where  $x = r \cos \theta$  and  $y = r \sin \theta$ , find  $\frac{\partial z}{\partial r}$  and  $\frac{\partial z}{\partial \theta}$ . Hence show that

$$(z_x)^2 + (z_y)^2 = (z_r)^2 + \frac{1}{r^2} (z_\theta)^2.$$

**Solution:** By the chain rule,

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r}, \quad \frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial \theta}.$$

The derivatives of the polar substitution are

$$\begin{aligned} \frac{\partial x}{\partial r} &= \frac{\partial}{\partial r} (r \cos \theta) = \cos \theta, & \frac{\partial x}{\partial \theta} &= -r \sin \theta, \\ \frac{\partial y}{\partial r} &= \sin \theta, & \frac{\partial y}{\partial \theta} &= r \cos \theta. \end{aligned}$$

Therefore

$$z_r = z_x \cos \theta + z_y \sin \theta, \quad (3.7.13)$$

$$z_\theta = z_x (-r \sin \theta) + z_y (r \cos \theta) = z_y r \cos \theta - z_x r \sin \theta. \quad (3.7.14)$$

Taking the right-hand side of the identity and substituting (3.7.13) and (3.7.14),

$$(z_r)^2 + \frac{1}{r^2} (z_\theta)^2 = (z_x \cos \theta + z_y \sin \theta)^2 + \frac{1}{r^2} (z_y r \cos \theta - z_x r \sin \theta)^2.$$

Expanding both squares,

$$\begin{aligned} &= z_x^2 \cos^2 \theta + z_y^2 \sin^2 \theta + 2z_x z_y \cos \theta \sin \theta \\ &\quad + \frac{1}{r^2} [z_y^2 r^2 \cos^2 \theta + z_x^2 r^2 \sin^2 \theta - 2z_x z_y r^2 \sin \theta \cos \theta]. \end{aligned}$$

The factor  $r^2$  cancels in the bracket, and the two cross terms cancel each other, leaving

$$= z_x^2 (\cos^2 \theta + \sin^2 \theta) + z_y^2 (\sin^2 \theta + \cos^2 \theta) = z_x^2 + z_y^2,$$

which is the left-hand side. Hence the identity is proved.

**Note:**

The quantity  $z_x^2 + z_y^2$  is the squared length of the gradient. The identity says that this length does not depend on whether it is measured in Cartesian or polar coordinates the factor  $\frac{1}{r^2}$  is exactly what is needed to compensate for  $\theta$  being an angle rather than a length.

**Problem 3.7.4.** If  $u = f(2x - 3y, 3y - 4z, 4z - 2x)$ , prove that  $6u_x + 4u_y + 3u_z = 0$ .

**Solution:** Write  $u = f(s(x, y), t(y, z), v(z, x))$  with

$$s(x, y) = 2x - 3y, \quad t(y, z) = 3y - 4z, \quad v(z, x) = 4z - 2x.$$

Each of  $s$ ,  $t$  and  $v$  involves only two of the three variables, so each of  $u_x$ ,  $u_y$ ,  $u_z$  has only two terms rather than three:

$$u_x = \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial v} \frac{\partial v}{\partial x}, \quad u_y = \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y}, \quad u_z = \frac{\partial u}{\partial t} \frac{\partial t}{\partial z} + \frac{\partial u}{\partial v} \frac{\partial v}{\partial z}. \quad (3.7.15)$$

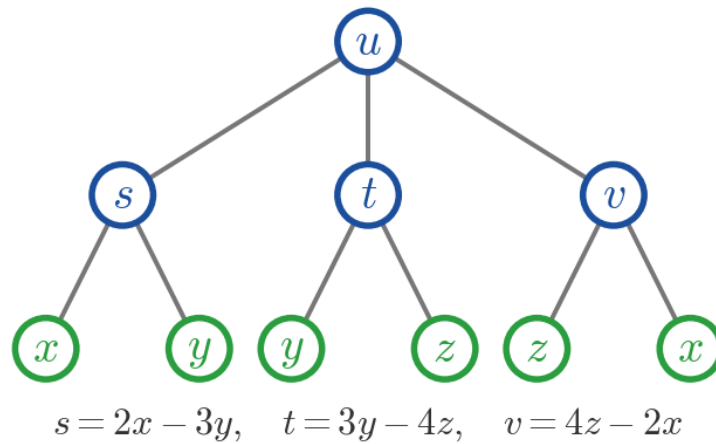


Figure 3.7.2: Each intermediate variable depends on only two of  $x$ ,  $y$ ,  $z$ . The branches that are absent are what removes a term from each derivative.

The required derivatives of the intermediate variables are

$$\begin{aligned} \frac{\partial s}{\partial x} &= 2, & \frac{\partial s}{\partial y} &= -3, & \frac{\partial t}{\partial y} &= 3, \\ \frac{\partial t}{\partial z} &= -4, & \frac{\partial v}{\partial z} &= 4, & \frac{\partial v}{\partial x} &= -2. \end{aligned}$$

Substituting into (3.7.15),

$$u_x = 2u_s - 2u_v, \quad u_y = -3u_s + 3u_t, \quad u_z = -4u_t + 4u_v.$$

Taking the left-hand side of the required identity,

$$\begin{aligned} 6u_x + 4u_y + 3u_z &= 6(2u_s - 2u_v) + 4(-3u_s + 3u_t) + 3(-4u_t + 4u_v) \\ &= 12u_s - 12u_v - 12u_s + 12u_t - 12u_t + 12u_v \\ &= 0, \end{aligned}$$

which is the right-hand side. Hence  $6u_x + 4u_y + 3u_z = 0$ .

**Note:**

The coefficients 6, 4, 3 are not arbitrary: they are chosen so that each of  $u_s$ ,  $u_t$ ,  $u_v$  cancels. Nothing about the form of  $f$  was used, so the identity holds for every differentiable  $f$ .

## 3.8 Maximum and Minimum Values

### 3.8.1 The one variable picture first

The two variable definitions are easier to read against the familiar one variable ones. For  $y = f(x)$ :

- $f(c)$  is a *local maximum* if  $f(x) \leq f(c)$  for all  $x$  near  $c$ , and a *local minimum* if  $f(x) \geq f(c)$  for all  $x$  near  $c$ .
- $f(c)$  is the *global* (or *absolute*) *maximum* if  $f(x) \leq f(c)$  for all  $x$  in the whole domain of  $f$ , and the *global minimum* if  $f(x) \geq f(c)$  throughout the domain.
- If  $f'(c) = 0$  and  $f''(c) > 0$ , then  $f(c)$  is a local minimum; if  $f'(c) = 0$  and  $f''(c) < 0$ , then  $f(c)$  is a local maximum.

The distinction is between *near  $c$*  and *everywhere*. A function may have several local maxima, each the highest point of its own neighbourhood, while only one of them is the highest point overall. Everything below is this same distinction transferred to a surface, with the interval around  $c$  replaced by a disk around  $(a, b)$ .

### 3.8.2 Local and absolute extrema of a surface

For a function of two variables  $f(x, y)$ :

- If  $f(x, y) \leq f(a, b)$  when  $(x, y)$  is near  $(a, b)$ , then  $f$  has a *local maximum value* at  $(a, b)$ , and the number  $f(a, b)$  is the local maximum value.
- If  $f(x, y) \geq f(a, b)$  when  $(x, y)$  is near  $(a, b)$ , then  $f$  has a *local minimum value* at  $(a, b)$ , and  $f(a, b)$  is the local minimum value.
- If  $f(x, y) \leq f(a, b)$  for *all*  $(x, y)$  in the domain of  $f$ , then  $f(a, b)$  is the *absolute* (or *global*) *maximum*; if  $f(x, y) \geq f(a, b)$  throughout the domain,  $f(a, b)$  is the *absolute minimum*.

**Note:**

“Near  $(a, b)$ ” means: there is a disk centred at  $(a, b)$  throughout which the inequality holds. The disk plays the part that an interval around  $c$  plays in one variable, and it must be a genuine two-dimensional disk — checking that  $f(a, b)$  is largest along the two coordinate directions is not enough, as the saddle points below show.

**Note:**

Every absolute extremum that occurs at an interior point is also a local one, so the tests below find absolute extrema too — but they cannot certify that what they have found is absolute. That needs a separate comparison of all the local values against

each other and against the behaviour on the boundary of the domain.

**Theorem 3.8.1: Necessary condition for a local extremum**

If  $f$  has a local maximum or minimum at  $(a, b)$  and the first order partial derivatives of  $f$  exist there, then

$$f_x(a, b) = 0 \quad \text{and} \quad f_y(a, b) = 0.$$

**Definition 3.8.2: Critical point**

A point  $(a, b)$  is called a *critical point* (or *stationary point*) of  $f$  if  $f_x(a, b) = 0$  and  $f_y(a, b) = 0$ , or if one of these partial derivatives does not exist.

**Note:**

Theorem 3.8.1 runs one way only. Every extremum is a critical point, but not every critical point is an extremum a saddle point satisfies both equations too. The next test is what separates the cases.

### 3.8.3 The second derivatives test

**Definition 3.8.3: Saddle point**

A critical point  $(a, b)$  of  $f$  is said to be a *saddle point* if  $D(a, b) < 0$ , where  $D$  is the discriminant defined in the theorem below.

Equivalently and more descriptively: a saddle point is a critical point that is neither a local maximum nor a local minimum. The tangent plane there is horizontal, but the surface rises in some directions away from the point and falls in others, so no disk around  $(a, b)$  has  $f(a, b)$  as its largest or its smallest value. The name comes from the shape of a horse's saddle, which curves up front-to-back and down side-to-side.

**Note:**

This is where the one variable intuition runs out. On a curve, a stationary point that is not a maximum or a minimum has to be a point of inflection, which is a degenerate case. On a surface, saddle points are entirely typical — the worked problems below produce more of them than extrema. It follows that  $f_x = f_y = 0$  is a much weaker piece of information in two variables than  $f' = 0$  is in one, which is exactly why the second derivatives test is needed.

**Theorem 3.8.4: Second derivatives test**

Suppose the second partial derivatives of  $f$  are continuous on a disk with centre  $(a, b)$ , and suppose that  $f_x(a, b) = 0$  and  $f_y(a, b) = 0$ , so that  $(a, b)$  is a critical point of  $f$ . Let

$$D = D(a, b) = f_{xx}(a, b) f_{yy}(a, b) - [f_{xy}(a, b)]^2.$$

- (i) If  $D > 0$  and  $f_{xx}(a, b) > 0$ , then  $f(a, b)$  is a local minimum.
- (ii) If  $D > 0$  and  $f_{xx}(a, b) < 0$ , then  $f(a, b)$  is a local maximum.

(iii) If  $D < 0$ , then  $(a, b)$  is a saddle point of  $f$ .

If  $D = 0$ , the test gives no information:  $f$  could have a local maximum or a local minimum at  $(a, b)$ , or  $(a, b)$  could be a saddle point of  $f$ .

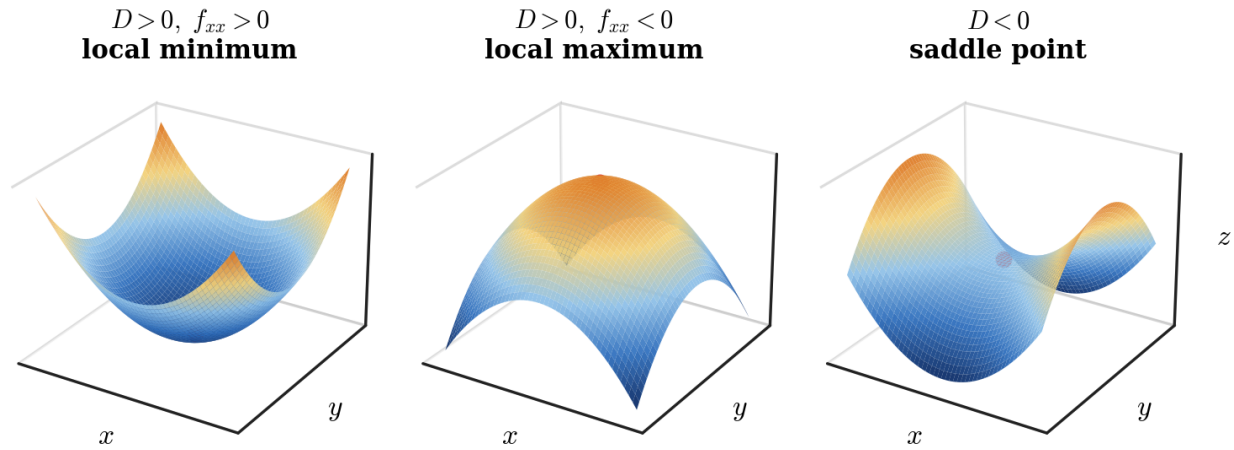


Figure 3.8.1: The three outcomes of the test at a critical point. When  $D > 0$  the surface curves the same way in every direction, and the sign of  $f_{xx}$  says which way; when  $D < 0$  it curves upward in one direction and downward in another.

### 3.8.4 Worked problems

**Problem 3.8.1.** Find the maximum and minimum values of  $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$ .

**Solution:** Differentiate  $f$  partially with respect to  $x$  and  $y$ :

$$f_x = \frac{\partial}{\partial x} [x^4 + y^4 - 2x^2 + 4xy - 2y^2] = 4x^3 - 4x + 4y,$$

$$f_y = \frac{\partial}{\partial y} [x^4 + y^4 - 2x^2 + 4xy - 2y^2] = 4y^3 + 4x - 4y.$$

**To find the critical points.** Set  $f_x = 0$  and  $f_y = 0$ :

$$4x^3 - 4x + 4y = 0, \quad (3.8.1)$$

$$4y^3 + 4x - 4y = 0. \quad (3.8.2)$$

Adding (3.8.1) and (3.8.2), the terms  $-4x + 4x$  and  $4y - 4y$  cancel, leaving

$$4x^3 + 4y^3 = 0 \implies x^3 + y^3 = 0 \implies (x + y)(x^2 - xy + y^2) = 0.$$

The second factor vanishes only at the origin, because  $x^2 - xy + y^2 = \left(x - \frac{y}{2}\right)^2 + \frac{3y^2}{4}$  is a sum of squares. So the useful factor is the first:

$$x + y = 0 \implies x = -y. \quad (3.8.3)$$

Putting (3.8.3) into (3.8.1),

$$4x^3 - 4x + 4(-x) = 0 \implies 4x^3 - 8x = 0 \implies 4x(x^2 - 2) = 0,$$

so that

$$x = 0 \quad \text{or} \quad x^2 - 2 = 0 \implies x = \pm\sqrt{2}.$$

Since  $y = -x$ , the critical points are

$$(0, 0), \quad (\sqrt{2}, -\sqrt{2}), \quad (-\sqrt{2}, \sqrt{2}).$$

The value of  $D$  at each is computed below.

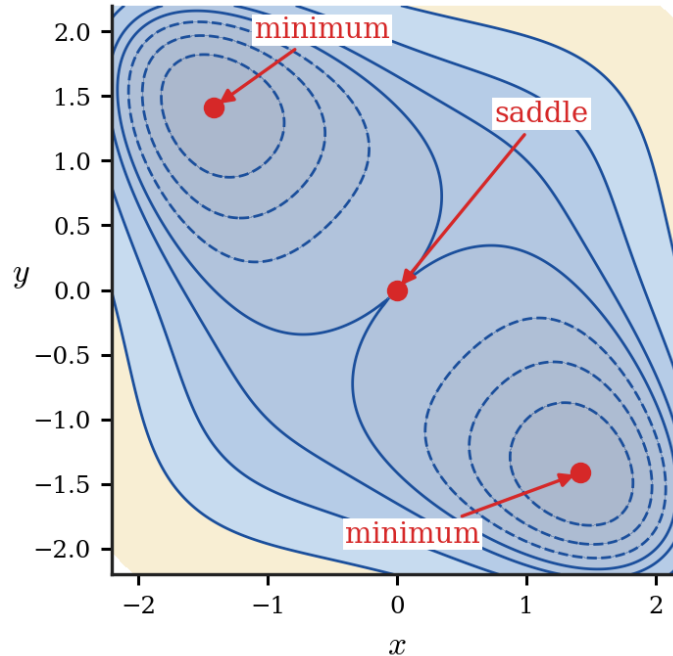


Figure 3.8.2: Level curves of  $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$  with its three critical points. The two off-origin points sit at the centres of closed curves; the origin lies where two level curves cross.

**Computing  $D$ .** The second order derivatives are

$$f_{xx} = \frac{\partial}{\partial x} (f_x) = \frac{\partial}{\partial x} [4x^3 - 4x + 4y] = 12x^2 - 4,$$

$$f_{yy} = \frac{\partial}{\partial y} (f_y) = \frac{\partial}{\partial y} [4y^3 + 4x - 4y] = 12y^2 - 4, \quad f_{xy} = \frac{\partial}{\partial y} [4x^3 - 4x + 4y] = 4.$$

Therefore

$$D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = (12x^2 - 4)(12y^2 - 4) - 16 = 144x^2y^2 - 48x^2 - 48y^2 + 16 - 16,$$

that is

$$D(x, y) = 48(3x^2y^2 - x^2 - y^2).$$

| Critical point          | $f(x, y)$ | $f_{xx} = 12x^2 - 4$ | $D$       | Conclusion    |
|-------------------------|-----------|----------------------|-----------|---------------|
| $(0, 0)$                | 0         | -4                   | 0         | No conclusion |
| $(\sqrt{2}, -\sqrt{2})$ | -8        | $20 > 0$             | $384 > 0$ | Local minimum |
| $(-\sqrt{2}, \sqrt{2})$ | -8        | $20 > 0$             | $384 > 0$ | Local minimum |

The two function values are

$$f(\sqrt{2}, -\sqrt{2}) = 4 + 4 - 4 - 8 - 4 = -8, \quad f(-\sqrt{2}, \sqrt{2}) = 4 + 4 - 4 - 8 - 4 = -8.$$

**Note:**

At the origin  $D = 0$ , so the second derivatives test says nothing. Direct inspection settles it: along  $y = x$ ,  $f(x, x) = 2x^4 > 0$  for  $x \neq 0$ , while along  $y = -x$ ,  $f(x, -x) = 2x^4 - 8x^2 < 0$  for small  $x \neq 0$ . Since  $f(0, 0) = 0$  and  $f$  takes both signs arbitrarily close to the origin,  $(0, 0)$  is a saddle point a fact the test could not detect.

**Problem 3.8.2.** Find the extreme values of the function  $f(x, y) = x^3 + y^3 - 3x - 12y + 20$ .

**Solution:** The first order partial derivatives are

$$f_x = \frac{\partial f}{\partial x} = 3x^2 - 3, \quad f_y = \frac{\partial f}{\partial y} = 3y^2 - 12.$$

**To find the critical points.** Set  $f_x = 0$  and  $f_y = 0$ :

$$3x^2 - 3 = 0 \implies x^2 = 1 \implies x = \pm 1,$$

$$3y^2 - 12 = 0 \implies y^2 = 4 \implies y = \pm 2.$$

The two conditions are independent, so every combination occurs and the critical points are

$$(1, 2), \quad (1, -2), \quad (-1, 2), \quad (-1, -2).$$

**Computing  $D$ .** The second order derivatives are

$$f_{xx} = \frac{\partial}{\partial x} [3x^2 - 3] = 6x, \quad f_{yy} = \frac{\partial}{\partial y} [3y^2 - 12] = 6y, \quad f_{xy} = \frac{\partial}{\partial y} [3x^2 - 3] = 0,$$

so that

$$D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = (6x)(6y) - 0^2 = 36xy.$$

| Critical point | $f(x, y)$ | $f_{xx} = 6x$ | $D = 36xy$ | Conclusion    |
|----------------|-----------|---------------|------------|---------------|
| $(1, 2)$       | 2         | $6 > 0$       | $72 > 0$   | Local minimum |
| $(1, -2)$      | 34        | $6 > 0$       | $-72 < 0$  | Saddle point  |
| $(-1, 2)$      | 6         | $-6 < 0$      | $-72 < 0$  | Saddle point  |
| $(-1, -2)$     | 38        | $-6 < 0$      | $72 > 0$   | Local maximum |

The function values are

$$\begin{aligned} f(1, 2) &= 1 + 8 - 3 - 24 + 20 = 2, & f(1, -2) &= 1 - 8 - 3 + 24 + 20 = 34, \\ f(-1, 2) &= -1 + 8 + 3 - 24 + 20 = 6, & f(-1, -2) &= -1 - 8 + 3 + 24 + 20 = 38. \end{aligned}$$

So the local minimum value is 2 at  $(1, 2)$  and the local maximum value is 38 at  $(-1, -2)$ ; the other two critical points are saddle points.

**Problem 3.8.3.** Find the extreme values of the function  $f(x, y) = x^3 + y^3 - 63(x + y) + 12xy$ .

**Solution:** The first order partial derivatives are

$$f_x = \frac{\partial f}{\partial x} = 3x^2 - 63 + 12y, \quad f_y = \frac{\partial f}{\partial y} = 3y^2 - 63 + 12x.$$

**To find the critical points.** Set  $f_x = 0$  and  $f_y = 0$ :

$$3x^2 + 12y - 63 = 0, \quad (3.8.4)$$

$$3y^2 + 12x - 63 = 0. \quad (3.8.5)$$

Subtracting (3.8.5) from (3.8.4), the constants cancel:

$$3(x^2 - y^2) + 12(y - x) = 0 \implies 3(x - y)(x + y) - 12(x - y) = 0,$$

and factoring out  $3(x - y)$ ,

$$3(x - y)[(x + y) - 4] = 0.$$

This gives two cases.

**Case 1:**  $x - y = 0$ , that is  $x = y$ . Dividing (3.8.4) by 3 gives  $x^2 + 4y - 21 = 0$ , and substituting  $y = x$ ,

$$x^2 + 4x - 21 = 0 \implies x = \frac{-4 \pm \sqrt{16 + 84}}{2} = \frac{-4 \pm 10}{2} = 3 \text{ or } -7.$$

This yields the critical points  $(3, 3)$  and  $(-7, -7)$ .

**Case 2:**  $x + y - 4 = 0$ , that is  $y = 4 - x$ . Substituting into  $x^2 + 4y - 21 = 0$ ,

$$x^2 + 4(4 - x) - 21 = 0 \implies x^2 - 4x - 5 = 0 \implies x = \frac{4 \pm \sqrt{16 + 20}}{2} = \frac{4 \pm 6}{2} = 5 \text{ or } -1.$$

With  $y = 4 - x$  this yields  $(5, -1)$  and  $(-1, 5)$ .

So the critical points are  $(3, 3)$ ,  $(-7, -7)$ ,  $(-1, 5)$  and  $(5, -1)$ .

**Computing  $D$ .** The second order derivatives are

$$f_{xx} = \frac{\partial}{\partial x} [3x^2 - 63 + 12y] = 6x, \quad f_{yy} = 6y, \quad f_{xy} = \frac{\partial}{\partial y} [3x^2 - 63 + 12y] = 12,$$

so that

$$D(x, y) = (6x)(6y) - 12^2 = 36xy - 144 = 36(xy - 4).$$

| Critical point | $f_{xx} = 6x$ | $D = 36(xy - 4)$ | $f(x, y)$ | Conclusion    |
|----------------|---------------|------------------|-----------|---------------|
| $(3, 3)$       | $18 > 0$      | $180 > 0$        | $-216$    | Local minimum |
| $(-7, -7)$     | $-42 < 0$     | $1620 > 0$       | $784$     | Local maximum |
| $(-1, 5)$      | $-6 < 0$      | $-324 < 0$       | $-188$    | Saddle point  |
| $(5, -1)$      | $30 > 0$      | $-324 < 0$       | $-188$    | Saddle point  |

The function values are

$$\begin{aligned} f(3, 3) &= 27 + 27 - 63(6) + 12(9) = 54 - 378 + 108 = -216, \\ f(-7, -7) &= -343 - 343 - 63(-14) + 12(49) = -686 + 882 + 588 = 784, \\ f(-1, 5) &= -1 + 125 - 63(4) + 12(-5) = -1 + 125 - 252 - 60 = -188, \\ f(5, -1) &= 125 - 1 - 63(4) + 12(-5) = -188. \end{aligned}$$

**Note:**

The two saddle points give the same value because  $f$  is symmetric under interchanging  $x$  and  $y$ , and  $(5, -1)$ ,  $(-1, 5)$  are exchanged by that swap. The same symmetry explains why the other two critical points lie on the line  $y = x$ .

**Problem 3.8.4.** Obtain the local maximum and local minimum of the function  $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ .

**Solution:** Differentiating partially with respect to  $x$  and to  $y$ ,

$$f_x = 3x^2 + 3y^2 - 30x + 72, \quad f_y = 6xy - 30y.$$

For critical points set both to zero:

$$3x^2 + 3y^2 - 30x + 72 = 0, \quad (3.8.6)$$

$$6xy - 30y = 0. \quad (3.8.7)$$

Equation (3.8.7) factorises,

$$6y(x - 5) = 0 \quad \implies \quad y = 0 \quad \text{or} \quad x = 5,$$

and the two cases are treated separately.

**Case  $y = 0$ .** Substituting into (3.8.6),

$$3x^2 - 30x + 72 = 0 \implies x^2 - 10x + 24 = 0 \implies (x - 4)(x - 6) = 0,$$

so  $x = 4$  or  $x = 6$ , giving the critical points  $(4, 0)$  and  $(6, 0)$ .

**Case  $x = 5$ .** Substituting into (3.8.6),

$$3(25) + 3y^2 - 150 + 72 = 0 \implies 3y^2 - 3 = 0 \implies y^2 = 1,$$

so  $y = \pm 1$ , giving the critical points  $(5, 1)$  and  $(5, -1)$ .

There are therefore four critical points in all. The second order partial derivatives are

$$f_{xx} = 6x - 30, \quad f_{yy} = 6x - 30, \quad f_{xy} = 6y,$$

so the discriminant is

$$D = f_{xx}f_{yy} - [f_{xy}]^2 = (6x - 30)^2 - 36y^2.$$

Evaluating at each critical point:

| Point     | $D$ | $f_{xx}$ | $f(x, y)$ | Conclusion    |
|-----------|-----|----------|-----------|---------------|
| $(4, 0)$  | 36  | $-6 < 0$ | 112       | Local maximum |
| $(6, 0)$  | 36  | $6 > 0$  | 108       | Local minimum |
| $(5, 1)$  | -36 | 0        | 110       | Saddle point  |
| $(5, -1)$ | -36 | 0        | 110       | Saddle point  |

Hence the local maximum value is  $f(4, 0) = 112$ , the local minimum value is  $f(6, 0) = 108$ , and the remaining two critical points are saddle points.

**Note:**

At the two saddle points  $f_{xx} = 6(5) - 30 = 0$ , so the surface has no curvature at all in the  $x$ -direction there. That is not an obstacle: once  $D < 0$  the classification is settled and the sign of  $f_{xx}$  is not consulted. It is only when  $D > 0$  that  $f_{xx}$  is needed, to distinguish a maximum from a minimum.

**Note:**

The two saddle values sit between the local maximum 112 and the local minimum 108, at 110. That is typical rather than accidental: a saddle point is a pass between higher and lower ground. Note also the symmetry —  $f$  is even in  $y$ , so critical points come in pairs  $(x, \pm y)$  with equal function values, which is why  $(5, 1)$  and  $(5, -1)$  behave identically and why the points with  $y = 0$  are unpaired.

### 3.9 Lagrange Multipliers

**Theorem 3.9.1: Method of Lagrange multipliers**

To find the maximum and minimum values of  $f(x, y, z)$  subject to the constraint  $g(x, y, z) = k$ , assuming that these extreme values exist and that  $\nabla g \neq 0$  on the surface  $g(x, y, z) = k$ :

1. Find all values of  $x, y, z$  and  $\lambda$  such that

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) \quad \text{and} \quad g(x, y, z) = k.$$

2. Evaluate  $f$  at all points  $(x, y, z)$  that result from step 1. The largest of these values is the maximum value of  $f$ ; the smallest is the minimum value of  $f$ .

Writing the vector equation  $\nabla f = \lambda \nabla g$  in terms of components, the equations in step 1 become

$$f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad f_z = \lambda g_z, \quad g(x, y, z) = k. \quad (3.9.1)$$

This is a system of four equations in the four unknowns  $x, y, z$  and  $\lambda$ , and we must find all possible solutions, although the explicit values of  $\lambda$  are not needed for the conclusion of the method.

**Note:**

The multiplier  $\lambda$  is a device, not part of the answer. In practice it is usually eliminated by comparing the first three equations of (3.9.1) rather than solved for.

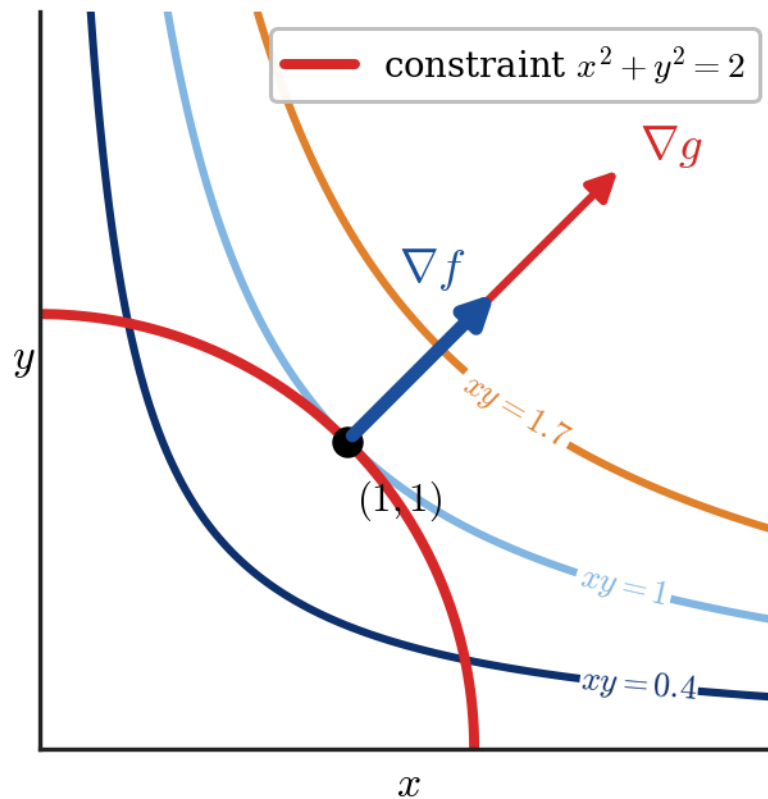


Figure 3.9.1: Why the method works, shown in two variables: maximising  $f = xy$  on the circle  $g = x^2 + y^2 = 2$ . Moving along the circle,  $f$  increases until its level curve just touches the circle, at  $(1, 1)$ . There the two curves share a tangent, so their normals  $\nabla f = (1, 1)$  and  $\nabla g = (2, 2)$  are parallel:  $\nabla f = \lambda \nabla g$  with  $\lambda = \frac{1}{2}$ .

### 3.9.1 Worked problems

**Problem 3.9.1.** A rectangular box open at the top is to have a volume of 32 units. Determine the dimensions of the box that require the least material for its construction.

**Solution:** Let  $x$ ,  $y$ ,  $z$  denote the length, breadth and height of the box. With the top face missing, the surface consists of the base and four sides, so the surface area is

$$f(x, y, z) = 2xz + 2yz + xy.$$

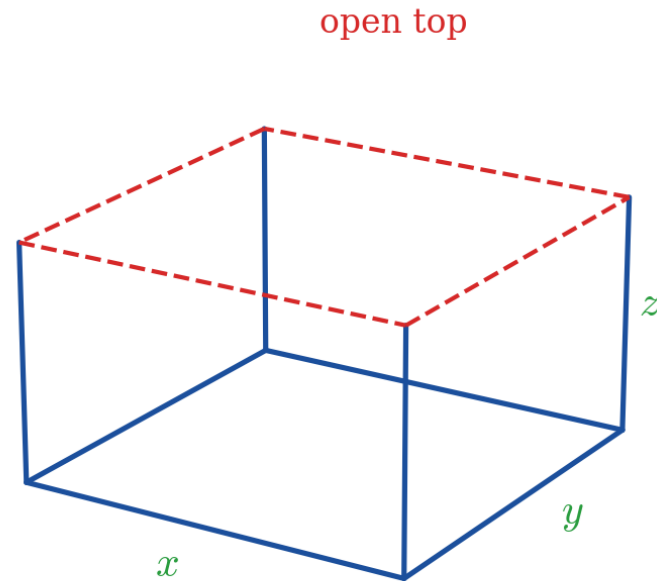


Figure 3.9.2: The box has a base  $xy$ , two sides  $xz$  and two sides  $yz$ . The missing top is why the area is  $2xz + 2yz + xy$  rather than  $2(xy + yz + zx)$ .

We want to minimise  $f$  subject to the condition

$$g(x, y, z) = xyz = 32.$$

Applying (3.9.1) with  $g_x = yz$ ,  $g_y = xz$  and  $g_z = xy$ ,

$$f_x = \lambda g_x : \quad 2z + y = \lambda yz, \quad (3.9.2)$$

$$f_y = \lambda g_y : \quad 2z + x = \lambda xz, \quad (3.9.3)$$

$$f_z = \lambda g_z : \quad 2x + 2y = \lambda xy, \quad (3.9.4)$$

together with  $xyz = 32$ .

To eliminate  $\lambda$ , multiply each equation by the variable missing from its right-hand side, so that all three right-hand sides become  $\lambda xyz$ :

$$(3.9.2) \times x : \quad 2xz + xy = \lambda xyz,$$

$$(3.9.3) \times y : \quad 2yz + xy = \lambda xyz,$$

$$(3.9.4) \times z : \quad 2xz + 2yz = \lambda xyz.$$

Comparing the first two,

$$2xz + xy = 2yz + xy \implies 2xz = 2yz \implies x = y.$$

Comparing the second and third,

$$2yz + xy = 2xz + 2yz \implies xy = 2xz \implies y = 2z.$$

Hence

$$x = y = 2z. \quad (3.9.5)$$

Putting (3.9.5) into the constraint  $xyz = 32$ ,

$$(2z)(2z)(z) = 32 \implies 4z^3 = 32 \implies z^3 = 8 \implies \boxed{z = 2}$$

and therefore  $x = y = 2z = 2(2) = 4$ .

The dimensions that minimise the surface area are  $x = 4$  units,  $y = 4$  units and  $z = 2$  units. The minimal surface area is therefore

$$f(4, 4, 2) = 2(4)(2) + 2(4)(2) + (4)(4) = 16 + 16 + 16 = 48 \text{ units}^2.$$

**Problem 3.9.2.** Find the maximum volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ .

**Solution:** Take the corner of the box in the first octant to be  $(x, y, z)$ . By symmetry the box is centred at the origin, so its edges have lengths  $2x$ ,  $2y$ ,  $2z$  and its volume is

$$V = (2x)(2y)(2z) = 8xyz.$$

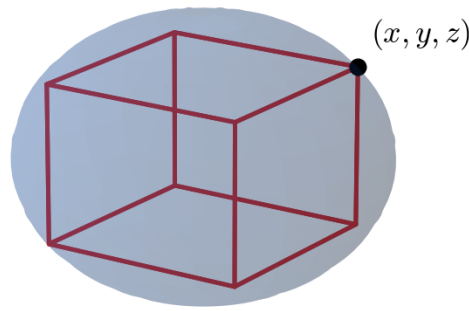


Figure 3.9.3: The inscribed box is centred at the origin with one corner  $(x, y, z)$  on the ellipsoid, so its edges measure  $2x$ ,  $2y$  and  $2z$ .

We want to maximise  $V$  subject to the condition that  $(x, y, z)$  lies on the ellipsoid. Let

$$f(x, y, z) = 8xyz, \quad g(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1. \quad (3.9.6)$$

The partial derivatives are

$$f_x = 8yz, \quad f_y = 8xz, \quad f_z = 8xy, \quad g_x = \frac{2x}{a^2}, \quad g_y = \frac{2y}{b^2}, \quad g_z = \frac{2z}{c^2},$$

so the Lagrange equations (3.9.1) read

$$8yz = \lambda \frac{2x}{a^2}, \quad 8xz = \lambda \frac{2y}{b^2}, \quad 8xy = \lambda \frac{2z}{c^2}.$$

Multiplying the first by  $x$ , the second by  $y$  and the third by  $z$  makes every left-hand side equal to  $8xyz$ :

$$4xyz = \lambda \frac{x^2}{a^2}, \quad (3.9.7)$$

$$4xyz = \lambda \frac{y^2}{b^2}, \quad (3.9.8)$$

$$4xyz = \lambda \frac{z^2}{c^2}. \quad (3.9.9)$$

Comparing (3.9.7), (3.9.8) and (3.9.9),

$$\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2}.$$

Substituting into the constraint in (3.9.6),

$$\frac{x^2}{a^2} + \frac{x^2}{a^2} + \frac{x^2}{a^2} = 1 \implies \frac{3x^2}{a^2} = 1 \implies \frac{x^2}{a^2} = \frac{1}{3} \implies x = \frac{a}{\sqrt{3}},$$

the positive root being taken because a length cannot be negative. In the same way  $y = \frac{b}{\sqrt{3}}$  and  $z = \frac{c}{\sqrt{3}}$ .

The maximum volume is therefore

$$V = 8xyz = 8 \cdot \frac{a}{\sqrt{3}} \cdot \frac{b}{\sqrt{3}} \cdot \frac{c}{\sqrt{3}} = \frac{8abc}{3\sqrt{3}}.$$

**Problem 3.9.3.** Find the shortest and longest distance from the point  $(1, 2, -1)$  to the sphere  $x^2 + y^2 + z^2 = 24$ .

**Solution:** Let  $(x, y, z)$  be a point on the given sphere. The distance between  $(x, y, z)$  and  $(1, 2, -1)$  is

$$d = \sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}.$$

Since the square root is increasing, minimising and maximising  $d$  is the same as minimising and maximising  $d^2$ , which avoids differentiating the root. So put

$$f(x, y, z) = d^2 = (x-1)^2 + (y-2)^2 + (z+1)^2,$$

subject to the constraint

$$g(x, y, z) = x^2 + y^2 + z^2 = 24. \quad (3.9.10)$$

The partial derivatives are

$$f_x = 2(x-1), \quad f_y = 2(y-2), \quad f_z = 2(z+1), \quad g_x = 2x, \quad g_y = 2y, \quad g_z = 2z.$$

The Lagrange equations give

$$2(x-1) = \lambda(2x) \implies 2x - 2 - 2\lambda x = 0 \implies 2x(1-\lambda) = 2 \implies x(1-\lambda) = 1,$$

and by the same working

$$y(1-\lambda) = 2, \quad z(1-\lambda) = -1.$$

Hence

$$x = \frac{1}{1-\lambda}, \quad y = \frac{2}{1-\lambda}, \quad z = \frac{-1}{1-\lambda}.$$

Substituting into (3.9.10),

$$\left(\frac{1}{1-\lambda}\right)^2 + \left(\frac{2}{1-\lambda}\right)^2 + \left(\frac{-1}{1-\lambda}\right)^2 = 24 \implies \frac{1+4+1}{(1-\lambda)^2} = 24,$$

so that

$$(1-\lambda)^2 = \frac{6}{24} = \frac{1}{4} \implies 1-\lambda = \pm \frac{1}{2} \implies \lambda = \frac{1}{2} \text{ or } \frac{3}{2}.$$

The two resulting points are found as follows. For  $\lambda = \frac{1}{2}$  we have  $1-\lambda = \frac{1}{2}$ , so

$$x = \frac{1}{\frac{1}{2}} = 2, \quad y = \frac{2}{\frac{1}{2}} = 4, \quad z = \frac{-1}{\frac{1}{2}} = -2,$$

giving the point  $P_1 = (2, 4, -2)$ . For  $\lambda = \frac{3}{2}$  we have  $1-\lambda = -\frac{1}{2}$ , so

$$x = \frac{1}{-\frac{1}{2}} = -2, \quad y = \frac{2}{-\frac{1}{2}} = -4, \quad z = \frac{-1}{-\frac{1}{2}} = 2,$$

giving  $P_2 = (-2, -4, 2)$ .

Finding the distance of each point from  $(1, 2, -1)$ ,

$$\sqrt{f(P_1)} = \sqrt{(2-1)^2 + (4-2)^2 + (-2+1)^2} = \sqrt{1+4+1} = \sqrt{6} \approx 2.449,$$

$$\sqrt{f(P_2)} = \sqrt{(-2-1)^2 + (-4-2)^2 + (2+1)^2} = \sqrt{9+36+9} = \sqrt{54} = 3\sqrt{6} \approx 7.348.$$

Therefore  $P_1(2, 4, -2)$  is the closest point and  $P_2(-2, -4, 2)$  is the farthest point on the sphere  $x^2 + y^2 + z^2 = 24$  from the point  $(1, 2, -1)$ , at distances  $\sqrt{6}$  and  $3\sqrt{6}$  respectively.

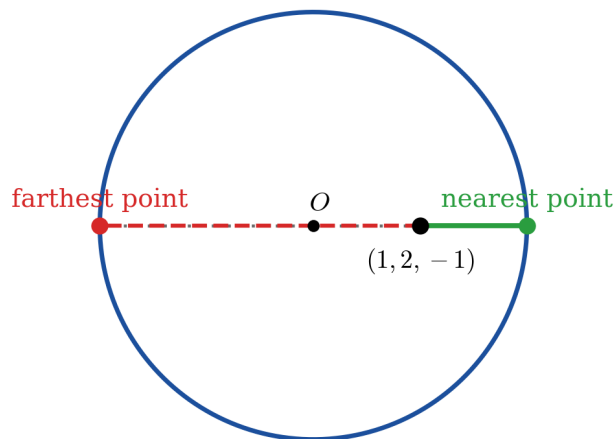


Figure 3.9.4: The plane section through the centre and the given point. Both extreme points lie on the line joining the centre to  $(1, 2, -1)$ , which is what the three Lagrange equations force.

**Note:**

Since  $1^2 + 2^2 + (-1)^2 = 6 < 24$ , the point  $(1, 2, -1)$  lies *inside* the sphere. The two answers are the near and far intersections of the ray through the centre with the surface,

and they differ by the diameter:  $3\sqrt{6} - \sqrt{6} = 2\sqrt{6} = 2\sqrt{24}/2$ , as expected.

**Problem 3.9.4.** The temperature  $T$  at any point  $(x, y, z)$  in space is  $T = kxyz^2$ , where  $k$  is a constant. Determine the highest temperature on the surface of the sphere  $x^2 + y^2 + z^2 = a^2$ .

**Solution:** The temperature at any point  $(x, y, z)$  is given by  $T(x, y, z) = kxyz^2$ , where  $k$  is a constant. To maximise  $T$  on the given spherical surface, take

$$f(x, y, z) = T(x, y, z) = kxyz^2, \quad g(x, y, z) = x^2 + y^2 + z^2 = a^2.$$

The Lagrange equations (3.9.1) give

$$f_x = \lambda g_x : \quad kyz^2 = 2\lambda x, \quad (3.9.11)$$

$$f_y = \lambda g_y : \quad kxz^2 = 2\lambda y, \quad (3.9.12)$$

$$f_z = \lambda g_z : \quad 2kxyz = 2\lambda z. \quad (3.9.13)$$

From (3.9.11) and (3.9.12), dividing each by the variable on the right,

$$2\lambda = \frac{kyz^2}{x} = \frac{kxz^2}{y} \implies y^2 = x^2.$$

$$x^2 = y^2 \quad (3.9.14)$$

From (3.9.13), cancelling  $z$  (which is non-zero at a maximum, since  $z = 0$  makes  $T = 0$ ),  $\lambda = kxy$ . Substituting into (3.9.11),

$$kyz^2 = 2kxy \cdot x \implies z^2 = 2x^2.$$

$$z^2 = 2x^2 \quad (3.9.15)$$

Substituting (3.9.14) and (3.9.15) into the constraint,

$$x^2 + x^2 + 2x^2 = a^2 \implies 4x^2 = a^2 \implies x^2 = \frac{a^2}{4},$$

so that

$$x = \pm \frac{a}{2}, \quad y^2 = x^2 = \frac{a^2}{4} \implies y = \pm \frac{a}{2}, \quad z^2 = 2x^2 = \frac{a^2}{2} \implies z = \pm \frac{a}{\sqrt{2}}.$$

Since  $T = kxyz^2$  and  $z^2 > 0$ , the temperature is largest when  $xy > 0$ , that is when  $x$  and  $y$  have the same sign. The temperature is therefore maximum at the four points

$$P_1\left(\frac{a}{2}, \frac{a}{2}, \frac{a}{\sqrt{2}}\right), \quad P_2\left(\frac{a}{2}, \frac{a}{2}, -\frac{a}{\sqrt{2}}\right), \quad P_3\left(-\frac{a}{2}, -\frac{a}{2}, \frac{a}{\sqrt{2}}\right), \quad P_4\left(-\frac{a}{2}, -\frac{a}{2}, -\frac{a}{\sqrt{2}}\right).$$

The highest temperature is

$$T\left(\frac{a}{2}, \frac{a}{2}, \frac{a}{\sqrt{2}}\right) = k\left(\frac{a}{2}\right)\left(\frac{a}{2}\right)\left(\frac{a^2}{2}\right) = \frac{ka^4}{8},$$

so  $T = \frac{ka^4}{8}$  is the highest temperature on the spherical surface.

**Note:**

The sign of  $z$  does not matter because  $z$  appears squared, which is why four points share the maximum rather than two. The four points where  $x$  and  $y$  have opposite signs give  $T = -\frac{ka^4}{8}$ , the minimum.

### 3.10 Unit 3 Summary

#### Functions of two variables

- Graph: the surface  $z = f(x, y)$ . Level curves:  $f(x, y) = k$ , crowded where the surface is steep.
- $\lim_{(x,y) \rightarrow (a,b)} f = L$  only if every path gives  $L$ ; two paths with different values prove the limit does not exist. Continuity:  $\lim f = f(a, b)$ .

#### Partial derivatives

- $f_x$ : differentiate in  $x$  with  $y$  held constant (slope of the slice  $y = b$ );  $f_y$  likewise.  $\nabla f = (f_x, f_y)$ .
- Clairaut:  $f_{xy} = f_{yx}$  when both are continuous. Harmonic:  $f_{xx} + f_{yy} = 0$  at every point.
- Tangent plane:  $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ ; linearisation  $f(x, y) \approx f(x_0, y_0) + f_x \Delta x + f_y \Delta y$ .

#### Chain rule

- $z = f(x, y)$ ,  $x = g(t)$ ,  $y = h(t)$ :  $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$ .
- $x = g(s, t)$ ,  $y = h(s, t)$ :  $\frac{\partial z}{\partial s} = z_x \frac{\partial x}{\partial s} + z_y \frac{\partial y}{\partial s}$ , and similarly for  $t$ . Draw the tree and multiply along every path.

#### Extreme values

- Critical points:  $f_x = f_y = 0$ .  $D = f_{xx}f_{yy} - f_{xy}^2$ .
- $D > 0$ ,  $f_{xx} > 0$ : local minimum;  $D > 0$ ,  $f_{xx} < 0$ : local maximum;  $D < 0$ : saddle;  $D = 0$ : no information.
- Lagrange: extremise  $f$  subject to  $g = k$  by solving  $\nabla f = \lambda \nabla g$ ,  $g = k$ ; compare the values of  $f$  at the solutions.

### 3.11 Practice Problems

The problems are arranged in three levels. Level A checks definitions and understanding, level B practises the standard methods, and level C asks you to analyse, justify or interpret a result, including in applied settings. Answers are given in brackets; full solutions are not.

## A. Check your understanding

- P1.** Describe the level curves of (a)  $f(x, y) = x^2 + y^2$  and (b)  $f(x, y) = x + y$ .  
[Ans. (a) circles about the origin (b) parallel lines  $x + y = k$ ]
- P2.** Find the domain of (a)  $\sqrt{9 - x^2 - y^2}$  and (b)  $\ln(x - y)$ .  
[Ans. (a) the disc  $x^2 + y^2 \leq 9$  (b) the half-plane  $x > y$ ]
- P3.** Show that  $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{x^2 + y^2}$  does not exist. [Ans. 1 along  $y = 0$ , 0 along  $x = 0$ ]
- P4.** Find  $\nabla f$  at  $(1, 2, 0)$  for  $f(x, y, z) = x^2y + e^z$ . [Ans.  $(4, 1, 1)$ ]
- P5.** True or false, with a reason: (a) if  $f_x = f_y = 0$  at  $(a, b)$  then  $f$  has a local extremum there; (b)  $D < 0$  at a critical point means a saddle point; (c)  $f_{xx} + f_{yy} = 0$  at one point makes  $f$  harmonic. [Ans. (a) F (b) T (c) F]

## B. Standard problems

- P6.** Find  $f_x$  and  $f_y$  for  $f(x, y) = x^2y^3 + \sin(xy)$ . [Ans.  $2xy^3 + y \cos xy$ ;  $3x^2y^2 + x \cos xy$ ]
- P7.** Find  $f_{xx}$ ,  $f_{yy}$  and  $f_{xy}$  for  $f(x, y) = x e^{2y}$ . [Ans. 0,  $4xe^{2y}$ ,  $2e^{2y}$ ]
- P8.** For  $u = \ln(x^2 + y^2)$ , verify  $u_{xy} = u_{yx}$  and show that  $u$  is harmonic away from the origin.  
[Ans.  $u_{xy} = u_{yx} = -\frac{4xy}{(x^2 + y^2)^2}$ ;  $u_{xx} + u_{yy} = 0$ ]
- P9.** Find the tangent plane to  $z = x^2 + xy$  at  $(1, 1, 2)$ . [Ans.  $z = 3x + y - 2$ ]
- P10.** Linearise  $f(x, y) = \sqrt{x^2 + y^2}$  at  $(3, 4)$  and estimate  $\sqrt{3.02^2 + 3.97^2}$ .  
[Ans.  $5 + \frac{3}{5}(x - 3) + \frac{4}{5}(y - 4)$ ; 4.988]
- P11.** Find  $\frac{dz}{dt}$  at  $t = \frac{\pi}{4}$  for  $z = x^2y$ ,  $x = \cos t$ ,  $y = \sin t$ . [Ans.  $-\frac{\sqrt{2}}{4}$ ]
- P12.** Find  $\frac{\partial z}{\partial s}$  for  $z = e^x \sin y$ ,  $x = st^2$ ,  $y = s^2t$ . [Ans.  $e^x (t^2 \sin y + 2st \cos y)$ ]
- P13.** Find and classify the critical points of  $f(x, y) = x^3 - 3x + y^2$ .  
[Ans. local minimum  $-2$  at  $(1, 0)$ ; saddle at  $(-1, 0)$ ]
- P14.** Maximise  $xy$  subject to  $x + y = 10$ . [Ans. 25 at  $(5, 5)$ ]
- P15.** Minimise  $x^2 + y^2 + z^2$  subject to  $x + 2y + 2z = 9$ . [Ans. 9 at  $(1, 2, 2)$ ]

## C. Analyse, justify and apply

- P16.** Prove that  $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^2 + y^2} = 0$ . (Checking paths is not enough.)  
[Ans.  $\left| \frac{x^2y}{x^2 + y^2} \right| \leq |y| \rightarrow 0$ ]
- P17.** The temperature on a plate is  $T(x, y) = 100 - x^2 - 2y^2$ . At  $(1, 1)$ , in which direction does  $T$  increase fastest, and at what rate?  
[Ans. direction  $(-2, -4)$ ; rate  $2\sqrt{5} \approx 4.47$ ]

- P18.** A cylinder of radius 5 cm and height 12 cm is measured with a possible error of 0.1 cm in each dimension. Use linear approximation to estimate the maximum error in its volume  $V = \pi r^2 h$ .  
[Ans.  $14.5\pi \approx 45.6 \text{ cm}^3$ , about 4.8%]
- P19.** For  $f(x, y) = x^4 + y^4$ , a student finds  $D = 0$  at  $(0, 0)$  and concludes that there is no extremum. Evaluate the conclusion.  
[Ans. wrong: the test is silent, but  $f \geq 0 = f(0, 0)$ , a minimum]
- P20.** A rectangular yard of area  $800 \text{ m}^2$  is fenced on three sides, a wall forming the fourth. Find the dimensions that use the least fencing.  
[Ans. 20 m by 40 m (the 40 m side parallel to the wall); 80 m]
- P21.** Show that  $z = f(x - ct)$  satisfies  $z_t + cz_x = 0$  for any differentiable  $f$ , and interpret the solution physically.  
[Ans. a profile moving right at speed  $c$  without changing shape]
- P22.** Construct a function with a saddle point at  $(1, 2)$  and justify it using the second derivatives test.  
[Ans. e.g.  $(x - 1)^2 - (y - 2)^2$ , with  $D = -4 < 0$ ]

# Unit 4 Matrix Algebra

## 4.1 Matrices

### Definition 4.1.1: Matrix

A *matrix* is a rectangular array of numbers arranged in rows and columns. Capital letters  $A$ ,  $B$ ,  $C$  and so on are used to denote matrices. A matrix with  $m$  rows and  $n$  columns,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix},$$

is said to be of *order*  $m \times n$ . The number  $a_{ij}$  in row  $i$  and column  $j$  is the  $(i, j)$  *entry* of  $A$ , and the matrix is written in short as

$$A = (a_{ij}), \quad 1 \leq i \leq m, \quad 1 \leq j \leq n.$$

### Note:

The order is always quoted rows first, columns second, and the subscripts follow the same convention:  $a_{23}$  is in row 2, column 3. Keeping this order straight matters as soon as matrices are multiplied, where the two dimensions play different roles.

### 4.1.1 Row and column vectors

#### Definition 4.1.2: Row vector and column vector

A matrix of order  $1 \times n$  is called a *row vector*, and a matrix of order  $n \times 1$  is called a *column vector*:

$$X = [a_1 \quad a_2 \quad a_3 \quad \cdots \quad a_n], \quad Y = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Any matrix can be regarded as being built from vectors. The *row vectors* of  $A$  are

$$\vec{a}_i = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}], \quad i = 1, 2, \dots, m,$$

and the *column vectors* of  $A$  are

$$a_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}, \quad j = 1, 2, \dots, n.$$

The matrix  $A$  can therefore be written in either of the two forms

$$A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} \vec{a}_1 \\ \vec{a}_2 \\ \vdots \\ \vec{a}_m \end{bmatrix}.$$

**Note:**

Both readings are used later in this unit and it is worth having them both to hand. The row picture is the one behind elementary row operations and echelon forms; the column picture is the one behind linear dependence and the rank of a matrix. They describe the same array.

## 4.2 Matrix Operations

### Definition 4.2.1: Equality of matrices

Let  $A$  and  $B$  be matrices of the same order  $m \times n$ . Then  $A = B$  if and only if  $a_{ij} = b_{ij}$  for all  $i$  and  $j$ .

### Definition 4.2.2: Addition of matrices

Let  $A$  and  $B$  be matrices of the same order. Their sum is the matrix of the same order given by

$$A + B = (a_{ij} + b_{ij}).$$

### Definition 4.2.3: Scalar multiplication

The product of a matrix  $A$  by a scalar  $\alpha$  is

$$\alpha A = (\alpha a_{ij}).$$

For example, with  $A = \begin{bmatrix} 1 & 5 & 6 \\ 2 & 3 & 1 \end{bmatrix}$ ,

$$2A = \begin{bmatrix} 2 & 10 & 12 \\ 4 & 6 & 2 \end{bmatrix}.$$

**Note:**

Equality and addition both require the two matrices to have the *same order*; there is no way to add a  $2 \times 3$  matrix to a  $3 \times 2$  one. Scalar multiplication has no such restriction, since it acts on each entry separately.

### Definition 4.2.4: Matrix multiplication

Let  $A$  be of order  $m \times n$  and  $B$  of order  $n \times k$ . Their product is the matrix  $AB = (c_{ij})$

of order  $m \times k$ , where

$$c_{ij} = \sum_{p=1}^n a_{ip} b_{pj} = \vec{a}_i \cdot \vec{b}_j,$$

that is, the  $(i, j)$  entry of  $AB$  is the  $i$ th row vector of  $A$  combined term by term with the  $j$ th column vector of  $B$ .

#### Note:

The product is defined only when the number of columns of  $A$  equals the number of rows of  $B$  — the two inner dimensions must match, and they are what disappears:

$$(m \times \underline{n}) \times (\underline{n} \times k) = (m \times k).$$

This is where the row and column pictures earn their keep:  $c_{ij}$  pairs the  $i$ th row of  $A$  with the  $j$ th column of  $B$ , and both must have  $n$  entries for the pairing to make sense.

#### Common mistake:

Matrix multiplication is not commutative. Even when both  $AB$  and  $BA$  are defined and of the same order, they are usually different matrices. The order in which factors are written can therefore never be changed. The same care is needed with transposes:  $(AB)^T = B^T A^T$ , with the order reversed.

#### Definition 4.2.5: Transpose

The *transpose* of an  $m \times n$  matrix  $A = (a_{ij})$  is the  $n \times m$  matrix  $A^T$  whose  $(i, j)$  entry is  $a_{ji}$ : the rows of  $A$  become the columns of  $A^T$ .

For example,

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

The transpose satisfies  $(A^T)^T = A$ ,  $(A+B)^T = A^T + B^T$ ,  $(\alpha A)^T = \alpha A^T$  and  $(AB)^T = B^T A^T$ . A column vector is the transpose of a row vector, which is why column vectors are often written  $(x_1, x_2, \dots, x_n)^T$  to save space.

#### Definition 4.2.6: Trace

The *trace* of a square matrix is the sum of its diagonal entries,  $\text{trace } A = a_{11} + a_{22} + \dots + a_{nn}$ .

## 4.3 Special Matrices

- (i) **Square matrix:** a matrix having  $n$  rows and  $n$  columns, that is of order  $n \times n$ . For example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 9 & 6 \\ -1 & 0 & 1 \end{bmatrix}.$$

- (ii) **Diagonal matrix:** a square matrix with  $a_{ij} = 0$  whenever  $i \neq j$ , so that only the entries on the main diagonal may be non-zero:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

- (iii) **Scalar matrix:** a diagonal matrix in which every diagonal entry is the same scalar  $c$ , that is  $a_{ii} = c$  for all  $i$ :

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

- (iv) **Identity matrix:** the scalar matrix with  $c = 1$ , denoted  $I_n = (\delta_{ij})$ , where

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad \text{so} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (v) **Null matrix (zero matrix):** a matrix with  $a_{ij} = 0$  for all  $i$  and  $j$ . It need not be square:

$$O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

- (vi) **Symmetric and skew-symmetric matrices.** A square matrix  $A$  is *symmetric* if

$$a_{ij} = a_{ji} \quad \text{for all } i, j, \quad \text{equivalently} \quad A = A^T \text{ (Definition 4.2.5),}$$

and *skew-symmetric* if

$$a_{ij} = -a_{ji} \quad \text{for all } i, j, \quad \text{equivalently} \quad A = -A^T.$$

For example,

$$A = \begin{bmatrix} 1 & 2 & -5 \\ 2 & 0 & 1 \\ -5 & 1 & 9 \end{bmatrix} \text{ is symmetric,} \quad B = \begin{bmatrix} 0 & -2 & 5 \\ 2 & 0 & -1 \\ -5 & 1 & 0 \end{bmatrix} \text{ is skew-symmetric.}$$

### Common mistake:

**A skew-symmetric matrix must have zeros all along its diagonal.** Taking  $i = j$  in  $a_{ij} = -a_{ji}$  gives  $a_{ii} = -a_{ii}$ , hence  $2a_{ii} = 0$  and  $a_{ii} = 0$ . It is a common slip to state the condition only for  $i \neq j$  and then write down an example with non-zero diagonal entries; such a matrix satisfies  $A = -A^T$  nowhere on the diagonal and is not skew-symmetric. A quick test is that  $A + A^T = O$  must hold exactly.

- (vii) **Triangular matrices.** A square matrix is *lower triangular* (LTM) if every entry above the main diagonal is zero, and *upper triangular* (UTM) if every entry below the main diagonal is zero:

$$\begin{bmatrix} 1 & 0 & 0 \\ -5 & 0 & 0 \\ 0 & 6 & 7 \end{bmatrix} \text{ is lower triangular,} \quad \begin{bmatrix} 1 & 9 & 6 \\ 0 & 7 & 5 \\ 0 & 0 & 0 \end{bmatrix} \text{ is upper triangular.}$$

The diagonal entries themselves are unrestricted and may be zero.

## 4.4 Determinants and the Inverse

Determinants and inverses are met at school for small matrices; this section collects what is used in the rest of the unit.

### Definition 4.4.1: Determinant

The determinant of a square matrix  $A$ , written  $|A|$  or  $\det A$ , is the number defined for  $2 \times 2$  and  $3 \times 3$  matrices by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

The same value results from expanding along any row or column, with the signs following the chessboard pattern  $\begin{smallmatrix} + & - & + \\ - & + & - \\ + & - & + \end{smallmatrix}$ .

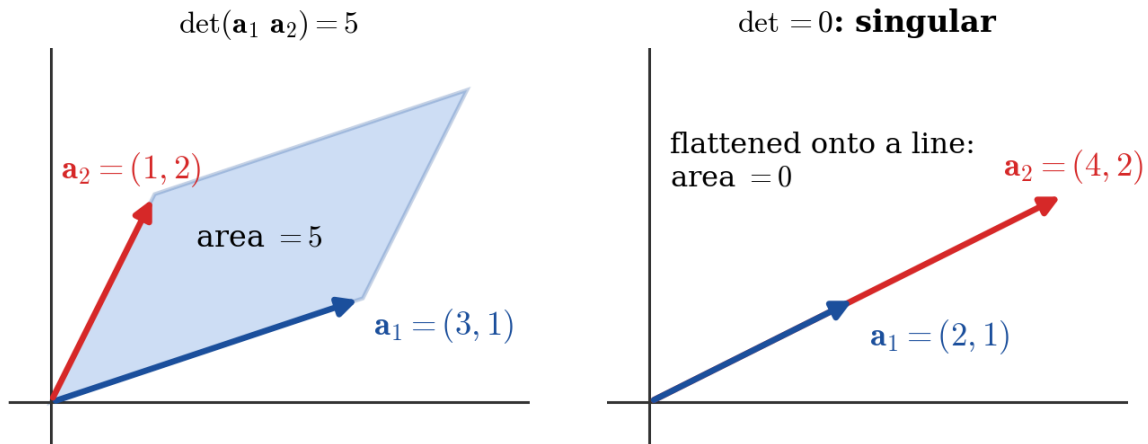


Figure 4.4.1: The absolute value of a  $2 \times 2$  determinant is the area of the parallelogram spanned by the columns. When the columns lie on one line the parallelogram collapses and the determinant is 0 — the matrix is singular.

### Theorem 4.4.2: Properties of determinants

For square matrices  $A$  and  $B$  of the same order:

- (i)  $|A^T| = |A|$ , so every property of rows holds for columns;
- (ii) interchanging two rows changes the sign of  $|A|$ ;
- (iii) multiplying one row by  $k$  multiplies  $|A|$  by  $k$ ;
- (iv) adding a multiple of one row to another leaves  $|A|$  unchanged;
- (v) if two rows are equal, or a row is zero, then  $|A| = 0$ ;
- (vi) the determinant of a triangular matrix is the product of its diagonal entries;
- (vii)  $|AB| = |A| |B|$ .

**Note:**

Properties (ii)–(iv) describe the effect of the three elementary row operations of [Section 4.6](#). Reducing a matrix to triangular form and multiplying the diagonal entries, property (vi), is usually the quickest way to evaluate a large determinant.

**Definition 4.4.3: Singular and non-singular matrices**

A square matrix  $B$  is *singular* if  $|B| = 0$ , and *non-singular* otherwise.

**Note:**

Only square matrices have determinants, so only square matrices can be called singular or non-singular. Non-singularity is the condition under which  $B$  has an inverse, and it will reappear in this unit as the condition under which a square system  $BX = C$  has exactly one solution.

**Definition 4.4.4: Cofactor and adjoint**

Deleting row  $i$  and column  $j$  of a square matrix  $A$  and taking the determinant of what remains gives the minor  $M_{ij}$ . The *cofactor* of  $a_{ij}$  is  $C_{ij} = (-1)^{i+j} M_{ij}$ , and the *adjoint* of  $A$  is the transpose of the matrix of cofactors,  $\text{adj } A = (C_{ij})^T$ .

**Theorem 4.4.5: Inverse of a matrix**

A square matrix  $A$  has an inverse  $A^{-1}$ , with  $AA^{-1} = A^{-1}A = I$ , if and only if  $|A| \neq 0$ , and then

$$A^{-1} = \frac{1}{|A|} \text{adj } A.$$

For a  $2 \times 2$  matrix this reads  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ .

**Example 4.4.6: A determinant and an inverse**

Find  $|A|$  and  $A^{-1}$  for  $A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ .

**Solution:** Expanding along the first row,  $|A| = 2(1 - 0) - 0 + 1(1 - 0) = 3 \neq 0$ , so  $A$  is non-singular. The cofactors are

$$\begin{aligned} C_{11} &= 1, & C_{12} &= -1, & C_{13} &= 1, \\ C_{21} &= 1, & C_{22} &= 2, & C_{23} &= -2, \\ C_{31} &= -1, & C_{32} &= 1, & C_{33} &= 2, \end{aligned}$$

so, transposing the cofactor matrix,

$$A^{-1} = \frac{1}{3} \text{adj } A = \frac{1}{3} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 2 & 1 \\ 1 & -2 & 2 \end{bmatrix}.$$

As a check,  $A(\text{adj } A) = 3I$ .

## 4.5 Minors and the Rank of a Matrix

### Definition 4.5.1: Minor of a matrix

Select  $r$  rows and  $r$  columns of a matrix  $A$  and delete all other rows and columns. The determinant of the resulting  $r \times r$  array is called a *minor* of  $A$  of order  $r$ .

For instance, selecting rows 1 and 3 and columns 1 and 3 of

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \text{gives the minor} \quad \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}.$$

There are in general several different minors of the same order — one for each choice of  $r$  rows and  $r$  columns.

### Definition 4.5.2: Rank of a matrix

The rank of a matrix  $A$  is said to be  $r$  when

- (i)  $A$  has at least one non-zero minor of order  $r$ , and
- (ii) every minor of  $A$  of order higher than  $r$  vanishes.

The rank of  $A$  is denoted  $\text{rank}(A)$ .

### Example 4.5.3: Rank from minors

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}.$$

The only minor of order 3 is  $|A|$  itself. Expanding along the first row,

$$|A| = 1(4 \cdot 5 - 2 \cdot 6) - 2(1 \cdot 5 - 2 \cdot 2) + 3(1 \cdot 6 - 4 \cdot 2) = 1(8) - 2(1) + 3(-2) = 0,$$

so condition (i) fails at  $r = 3$  and  $\text{rank}(A) \neq 3$ . Now try order 2, taking the minor in the top left corner:

$$\begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 4 - 2 = 2 \neq 0.$$

There is a non-zero minor of order 2, and the only higher order minor vanishes. Hence

$$\boxed{\text{rank}(A) = 2}$$

### Note:

The search runs *downwards*. Start at the largest possible order — the smaller of the number of rows and the number of columns — and work down until a non-zero minor appears. The first order at which one is found is the rank, because condition (ii) is then automatic: every larger minor has already been shown to vanish.

**Common mistake:**

Only *one* non-zero minor of order  $r$  is needed, but *all* minors of order greater than  $r$  must vanish. The asymmetry is easy to lose. Here there was only one minor of order 3 so the check was short; for a  $3 \times 4$  matrix there would be four of them, and every one would have to be zero before the rank could be declared less than 3.

**Note:**

This determinant definition is the reason rank is well defined, but it is laborious — a  $4 \times 4$  matrix has 16 minors of order 3 alone. The row echelon method ([Theorem 4.7.5](#)) computes the same number by counting non-zero rows, and is what is used in practice throughout the rest of this unit. The two definitions always agree.

## 4.6 Elementary Row Operations

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{matrix}$$

(i) **Interchange two rows:**  $R_1 \leftrightarrow R_2$ ,

$$A \sim \begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

(ii) **Multiply a row by a non-zero real number:**  $R_2 \rightarrow \alpha R_2$ ,

$$A \sim \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ \alpha a_{21} & \alpha a_{22} & \alpha a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

(iii) **Replace a row by its sum with a multiple of another row:**  $R_3 \rightarrow R_3 + \alpha R_1$ ,

$$A \sim \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} + \alpha a_{11} & a_{32} + \alpha a_{12} & a_{33} + \alpha a_{13} \end{bmatrix}.$$

**Note:**

The multiplier in (ii) must be non-zero, otherwise the row is destroyed and the operation cannot be undone. Every elementary row operation is reversible, which is what makes the resulting matrix equivalent to the original.

### Definition 4.6.1: Equivalent matrices

Two matrices  $A$  and  $B$  are said to be *equivalent* if one can be obtained from the other by a sequence of elementary row operations. If  $A$  is equivalent to  $B$ , it is denoted by  $A \sim B$ .

- Two equivalent matrices have the same rank.

## 4.7 Echelon Forms

**Definition 4.7.1: Row echelon form**

A matrix is said to be in *row echelon form* if:

- (i) the first non-zero entry in each non-zero row is one;
- (ii) if row  $k$  does not consist entirely of zeroes, the number of leading zero entries in row  $k + 1$  is greater than the number of leading zero entries in row  $k$ ;
- (iii) if there are rows whose entries are all zero, they are below the rows having non-zero entries.

**Example 4.7.2: Matrices in row echelon form**

$$(i) \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (ii) \begin{bmatrix} 1 & 0 & 0 & 5 & 6 \\ 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

**Definition 4.7.3: Reduced row echelon form**

A matrix is said to be in *reduced row echelon form* (rref) if:

- (i) the matrix is in row echelon form;
- (ii) the first non-zero entry in each row is the only non-zero entry in its column.

The leading ones are called the *pivots*.

**Example 4.7.4: Recognising reduced row echelon form**

$$(i) \begin{bmatrix} 1 & 5 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (ii) \begin{bmatrix} 1 & 0 & 5 & 0 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (iii) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The first two are in reduced row echelon form. The third is *not*: the pivot in row 4 sits in column 4, but that column also contains a 1 in row 3, so the pivot is not the only non-zero entry in its column.

**Theorem 4.7.5: Rank from the echelon form**

The rank of a matrix (Definition 4.5.2) equals the number of non-zero rows in its row echelon form, or in its reduced row echelon form.

This holds because elementary row operations do not change the rank (Definition 4.6.1), and in an echelon form the non-zero rows visibly carry a non-zero minor of that order while every larger minor contains a zero row.

**4.7.1 Worked problems**

**Problem 4.7.1.** Determine the row echelon form of the following matrices.

$$(a) A = \begin{pmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{pmatrix}$$

**Solution:** Use the entry 1 in the top left as the pivot and clear the column below it:

$$A \xrightarrow[R_3 \rightarrow R_3 + (-1)R_1]{R_2 \rightarrow R_2 + (-2)R_1} \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 3 \end{bmatrix} \xrightarrow{R_2 \rightarrow (-1)R_2} \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & 1 & 3 \end{bmatrix}.$$

Now clear below the new pivot:

$$\xrightarrow{R_3 \rightarrow R_3 + (-1)R_2} \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

This is in row echelon form and has two non-zero rows, so  $\text{rank}(A) = 2$ .

$$(b) A = \begin{pmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 7 & 22 \end{pmatrix}$$

**Solution:**

$$A \xrightarrow[R_3 \rightarrow R_3 + (-3)R_1]{R_2 \rightarrow R_2 + (-2)R_1} \begin{bmatrix} 1 & 4 & 5 \\ 0 & -2 & -2 \\ 0 & -5 & 7 \end{bmatrix} \xrightarrow{R_2 \rightarrow -\frac{1}{2}R_2} \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & -5 & 7 \end{bmatrix}.$$

Clearing below the second pivot and scaling the last row,

$$\xrightarrow{R_3 \rightarrow R_3 + 5R_2} \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 12 \end{bmatrix} \xrightarrow{R_3 \rightarrow \frac{1}{12}R_3} \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

All three rows are non-zero, so  $\text{rank}(A) = 3$ .

$$(c) A = \begin{pmatrix} 2 & 3 & 4 \\ -1 & 2 & 3 \\ 1 & 5 & 7 \end{pmatrix}$$

**Solution:** The leading entry is 2 rather than 1, so start by swapping in a row that begins with 1:

$$A \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 5 & 7 \\ -1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 + (-2)R_1]{R_2 \rightarrow R_2 + R_1} \begin{bmatrix} 1 & 5 & 7 \\ 0 & 7 & 10 \\ 0 & -7 & -10 \end{bmatrix}.$$

Then

$$\xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 1 & 5 & 7 \\ 0 & 7 & 10 \\ 0 & 0 & 0 \end{bmatrix},$$

which has two non-zero rows, so  $\text{rank}(A) = 2$ .

**Note:**

Swapping rows to bring a 1 into the pivot position avoids fractions later. It is always allowed, being an elementary row operation of the first kind.

$$(d) \ A = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 2 & 4 & 3 & 1 \\ 3 & 6 & 4 & -2 \end{pmatrix}$$

**Solution:**

$$A \xrightarrow[R_3 \rightarrow R_3 + (-3)R_1]{R_2 \rightarrow R_2 + (-2)R_1} \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + (-1)R_2} \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & -2 \end{bmatrix}.$$

Scaling the last row,

$$\xrightarrow{R_3 \rightarrow -\frac{1}{2}R_3} \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

so  $\text{rank}(A) = 3$ .

**Note:**

The second column is entirely zero below the first row. That is permitted: the definition of echelon form counts *leading* zeroes, and each row here starts further right than the one above it.

**Problem 4.7.2.** Determine the reduced row echelon form of the following matrices.

$$(a) \ A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{pmatrix}$$

**Solution:** Clearing the first column,

$$A \xrightarrow[R_3 \rightarrow R_3 + (-2)R_1]{R_2 \rightarrow R_2 + (-1)R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + (-1)R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Making the second pivot a 1 and clearing *above* it as well,

$$\xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + (-2)R_2} \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix},$$

which is the reduced row echelon form of  $A$ . Hence  $\text{rank}(A) = 2$ , the same value found from minors in [Example 4.5.3](#).

**Note:**

Reaching row echelon form only requires clearing *below* each pivot. The extra step for reduced form is clearing above them as well, which is what makes each pivot the only non-zero entry in its column.

$$(b) \ A = \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{pmatrix}$$

**Solution:**

$$A \xrightarrow[R_3 \rightarrow R_3 + (-1)R_1]{R_2 \rightarrow R_2 + (-3)R_1} \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} \xrightarrow[R_3 \rightarrow -\frac{1}{2}R_3]{R_2 \rightarrow -\frac{1}{6}R_2} \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Continuing,

$$\xrightarrow{R_3 \rightarrow R_3 + (-1)R_2} \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + (-3)R_2} \begin{bmatrix} 1 & 3 & 4 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

which is the reduced row echelon form of  $A$ . Hence  $\text{rank}(A) = 2$ .

$$(c) \ A = \begin{pmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 7 & 22 \end{pmatrix}$$

**Solution:** The row echelon form was found in problem 1(b):

$$A \sim \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Clearing above each pivot, starting with the second column,

$$\xrightarrow{R_1 \rightarrow R_1 + (-4)R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_2 \rightarrow R_2 + (-1)R_3]{R_1 \rightarrow R_1 + (-1)R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which is the reduced row echelon form of  $A$ . Hence  $\text{rank}(A) = 3$ .

$$(d) \ A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{pmatrix}$$

**Solution:** Clearing the first column,

$$A \xrightarrow[R_3 \rightarrow R_3 + 2R_1]{R_2 \rightarrow R_2 + (-1)R_1} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & -6 \\ 0 & -2 & 2 \end{bmatrix} \xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -3 \\ 0 & -2 & 2 \end{bmatrix}.$$

Then

$$\xrightarrow{R_3 \rightarrow R_3 + 2R_2} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -4 \end{bmatrix} \xrightarrow{R_3 \rightarrow -\frac{1}{4}R_3} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Clearing above the pivots,

$$\xrightarrow{R_1 \rightarrow R_1 + (-1)R_2} \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_2 \rightarrow R_2 + 3R_3]{R_1 \rightarrow R_1 + (-6)R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which is the reduced row echelon form of  $A$ . Hence  $\text{rank}(A) = 3$ .

## 4.8 Systems of Linear Equations

### 4.8.1 Homogeneous linear equations

#### Definition 4.8.1: Homogeneous system

The system of  $m$  homogeneous linear equations in  $n$  unknowns  $x_i$  is given by

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0, \end{aligned}$$

where the coefficients  $a_{ij}$  are scalars,  $i = 1, 2, \dots, m$  and  $j = 1, 2, \dots, n$ .

### 4.8.2 Matrix representation of homogeneous linear equations

The system can be written in matrix form as  $Ax = 0$ , where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}, \quad 0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{m \times 1}.$$

Here  $A$  is called the *coefficient matrix* of the system of equations,  $x$  is called the *matrix of unknowns* and  $0$  is the zero matrix.

#### Note:

$m$  is the number of equations, and gives the number of rows.  $n$  is the number of unknowns, and gives the number of columns.

### 4.8.3 Non-homogeneous linear equations

#### Definition 4.8.2: Non-homogeneous system

The system of  $m$  non-homogeneous linear equations in  $n$  unknowns  $x_i$  is given by

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

where the coefficients  $a_{ij}$  are scalars and at least one of the  $b_i$  is not zero.

### 4.8.4 Matrix representation and the augmented matrix

The non-homogeneous system is written  $Ax = b$ , where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m \times 1}.$$

The *augmented matrix* is written

$$[A : B] = \left[ \begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right].$$

### 4.8.5 Solution of a linear system

- A *solution* of the system is a list  $(s_1, s_2, \dots, s_n)$  of numbers that makes each equation a true statement when the values  $s_1, \dots, s_n$  are substituted for  $x_1, \dots, x_n$  respectively.
- The set of all possible solutions is called the *solution set* of the linear system.
- Two linear systems are called *equivalent* if they have the same solution set.
- A system of linear equations has either (1) no solution, or (2) exactly one solution, or (3) infinitely many solutions.
- A system is said to be *consistent* if it has either one solution or infinitely many solutions.
- A system is said to be *inconsistent* if it has no solution.

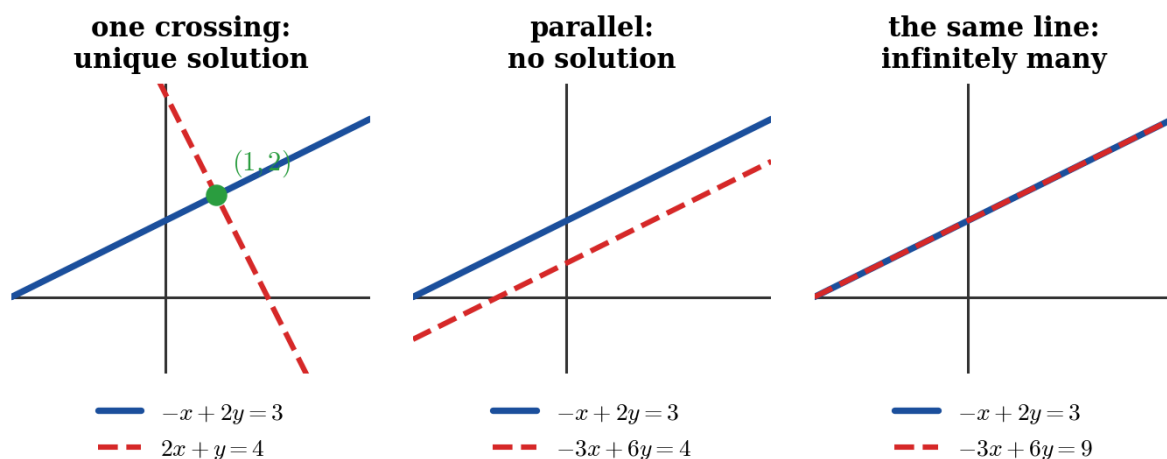


Figure 4.8.1: The three possibilities, drawn for two equations in two unknowns. Each equation is a line, and the solution set is the intersection.

**Theorem 4.8.3: Rank criterion for a non-homogeneous system**

For the system  $Ax = b$  in  $n$  unknowns:

- If  $\text{rank}[A] < \text{rank}[A : B]$ , the system is *inconsistent* and has no solution.
- If  $\text{rank}[A] = \text{rank}[A : B] = n$ , the system is consistent with a *unique* solution.
- If  $\text{rank}[A] = \text{rank}[A : B] < n$ , the system is consistent with *infinitely many* solutions.

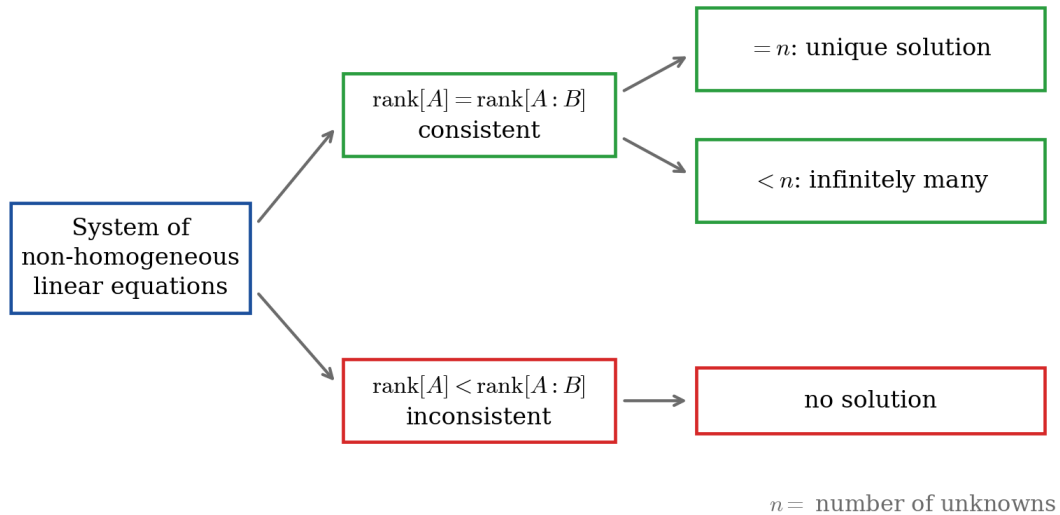


Figure 4.8.2: The rank test decides consistency first, then counts solutions.

### 4.8.6 Solving a system: the method

**Note:**

A matrix is said to be *singular* if it is not invertible. Equivalently, its determinant is zero, or its rank is less than the number of columns.

A homogeneous system  $Ax = 0$  is always consistent, since  $x = 0$  is always a solution.

- If  $\text{rank}(A)$  equals the number of unknowns, the only solution is the zero solution.
- If  $r = \text{rank}(A) < \text{the number of unknowns } n$ , there are infinitely many solutions, and  $n - r$  variables are chosen as free variables.

**Steps to solve a homogeneous system.**

1. Write the matrix-vector form of the equation,  $Ax = 0$ .
2. Obtain the row echelon form of  $A$  and denote it by  $R$ .
3. If  $\text{rank}(A) = n$ , the system has a unique solution; if  $\text{rank}(A) < n$ , it has infinitely many solutions.
4. To obtain the solutions, write  $Rx = 0$  and use back substitution.

**Steps to solve a non-homogeneous system.**

1. Write the matrix-vector form,  $Ax = b$ .
2. Form the augmented matrix  $[A : b]$ .
3. Find the row echelon form of  $[A : b] = [R : b']$ , where  $R$  is the row echelon form of  $A$ .
  - If  $\text{rank}[A : b] = \text{rank}(A) = r$ , the system is consistent and has a solution. Otherwise it is inconsistent and has no solution.
  - If consistent and  $r = n$ , the number of unknowns, the system has a unique solution.
  - If consistent and  $r < n$ , the system has infinitely many solutions, and  $n - r$  free variables are chosen.
4. Solve the system  $Rx = b'$  using back substitution.

#### 4.8.7 Gauss elimination and Gauss–Jordan methods

**Note:**

The method described above — finding the row echelon form of  $[A : b]$  and then using back substitution — is known as the *Gauss elimination method*. If instead the reduced row echelon form of  $[A : b]$  is found and the system solved from it, the method is known as the *Gauss–Jordan method*.

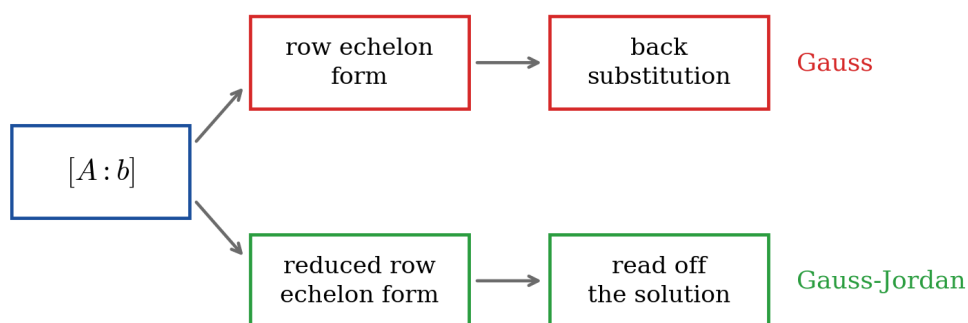


Figure 4.8.3: The two methods differ only in how far the reduction is carried. The extra elimination above the pivots replaces the back substitution.

#### Example 4.8.4: Two equations in two unknowns

For  $ax + by = c$ , each equation is a line in the plane, and the three cases of [Theorem 4.8.3](#) are the three ways two lines can meet.

|                 |                             |                            |
|-----------------|-----------------------------|----------------------------|
| No solution     | $-x + 2y = 3, -3x + 6y = 4$ | parallel lines             |
| Infinitely many | $-x + 2y = 3, -3x + 6y = 9$ | the same line              |
| Unique solution | $-x + 2y = 3, 2x + y = 4$   | lines crossing at $(1, 2)$ |

#### 4.8.8 Worked problems

**Problem 4.8.1.** Solve the following systems of homogeneous equations.

(a)  $x_1 + 2x_2 + 3x_3 = 0; \quad 4x_1 + 5x_2 + 6x_3 = 0; \quad 7x_1 + 8x_2 + 10x_3 = 0.$

**Solution:** The matrix-vector form of the given system is

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}_{3 \times 3} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \text{that is} \quad Ax = 0 \quad \text{with} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}.$$

Reducing  $A$  to row echelon form,

$$A \xrightarrow[R_3 \rightarrow R_3 - 7R_1]{R_2 \rightarrow R_2 - 4R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -11 \end{bmatrix} \xrightarrow{R_2 \rightarrow -\frac{1}{3}R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -6 & -11 \end{bmatrix},$$

and then

$$\xrightarrow{R_3 \rightarrow R_3 + 6R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = R.$$

Hence  $\text{rank}(A) = 3$ , which equals the number of unknowns, so the system has a unique solution. The corresponding system is  $Rx = 0$ :

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \implies \quad \begin{aligned} x_1 + 2x_2 + 3x_3 &= 0, \\ x_2 + 2x_3 &= 0, \\ x_3 &= 0. \end{aligned}$$

Using back substitution,  $x_3 = 0$  gives  $x_2 + 2(0) = 0$ , so  $x_2 = 0$ , and then  $x_1 + 2(0) + 3(0) = 0$ , so  $x_1 = 0$ . The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

#### Note:

A homogeneous system always has the zero solution, so the only question is whether it has others. Here it does not.

(b)  $x_1 + 2x_2 + x_3 - 3x_4 = 0; \quad 2x_1 + 4x_2 + 3x_3 + x_4 = 0; \quad 3x_1 + 6x_2 + 4x_3 - 2x_4 = 0.$

**Solution:** The matrix-vector form is  $Ax = 0$  with

$$A = \begin{bmatrix} 1 & 2 & 1 & -3 \\ 2 & 4 & 3 & 1 \\ 3 & 6 & 4 & -2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}.$$

Reducing  $A$ ,

$$A \xrightarrow[R_3 \rightarrow R_3 - 3R_1]{R_2 \rightarrow R_2 + (-2)R_1} \begin{bmatrix} 1 & 2 & 1 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 1 & 7 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 2 & 1 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R.$$

Hence  $\text{rank}(A) = 2 < 4$ , the number of unknowns, so the given system has infinitely many solutions. The number of free variables is

$$n - r = 4 - 2 = 2,$$

and we choose  $x_2$  and  $x_4$  as the free variables, the two whose columns carry no pivot.

The corresponding system  $Rx = 0$  is

$$x_1 + 2x_2 + x_3 - 3x_4 = 0, \quad (4.8.1)$$

$$x_3 + 7x_4 = 0. \quad (4.8.2)$$

From (4.8.2),

$$x_3 = -7x_4,$$

and substituting into (4.8.1),

$$x_1 + 2x_2 - 7x_4 - 3x_4 = 0 \implies x_1 = 10x_4 - 2x_2.$$

Therefore the solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2x_2 + 10x_4 \\ x_2 \\ -7x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 10 \\ 0 \\ -7 \\ 1 \end{bmatrix}, \quad x_2, x_4 \in \mathbb{R}.$$

**Note:**

The two column vectors are independent, and every solution is a combination of them. The number of vectors needed is exactly  $n - r$ , the number of free variables.

**Problem 4.8.2.** Check the consistency of the following systems of equations and solve.

(a)  $x_1 + x_2 + x_3 = 6; \quad x_1 - x_2 + 2x_3 = 5; \quad 3x_1 + x_2 + x_3 = 8.$

**Solution:** The matrix-vector form is  $Ax = b$ , that is

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ 8 \end{bmatrix}.$$

Consider the augmented matrix and reduce it:

$$[A : b] = \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \\ 3 & 1 & 1 & 8 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 3R_1]{R_2 \rightarrow R_2 - R_1} \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & -2 & -2 & -10 \end{array} \right].$$

Scaling and clearing the second column,

$$\xrightarrow{R_2 \rightarrow -\frac{1}{2}R_2} \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & -2 & -2 & -10 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + 2R_2} \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -3 & -9 \end{array} \right],$$

and finally

$$\xrightarrow{R_3 \rightarrow -\frac{1}{3}R_3} \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 3 \end{array} \right] = [R : b'].$$

Hence  $\text{rank}[A : b] = \text{rank}(A) = 3$ , the number of unknowns, so by [Theorem 4.8.3](#) the given system is consistent and has a unique solution. Solving  $Rx = b'$ ,

$$x_1 + x_2 + x_3 = 6, \quad (4.8.3)$$

$$x_2 - \frac{x_3}{2} = \frac{1}{2}, \quad (4.8.4)$$

$$x_3 = 3. \quad (4.8.5)$$

Using back substitution, (4.8.5) in (4.8.4) gives  $x_2 - \frac{3}{2} = \frac{1}{2}$ , so  $x_2 = 2$ ; then (4.8.3) gives  $x_1 + 2 + 3 = 6$ , so  $x_1 = 1$ . The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

(b)  $x_1 - x_2 + 2x_3 = 4; \quad 2x_1 + 3x_2 - x_3 = 1; \quad 3x_1 + 2x_2 + x_3 = 5.$

**Solution:** Reducing the augmented matrix,

$$[A : b] = \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 2 & 3 & -1 & 1 \\ 3 & 2 & 1 & 5 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 3R_1]{R_2 \rightarrow R_2 - 2R_1} \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 5 & -5 & -7 \\ 0 & 5 & -5 & -7 \end{array} \right],$$

and then

$$\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 5 & -5 & -7 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow \frac{1}{5}R_2} \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & -\frac{7}{5} \\ 0 & 0 & 0 & 0 \end{array} \right] = [R : b'].$$

Here  $\text{rank}[A : b] = \text{rank}(A) = 2 < 3$ , the number of unknowns, so the given system is consistent and has infinitely many solutions. We choose  $n - r = 3 - 2 = 1$  variable as free, namely  $x_3$ , whose column carries no pivot. The corresponding system  $Rx = b'$  is

$$x_1 - x_2 + 2x_3 = 4, \quad (4.8.6)$$

$$x_2 - x_3 = -\frac{7}{5}. \quad (4.8.7)$$

From (4.8.7),  $x_2 = -\frac{7}{5} + x_3$ , and substituting into (4.8.6),

$$x_1 = 4 - 2x_3 - \frac{7}{5} + x_3 = \frac{13}{5} - x_3.$$

The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{13}{5} - x_3 \\ -\frac{7}{5} + x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{13}{5} \\ -\frac{7}{5} \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R}.$$

#### Note:

For a non-homogeneous system the solution set is a particular solution plus any multiple of a solution of the associated homogeneous system. Setting  $x_3 = 0$

recovers the particular solution shown first.

$$(c) \quad x_1 - 4x_2 + 7x_3 = 14; \quad 3x_1 + 8x_2 - 2x_3 = 13; \quad 7x_1 - 8x_2 + 26x_3 = 5.$$

**Solution:** Reducing the augmented matrix,

$$[A : b] = \left[ \begin{array}{ccc|c} 1 & -4 & 7 & 14 \\ 3 & 8 & -2 & 13 \\ 7 & -8 & 26 & 5 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 7R_1]{R_2 \rightarrow R_2 - 3R_1} \left[ \begin{array}{ccc|c} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 20 & -23 & -93 \end{array} \right],$$

and then

$$\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{ccc|c} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 0 & 0 & -64 \end{array} \right] \xrightarrow[R_3 \rightarrow -\frac{1}{64}R_3]{R_2 \rightarrow \frac{1}{20}R_2} \left[ \begin{array}{ccc|c} 1 & -4 & 7 & 14 \\ 0 & 1 & -\frac{23}{20} & -\frac{29}{20} \\ 0 & 0 & 0 & 1 \end{array} \right].$$

Here  $\text{rank}[A : b] = 3$  while  $\text{rank}(A) = 2$ , so  $\text{rank}[A : b] \neq \text{rank}(A)$  and the given system is *inconsistent*: it has no solution.

**Note:**

The last row reads  $0x_1 + 0x_2 + 0x_3 = 1$ , which no values of the unknowns can satisfy. That single impossible row is what the rank comparison detects.

**Problem 4.8.3.** Find the value of  $\lambda$  for which the system

$$x_1 + 2x_2 + 3x_3 = 14; \quad x_1 + 4x_2 + 7x_3 = 30; \quad x_1 + x_2 + x_3 = \lambda$$

is consistent. Determine the solution for this value of  $\lambda$ .

**Solution:** The matrix-vector form is  $Ax = b$ , that is

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 7 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 14 \\ 30 \\ \lambda \end{bmatrix}.$$

Reducing the augmented matrix,

$$[A : b] = \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 1 & 4 & 7 & 30 \\ 1 & 1 & 1 & \lambda \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 2 & 4 & 16 \\ 0 & -1 & -2 & \lambda - 14 \end{array} \right],$$

then

$$\xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 2 & 8 \\ 0 & -1 & -2 & \lambda - 14 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_2} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & \lambda - 6 \end{array} \right].$$

Everything now turns on the last entry, and there are two cases.

**Case (i):**  $\lambda \neq 6$ .

Then  $\lambda - 6 \neq 0$ , so the last row may be scaled:

$$\xrightarrow{R_3 \rightarrow \frac{1}{\lambda-6}R_3} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

Hence  $\text{rank}[A : b] = 3 \neq \text{rank}(A) = 2$ , so the system is inconsistent and has no solution.

**Case (ii):**  $\lambda = 6$ .

Then  $\lambda - 6 = 0$  and the augmented matrix becomes

$$[A : b] \sim \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Hence  $\text{rank}[A : b] = \text{rank}(A) = 2 = r$ . But  $r = 2 < 3 = n$ , the number of unknowns, so the given system is consistent and has infinitely many solutions.

We choose  $n - r = 3 - 2 = 1$  free variable, namely  $x_3$ . The corresponding system  $Rx = b'$  is

$$x_1 + 2x_2 + 3x_3 = 14, \quad (4.8.8)$$

$$x_2 + 2x_3 = 8. \quad (4.8.9)$$

From (4.8.9),

$$\boxed{x_2 = 8 - 2x_3}$$

and substituting into (4.8.8),

$$x_1 + 2(8 - 2x_3) + 3x_3 = 14 \implies x_1 + 16 - 4x_3 + 3x_3 = 14 \implies \boxed{x_1 = x_3 - 2}$$

The solution is therefore

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 - 2 \\ 8 - 2x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 8 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R}.$$

#### Note:

The value  $\lambda = 6$  is exactly the one for which the third equation is a combination of the first two. Taking  $\frac{3}{2}$  of the first equation minus  $\frac{1}{2}$  of the second gives

$$\frac{3}{2}(x_1 + 2x_2 + 3x_3) - \frac{1}{2}(x_1 + 4x_2 + 7x_3) = x_1 + x_2 + x_3, \quad \frac{3}{2}(14) - \frac{1}{2}(30) = 6.$$

So the left-hand side of the third equation is already forced to equal 6. For any other  $\lambda$  the third equation contradicts the first two, which is why the system fails.

**Problem 4.8.4.** Solve the following systems by the Gauss–Jordan method.

(a)  $x_1 + x_2 + x_3 = 6; \quad x_1 - x_2 + 2x_3 = 5; \quad 3x_1 + x_2 + x_3 = 8.$

**Solution:** From problem 4(a), the row echelon form of the augmented matrix is

$$[A : b] \sim \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 3 \end{array} \right].$$

Continuing to the reduced row echelon form, first clear above the second pivot,

$$\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[ \begin{array}{ccc|c} 1 & 0 & \frac{3}{2} & \frac{11}{2} \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 3 \end{array} \right],$$

and then above the third,

$$\xrightarrow[R_2 \rightarrow R_2 + \frac{1}{2}R_3]{R_1 \rightarrow R_1 - \frac{3}{2}R_3} \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] = [R : b'],$$

which is the reduced row echelon form of  $[A : b]$ . Hence  $\text{rank}[A : b] = \text{rank}(A) = 3$ , the number of unknowns, so the system has a unique solution given by  $Rx = b'$ :

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \Rightarrow \quad x_1 = 1, \quad x_2 = 2, \quad x_3 = 3.$$

The solution is  $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ , agreeing with the answer found earlier by back substitution.

(b)  $x_1 - x_2 + 2x_3 = 4; \quad 2x_1 + 3x_2 - x_3 = 1; \quad 3x_1 + 2x_2 + x_3 = 5.$

**Solution:** From problem 4(b), the row echelon form of the augmented matrix is

$$[A : b] \sim \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & -\frac{7}{5} \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Clearing above the second pivot,

$$\xrightarrow{R_1 \rightarrow R_1 + R_2} \left[ \begin{array}{ccc|c} 1 & 0 & 1 & \frac{13}{5} \\ 0 & 1 & -1 & -\frac{7}{5} \\ 0 & 0 & 0 & 0 \end{array} \right] = [R : b'].$$

Hence  $\text{rank}[A : b] = \text{rank}(A) = 2 < 3$ , the number of unknowns, so the system has infinitely many solutions. Take  $x_3$  as the free variable, since  $n - r = 3 - 2 = 1$ . The system  $Rx = b'$  is

$$x_1 + x_3 = \frac{13}{5}, \quad (4.8.10)$$

$$x_2 - x_3 = -\frac{7}{5}. \quad (4.8.11)$$

From (4.8.10),  $x_1 = \frac{13}{5} - x_3$ , and from (4.8.11),  $x_2 = -\frac{7}{5} + x_3$ . The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{13}{5} \\ -\frac{7}{5} \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R},$$

which again agrees with the answer found earlier.

#### Note:

In the reduced form each equation gives one unknown directly in terms of the free variables, with no back substitution needed. That is the whole advantage of Gauss–Jordan over Gauss elimination, paid for by the extra row operations.

## 4.9 Linear Dependence and Independence

**Definition 4.9.1: Linear independence**

The vectors  $v_1, v_2, v_3, \dots, v_n$  in  $\mathbb{R}^m$  are said to be *linearly independent* if no vector  $v_i$ ,  $i \in \{1, 2, \dots, n\}$ , can be written as a linear combination of the others.

**Definition 4.9.2: Linear dependence**

The vectors  $v_1, v_2, \dots, v_n$  in  $\mathbb{R}^m$  are said to be *linearly dependent* if one of them can be written as a linear combination of the others.

Both notions are easier to use in the following equivalent form, which replaces “can be written in terms of the others” by a single equation.

- (i) The vectors  $v_1, v_2, \dots, v_n$  are linearly **dependent** if

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$$

has a *non-trivial* solution, that is, a choice of scalars  $\alpha_1, \alpha_2, \dots, \alpha_n$  that are not all zero.

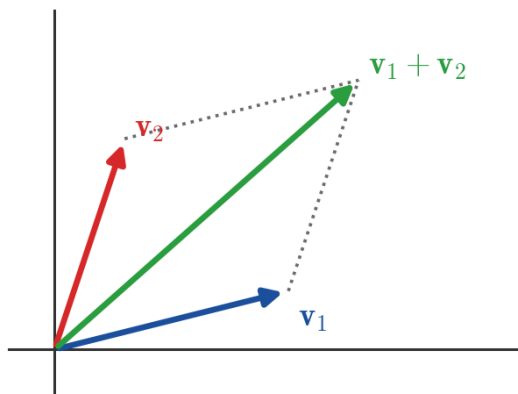
- (ii) The vectors are linearly **independent** if

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0 \quad \text{implies} \quad \alpha_1 = \alpha_2 = \dots = \alpha_n = 0.$$

**Note:**

The equation  $\sum \alpha_i v_i = 0$  always has the solution  $\alpha_1 = \dots = \alpha_n = 0$ , whatever the vectors are; that is the *trivial* solution and it carries no information. The whole question is whether there is *another* one. This is why the two definitions above are equivalent to the original ones: if some  $\alpha_k \neq 0$ , the equation can be divided by  $\alpha_k$  and solved for  $v_k$  in terms of the rest.

**independent: every vector of the plane  
is a combination**  $c_1 v_1 + c_2 v_2$



**dependent: both lie on one line,**  
 $3v_1 - 2v_2 = 0$

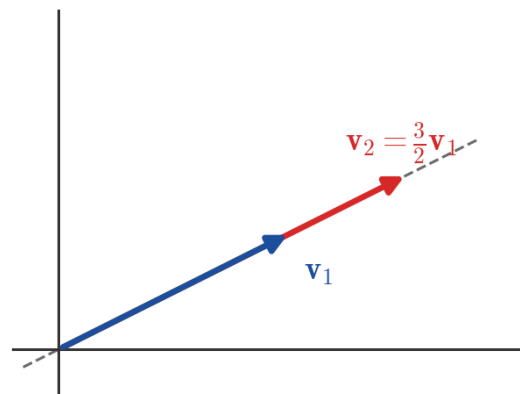


Figure 4.9.1: Two vectors in the plane. Independent vectors point in genuinely different directions and together reach every point of the plane; dependent ones lie along a single line, so one is a multiple of the other.

### 4.9.1 The rank test

Writing the vectors as the rows of a matrix  $A$  turns the question into one about rank, which the row echelon method already answers.

**Note:**

Let  $A$  be the matrix whose rows are the given vectors,  $n$  of them in all. Then

$$\text{rank}(A) = n \implies \text{linearly independent}, \quad \text{rank}(A) < n \implies \text{linearly dependent}.$$

The rank counts how many of the vectors are genuinely contributing; anything less than  $n$  means at least one is redundant.

**Note:**

An immediate consequence is worth having as a shortcut. Since a matrix with  $m$  rows and  $n$  columns has  $\text{rank}(A) \leq \min(m, n)$ , **any set of more than  $n$  vectors in  $\mathbb{R}^n$  is automatically linearly dependent**. Five vectors in  $\mathbb{R}^4$ , for instance, cannot be independent: the matrix has only four columns, so its rank is at most 4, which is less than 5. No computation is needed at all.

**4.9.2 Worked problems**

**Problem 4.9.1.** Determine whether the vectors  $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ,  $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ ,  $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$  are linearly independent in  $\mathbb{R}^3$ .

**Solution:** Consider

$$\alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 0,$$

which written out row by row is the homogeneous system

$$\alpha_1 + \alpha_2 + \alpha_3 = 0, \quad \alpha_1 + \alpha_2 = 0, \quad \alpha_1 = 0.$$

Solving upwards from the last equation:  $\alpha_1 = 0$ ; substituting into the second gives  $\alpha_2 = 0$ ; substituting both into the first gives  $\alpha_3 = 0$ . Only the trivial solution exists, so the vectors are **linearly independent**.

**Note:**

The rank test agrees and is quicker. The matrix formed from the three vectors is already lower triangular, with determinant  $1 \cdot 1 \cdot 1 = -1 \neq 0$  up to sign, so its rank is 3, which equals the number of vectors.

**Problem 4.9.2.** Determine whether the five vectors

$$\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 11 \\ 9 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

in  $\mathbb{R}^4$  are linearly dependent or independent.

**Solution:** Form the matrix  $A$  having these five vectors as its columns:

$$A = \begin{bmatrix} -1 & 2 & 1 & -5 & 1 \\ 0 & 4 & 0 & 3 & 0 \\ 1 & 5 & 0 & 11 & 0 \\ 0 & 1 & 1 & 9 & 0 \end{bmatrix},$$

a matrix of order  $4 \times 5$ . Since  $A$  has only four rows,

$$\text{rank}(A) \leq 4 < 5 = \text{number of vectors}.$$

The rank is strictly less than the number of vectors, so they are **linearly dependent**.

**Note:**

No row reduction was needed. The conclusion follows from the shape of the matrix alone and would hold whatever the entries were — this is the shortcut noted above, that in  $\mathbb{R}^4$  at most four vectors can be independent. Reducing  $A$  does confirm  $\text{rank}(A) = 4$ , so in fact exactly one of the five is redundant, but that extra detail is not needed to answer the question asked.

**Problem 4.9.3.** Check whether  $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ ,  $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ ,  $\begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix}$  are linearly dependent or independent.

**Solution:** Take the three vectors as the columns of

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$$

and reduce to row echelon form. Applying  $R_2 \rightarrow R_2 - 2R_1$  and  $R_3 \rightarrow R_3 + R_1$ ,

$$A \sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -9 \\ 0 & 2 & 5 \end{bmatrix}.$$

Applying  $R_2 \rightarrow -\frac{1}{3}R_2$ ,

$$\sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 2 & 5 \end{bmatrix}.$$

Applying  $R_3 \rightarrow R_3 - 2R_2$ ,

$$\sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & -1 \end{bmatrix},$$

which is in row echelon form. All three rows are non-zero, so

$$\text{rank}(A) = 3 = \text{number of vectors},$$

and the vectors are **linearly independent**.

**Note:**

For a square matrix the determinant gives the same answer in one step:  $|A| = 3 \neq 0$ , so  $A$  is non-singular and its rank is full. Row reduction is the method to use when the matrix is not square, or when the row echelon form is wanted for some other purpose as well.

## 4.10 Eigenvalues and Eigenvectors

**Definition 4.10.1: Eigenvalue and eigenvector**

Let  $A$  be an  $n \times n$  matrix. A real number  $\lambda$  is said to be an *eigenvalue* of the matrix  $A$  if there exists a non-zero vector  $x$  in  $\mathbb{R}^n$  such that

$$Ax = \lambda x.$$

Here  $x$  is called an *eigenvector* of  $A$  corresponding to the eigenvalue  $\lambda$ .

**Example 4.10.2: An eigenvector**

$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$  is an eigenvector of  $\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$  corresponding to  $\lambda = 2$ , since

$$\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

**Note:**

For an eigenvector, the effect of the matrix is the same as multiplying by a scalar: the vector is stretched or shrunk but its direction is unchanged. A general vector is turned as well.

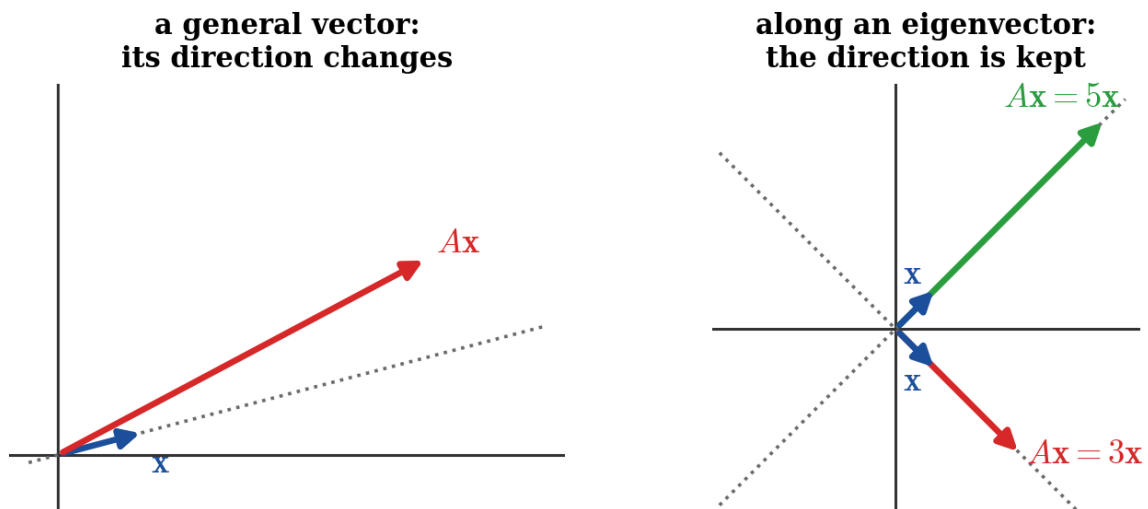


Figure 4.10.1: The action of  $A = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$ . On the left the image leaves the dotted line through  $x$ ; on the right, along either eigen-direction, it stays on it.

**4.10.1 Computing eigenvalues and eigenvectors**

Given  $Ax = \lambda x$ , we have

$$Ax - \lambda Ix = 0 \implies (A - \lambda I)x = 0,$$

where  $I$  is the  $n \times n$  identity matrix. This represents a system of homogeneous equations. Since  $x \neq 0$ , this system must have a non-zero solution, so  $A - \lambda I$  must have rank less than  $n$ , that is

$$\det(A - \lambda I) = 0 \quad \text{or} \quad |A - \lambda I| = 0,$$

which represents a polynomial of degree  $n$  in  $\lambda$  on the left-hand side. The roots of this equation give the eigenvalues. The corresponding eigenvectors are obtained by solving  $(A - \lambda I)x = 0$ .

**Note:**

The equation  $|A - \lambda I| = 0$  is known as the *characteristic equation* of  $A$ , and its roots are the eigenvalues, also called the *characteristic roots*.

### 4.10.2 Diagonalisation

**Definition 4.10.3: Diagonalisable matrix**

A matrix  $A$  is said to be *diagonalisable* if there exists a diagonal matrix  $D$  and an invertible matrix  $P$  such that

$$P^{-1}AP = D.$$

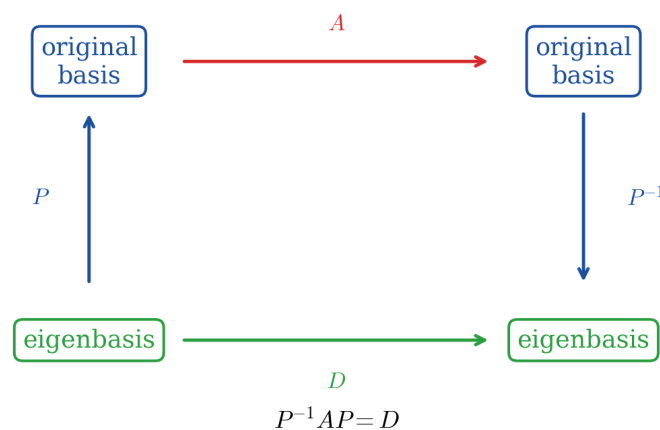


Figure 4.10.2: Diagonalisation as a change of basis. In the eigenbasis the map is the diagonal matrix  $D$ ;  $P$  and  $P^{-1}$  translate between the two bases.

**Theorem 4.10.4: A test for diagonalisability**

If an  $n \times n$  matrix  $A$  has  $n$  linearly independent eigenvectors  $v_1, \dots, v_n$ , with eigenvalues  $\lambda_1, \dots, \lambda_n$ , then

$$P = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \quad \text{and} \quad D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

satisfy  $AP = PD$ , and so  $P^{-1}AP = D$ . This always happens when the  $n$  eigenvalues are distinct.

**Note:**

Two checks come free with every eigenvalue calculation: the eigenvalues add up to trace  $A$  and multiply to  $\det A$ . The worked problems below use them to confirm their answers.

### 4.10.3 Worked problems

**Problem 4.10.1.** Determine the eigenvalues and eigenvectors of the given matrices.

(a)  $A = \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$

**Solution:** To find the eigenvalues, consider  $|A - \lambda I| = 0$ . Now

$$A - \lambda I = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 - \lambda & 3 \\ 4 & 5 - \lambda \end{bmatrix},$$

so the characteristic equation is

$$\begin{vmatrix} 1 - \lambda & 3 \\ 4 & 5 - \lambda \end{vmatrix} = 0 \implies (1 - \lambda)(5 - \lambda) - 12 = 0.$$

Expanding,

$$\lambda^2 - 6\lambda - 7 = 0 \implies \lambda^2 - 7\lambda + \lambda - 7 = 0 \implies (\lambda - 7)(\lambda + 1) = 0,$$

giving  $\lambda = 7$  and  $\lambda = -1$ . Written in decreasing order, the eigenvalues of  $A$  are  $\lambda_{1,2} = 7, -1$ .

**Note:**

For a  $2 \times 2$  matrix the characteristic equation is always  $\lambda^2 - (\text{trace } A)\lambda + \det A = 0$ . Here  $\text{trace } A = 1 + 5 = 6$  and  $\det A = 5 - 12 = -7$ , which reproduces  $\lambda^2 - 6\lambda - 7 = 0$  and checks the expansion.

**Eigenvector for  $\lambda = 7$ .** Solve  $(A - 7I)x = 0$ . Now

$$A - 7I = \begin{bmatrix} -6 & 3 \\ 4 & -2 \end{bmatrix} \xrightarrow{R_1 \rightarrow -\frac{1}{6}R_1} \begin{bmatrix} 1 & -\frac{1}{2} \\ 4 & -2 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 4R_1} \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 0 \end{bmatrix} = R.$$

The corresponding system  $Rx = 0$  is

$$x_1 - \frac{x_2}{2} = 0 \implies x_1 = \frac{x_2}{2},$$

with  $x_2$  as the free variable, so the solution is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2/2 \\ x_2 \end{bmatrix}, \quad x_2 \in \mathbb{R}.$$

Taking  $x_2 = 2$  gives

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

the eigenvector corresponding to  $\lambda = 7$ .

**Eigenvector for  $\lambda = -1$ .** Solve  $(A - \lambda I)x = 0$ , that is  $(A + I)x = 0$ . Now

$$A + I = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \begin{bmatrix} 1 & \frac{3}{2} \\ 4 & 6 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 4R_1} \begin{bmatrix} 1 & \frac{3}{2} \\ 0 & 0 \end{bmatrix}.$$

The corresponding system is

$$x_1 + \frac{3}{2}x_2 = 0 \implies x_1 = -\frac{3}{2}x_2,$$

with  $x_2$  as the free variable, so

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2}x_2 \\ x_2 \end{bmatrix}, \quad x_2 \in \mathbb{R}.$$

Taking  $x_2 = 2$ ,

$$v_2 = \begin{bmatrix} -\frac{3}{2}(2) \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix},$$

the eigenvector corresponding to  $\lambda = -1$ .

**Diagonalisation.** Let  $P = [v_1 \ v_2] = \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix}$  and let  $D$  be the diagonal matrix of eigenvalues. Then

$$AP = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 14 & -2 \end{bmatrix},$$

$$PD = \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 7 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 14 & -2 \end{bmatrix}.$$

Hence  $AP = PD$ . This is called the *diagonalisation* of  $A$ , where  $D$  is the diagonal matrix with the eigenvalues as its elements.

(b)  $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

**Solution:** Consider

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 1 \\ 1 & 4 - \lambda \end{bmatrix},$$

so the characteristic equation  $|A - \lambda I| = 0$  reads

$$(4 - \lambda)^2 - 1 = 0 \implies 16 - 8\lambda + \lambda^2 - 1 = 0 \implies \lambda^2 - 8\lambda + 15 = 0,$$

that is  $(\lambda - 5)(\lambda - 3) = 0$ , giving  $\lambda_{1,2} = 5, 3$ .

**Eigenvector for  $\lambda_1 = 5$ .** Solve  $(A - 5I)x = 0$ :

$$A - 5I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + R_1} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = R.$$

The corresponding system  $Rx = 0$  is  $x_1 - x_2 = 0$ , so  $x_1 = x_2$  with  $x_2$  free, and the solution is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \end{bmatrix}, \quad x_2 \in \mathbb{R} \setminus \{0\}.$$

Taking  $x_2 = 1$ ,

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is the eigenvector corresponding to  $\lambda_1 = 5$ .

**Eigenvector for  $\lambda_2 = 3$ .** Solve  $(A - 3I)x = 0$ :

$$A - 3I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = R.$$

The corresponding system is  $x_1 + x_2 = 0$ , so  $x_2 = -x_1$  with  $x_1$  free, and the solution is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix}, \quad x_1 \in \mathbb{R} \setminus \{0\}.$$

Taking  $x_1 = 1$ ,

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

is the eigenvector corresponding to  $\lambda_2 = 3$ .

**Note:**

The zero vector is excluded because the definition requires an eigenvector to be non-zero. Any non-zero multiple of  $v_1$  or  $v_2$  serves equally well; the convenient one is chosen.

The eigenvectors are  $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  and  $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ . Here

$$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad |P| = -1 - 1 = -2.$$

Since  $|P| \neq 0$  the matrix  $P$  is invertible, and

$$P^{-1} = \frac{1}{|P|} \text{adj } P = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

Now

$$AP = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 5 & -3 \end{bmatrix},$$

and therefore

$$P^{-1}AP = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 5 & 3 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} = D.$$

Since  $P^{-1}AP = D$ , the matrix  $A$  is diagonalisable.

(c)  $A = \begin{pmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{pmatrix}$

**Solution:** Consider

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & -3 & 1 \\ 1 & -2 - \lambda & 1 \\ 1 & -3 & 2 - \lambda \end{bmatrix},$$

so the characteristic equation  $|A - \lambda I| = 0$  is

$$\begin{vmatrix} 2 - \lambda & -3 & 1 \\ 1 & -2 - \lambda & 1 \\ 1 & -3 & 2 - \lambda \end{vmatrix} = 0.$$

Expanding along the first row,

$$(2 - \lambda)[(-2 - \lambda)(2 - \lambda) + 3] + 3[(2 - \lambda) - 1] + 1[-3 - (-2 - \lambda)] = 0,$$

that is

$$(2 - \lambda)(\lambda^2 - 4 + 3) + 3(1 - \lambda) + (\lambda - 1) = 0 \implies (2 - \lambda)(\lambda^2 - 1) + 3 - 3\lambda + \lambda - 1 = 0.$$

Expanding and collecting terms,

$$2\lambda^2 - 2 - \lambda^3 + \lambda - 2\lambda + 2 = 0 \implies -\lambda^3 + 2\lambda^2 - \lambda = 0,$$

and factoring out  $-\lambda$ ,

$$-\lambda(\lambda^2 - 2\lambda + 1) = 0 \implies -\lambda(\lambda - 1)^2 = 0.$$

Hence  $\lambda = 0$  or  $\lambda = 1$  (twice), so the eigenvalues of  $A$  are

$$\lambda_{1,2,3} = 1, 1, 0.$$

**Note:**

As a check, the eigenvalues must sum to the trace,  $2 + (-2) + 2 = 2 = 1 + 1 + 0$ , and their product must equal  $\det A = 0 = 1 \cdot 1 \cdot 0$ . Both hold.

**Eigenvectors for  $\lambda_1 = \lambda_2 = 1$ .** Solve  $(A - I)x = 0$ :

$$A - I = \begin{bmatrix} 1 & -3 & 1 \\ 1 & -3 & 1 \\ 1 & -3 & 1 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 1 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = R.$$

The corresponding system  $Rx = 0$  is

$$x_1 - 3x_2 + x_3 = 0. \quad (4.10.1)$$

Here  $\text{rank}(A - I) = 1$ , so the number of free variables is  $n - r = 3 - 1 = 2$ ; take  $x_2$  and  $x_3$  as the free variables. From (4.10.1),  $x_1 = 3x_2 - x_3$ , and the solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad x_2, x_3 \in \mathbb{R}.$$

Taking  $(x_2, x_3) = (1, 0)$  and  $(0, 1)$  in turn gives the two independent eigenvectors

$$v_1 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

**Note:**

The eigenvalue 1 is repeated, and here it produces two independent eigenvectors as many as its multiplicity. That is what will make  $A$  diagonalisable; a repeated eigenvalue does not always manage it.

**Eigenvector for  $\lambda_3 = 0$ .** Solve  $(A - 0I)x = 0$ , that is  $Ax = 0$ :

$$A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 1 \\ 2 & -3 & 1 \\ 1 & -3 & 2 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix},$$

and then

$$\xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} = R.$$

Here  $\text{rank}(A) = 2 < 3$ , the number of unknowns, so there is  $n - r = 3 - 2 = 1$  free variable; take  $x_3$ . The corresponding system  $Rx = 0$  is

$$x_1 - 2x_2 + x_3 = 0, \quad (4.10.2)$$

$$x_2 - x_3 = 0. \quad (4.10.3)$$

From (4.10.3),  $x_2 = x_3$ , and substituting into (4.10.2) gives  $x_1 = 2x_3 - x_3 = x_3$ .

Taking  $x_3$  as the free variable, the solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R} \setminus \{0\},$$

so taking  $x_3 = 1$ ,

$$v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

is the eigenvector corresponding to  $\lambda = 0$ . The three eigenvectors are

$$v_1 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

### Diagonalisation.

Here

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad P = \begin{bmatrix} 3 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad |P| = -3 + 1 + 1 = -1.$$

Since  $|P| \neq 0$ , the matrix  $P$  is invertible, and

$$P^{-1} = \frac{1}{|P|} \text{adj } P = (-1) \begin{bmatrix} -1 & 2 & -1 \\ -1 & 3 & -2 \\ 1 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 1 \\ 1 & -3 & 2 \\ -1 & 3 & -1 \end{bmatrix}.$$

Therefore

$$P^{-1}AP = \begin{bmatrix} 1 & -2 & 1 \\ 1 & -3 & 2 \\ -1 & 3 & -1 \end{bmatrix} \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = D,$$

so  $A$  is diagonalisable.

**Problem 4.10.2.** Determine the eigenvalues and eigenvectors of  $A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$  and diagonalise it.

**Solution:** Consider  $|A - \lambda I| = 0$ :

$$\begin{vmatrix} 1 - \lambda & 1 & -2 \\ -1 & 2 - \lambda & 1 \\ 0 & 1 & -1 - \lambda \end{vmatrix} = 0.$$

Expanding along the first row,

$$(1 - \lambda)[(2 - \lambda)(-1 - \lambda) - 1] - 1[(-1)(-1 - \lambda) - 0] - 2[-1 - 0] = 0.$$

The first bracket is  $\lambda^2 - \lambda - 3$ , so

$$(1 - \lambda)(\lambda^2 - \lambda - 3) - (1 + \lambda) + 2 = 0,$$

and expanding,

$$\lambda^2 - \lambda - 3 - \lambda^3 + \lambda^2 + 3\lambda + 1 - \lambda = 0 \implies -\lambda^3 + 2\lambda^2 + \lambda - 2 = 0.$$

Multiplying through by  $-1$  and factoring,

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0 \implies (\lambda - 1)(\lambda - 2)(\lambda + 1) = 0,$$

so the eigenvalues of  $A$  are

$$\lambda_{1,2,3} = 2, 1, -1.$$

**Note:**

Check: the eigenvalues sum to  $2 + 1 - 1 = 2$ , which is  $\text{trace } A = 1 + 2 - 1 = 2$ , and their product is  $2 \cdot 1 \cdot (-1) = -2 = \det A$ . Both hold.

**Eigenvector for  $\lambda = 2$ .** Solve  $(A - 2I)x = 0$ :

$$A - 2I = \begin{bmatrix} -1 & 1 & -2 \\ -1 & 0 & 1 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow{R_1 \rightarrow (-1)R_1} \begin{bmatrix} 1 & -1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + R_1} \begin{bmatrix} 1 & -1 & 2 \\ 0 & -1 & 3 \\ 0 & 1 & -3 \end{bmatrix},$$

and then

$$\xrightarrow[R_3 \rightarrow R_3 + R_2]{R_2 \rightarrow (-1)R_2} \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix} = R.$$

Here  $\text{rank}(A - 2I) = 2 < 3$ , the number of unknowns, so we choose  $n - r = 3 - 2 = 1$  variable as free, namely  $x_3$ . The system  $Rx = 0$  is

$$x_1 - x_2 + 2x_3 = 0, \tag{4.10.4}$$

$$x_2 - 3x_3 = 0. \tag{4.10.5}$$

From (4.10.5),  $x_2 = 3x_3$ , and substituting into (4.10.4),  $x_1 - 3x_3 + 2x_3 = 0$ , so  $x_1 = x_3$ . The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ 3x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R} \setminus \{0\},$$

so

$$v_1 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

is the required eigenvector.

**Eigenvector for  $\lambda = 1$ .** Solve  $(A - I)x = 0$ :

$$A - I = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_2]{R_1 \rightarrow (-1)R_1} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} = R.$$

Again rank = 2 < 3, so take  $x_3$  as the free variable. The system  $Rx = 0$  is

$$x_1 - x_2 - x_3 = 0, \quad (4.10.6)$$

$$x_2 - 2x_3 = 0. \quad (4.10.7)$$

From (4.10.7),  $x_2 = 2x_3$ , and then (4.10.6) gives  $x_1 = 2x_3 + x_3 = 3x_3$ . The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3x_3 \\ 2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R} \setminus \{0\},$$

so

$$v_2 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

is the eigenvector corresponding to  $\lambda = 1$ .

**Eigenvector for  $\lambda = -1$ .** Solve  $(A + I)x = 0$ :

$$A + I = \begin{bmatrix} 2 & 1 & -2 \\ -1 & 3 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{bmatrix} 1 & 4 & -1 \\ -1 & 3 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + R_1} \begin{bmatrix} 1 & 4 & -1 \\ 0 & 7 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

and then

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 4 & -1 \\ 0 & 1 & 0 \\ 0 & 7 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 7R_2} \begin{bmatrix} 1 & 4 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = R.$$

Here rank = 2 < 3, so we choose  $n - r = 1$  free variable. The system  $Rx = 0$  is

$$x_1 + 4x_2 - x_3 = 0, \quad (4.10.8)$$

$$x_2 = 0. \quad (4.10.9)$$

Taking  $x_1$  as the free variable, (4.10.9) gives  $x_2 = 0$ , and (4.10.8) then gives  $x_3 = x_1$ . The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad x_1 \in \mathbb{R} \setminus \{0\},$$

so

$$v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

is the eigenvector corresponding to  $\lambda = -1$ .

**Verification.**

Here

$$P = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad |P| = 2 - 9 + 1 = -6.$$

Since  $|P| \neq 0$ , the matrix  $P$  is invertible, and

$$P^{-1} = \frac{1}{|P|} \text{adj } P = \frac{1}{-6} \begin{bmatrix} 2 & -2 & -2 \\ -3 & 0 & 3 \\ 1 & 2 & -7 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{6} & -\frac{1}{3} & \frac{7}{6} \end{bmatrix}.$$

Then

$$P^{-1}AP = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{6} & -\frac{1}{3} & \frac{7}{6} \end{bmatrix} \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 3 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = D,$$

so  $A$  is diagonalisable.

**Problem 4.10.3.** Determine the eigenvalues and eigenvectors of  $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$  and diagonalise it.

**Solution:** Consider  $|A - \lambda I| = 0$ :

$$\begin{vmatrix} 8 - \lambda & -6 & 2 \\ -6 & 7 - \lambda & -4 \\ 2 & -4 & 3 - \lambda \end{vmatrix} = 0.$$

Expanding along the first row,

$$(8 - \lambda)[(7 - \lambda)(3 - \lambda) - 16] + 6[-6(3 - \lambda) + 8] + 2[24 - 2(7 - \lambda)] = 0,$$

that is

$$(8 - \lambda)(\lambda^2 - 10\lambda + 5) + 6(6\lambda - 10) + 2(10 + 2\lambda) = 0.$$

Expanding and collecting terms,

$$8\lambda^2 - 80\lambda + 40 - \lambda^3 + 10\lambda^2 - 5\lambda + 36\lambda - 60 + 20 + 4\lambda = 0 \implies -\lambda^3 + 18\lambda^2 - 45\lambda = 0.$$

Factoring out  $-\lambda$ ,

$$\lambda(\lambda^2 - 18\lambda + 45) = 0 \implies \lambda(\lambda - 15)(\lambda - 3) = 0,$$

so the eigenvalues of  $A$  are

$$\lambda_{1,2,3} = 15, 3, 0.$$

**Note:**

Check: the eigenvalues sum to  $15 + 3 + 0 = 18 = \text{trace } A = 8 + 7 + 3$ , and their product is  $0 = \det A$ . Note also that  $A$  is *symmetric* and its three eigenvalues are distinct, so it is certainly diagonalisable – a useful thing to know before starting.

**Eigenvector for  $\lambda = 15$ .** Solve  $(A - 15I)x = 0$ :

$$A - 15I = \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \xrightarrow{R_1 \rightarrow -\frac{1}{7}R_1} \begin{bmatrix} 1 & \frac{6}{7} & -\frac{2}{7} \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix},$$

then

$$\xrightarrow[\substack{R_2 \rightarrow R_2 + 6R_1 \\ R_3 \rightarrow R_3 - 2R_1}]{\substack{R_2 \rightarrow -\frac{7}{20}R_2 \\ R_3 \rightarrow R_3 - 2R_2}} \begin{bmatrix} 1 & \frac{6}{7} & -\frac{2}{7} \\ 0 & -\frac{20}{7} & -\frac{40}{7} \\ 0 & -\frac{40}{7} & -\frac{80}{7} \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow -\frac{7}{20}R_2 \\ R_3 \rightarrow R_3 - 2R_2}} \begin{bmatrix} 1 & \frac{6}{7} & -\frac{2}{7} \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} = R.$$

Here rank = 2, so there is  $n - r = 3 - 2 = 1$  free variable; take  $x_3$ . The system  $Rx = 0$  is

$$x_1 + \frac{6}{7}x_2 - \frac{2}{7}x_3 = 0, \quad (4.10.10)$$

$$x_2 + 2x_3 = 0. \quad (4.10.11)$$

From (4.10.11),  $x_2 = -2x_3$ , and substituting into (4.10.10),

$$x_1 = \frac{2}{7}x_3 + \frac{12}{7}x_3 = 2x_3.$$

The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \implies v_1 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}.$$

**Eigenvector for  $\lambda = 3$ .** Solve  $(A - 3I)x = 0$ :

$$A - 3I = \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{5}R_1} \begin{bmatrix} 1 & -\frac{6}{5} & \frac{2}{5} \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix},$$

then

$$\xrightarrow[\substack{R_2 \rightarrow R_2 + 6R_1 \\ R_3 \rightarrow R_3 - 2R_1}]{\substack{R_2 \rightarrow -\frac{5}{16}R_2 \\ R_3 \rightarrow R_3 + \frac{8}{5}R_2}} \begin{bmatrix} 1 & -\frac{6}{5} & \frac{2}{5} \\ 0 & -\frac{16}{5} & -\frac{8}{5} \\ 0 & -\frac{8}{5} & -\frac{4}{5} \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow -\frac{5}{16}R_2 \\ R_3 \rightarrow R_3 + \frac{8}{5}R_2}} \begin{bmatrix} 1 & -\frac{6}{5} & \frac{2}{5} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} = R.$$

Taking  $x_3$  as the free variable, the system  $Rx = 0$  is

$$x_1 - \frac{6}{5}x_2 + \frac{2}{5}x_3 = 0, \quad (4.10.12)$$

$$x_2 + \frac{1}{2}x_3 = 0. \quad (4.10.13)$$

From (4.10.13),  $x_2 = -\frac{1}{2}x_3$ , and substituting into (4.10.12),

$$x_1 = \frac{6}{5} \left( -\frac{1}{2}x_3 \right) - \frac{2}{5}x_3 = -\frac{3}{5}x_3 - \frac{2}{5}x_3 = -x_3.$$

The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ -\frac{1}{2} \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R} \setminus \{0\},$$

and taking  $x_3 = 2$  to clear the fraction,

$$v_2 = \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}.$$

Checking directly,  $Av_2 = (-6, -3, 6)^T = 3v_2$ , as required. **Eigenvector for  $\lambda = 0$ .** Solve  $(A - 0I)x = 0$ , that is  $Ax = 0$ :

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{8}R_1} \begin{bmatrix} 1 & -\frac{3}{4} & \frac{1}{4} \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix},$$

then

$$\xrightarrow[\substack{R_2 \rightarrow R_2 + 6R_1 \\ R_3 \rightarrow R_3 - 2R_1}]{\substack{R_2 \rightarrow \frac{2}{5}R_2 \\ R_3 \rightarrow R_3 + R_2}} \begin{bmatrix} 1 & -\frac{3}{4} & \frac{1}{4} \\ 0 & \frac{5}{2} & -\frac{5}{2} \\ 0 & -\frac{5}{2} & \frac{5}{2} \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow \frac{2}{5}R_2 \\ R_3 \rightarrow R_3 + R_2}} \begin{bmatrix} 1 & -\frac{3}{4} & \frac{1}{4} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} = R.$$

Taking  $x_3$  as the free variable, the system  $Rx = 0$  gives  $x_2 = x_3$  from the second row, and then

$$x_1 = \frac{3}{4}x_2 - \frac{1}{4}x_3 = \frac{3}{4}x_3 - \frac{1}{4}x_3 = \frac{x_3}{2}.$$

The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{1}{2} \\ 1 \\ 1 \end{bmatrix}, \quad x_3 \in \mathbb{R} \setminus \{0\},$$

and taking  $x_3 = 2$ ,

$$v_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

### Diagonalisation.

Here

$$P = \begin{bmatrix} 2 & -2 & 1 \\ -2 & -1 & 2 \\ 1 & 2 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} 15 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad |P| = -27 \neq 0,$$

so  $P$  is invertible and  $P^{-1}AP = D$ . Hence  $A$  is diagonalisable.

#### Note:

The columns of  $P$  are mutually orthogonal here, since  $A$  is symmetric and the three eigenvalues are distinct. Dividing each column by its length 3 turns  $P$  into an orthogonal matrix  $Q$ , for which  $Q^{-1} = Q^T$  and the inverse costs nothing to compute.

## 4.11 Unit 4 Summary

### Matrices and determinants

- $AB$  needs (columns of  $A$ ) = (rows of  $B$ ); in general  $AB \neq BA$ ;  $(AB)^T = B^T A^T$ .
- Symmetric  $A = A^T$ ; skew-symmetric  $A = -A^T$  (zero diagonal).

- Row swap changes the sign of  $|A|$ ; scaling a row scales  $|A|$ ; adding a multiple of a row leaves  $|A|$  unchanged;  $|AB| = |A||B|$ .
- $A^{-1} = \frac{\text{adj } A}{|A|}$  exists iff  $|A| \neq 0$ .

### Rank and echelon forms

- Rank: largest order of a non-zero minor = number of non-zero rows in any echelon form. Row operations do not change it.
- REF: clear below each pivot. RREF: also clear above, pivots equal to 1.

### Linear systems ( $n$ unknowns)

- $\text{rank } A < \text{rank}[A : b]$ : inconsistent. Equal ranks  $r$ : consistent; unique solution if  $r = n$ , otherwise  $n - r$  free variables.
- $Ax = 0$  is always consistent; non-trivial solutions exist iff  $\text{rank } A < n$  (for square  $A$ : iff  $|A| = 0$ ).
- Gauss elimination: REF of  $[A : b]$  then back substitution. Gauss–Jordan: RREF, then read off.
- Vectors are independent iff the matrix they form has rank equal to their number; more than  $n$  vectors in  $\mathbb{R}^n$  are always dependent.

### Eigenvalues and diagonalisation

- Characteristic equation  $|A - \lambda I| = 0$ ; eigenvectors: non-zero solutions of  $(A - \lambda I)x = 0$ .
- $\sum \lambda_i = \text{trace } A$  and  $\prod \lambda_i = \det A$ .
- $n$  independent eigenvectors (e.g.  $n$  distinct eigenvalues)  $\Rightarrow P^{-1}AP = D$  with  $P = [v_1 \cdots v_n]$ .

## 4.12 Practice Problems

The problems are arranged in three levels. Level A checks definitions and understanding, level B practises the standard methods, and level C asks you to analyse, justify or interpret a result, including in applied settings. Answers are given in brackets; full solutions are not.

### A. Check your understanding

**P1.**  $A$  is  $2 \times 3$  and  $B$  is  $3 \times 2$ . Which of  $A + B$ ,  $AB$  and  $BA$  are defined, and what are their orders? [Ans.  $A + B$  undefined;  $AB$  is  $2 \times 2$ ;  $BA$  is  $3 \times 3$ ]

**P2.** Is  $\begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{bmatrix}$  skew-symmetric? [Ans. yes]

**P3.** True or false, with a reason: (a) a  $3 \times 5$  matrix can have rank 4; (b) a homogeneous system is always consistent; (c) if  $|A| = 0$  then  $A$  has an inverse. [Ans. (a) F (b) T (c) F]

**P4.** Without calculation, decide whether  $(1, 2)$ ,  $(3, 4)$ ,  $(5, 6)$  are linearly independent in  $\mathbb{R}^2$ .  
 [Ans. dependent: three vectors in  $\mathbb{R}^2$ ]

**P5.** A  $3 \times 3$  matrix has eigenvalues 1, 2, 3. Find its trace and determinant. [Ans. 6 and 6]

## B. Standard problems

**P6.** For  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , find  $AB$  and  $BA$ . [Ans.  $\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ ,  $\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ ]

**P7.** Find  $|A|$  and  $A^{-1}$  for  $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$ . [Ans. 1;  $\begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$ ]

**P8.** Find the rank of  $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$  by minors and by the echelon form. [Ans. 2]

**P9.** Find the reduced row echelon form of  $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$ . [Ans.  $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ ]

**P10.** Solve  $x + y + z = 0$ ,  $2x - y + z = 0$ ,  $x - 2y = 0$ . [Ans.  $(x, y, z) = t(-2, -1, 3)$ ]

**P11.** Solve by Gauss elimination:  $x + 2y + z = 8$ ,  $2x + 3y + 4z = 20$ ,  $4x + 3y + 2z = 16$ .  
 [Ans.  $(1, 2, 3)$ ]

**P12.** For which  $k$  is  $x + y + z = 1$ ,  $x + 2y + 3z = k$ ,  $x + 5y + 9z = k^2$  consistent?  
 [Ans.  $k = 1$  or  $k = 3$ ]

**P13.** Are  $(1, 2, 3)$ ,  $(2, 3, 4)$ ,  $(3, 5, 7)$  linearly independent?  
 [Ans. no: the third is the sum of the first two]

**P14.** Find the eigenvalues and eigenvectors of  $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ . [Ans. 3:  $(1, 1)$ ; 1:  $(1, -1)$ ]

**P15.** Diagonalise  $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ . [Ans.  $P = \begin{bmatrix} 4 & -1 \\ 1 & 1 \end{bmatrix}$ ,  $D = \text{diag}(6, 1)$ ]

## C. Analyse, justify and apply

**P16.** For  $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ , verify  $(AB)^T = B^T A^T$  and show that  $(AB)^T \neq A^T B^T$ .  
 [Ans.  $(AB)^T = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix}$ ,  $A^T B^T = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ ]

**P17.** A student row-reduces  $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$  to echelon form and reads the eigenvalues off its diagonal. Evaluate the method.  
 [Ans. invalid: row operations change eigenvalues; the true ones are 4 and  $-1$ ]

**P18.** Kirchhoff's laws for a circuit give  $I_1 - I_2 - I_3 = 0$ ,  $2I_1 + 3I_2 = 12$  and  $3I_2 - 6I_3 = 0$  (currents in amperes). Check consistency and find the currents.

[**Ans.** unique solution  $I_1 = 3$ ,  $I_2 = 2$ ,  $I_3 = 1$ ]

**P19.** Each year 90% of a city's population stays in the city and 10% moves to the suburbs, while 20% of the suburban population moves to the city. With  $A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$ , find the eigenvector for  $\lambda = 1$  and interpret it.

[**Ans.** (2, 1): in the long run  $\frac{2}{3}$  live in the city]

**P20.** Prove that if  $|A| \neq 0$  then  $Ax = 0$  has only the trivial solution.

[**Ans.**  $x = A^{-1}0 = 0$ ]

**P21.** Construct a  $2 \times 2$  matrix, not diagonal, with eigenvalues 2 and 5.

[**Ans.** e.g.  $\begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix}$ ]

**P22.** Explain why a  $3 \times 3$  matrix with eigenvalues 1, 1, 2 need not be diagonalisable, and give an example.

[**Ans.** e.g.  $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$  has only two independent eigenvectors]

## References

- [1] J. Stewart, D. K. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th edition, Cengage Learning, Boston, 2021.
- [2] S. J. Leon, *Linear Algebra with Applications*, 9th edition, Pearson, New Jersey, 2014.